
Classical Algebraic Geometry

— *EYNTKA* Series —

NATHANAEL CHWOJKO-SRAWLEY

July 18, 2026

Contents

Contents	3
1 Relating Algebra to Geometry	5
1.1 Introduction to Algebraic Sets	7
1.1.1 Affine Sets	8
1.2 Zariski Topology and Duality	20
1.2.1 Zariski Topology	20
1.2.2 Regular Functions and Morphisms	23
1.2.3 Duality: Affine Sets and Coordinate Rings	27
1.2.4 Closed Sets and Radical Ideals	32
1.2.5 Affine Variety and Prime Ideals	35
1.2.6 Computation of Algebraic Sets and k -algebras	39
1.2.7 Hilbert's Nullstellensatz	40
1.3 Rational Maps: Classifying Varieties	46
1.4 Global Dimension	57
1.4.1 Krull dimension and Transcendental Degree	59
1.5 Local Dimension	62
1.5.1 Hilbert Function	62
1.5.2 Application: Dimension of Noetherian Local Ring	69
1.5.3 Smoothness: Regular Local Rings	74
1.5.4 Relating local dimension and transcendental dimension	77
1.5.5 Nonsingular Curves and Surfaces	79

1.6	Projective Variety	81
1.6.1	Definitions and Properties	84
1.6.2	Properties and Quasi-Projective Varieties	95
1.6.3	Regular Maps	104
1.6.4	Products	108
1.6.5	Finite Maps and Normalization	114
1.7	Application: Curves	119
1.7.1	Intersection Number	126
1.7.2	Bézout Theorem	132
1.7.3	Divisors and Groups on Cubic Curves	135
1.8	Blowing Up	140
1.9	*Prelude to Algebraic Geometry	143

Relating Algebra to Geometry

For over two millennia, a central domain of study in mathematics has been geometry, which studies shapes, spaces, and their geometric properties. Let us use the term “geometrical object” as a catch-all for what would usually be considered a shape or a space. Examples of geometrical objects studied throughout history include:

- Circles, more generally ellipsoids and cones
- Spheres and tori
- Curves on a plane
- Möbius bands
- The Platonic solids
- Graphs of functions
- The Cartesian plane

and so forth, the list stretching out endlessly. Historically, much of this study concentrated on the *Euclidean approach* or *ruler and compass approach*, taking most of its inspiration from the continuation of Euclid’s book *Elements* written around 300 BC. From just 5 axioms, Euclid managed to deduce a large portion of classical geometry. This was a *synthetic* approach, building geometric facts purely from logical deductions about primitive objects like points and lines, without any underlying reference frame. However, there are fundamental limitations to the Euclidean method. Its rigid axiomatic framework – specifically the infamous 5th axiom, known as the *parallel postulate* – eliminates from consideration many important geometric arenas that modern mathematicians and physicists work with, such as the curved geometry on the surface of a sphere.

Today, the standard starting point for many geometric fields involves explicitly defining a space, such as the real vector space $\mathbb{R}^n$, and equipping it with continuous or smooth local structures. This modern transition from synthetic to analytic geometry relies on the idea of an origin and a coordinate system, introduced by *René Descartes* in the 17th century to quantify geometric problems. Descartes’ insight created the first formal bridge between the spatial world of geometry and the symbolic world of algebra.

We now recognize that there are many different paradigms in which we can pursue the study of geometry, each focusing on a different area of interest. These paradigms fall under broad, heavily overlapping umbrellas: *Differential Geometry*, *Topology*, *Discrete Geometry*, and *Classical Geometry*¹. Yet, a major consequence of Descartes' initial bridge lies in the study of *Algebraic Geometry*.

Algebraic geometry is the study of shapes defined by *algebraic equations* – mostly polynomials or ratios of polynomials. But why polynomials? Because they are constructed from the most fundamental arithmetic operations (addition and multiplication), making them purely algebraic and straightforward to manipulate; yet, they are remarkably expressive, capable of capturing a wide array of geometric forms. When we say “shapes given by polynomials,” we are referring to the *algebraic sets*, or the *zero-sets* (the vanishing points) of a collection of polynomials. Famously, the polynomial $x^2 + y^2 - 1 \in \mathbb{R}[x, y]$ is geometrically interpreted as the unit circle by considering all points where $x^2 + y^2 - 1 = 0$:

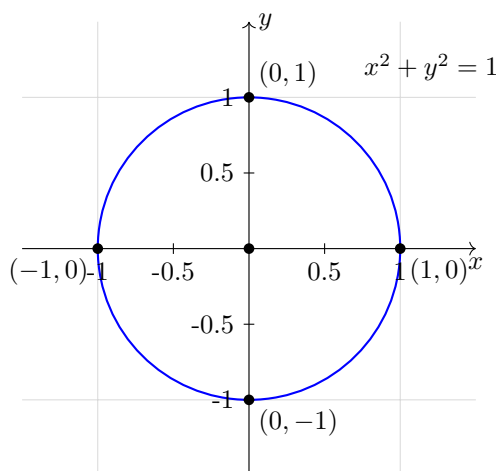


Figure 1.1: Circle represented by the equation $x^2 + y^2 - 1 = 0$

A central theme of classical algebraic geometry is that these shapes can be characterized by the algebraic functions that “live” on them. We will see that there is a symbiotic relationship – a “dictionary” – between polynomials and their geometry.^{2,3} The properties of quotient polynomial rings, ideals, and dimensions offer information about the geometric zero-sets. Conversely, looking at the geometry of these zero-sets guides our algebraic intuition. Through theorems like Hilbert’s *Nullstellensatz* (theorem 1.2.46), we will translate geometric spaces (via the Zariski topology) into algebraic rings, and see how abstract concepts like Krull dimension and regular local rings formalize geometric dimension and smoothness.

Furthermore, classical algebraic geometry provides a framework to resolve the exceptions and

¹which includes Euclidean, Affine, Conformal, and Analytic geometry, among other fields

²While this dictionary is a highly effective tool, it should not be overstated. There are inherently geometric phenomena that transcend pure algebra. For instance, differential geometry routinely utilizes smooth “bump functions,” and topological studies may rely on soft sheaves, which possess analytic or infinite properties that lack direct algebraic analogues.

³Determining the exact limits of this relationship remains an active area of modern research. For example, the *Hodge Conjecture*, one of the Millennium Prize Problems, explores to what extent the topological spaces of certain complex algebraic varieties are determined entirely by their algebraic subvarieties.

anomalies that arise in standard affine geometry, such as the behavior of parallel lines. During the 19th and early 20th centuries, mathematicians – particularly those of the Italian school of algebraic geometry – sought a setting where intersection theory could be studied systematically without the need for special cases. By extending our familiar affine spaces into *Projective Varieties*, we introduce “points at infinity.” In projective geometry, parallel lines finally meet, smoothing out exceptions and allowing for uniform intersection theorems. For instance, Bézout’s Theorem (theorem 1.7.18), which counts the number of intersection points between curves, was later realized to naturally exist in projective space, where the theorem holds universally.

The goal of this text is to systematically build up this dictionary between geometry and algebra so that the reader can convert problems back and forth between the two fields. We will explore how to classify these shapes (rational maps), how to study their intersections, and how to untangle their singularities (blowing up). We conclude by briefly outlining modern algebraic geometry, which takes our classical dictionary of polynomial zero-sets and generalizes it to the abstract framework of schemes and commutative rings.

1.0.1 Quick Side-Note A common misconception, born from a strong reliance on graphs in most first-year undergraduate calculus/analysis classes, is to intuitively associate the zero-set of a function with the *graph* of a function.

For example, $f(x) = x$ has a zero-set containing only the point 0, while the function $F(x, y) = x - y$ has a zero-set that actually forms the graph $\Gamma(f)$. This distinction may seem trivial, but it is surprisingly easy to conflate the two concepts when moving into higher dimensions.

Graphs can be thought of as the collection of the solutions of $f(x_1, \dots, x_n) = y_0$ as the constant term y_0 varies.

1.1 Introduction to Algebraic Sets

In this whole chapter, R is a commutative ring and k be a field.

So far when polynomials were encountered, they were treated as a commutative ring with an indeterminate extension (or equivalently, they can be thought of as a ring over a free monoid). There is a natural way of thinking about any such polynomial as a function (which is probably how most first encounter polynomials): if $f \in k[x_1, x_2, \dots, x_n]$, then if $(a_1, a_2, \dots, a_n) \in k^n$, let $f(a_1, a_2, \dots, a_n) \in k$ be the element which you get when replacing each x_i with a_i . Then we can make f into a function through the following map^{4,5}:

$$k[x_1, \dots, x_n] \rightarrow \text{Hom}_{\mathbf{Set}}(k^n, k) \quad f \mapsto ((a_1, \dots, a_n) \mapsto f(a_1, \dots, a_n)) \quad (1.1)$$

where we shall consider the constants $c \in k[x_1, \dots, x_n]$ to map to the functions that shall return the value of $c \in k$:

$$c \mapsto ((a_1, \dots, a_n) \mapsto c)$$

⁴For those familiar with algebraic geometry, this map can be thought of as selecting the global sections of $\mathbb{A}_k^n$. By the Nullstellensatz, $k^n = \text{Spec}(k[x_1, \dots, x_n])$, and by the representability of affine schemes $\Gamma(\text{Spec}(R)) \cong \text{Hom}_{\mathbf{Reg}}(\text{Spec}(R), \mathbb{A}_k^1)$

⁵advanced comment: if k is not algebraically closed, we are missing points, and hence this technically is *not* a representable functor and is not the sheaf obtained from embedding Var_k into $\mathbf{Sch}_k$

Note that the codomain of the function in equation 1.1 is be the hom-set over the category **Set**, and not any other category we've extensively worked over so far (**Grp**, **Ring**, **Alg**, etc.); this was done since there are many non-linear functions representing non-linear polynomials (a simple example would be $x^2 \in k[x]$). On the other hand, the image of the above function is not just a collection of all set functions as they must all satisfy polynomials relations (can you find a set-function that is not given/representable by a polynomial?⁶). The subset given by the function in equation (1.1) shall be called the set of *regular functions*: $\text{Hom}_{\mathbf{Reg}}(k^n, k)$. Before properly boxing this definition, we shall take some more time pondering what the domain of the hom-set should be, namely we shall want the domain to represent Cartesian space without the extra algebraic structure that is usually implied by the notation k^n . For example, k^n usually denotes the n -dimensional vector-space over k with an origin, and more generally we think of k^n as having addition and scalar multiplication satisfying the axioms of a vector-space. Vector-spaces are "defined" for linear polynomials, with subspaces that are naturally "linear" (recall that subspaces are given by solutions of linear polynomials, see ??)⁷. We shall want a new type of space for which subspaces are solutions to general polynomial equations.

1.1.1 Affine Sets

Definition 1.1.1: Affine Space

Let $(k, +, \cdot)$ be any field. Then the collection of the n -tuples of k^n as a set shall be called *affine space*. It's elements shall be called *points* and the space shall be denoted $\mathbb{A}_k^n$.

Hence, a field can be thought of as $(\mathbb{A}_k^1, +, \cdot)$. The space $\mathbb{A}_k^n$ is called *affine space* rather than just "space" as there shall be an equally important (or arguably more important) space called *projective space* that shall be introduced in section 1.6⁸. The term affine can also be thought of as emphasizing the lack of importance of the origin (recall that a linear transformation that doesn't need to map the origin to itself was called affine).

With affine space established, let us now define the image of the function in equation (1.1):

Definition 1.1.2: Regular Function [on Affine Space]

Let F represent the function in equation (1.1):

$$F : k[x_1, \dots, x_n] \rightarrow \text{Hom}_{\mathbf{Set}}(k^n, k) \quad f \mapsto ((a_1, \dots, a_n) \mapsto f(a_1, \dots, a_n))$$

Then the collection $\text{im}(F)$ is called the *regular functions*, and is usually denoted $\Gamma(\mathbb{A}_k^n, k[\mathbb{A}_k^n])$, or $\text{Hom}_{\mathbf{Reg}}(\mathbb{A}_k^n, \mathbb{A}_k^1)$.

Each of these notations for the collection of regular functions have its use. The $\text{Hom}_{\mathbf{Reg}}(\mathbb{A}_k^n, \mathbb{A}_k^1)$ notion makes it clear what the domain and codomain is, which shall be particularly important once we take different codomain. The $k[\mathbb{A}_k^n]$ notation is particularly useful when we shall take subsets $V \subseteq \mathbb{A}_k^n$ and denote all regular functions on V as $k[V]$; in this case this notation emphasizes that we

⁶Over finite fields (and only over finite fields) all set functions are polynomials, see exercise ref:HERE

⁷In [4], we shall show that the vector-space structure can be recovered using Hopf Algebras, but this adds more structure to the algebraic set

⁸More generally, we shall see that affine space will be the representation space for schemes which generalizes affine space, something we shall touch on briefly in chapter 1.9

are working over the particular field k . This shall be useful for there are circumstances where we change which field we are working over⁹. The $\Gamma(\mathbb{A}_k^n)$ notation comes from modern algebraic geometry where gamma is used hint that these are the *global section* over $\mathbb{A}_k^n$ (see [1] for a full explanation). As this chapter is gearing up the reader for modern algebraic geometry, we shall preferentially use the $\Gamma(\mathbb{A}_k^n)$ notation to acclimate the reader. It should be pointed out that another common notation is $\mathcal{O}_{\mathbb{A}_k^n}(\mathbb{A}_k^n)$ which is more useful when it is important to keep track of regular functions on open subsets of $\mathbb{A}_k^n$ (once we define a topology on $\mathbb{A}_k^n$) in which case we shall have the collection of regular function on $U \subseteq \mathbb{A}_k^n$ be written as $\mathcal{O}_{\mathbb{A}_k^n}(U)$. This is not useful to us in this chapter, and so shall not be used.

The set $\Gamma(\mathbb{A}_k^n)$ forms a natural k -algebra structure given by point-wise operations:

$$f + g \equiv f(a) + g(a) \quad fg \equiv f(a)g(a) \quad c \cdot f = cf(a) \quad \forall a \in k \quad (1.2)$$

With this k -algebra structure, it can be shown that $k[x_1, \dots, x_n] \cong_k \text{Hom}_{\mathbf{Reg}}(\mathbb{A}_k^n, \mathbb{A}_k^1)$, and so the two sets are often used interchangeably. If the distinction needs to be made clear between the element x_i and the function x_i , we shall write

$$k[X_1, \dots, X_n] = \text{Hom}_{\mathbf{Reg}}(\mathbb{A}_k^n, \mathbb{A}_k^1) = k[\mathbb{A}_k^n]$$

where each X_i is a regular function that is the projection onto the i th coordinate, and call an element of $k[X_1, \dots, X_n]$ a *polynomial function* or *polynomial map*. This distinction shall become more important once there can be multiple polynomials that will be associated to the same regular function, which shall be the case in section 1.2.3. When it is clear from context, we may denote a polynomial or a regular function by $f(x)$ and $f(X)$ respectively where x is understood to be the indeterminate variables and X is understood to be the regular functions $X_1, \dots, X_n$.

Due to this isomorphism, we can define a function from $k[x_1, \dots, x_n]$ to k directly by directly substituting the values for the indeterminates with a point in the affine space. Namely, for each $a \in \mathbb{A}_k^n$, we take:

$$\text{ev}_a^k : k[x_1, \dots, x_n] \rightarrow k \quad f(x) \mapsto f(a)$$

where x and a is understood to be n -tuples of indeterminates and values respectively. These are all k -algebra homomorphisms, for

$$\text{ev}_a^k(f(x) + cg(x)) = \text{ev}_a^k(f(x)) + c \cdot \text{ev}_a^k(g(x)) \quad \text{ev}_a^k(f(x)g(x)) = \text{ev}_a^k(f(x))\text{ev}_a^k(g(x))$$

The subscript of the evaluation function represents the point of evaluation, while the superscript represents the values we choose to plug for every indeterminate, where the values are usually the same as the base field, however they may differ, for example it is common to take $\mathbb{Z}$ or $\mathbb{Q}$ ¹⁰. Thus, a regular function can be thought of as the collection of evaluation functions on a polynomial.

⁹A good example of the use of group cohomology is measure how much an isomorphism $k[V] \cong k[W]$ is preserved/fails when consider $\bar{k}[V]$ and $\bar{k}[W]$

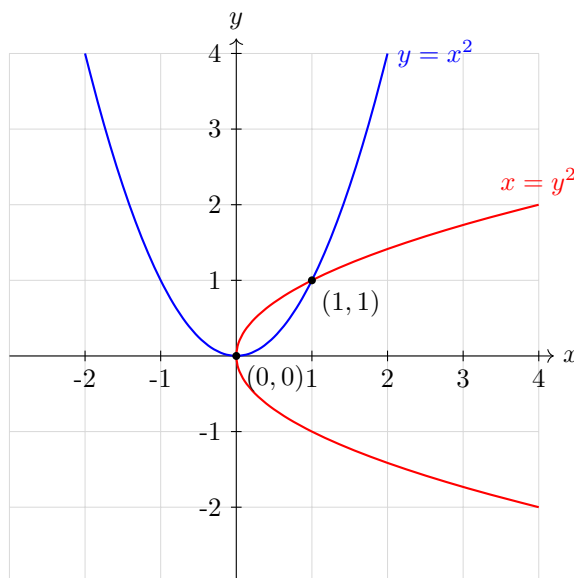
¹⁰These shall be called $\mathbb{Z}$ -valued and $\mathbb{Q}$ -valued points of the polynomial, or more precisely the variety (as we shall soon see)

1.1.3 Foreshadowing If you know elementary analysis or calculus, then you are probably aware that the types of functions we most commonly use are *continuous*. Continuous functions are a very important tool in analysing the properties of the space we're studying (ex. $\mathbb{R}$, $\mathbb{R}^n$, $L^2(\mathbb{R})$, etc.), continuous functions can be thought of as the natural types of functions that preserve some local properties of the space. In Algebra, we can think of polynomial functions as the natural types of functions. This comparison between continuous functions mapping to $\mathbb{R}$ and polynomial functions mapping to k will in fact be incredibly precise, as we will see in the last section of this chapter.

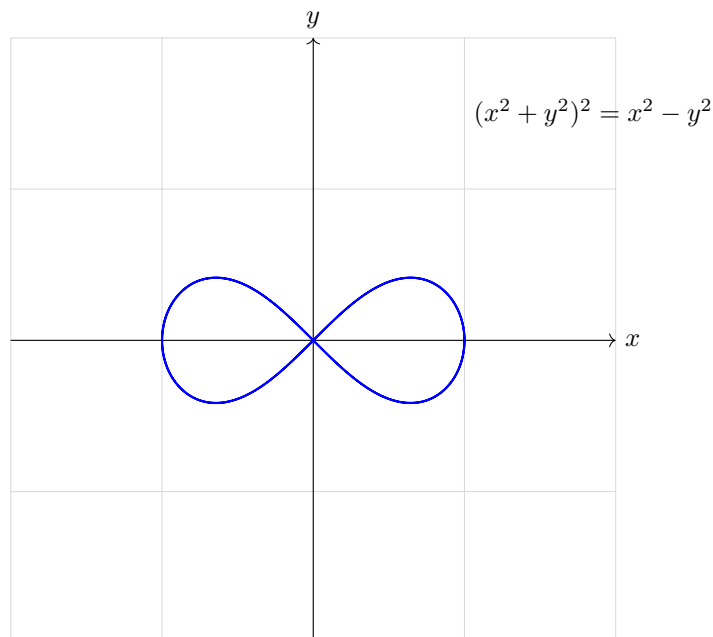
From such polynomial functions, we can create many interesting shapes depending on the zeros', or locus, of polynomial:

Example 1.1.4: Zeros' of Polynomials

1. As we saw, $x^2 + y^2 - 1 \in \mathbb{R}[x, y]$ gives a circle, see figure (1.1). More precisely, the polynomial shall be associated to an element of $\Gamma(\mathbb{A}_k^n)$, where plugin in any point on from $\mathbb{A}_{\mathbb{R}}^2$ that is on the circle (ex. $(1, 0)$ or $(\sqrt{2}/2, \sqrt{2}/2)$ shall give 0.
2. Take $y - x^2 \in \mathbb{R}[x, y]$. Then this shall give the usual parabola $y = x^2$. If we take $x - y^2 \in \mathbb{R}[x, y]$, we shall get a parabola rotated 90 degree's (or $\pi/2$ radians) clockwise. If we take the polynomial $(y - x^2)(x - y^2) \in \mathbb{R}[x, y]$, then the zero set of this new polynomial shall be the zero-set of the two multiplied polynomials:



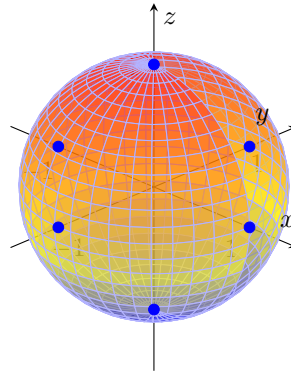
3. Take $(x^2 + y^2)^2 - (x^2 - y^2) \in \mathbb{R}[x, y]$. This shape is called the *Bernoulli Lemniscate*:



Notice that the shape crosses over itself. This shows that zero-sets of polynomials differ from graphs, and also differ from the usual smooth manifolds from differential geometry.

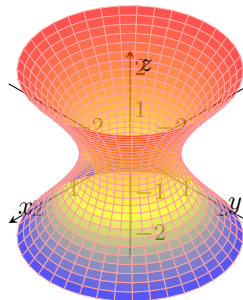
4. Take $p(x) \in k[x]$ over any field k . Then the shape associated to $p(x)$ are the *roots* of the polynomials. For example, taking $x^2 - 2 \in \mathbb{Q}(\sqrt{2})[x]$ shall give $\{\sqrt{2}, -\sqrt{2}\} \subseteq \mathbb{A}_{\mathbb{Q}(\sqrt{2})}^2$ as these are the solution to $x^2 - 2 = 0$. More generally, the shape given by a 1-dimensional polynomial shall always be the collection of roots that is present in the underlying field. This means that $x^2 - 2 \in \mathbb{Q}[x]$ would give the empty set, as there are no rational roots. By the fundamental theorem of algebra, any polynomial $p(x) \in \mathbb{C}[x]$ of degree d shall have d roots, and hence the shape associated to $p(x)$ shall have d points.
5. Take $x^2 + y^2 + z^2 - 1 \in \mathbb{R}[x, y, z]$. Then this is the classical unit sphere centered at the origin:

$$x^2 + y^2 + z^2 = 1$$

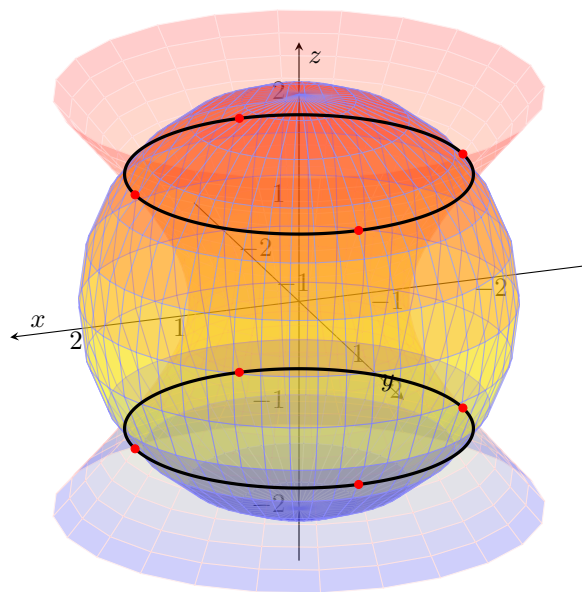


What polynomial would you choose to center the unit sphere at different points of $\mathbb{A}_{\mathbb{R}}^3$? How would you change the radius of the sphere?

6. Let us now consider taking two shapes: the sphere given by $x^2 + y^2 + z^2 = 4$ and the hyperboloid given by $x^2 + y^2 - z^2 = 1$. The hyperboloid looks like so:

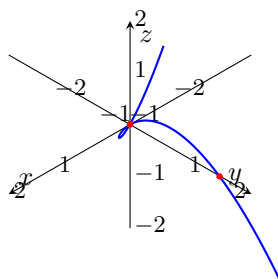


Then the intersection of these two shapes shall produce two circles, visualized like so:



This produce *two disconnected components* (in the usual euclidean topology). We shall show later that no single polynomial can create these two circles, hence we shall often have the case where we get interesting shapes by interesting or taking the union of two zero-sets.

7. The following is an infamous example of a shape that looks like it could be the zero-set of a single polynomial, but can't be: consider the following shape given by $t \mapsto (t, t^2, t^3)$:



It can be proven that this shape is *not* the zero-set of a single polynomial $p(x, y, z) \in k[x, y, z]$. On the other hand, this shape, is the intersection of two zero-sets of the surfaces $y = x^2$ and $z = xy$. This shape is called the *twisted cubic*.

More generally, given any polynomial in $k[x_1, x_2, \dots, x_n]$, we can associate with it a geometric shape based on it's zero set!

Definition 1.1.5: [Affine] Algebraic Set

Let $\mathbb{A}^n$ be an affine n -space over a field k , and let $S \subseteq \Gamma(\mathbb{A}^n)$. Then:

$$\mathcal{Z}(S) \subseteq \mathbb{A}^n \quad \mathcal{Z}(S) := \{(a_1, a_2, \dots, a_n) \mid f(a_1, a_2, \dots, a_n) = 0, \text{ for all } f \in S\}$$

If for $V \subseteq \mathbb{A}^n$, there exists an $S \subseteq \Gamma(\mathbb{A}^n)$ such that $V = \mathcal{Z}(S)$, then V is called an *affine algebraic set*, *locus*, or *zero set*. If it is unambiguous, the set may be called an *algebraic set*^a.

^aIn some textbooks, this set is called a *variety*. The term variety shall be reserved for definition 1.2.29

From now on, when taking a subset of $\mathbb{A}^n$, $A \subseteq \mathbb{A}^n$, it will be assumed that A is an algebraic set unless specified otherwise (i.e. it would be stated “for some subset $A \subseteq \mathbb{A}^n$ ”)

Example 1.1.6: Affine Algebraic Sets

All the above are algebraic sets. Let's go over a few more examples:

1. If $n = 1$, then the zero's of a polynomial will always be a points, that is, the roots of the polynomial. The only polynomial with infinitely many points (if working over a field of characteristic 0) is 0 (more precisely, the regular function associated to 0), which has all of k as its vanishing locus. Furthermore, the polynomial 1 vanishes nowhere, and hence the corresponding algebraic set is $\emptyset$. Therefore, in $\mathbb{A}^1$, the affine sets are

$$\emptyset \quad \text{finite sets} \quad k$$

This also shows that not all sets are algebraic sets (for example, if $k = \mathbb{R}$, then $[0, 1]$ is not an algebraic set in $\mathbb{A}_{\mathbb{R}}^1$)

2. If $n = 2$, Then take $f \in \Gamma(\mathbb{A}^2)$ such that $f(x, y) = x^2 - y^2 - 1$. Then

$$\mathcal{Z}(f) = \{(x, y) \in \mathbb{A}^2 \mid x^2 + y^2 = 1\}$$

If $k = \mathbb{R}$, Notice that $\mathcal{Z}(f)$ is not finite, and so cannot be a product of two algebraic sets from $K[\mathbb{A}^1]$.

Furthermore, this examples shows that there are lot's more zero sets in $\mathbb{A}^2$ than there are in $\mathbb{A}^1 \times \mathbb{A}^1$. This is in stark contrast to how every element in the vector space $\mathbb{R}^2$ can be written as an element in $\mathbb{R}$ and an element in $\mathbb{R}$, i.e. $\mathbb{R}^2 \cong \mathbb{R} \times \mathbb{R}$ as vector spaces.

3. For any $\mathbb{A}^n$, can you always find a set of functions $F \subseteq \Gamma(\mathbb{A}^n)$ such that $|V(F)| = 1$ (i.e. is every singleton an algebraic set for every $\mathbb{A}^n$ over any field?) This shall be answered in section 1.2.7.
4. What is the vanishing set of linear polynomials?
5. In the first example, we produced the entire space using the zero polynomial. Is it necessary that we use the zero-polynomial to make the zero set the entire space? The answer is no: simply take $k = \mathbb{Z}/2\mathbb{Z}$ and $p(x) = x^2 + x$. However, this is a problem exclusive to finite fields. Can you prove that if k is an infinite field, $p \in k[x_1, x_2, \dots, x_n]$ and $p(a_1, a_2, \dots, a_n) = 0$ for all $a_i \in k$ then $p \equiv 0$? [exercise ref:HERE (freely use any polynomial has at most finitely many roots, and then induction on the number of terms)]

6. Let k be a finite field and take any subset $S \subseteq \mathbb{A}_k^n$. Show that S is always an algebraic set. Hence, being algebraic gives no additional information in the finite case.
7. Let $f \in k[x_1, x_2, \dots, x_n]$. Then $\mathcal{Z}(f)$ is called a *hyper-surface*. For example $S^1 = \mathcal{Z}(x^2 + y^2 - r^2)$ represents a circle of radius r . Another example would be $\mathcal{Z}(xy - 1)$ which represents the hyperbola $y = 1/x$. More generally, any conic section is a hyper-surface. In higher dimensions, $\mathcal{Z}(x^2 + y^2/4 + z^2/9 - 1)$ represents an ellipsoid in $\mathbb{R}^3$. The twisted cubic is *not* a hyper-surface.
8. For any polynomial function $f : k \rightarrow k$, we can define $F : k^2 \rightarrow k$ where $F(x, y) = f(x) - y$. Then $\mathcal{Z}(F)$ consists of all points where $f(x) - y = 0$, i.e. $f(x) = y$, which is the pair of points $\{(a, b) \mid f(a) = b\}$, i.e. the graph of f ! More generally, if $f : k^n \rightarrow k$ is a polynomial, the polynomial $F : k^{2n} \rightarrow k$ gives rise to the polynomial $F : (x, y) = f(x) - y$ whose zero set is the graph of f (recall the convention that x, y represent multiple variables when it is clear from context).

Proposition 1.1.7: Properties of Affine Algebraic Sets

Let $S, T \subseteq \Gamma(\mathbb{A}^n)$. Then

1. If $S \subseteq T$ then $\mathcal{Z}(T) \subseteq \mathcal{Z}(S)$ (i.e. $\mathcal{Z}$ is *contravariant* in respect to inclusion)
2. If $(S) = I \subseteq \Gamma(\mathbb{A}^n)$ is the ideal of S , then $\mathcal{Z}(I) = \mathcal{Z}(S)$.
3. For any arbitrary collection of algebraic sets $\{S_i\}_{i \in I}$,

$$\bigcap_{i \in I} \mathcal{Z}(S_i) = \mathcal{Z}\left(\bigcup_{i \in I} S_i\right)$$

that is, the arbitrary intersection of algebraic sets is an algebraic set, and is equal to the algebraic set of the intersections of the $\{S_i\}_{i \in I}$

4. For any finite collection of algebraic sets $\{S_i\}_{i=1}^n$,

$$\bigcup_{i=1}^n \mathcal{Z}(S_i) = \mathcal{Z}\left(\prod_{i=1}^n S_i\right)$$

that is, the finite union of algebraic sets is an algebraic set, and is equal to the product of the S_i as ideals (from part 2, we can assume $(S_i) = S_i$ without loss of generality).

5. If $0, 1 \in \Gamma(\mathbb{A}^n)$ denote the zero and constant function, then

$$\mathcal{Z}(0) = \mathbb{A}^n \quad \mathcal{Z}(1) = \emptyset$$

6. If I is an ideal of $\Gamma(\mathbb{A}^n)$, then

$$\mathcal{Z}(I) = \mathcal{Z}(I^m) \quad \forall m \in \mathbb{N}_{>0}$$

that is, $\mathcal{Z}$ is invariant to taking higher powers of the ideals

These are all straight-forwards proofs, and so should all be done before reading these proofs for the best learning experience.

Proof:

1. Let $S \subseteq T$, and consider $x \in \mathcal{Z}(T)$. Then for all $f \in T$, $f(x) = 0$. Since $S \subseteq T$, for all $g \in T$, $g(x) = 0$. Thus, $x \in \mathcal{Z}(S)$, establishing $\mathcal{Z}(T) \subseteq \mathcal{Z}(S)$.
2. Let $S \subseteq \Gamma(\mathbb{A}^n)$ be any set and let $I = (S)$. We want to show that $\mathcal{Z}(I) = \mathcal{Z}(S)$. Since $S \subseteq I$, the $\mathcal{Z}(I) \subseteq \mathcal{Z}(S)$ conclusion comes from the first part. For the reverse, consider $x \in \mathcal{Z}(S)$. Then for all $t \in S$, $t(x) = 0$. If we pick any two $f, g \in S$, then $(f + g)(x) = f(x) + g(x) = 0$. If we pick any $c(x) \in \Gamma(\mathbb{A}^n)$ and any $f \in S$, then $(cf)(x) = c(x)f(x) = c(x)0 = 0$. Thus, if we enlarge S to be closed under addition and multiplication by elements of $k[\mathbb{A}^n]$, those polynomial will also be zero on x , and so $\mathcal{Z}(S) \subseteq \mathcal{Z}(I)$, establishing $\mathcal{Z}(S) = \mathcal{Z}(I)$.
3. We want to show that $\bigcap_{i \in I} \mathcal{Z}(S_i) = \mathcal{Z}(\bigcup_{i \in I} S_i)$ for an arbitrary indexing set I . This is just some set theory: by definition:

$$x \in \mathcal{Z}(\bigcup_{i \in I} S_i) = \{y \in \mathbb{A}^n \mid f(y) = 0, \forall f \in \bigcup_{i \in I} S_i\} = \mathcal{Z}\left(\sum_{\alpha} I_{\alpha}\right)$$

which is true if and only if x is the solution for polynomials in every set S_i . The collection of such elements is:

$$\bigcap_{i \in I} \mathcal{Z}(S_i)$$

4. We want to show that $\bigcup_{i=1}^n \mathcal{Z}(S_i) = \mathcal{Z}(\prod_{i=1}^n S_i)$. By definition:

$$x \in \mathcal{Z}\left(\prod_{i=1}^n S_i\right) = \left\{y \in \mathbb{A}^n \mid f(y) = 0 \forall f \in \prod_{i=1}^n S_i\right\}$$

where by definition $f(x) = \sum_{j=1}^m (f_{1j} f_{2j} \cdots f_{nj})(x) = \sum_{j=1}^m f_{1j}(x) f_{2j}(x) \cdots f_{nj}(x) = 0$. Notice that for a particular term in the sum to be zero, we only need one of the f_{i_j} to be zero. Therefore, the sum is zero if and only if one polynomial in every term of the sum is zero. Since this also applies to sums with only one term in it, then if x is the solution to one polynomial in S_i , then it will be part of the zero's of $\prod_{i=1}^n S_i$. But this is just another way of saying:

$$x \in \bigcup_{i=1}^n \mathcal{Z}(S_i)$$

5. Notice that $0(x) = 0$ for all $x \in \mathbb{A}^n$ and $1(x) = 1 \neq 0$ for all $x \in \mathbb{A}^n$, and so

$$\mathcal{Z}(0) = \mathbb{A}^n \quad \mathcal{Z}(1) = \emptyset$$

6. Since $I^m \subseteq I$ (recall that $1 \notin I \subsetneq R$), then by the first part $\mathcal{Z}(I) \subseteq \mathcal{Z}(I^m)$. For the $\supseteq$ direction, if $x \in \mathcal{Z}(I^m)$, then for every $f \in I^m$, $f(x) = 0$. By definition of I^m , $f = \sum_{i=1}^n f_i^m$ for some $n \in \mathbb{N}$. In order for $\sum_{i=1}^n f_i^m(x) = 0$, it must be that every $f_i(x) = 0$. But since every $f_i \in I$, and the summand can express any linear combination of elements of I raised to the power of m , it must be that $g(x) = 0$ for all $g \in I$, and so $\mathcal{Z}(I^m) \subseteq \mathcal{Z}(I)$. Combining with the previous direction we get $\mathcal{Z}(I) = \mathcal{Z}(I^m)$.

A couple of remarks are in order. From (2), we can without loss of generality restrict $\mathcal{Z}$ be a function from the ideals of $k[x_1, \dots, x_n]$ to the algebraic sets in $\mathbb{A}^n$:

$$\mathcal{Z} : \{\text{ideals in } k[x_1, \dots, x_n]\} \rightarrow \{\text{affine algebraic sets in } \mathbb{A}^n\}$$

Furthermore, $\mathcal{Z}$ is a surjection from the ideals of $k[x_1, x_2, \dots, x_n]$ and the algebraic sets (the ideals I^m for each $m \geq 1$ map to the same algebraic set). Next, the finiteness condition on the product of ideals is a necessary condition for closure of algebraic sets:

Example 1.1.8: Not Closed Under Arbitrary Unions

Take $\mathbb{A}_{\mathbb{R}}^1$ and the sets $V(x - a)$ for $a \in \mathbb{Z}$. Then prove that the union of these sets is *not* an algebraic set (you may freely use the fact that polynomials have at most finite roots, and recall that $\mathbb{R}[x]$ is Noetherian).

Next, we took the product of ideal $\prod_i^n I_i$ instead of the intersection. Since it always the case that $IJ \subseteq I \cap J$ then in the cases where $IJ \subsetneq I \cap J$, $\mathcal{Z}(IJ) \subsetneq \mathcal{Z}(I) \cup \mathcal{Z}(J)$ ¹¹. However, since $\mathcal{Z}(I^2) = \mathcal{Z}(I)$, there is a trick we can use in which $\mathcal{Z}(IJ) = \mathcal{Z}(I \cap J)$, making the distinction obsolete (see exercise ref:HERE)

Perhaps most importantly for this chapter, notice that these properties are the properties of a topology (see appendix ??):

Definition 1.1.9: Zariski Topology

Let $\mathbb{A}^n$ be an n -affine space over k . Then the set

$$\mathcal{T}_{\mathcal{Z}} = \{\mathcal{Z}(I) \mid I \text{ is an ideal of } \Gamma(\mathbb{A}^n)\}$$

from the set of closed sets in the *Zariski topology* $(\mathbb{A}^n, \mathcal{T}_{\mathcal{Z}})$

We can therefore try and study this topological space! This is essentially the foundation for studying geometrical concepts in algebra, and vice-versa, it is the foundation for studying algebraic concepts in geometry, and so we will spend a lot of time establishing the link between the ideals of $\Gamma(\mathbb{A}^n)$ and the Zariski topology.

Before such topological considerations, we shall first see if we can find some function that is the inverse of $\mathcal{Z}$, that is we start from an algebraic set in $\mathbb{A}^n$ (or more generally any subset of $\mathbb{A}^n$) and associate it to an ideal in $\Gamma(\mathbb{A}^n)$, letting us strive towards some way of corresponding the two in a manner similar to Galois Theory¹². Towards the goal of finding an inverse for $\mathcal{Z}$, notice that we already have that $\mathcal{Z}$ is a surjection of ideals of $\Gamma(\mathbb{A}^n)$ to the algebraic sets. Thus, it is meaningful to ask about having a function from the set of algebraic sets to the set of ideals of $k[x_1, x_2, \dots, x_n]$. To understand this question, we start off with a slightly more general question (just like we started with subsets of $k[x_1, \dots, x_n]$ instead of ideals):

¹¹The condition that $IJ = I \cap J$ is surprisingly technical, requiring that $\text{Tor}^1(R/I, R/J) = 0$, see ref:HERE

¹²In fact, the order reversing nature of this correspondence will end up making this correspondence a *Galois correspondence*, which name is inspired by how the Galois correspondence between intermediate fields and subgroups was one of the first well-known order-reversing correspondences

Definition 1.1.10: Vanishing Ideal

Let $V \subseteq \mathbb{A}^n$ be a subset of the affine n -space. Then

$$\mathcal{I}(V) := \{f \in \Gamma(\mathbb{A}^n) \mid f(v) = 0, \forall v \in V\}$$

is the *vanishing ideal* of V .

Notice that $\mathcal{I}(V)$ was not defined as an ideal, however it is indeed an ideal; it is important for the sake of intuition that the reader should prove this at least once. Even better, $\mathcal{I}(V)$ is the *largest* ideal of $\Gamma(\mathbb{A}^n)$ that corresponds to the set V . Just like $\mathcal{Z}$, $\mathcal{I}$ can be interpreted as a function

$$\mathcal{I} : \{V \subseteq \mathbb{A}_k^n\} \rightarrow \{\text{ideals of } \Gamma(\mathbb{A}^n)\}$$

Example 1.1.11: Vanishing Ideal

1. Let $\mathbb{A}^2$ be an affine space over the field $\mathbb{R}$ and let $X \subseteq \mathbb{A}^2$ be the x -axis. Then $\mathcal{I}(X) = (y) \subseteq \mathbb{R}[x, y]$
2. Let k be infinite. Then $\mathcal{I}(\mathbb{A}^n) = 0$.
3. Find the vanishing ideals if of subsets $\mathbb{A}_k^n$ if k is finite.

Just like $\mathcal{Z}$, $\mathcal{I}$ shares similar properties:

Proposition 1.1.12: Properties of Vanishing Ideals

let $A, B \subseteq \mathbb{A}^n$ be two algebraic sets. Then

1. If $A \subseteq B$, then $\mathcal{I}(B) \subseteq \mathcal{I}(A)$ (i.e. $\mathcal{I}$ is *contravariant* with respect to inclusion)
2. $\mathcal{I}(A \cup B) = \mathcal{I}(A) \cap \mathcal{I}(B)$
3. $\mathcal{I}(\emptyset) = \Gamma(\mathbb{A}^n)$. If k is infinite, $\mathcal{I}(\mathbb{A}^n) = 0$

Proof:

These are similar to proposition 1.1.7.

Notice that we have omitted showing what $\mathcal{I}(A \cap B)$ is. One problem is that $\mathcal{I}(A) \cup \mathcal{I}(B)$ is not necessarily an ideal (since the union of two ideals is not necessarily an ideal). It might be tempting to say that since

$$\langle \mathcal{I}(A) \cup \mathcal{I}(B) \rangle = \mathcal{I}(A) + \mathcal{I}(B)$$

we can conclude that:

$$\mathcal{I}(A \cap B) = \mathcal{I}(A) + \mathcal{I}(B)$$

However, this is not generally the case (consider the algebraic sets given by the ideals $(x^2 - y)$ and (x)). We will require some more machinery to prove the appropriate equality (i.e. the Nullstellensatz), but it can already be shown with the current definitions that $\mathcal{I}(A \cap B) \supseteq \mathcal{I}(A) + \mathcal{I}(B)$ (see exercise

ref:HERE). With the properties of $\mathcal{I}$ established, we can start getting an idea of what ideals of $\Gamma(\mathbb{A}^n)$ are necessary to restrict $\mathcal{Z}$ to make it a bijection:

Proposition 1.1.13: Zero Sets and Coordinate Ideals

Let $\Gamma(\mathbb{A}^n)$ be the set of polynomial functions from $\mathbb{A}^n$ to k . Then

1. If $V \subseteq \mathbb{A}^n$, then $V \subseteq \mathcal{Z}(\mathcal{I}(V))$. If $I \subseteq \Gamma(\mathbb{A}^n)$ is an ideal of $\Gamma(\mathbb{A}^n)$, then $I \subseteq \mathcal{I}(\mathcal{Z}(I))$.
2. If $V = \mathcal{Z}(I)$ is an algebraic set, then $V = \mathcal{Z}(\mathcal{I}(V))$. Similarly, if $I = \mathcal{I}(V)$, then $I = \mathcal{I}(\mathcal{Z}(I))$. In other words,

$$\mathcal{I}(\mathcal{Z}(\mathcal{I}(V))) = \mathcal{I}(V) \quad \mathcal{Z}(\mathcal{I}(\mathcal{Z}(V))) = \mathcal{Z}(V)$$

Proof:

1. For $V \subseteq \mathcal{Z}(\mathcal{I}(V))$, since $\mathcal{I}(V)$ is the set of all functions such that $f(V) \equiv 0$ (including at least the zero polynomial), then V is certainly part of its zero set.
Conversely, for $I \subseteq \mathcal{I}(\mathcal{Z}(I))$, the coordinate ideal of $\mathcal{Z}(I)$ certainly contains I since for any $f \in I$ and $x \in \mathcal{Z}(I)$, $f(x) = 0$. Thus, $I \subseteq \mathcal{I}(\mathcal{Z}(I))$.
2. Let $V = \mathcal{Z}(I)$. Using part 1, it suffices to prove $V \supseteq \mathcal{Z}(\mathcal{I}(V))$. If $x \in \mathcal{Z}(\mathcal{I}(V))$, then for every $f \in \mathcal{I}(V)$, $f(x) = 0$. Since $V = \mathcal{Z}(I)$, f is defined as the function such that

What the 2nd part of proposition 1.1.13 tells us is that if we restrict to just the algebraic sets and vanishing ideals of $\mathbb{A}^n$ and $\Gamma(\mathbb{A}^n)$ respectively, then $\mathcal{Z}$ and $\mathcal{I}$ are indeed inverses of each other. The goal then would be to be able to characterize vanishing ideals, that is ideals that are of the form $I = \mathcal{I}(V)$ for some algebraic set V . It is in fact a rather involved process to find the nature of the ideal $\mathcal{I}(V)$; a lot of section 1.2.4 will be building up to get to the theorem which will tell us the answer: *Hilbert's Nullstellensatz*. In the process of building up the theory to answer this question, there are a few other loose ends that we will need to tie up. For example, what properties must the Zariski topology have, and what does that imply for the corresponding ring $\Gamma(\mathbb{A}^n)$? Another important point to address is that as $k[x_1, \dots, x_n]$ is Noetherian, hence every ideal is finitely generated: $I = (f_1, \dots, f_m) = (f_1) \cup (f_2) \cup \dots \cup (f_m)$. By the properties of zero sets:

$$\mathcal{Z}(I) = \mathcal{Z}(f_1) \cap \mathcal{Z}(f_2) \cap \dots \cap \mathcal{Z}(f_m)$$

In other words, there seem to be a natural theory of decomposition of algebraic sets. Furthermore, we have been shown theoretically what properties algebraic sets have, but how would we go about actually computing one? These questions will receive answers in the following section.

Nuances

1. I want to mention how $\text{Hom}(V, \mathbb{A}_k^1) = \text{Hom}(V, k)$ works since closed points shall be associated to k -points when k is algebraically closed.
2. The $k[\mathbb{A}_k^n]$ notation has its flaw when working with algebraic groups, namely if G is an algebraic group then $k[G]$ can be thought of as the *group ring*, which is patently not the case.

Exercise 1.1.1

1. Show that $z \mapsto \bar{z}$ is not a complex polynomial.
2. Show that $V = \{(z, w) \in \mathbb{A}_{\mathbb{C}}^2 \mid |z|^2 + |w|^2 = 1\}$ is not an algebraic set. However, it is an algebraic set over $\mathbb{R}$.
3. Let $\text{Gal}_k(\bar{k})$ be the galois group of $\bar{k}/k$ where $\bar{k}$ is the algebraic closure. Show that $\mathbb{A}_k^n$ is isomorphic to the stable points of A_k^n , namely

$$\mathbb{A}_k^n \cong \left\{ p \in \mathbb{A}_{\bar{k}}^n \mid p^\sigma = p \quad \forall \sigma \in \text{Gal}_k(\bar{k}) \right\}$$

4. Let $V = \mathcal{Z}(x^2 - y)$. Show that V is irreducible over $\mathbb{Q}$, but is reducible over $\bar{\mathbb{Q}}$ (hence, irreducibility is dependent over which field we are working over).
5. The following is a natural way of recovering $A = k[x_1, \dots, x_n]$ from $k^n = V$. We know that $V^\vee := \text{Hom}(V, k)$. Recall ??:

$$\text{Sym}^\bullet(V^*) = k \oplus V^* \oplus \text{Sym}^2(V^*) \oplus \text{Sym}^3(V^*) \oplus \dots$$

Show that $\text{Sym}^\bullet(V^*) \cong A$. Next, let $\mathfrak{m} \subseteq A$ be a maximal ideal. Show that:

$$\mathfrak{m}/\mathfrak{m}^2 \cong V^*$$

showing that the linear structure is recovered at each point; this is the *Zariski tangent space*.

1.2 Zariski Topology and Duality

In this section, we will be assuming basic topological knowledge from the reader (see appendix ??)

1.2.1 Zariski Topology

As the reader (ought to have) showed in in example 1.1.6, if $k = \mathbb{F}_q$ is finite, then every subset of $\mathbb{A}_{\mathbb{F}_q}^n$ is algebraic, hence the Zariski topology is isomorphic to the discrete topology, showing us that there is no “interesting” behaviour in the finite case. Thus, we shall usually focus on the case where k is not-finite. This doesn’t mean that finite fields don’t play an important role in algebraic geometry, however their importance shall not be visible until the development of schemes¹³ or Class Field Theory¹⁴ and so they shall have negligible importance this chapter.

As the topology was defined using closed sets (sets for which a collection of polynomials is zero on that set), a moment should be given to give a geometric interpretation to open sets. Open sets are those for which a collection of polynomials are *not all zero* at once on those points. If $S = \{f\} \in \Gamma(\mathbb{A}_k^n)$ is a

¹³In particular, for any Scheme X , there is always a scheme morphism $X \rightarrow \text{Spec}(\mathbb{Z})$ and there is always a ring homomorphism $\mathbb{Z} \rightarrow \mathcal{O}_X(X)$; the fibers of this scheme X_p are $\mathbb{F}_p$ -schemes. Furthermore, $\mathbb{F}_q$ -schemes shall be important in number theory

¹⁴In particular, the behaviour of unramified field extensions between local fields will be captured via the automorphisms of finite fields

singleton, then it will be precisely the set of points where the function f does not vanish. This makes all open sets of $\mathbb{A}_k^1$ (where k is not finite) the *cofinite sets* and hence it has the cofinite topology. If $n > 1$, the topology is certainly no longer cofinite. Open sets shall become importance when we look to define *rational polynomials* and *rational functions* for they give us the domain on which a function is non-zero. (see definition 1.2.5).

Let us now look at the separatedness of the Zariski topology. Notice that $\mathcal{T}_Z$ is rarely Hausdorff (also known as T_2): if $\mathbb{A}^1 = k$ is any infinite field, as we've established that the only algebraic sets (and hence closed sets) are the empty set, finite sets, and $\mathbb{A}_k^1$ and so the only non-trivial open sets in $\mathcal{T}_Z$ would be co-finite sets (i.e. $\mathcal{T}_Z$ is the co-finite topology). Then as we can see, if we pick any two points $a, b \in \mathbb{A}^1$, we cannot separate them by two open disjoint sets. However, over any $\mathbb{A}^1 = k$, the singleton's are always algebraic sets (giving a hint to those who were working on example 1.1.6), and so it is always a Fréchet space (or T_1)¹⁵.

This result generalizes for any k^n

Proposition 1.2.1: Zariski Topology Always T_1

Let k be an infinite field and $\mathbb{A}_k^n$ have the Zariski topology. Then $\mathbb{A}_k^n$ is T_1 , i.e. every singleton is always closed, and $\mathbb{A}_k^n$ is not Hausdorff.

Proof:

For any $a = (a_1, a_2, \dots, a_n) \in k^n$, Let $I = (x_1 - a_1, \dots, x_n - a_n)$. Then $\mathcal{Z}(I) = \{a\}$, showing that singleton's are closed.

To show's it not Hausdorff, we already showed it in the case of $n = 1$ in example 1.1.6, so we'll do k^n with $n > 1$. Take $\mathbb{A}_k^1 \times \{0\}^{n-1}$ with the sub-space topology. Then any closet subset must be finite. Since this is true for any arbitrary axis we choose, the closed set that intersects each of these subspaces can have at most finitely many points, and so the compliment has co-finitely many points. It is now easy to verify that the space is not Hausdorff.

Two topological concepts which should be immediately introduced to be explored as the chapter develops are the following:

Definition 1.2.2: Zariski Dense And Closure

Let $A \subseteq \mathbb{A}^n$ be any subset of the affine space $\mathbb{A}^n$. Then the *zariski closure* of A is the smallest algebraic set containing A , denoted $\bar{A}$. Geometrically, it is the smallest set $\mathcal{A}$ for which every point in A is the zero set of a collection of polynomials.

If $\bar{A} = \mathbb{A}^n$, then A is said to be *zariski dense*. More generally if $A \subseteq V \subseteq \mathbb{A}^n$ such that $\bar{A} = V$, then A is said to be *zariski dense* in V .

As an exmple, if we give $\mathbb{R}$ the Zariski topology, the closure of any finite set is itself, while the closure of any infinite set is all of $\mathbb{R}$. The Zariski closure of a set is in fact quite easy to characterize using $\mathcal{Z}$ and $\mathcal{I}$ and follows quickly using proposition 1.1.13:

¹⁵For those familiar with algebraic topology, schemes are almost never T_1 , as prime ideals that are not maximal are not closed points

Corollary 1.2.3: Zariski Closure

Let $A \subseteq \mathbb{A}^n$. Then:

$$\overline{A} = \mathcal{Z}(\mathcal{I}(A))$$

Proof:

Hint: $\mathcal{I}(A)$ is the largest ideal on which every element of A is in the zero set, and so $\mathcal{Z}(\mathcal{I}(A))$ is the smallest zero set containing A .

Sticking to $k = \mathbb{R}$ or $k = \mathbb{C}$ for a moment, we can compare the euclidean topology on $\mathbb{R}^n$ with the Zariski topology on $\mathbb{A}_{\mathbb{R}}^n$. (Coarser, and I think is the coarsest topology on which all polynomial functions are continuous)

Another important topological consideration would be to find a topological basis for the Zariski topology. Recall a subset of the topology on a space X is a basis B if any open set of X is the union of a family of sets from B .

Proposition 1.2.4: Basis For Zariski Topology

Let $A \subseteq \Gamma(\mathbb{A}^n)$. Define

$$D(A) = \mathbb{A}_k^n \setminus \mathcal{Z}(A) = \{a \in \mathbb{A}_k^n \mid f(a) \neq 0, \forall f \in A\}$$

Then the collection $\{D(f)\}_{f \in k[A^n]}$ forms a [topological] basis for the Zariski topology on $\mathbb{A}^n$.

The letter D can be thought of as a mnemonic for *doesn't* vanish.

Proof:

It suffice to show that for any $a \in U$ for an open set U , there exists an f such that $a \in D(f) \subseteq U$, since then U is the union of $D(f)$'s. Since U is open, there exists an ideal $I \subseteq \Gamma(\mathbb{A}^n)$ such that $U = \mathbb{A}_k^n \setminus \mathcal{Z}(I)$. Since $a \in U$, for any $f \in I$, $f(a) \neq 0$, so $a \in D(f)$. Since $(f) \subseteq I$,

$$\mathcal{Z}(I) \subseteq \mathcal{Z}(f)$$

which, after taking the compliment of these sets, we get:

$$a \in D(f) \subseteq U$$

and since a was arbitrary, we see that U is the union of sets $D(f)$, as we sought to show.

The collection of $D(f)$ for each $f \in k[x_1, \dots, x_n]$ forms the coarsest basis:

Definition 1.2.5: Distinguished Open Sets

Let $f \in k[x_1, \dots, x_n]$. Then the set

$$D(f) = \mathbb{A}_k^n \setminus \mathcal{Z}(f) = \{a \in \mathbb{A}_k^n \mid f(a) \neq 0\}$$

is called the distinguished open set.

The next property we get about the Zariski topology is that it's always Noetherian, that is, if there is an ascending chain of open sets, it must stabilize:

Proposition 1.2.6: Zariski Topology Is Noetherian

Let $\mathbb{A}_k^n$ have the Zariski topology. Then $\mathbb{A}_k^n$ is Noetherian.

Proof:

Let $U_1 \subsetneq U_2 \subsetneq \dots$ be an ascending chain of open sets, so there is a corresponding

$$X_1 \supsetneq X_2 \supsetneq \dots$$

descending chain of closed sets. Then there is a corresponding

$$I(X_1) \subsetneq I(X_2) \subsetneq \dots$$

ascending chain of ideals. Then we know that $\Gamma(\mathbb{A}^n)$ is *noetherian*, and so this chain must stabilize at some n :

$$I(X_n) = I(X_{n+1}) = \dots$$

meaning the respective closed chain, and hence open chain, must stabilize, as we sought to show.

We will soon show that the Zariski topology on $\mathbb{A}_k^n$ is in fact always *compact*, but we require the Nullstellensatz to prove this result ¹⁶.

1.2.2 Regular Functions and Morphisms

The next thing we should consider the continuous functions. As the topology is the weakest in which the zero-sets is closed, it may not surprise the reader that it is the coarsest topology in which polynomials are continuous

Definition 1.2.7: Regular Map on Affine Space

A map $\varphi : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^m$ is called a *regular map* if there exists polynomials $\varphi_1, \varphi_2, \dots, \varphi_m \in k[x_1, x_2, \dots, x_n]$ such that

$$\varphi(a_1, a_2, \dots, a_n) = (\varphi_1(a_1, a_2, \dots, a_n), \varphi_2(a_1, a_2, \dots, a_n), \dots, \varphi_m(a_1, a_2, \dots, a_n))$$

If $m = 1$, a regular map is often called a *regular function*.

Lemma 1.2.8: Coordinate Ring and Regular Map

The map $\varphi : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^m$ is regular if and only if the induced map $\varphi^\# : \Gamma(\mathbb{A}_k^m) \rightarrow \Gamma(\mathbb{A}_k^n)$ given by $\varphi^\#(f) = f \circ \varphi$ is a k -algebra homomorphism

¹⁶For those familiar with algebraic geometry, almost all schemes are compact, making the notion much less useful. Instead, the term the notion of *properness* shall be the appropriate replacement

Proof:

Let $\varphi : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^m$ be a regular map so that each component is given by a polynomial. Then $\varphi^\#$ maps a function $f \in \Gamma(\mathbb{A}_k^m)$ to $f \circ \varphi \in \Gamma(\mathbb{A}_k^n)$, namely $f \circ \varphi$ is a polynomial map. If $f, g \in \Gamma(\mathbb{A}_k^m)$, then:

$$\begin{aligned}(f + g) \circ \varphi &= f \circ \varphi + g \circ \varphi \\ (f \cdot g) \circ \varphi &= f \circ \varphi \cdot g \circ \varphi \\ (k \cdot f) \circ \varphi &= k \cdot (f \circ \varphi)\end{aligned}$$

thus it is a k -algebra homomorphism.

Conversely, let $\varphi^\# : \Gamma(\mathbb{A}_k^m) \rightarrow \Gamma(\mathbb{A}_k^n)$ is a k -algebra homomorphism that maps:

$$f \mapsto \varphi^\#(f) = f \circ \varphi \in \Gamma(\mathbb{A}_k^n)$$

Let $y_1, \dots, y_n : \mathbb{A}_k^m \rightarrow k$ be the coordinate functions in $\Gamma(\mathbb{A}_k^m)$. If one of the components of φ were not a polynomial, then take $f = x_i$, so that the composition $x_i \circ \varphi = \varphi_i$ is not a polynomial, giving a contradiction. Hence φ must be a regular map, completing the proof.

Lemma 1.2.9: Ring homomorphism Induces Regular map

Let $f : \Gamma(\mathbb{A}_k^m) \rightarrow \Gamma(\mathbb{A}_k^n)$ be a k -algebra homomorphism. Then it induces a map $\varphi : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^m$ such that $\varphi^\# = f$.

Proof:

Let $k[y_1, \dots, y_m] \cong \Gamma(\mathbb{A}_k^m)$ and $k[x_1, \dots, x_n] \cong \Gamma(\mathbb{A}_k^n)$. Then as f is a k -algebra homomorphism, it is enough to map each y_i to some polynomial $f(y_i) = p(x_1, \dots, x_m)$ and extend to form a map on all of $\Gamma(\mathbb{A}_k^m)$. Let φ_i be the function determine by $f(y_i)$. Then it is a matter of course to check that $\varphi^\# = f$, completing the proof.

Note that the maps must be k -algebra homomorphism, not just ring homomorphisms. For example, if $n = m = 1$, then $f : \mathbb{C}[\mathbb{A}_k^1] \rightarrow \mathbb{C}[\mathbb{A}_k^1]$ sending $p(x) \mapsto \overline{p(x)}$ (i.e. the complex conjugate) is not a $\mathbb{C}$ -algebra homomorphism, and the reader could check there can't be a corresponding φ such that $\varphi^\# = f$.

Lemma 1.2.10: Regular Map is Continuous

A regular map $\varphi : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^m$ is continuous.

Proof:

It suffices to check that the pre-image of every closed set is closed. If $V = \mathcal{Z}(I) \subseteq \mathbb{A}_k^m$ is an algebraic set, we need to show there exists an ideal J such that

$$\mathcal{Z}(J) = \varphi^{-1}(V)$$

To that end:

$$\begin{aligned}\varphi^{-1}(V) &= \{a \in \mathbb{A}_k^n \mid \varphi(a) \in V\} \\ &= \{a \in \mathbb{A}_k^n \mid f(\varphi(a)) = 0 \ \forall f \in I\} \\ &= \{a \in \mathbb{A}_k^n \mid \varphi^\#(f)(a) = 0 \ \forall f \in I\}\end{aligned}$$

Consider the set $\varphi^\#(I) = \{g \in \Gamma(\mathbb{A}_k^n) \mid g = f \circ \varphi\}$. This set would be the desired pre-image ideal, except it is in general not an ideal as it need not be closed under multiplication (recall the image of an ideal need not be an ideal). Thus, take $J = \langle \varphi^\#(I) \rangle$. Then it is clear that

$$\mathcal{Z}(J) = \varphi^{-1}(V)$$

and as V was an arbitrary closed set, φ is continuous, as we sought to show.

Note that *not all* continuous maps arise from a ring homomorphism:

Example 1.2.11: Continuous, No Associated Ring Homomorphism

Take $f : \mathbb{A}_{\mathbb{R}}^1 \rightarrow \mathbb{A}_{\mathbb{R}}^1$ given by:

$$f(a) = \begin{cases} 2 & a = 1 \\ 1 & a = 2 \\ a & \text{otherwise} \end{cases}$$

Then note that the pre-image of every closed set is closed, and hence it is continuous. However, there are no real polynomial functions that only permute two elements whose only root is 0.

Another such example would be with $k = \mathbb{C}$ and the conjugation map presented earlier.

There needs to be more information for the full and faithful correspondence between regular maps and ring homomorphisms. This extra information is a *sheaf* and a *locally ringed morphism*. We shall not define these in this book, instead the reader may refer to [1].

Example 1.2.12: Regular Maps

1. Take $V = \mathcal{Z}(y - x^2, z - x^3)$. Show that the map $f : \mathbb{A}_k^1 \rightarrow V$ given by $t \mapsto (t, t^2, t^3)$ is a regular map. Show it is not an isomorphism^a.
2. Any straight line $\mathbb{A}_k^n$ is isomorphic to any other straight-line. Similarly for any k -dimensional linear subspace.
3. The map $\mathbb{A}_k^1 \rightarrow \mathcal{Z}(y - x^2)$ given by $t \mapsto (t, t^2)$ is a regular map with inverse $(x, y) \mapsto x$, and hence is an isomorphism. Note that it was important which coordinate we project to.
4. Here is another version of the above map. Consider the global sections of $\mathbb{A}_k^1 \rightarrow \mathbb{A}_k^1$ given by

$$k[y] \rightarrow k[x] \quad y \mapsto x^2$$

(note that the k -homomorphism is the contravariant direction of the regular morphism). As a map between varieties, this gives the map $a \mapsto a^2$ (the reader should verify this!).

5. Take $f : \mathbb{A}_k^2 \rightarrow \mathbb{A}_k^2$ given by $(x, y) \mapsto (x, xy)$. Show that $\text{im}(f)$ is *not* an algebraic set. The image of regular maps shall always be something called a *constructible sets* (see ref:HERE).

6. The reader should be careful with their geometric intuitions coming from the real numbers. Consider the map:

$$\mathcal{Z}(x^2 + y^2 - 1) \rightarrow \mathbb{A}_{\mathbb{C}}^1 \quad (x, y) \mapsto x$$

Show that this map is *surjective* (in contrast to the map certainly being *injective* when over $\mathbb{R}$). In fact, by taking different cross sections of the resulting shape interpreted as a 2-real dimensional shape living in 4-real dimensional space, you will see that $\mathcal{Z}(x^2 + y^2 - 1)$ will have both the parabola and the hyperbolic curve as sections. The hyperbolic sections demonstrate why the map is surjective and 2-to-1.

^aIt shall later be shown to be a *birational equivalence*

Let us now take a moment to look at isomorphism of algebraic sets:

Definition 1.2.13: Isomorphism of Varieties

let X, Y be two algebraic sets. Then if there exists a regular map $f : X \rightarrow Y$ with regular inverse, then the two varieties are said to be *isomorphic*

Note that this was not called a homeomorphism between two varieties. All isomorphisms are homeomorphisms, but not all homeomorphisms are isomorphisms

Example 1.2.14: Isomorphism of Varieties

1. Consider $X = \mathcal{Z}(y - x^n) \subseteq \mathbb{A}_k^2$. Then $X \cong \mathbb{A}_k^1$ with $\varphi : (x, y) \mapsto x$ that has inverse $\psi(t) = (t, t^n)$
2. Take $X = \mathcal{Z}(x^3 - y^2) \subseteq \mathbb{A}_k^2$. Then X can be parameterized by the regular map $f : \mathbb{A}_k^1 \rightarrow X$ by $f(t) = (t^2, t^3)$. However, this is not an isomorphism: the inverse map is:

$$g(x, y) = \frac{y}{x}$$

however, there is no polynomial associated to g (can the reader see/prove why?), and hence it is not an isomorphism.

3. The reader may have recalled that $\mathcal{Z}(x^3 - y^2)$ has a cusp at 0, and believe this is the obstruction to the inverse. It is in fact a bit more subtle. Consider the map $\varphi : \mathbb{A}_k^1 \rightarrow \mathbb{A}_k^2$ given by $\varphi(t) = (t^2, t^3)$. This is a bijective regular map, however it certainly has no regular inverse, namely $\psi(t) = t^{1/3}$ is not a regular function.

On the other hand, the map given in the previous example had a rational polynomial inverse. Rational functions appear as localization of the coordinate ring, and so they offer a different class of functions and a different classification problem that turns out to be much easier, as we shall demonstrate in section 1.3.

It is in general difficult to ascertain isomorphic classes of algebraic sets or varieties. Some isomorphism classes that we shall state here but not prove are:

1. Smooth genus 0 curves are isomorphic to conic curves in $\mathbb{P}_k^2$
2. classes of elliptic curves can be classified by their j -invariant

- Algebraic surfaces are birationally equivalent (defined in section 1.3), but have different isomorphism classes given their Kodaira dimension.

Exercise 1.2.1

- (if you are interested in categories and topology) Show that the Zariski topology is the coarsest topology where singletons are closed.
- Prove that $(I \cap J)^2 \subseteq IJ \subseteq (I \cap J)$, and conclude that $\mathcal{Z}(I \cap J) = \mathcal{Z}(IJ)$.
- Let $k = \mathbb{F}_q$ be a finite field. Show that each set-function $\text{Hom}_{\mathbf{Set}}(k^n, k)$ is represented by a polynomial (hint: find a polynomial that is a characteristic function $\chi_{\bar{a}}(\bar{a}) = 1$ and 0 otherwise and show these generate all the set functions). Conclude the Zariski topology is discrete over finite fields. Thus, the Zariski topology over finite fields reduces to set-theoretic counting arguments.
- Show that the Zariski topology on $\mathbb{A}_k^n$ is Hausdorff if and only if k is finite.
- Let $V \subseteq \mathbb{A}_k^n$ and $W \subseteq \mathbb{A}_k^m$ be algebraic sets. Show that $V \times W$ is an algebraic set in $\mathbb{A}_k^{n+m}$. Conclude that $\mathbb{A}_k^n \times \mathbb{A}_k^m \cong \mathbb{A}_k^{n+m}$ ¹⁷.
- Let $f, g \in k[x, y]$ be two polynomials with no common roots. Show that $V(f, g)$ is finite (hint: factorizing over $k[x, y]$ is the same as factorizing over $k(x)[y]$ by an argument similar to Gauss lemma)
- Show that if $V \subseteq \mathbb{A}_k^n$, $W \subseteq \mathbb{A}_k^m$ are varieties (not just algebraic sets), then $A \times B \subseteq \mathbb{A}_k^{n+m}$ is a variety (namely that it is irreducible).

1.2.3 Duality: Affine Sets and Coordinate Rings

In this section, we explore more properties of affine sets. As affine sets correspond to ideals, we shall see that there is a correspondence between the two objects. In particular it shall will show that properties of algebraic sets will often have corresponding properties in their coordinate rings, and vice-versa. This theme shall play out often in this chapter, where a property proven in a geometric or algebraic setting shall have a corresponding result in the respective dual.

By lemma 1.2.8, the collection of regular function $\text{Hom}_{\mathbf{Reg}}(\mathbb{A}_k^n, \mathbb{A}_k^1)$ can be thought of as the generalization of the dual vector-space, where polynomial functions are taken instead of linear functions. We saw that for each vector subspace $W \subseteq V$, there is a dual space $W^\vee = \text{Hom}_k(W, k)$ which it is isomorphic to. It is thus natural to ask what is the appropriate dual space for an algebraic set $V \subseteq \mathbb{A}_k^n$ corresponding to the *quotient* of $\Gamma(\mathbb{A}_k^n)$ ¹⁸.

In order to find the dual, we must consider more the subspace topology put on $V \subseteq \mathbb{A}_k^n$. Note that all of the properties of the Zariski topology apply to A since $\mathbb{A}_k^n$ is T1 (proposition 1.2.1), from which we can conclude that closed sets are preserved in the subspace topology. We will always limit ourselves to taking A to be an algebraic set, since algebraic set are the object of interest. The dual space we wish to establish is a corresponding “polynomial ring” which will generate all of the closed

¹⁷This seemingly simple result shall be shown to be greatly complicated over projective algebraic sets, see section 1.6.4

¹⁸For the categorically inclined, since the arrows in the dual space are reversed, sub-objects correspond to quotient-objects

sets of A , just like how $k[x_1, x_2, \dots, x_n]$ is needed to generate the closed sets of $\mathbb{A}^n$. Without looking at the next definition, can you see what the definition of such a ring may be?

1.2.15 taking a moment

To elucidate the coming definition, let us come back to the algebraic set V defined given by the vanishing set of $x^2 + y^2 - 1$. Then on V , the function $X^2 + Y^2 - 1$ on V will evaluate to 0, hence these two polynomials are equivalent as functions:

$$X^2 + Y^2 - 1 \equiv 0 \quad \text{on } V$$

If g is any other polynomial corresponding to the function G , then $(X^2 + Y^2 - 1)G$ evaluates to 0 everywhere as well, showing the set of regular functions taking inputs on V is closed under multiplication by regular functions $\Gamma(\mathbb{A}_k^n)$. Furthermore, if two regular functions f, g which evaluate to zero everywhere are added, the function $f + g$ will also evaluate to zero, showing it is closed under addition. In other words, the collection of functions which evaluates to 0 forms an ideal of $\Gamma(\mathbb{A}_k^n)$. What is this ideal? If we let F be the function

$$F : k[x, y] \rightarrow \text{Hom}_{\mathbf{Reg}}(V, k) \quad f \mapsto ((a, b) \rightarrow f(a, b)) \quad (1.3)$$

where $\text{Hom}_{\mathbf{Reg}}(V, k)$ is the set of regular functions which evaluate only on the points V , then we get that this map is surjective and that:

$$\frac{k[x, y]}{\ker(F)} \cong_k \text{Hom}_{\mathbf{Reg}}(V, k)$$

The reader should clue in that $\ker(F)$ is by definition $\mathcal{I}(V)$ as defined in definition 1.1.10! This leads to the following definition:

Definition 1.2.16: [Affine] Coordinate Ring

Let $k[x_1, \dots, x_n]$ be a polynomial ring over n variables, V an algebraic set, and $\mathcal{I}(V)$ the corresponding vanishing ideal. Then:

$$\Gamma(V) = k[V] = \frac{k[x_1, \dots, x_n]}{\mathcal{I}(V)}$$

is called the *coordinate ring* of V . The corresponding regular functions shall be denoted $k[V]$ or $\Gamma(V)$.

Notice how this notation is the same as for regular functions: the coordinate ring of $V = \mathbb{A}_k^n$ is isomorphic to $k[X_1, \dots, X_n]$ since $\mathcal{I}(\mathbb{A}^n) = (0)$. In $\Gamma(V)$, we have that $\mathcal{I}(V) = \overline{(0)}$, showing the notation is consistent.

We can naturally bring the notion of regular function to algebraic sets:

Definition 1.2.17: Regular Maps on Algebraic Sets

Let $V \subseteq \mathbb{A}_k^n$, $W \subseteq \mathbb{A}_k^m$ be algebraic sets. Then a map $\varphi : V \rightarrow W$ is called a *regular map*, or *regular morphism*, if there exists polynomials $\varphi_1, \varphi_2, \dots, \varphi_m \in k[x_1, x_2, \dots, x_n]$ such that

$$\varphi(a_1, a_2, \dots, a_n) = (\varphi_1(a_1, a_2, \dots, a_n), \varphi_2(a_1, a_2, \dots, a_n), \dots, \varphi_m(a_1, a_2, \dots, a_n))$$

If $m = 1$, then φ is often called a *regular function*.

This definition also allows us to upgrade definition 1.1.2 and definition 1.2.7. To check that $\varphi : V \rightarrow W$ is well-defined, it suffices to check that $\varphi_1, \dots, \varphi_m$ as elements of $k[V]$ satisfy the equations of W .

Proposition 1.2.18: Regular Maps Are Continuous

Let $\varphi : V \rightarrow W$ be a regular map where V, W are equipped with the Zariski topology. Then φ is continuous.

Proof:

Let $S \subseteq W$ be a closed subset. It suffices to show that $\varphi^{-1}(S)$ is closed. If $\varphi^{-1}(S) = \emptyset$, it is closed by definition, so assume $\varphi^{-1}(S) \neq \emptyset$. As S is closed, there exists polynomials $f_1, \dots, f_n$ such that for each $x \in S$, $g_1(x) = \dots = g_m(x) = 0$. If $a \in \varphi^{-1}(S)$, then $\varphi(a) \in S$, so

$$g_1(\varphi(a)) = \dots = g_m(\varphi(a)) = 0 \quad (1.4)$$

As $\varphi = (\varphi_1, \dots, \varphi_n)$ is a tuple of polynomials, notice that equation (1.4) is a collection of polynomials in $k[V]$ where a vanishes. As this is true for all points in $\varphi^{-1}(S)$, we see that:

$$\varphi^{-1}(S) = Z(g_1 \circ \varphi, \dots, g_m \circ \varphi)$$

showing it is closed, as we sought to show.

As before, this is not an iff as not all continuous functions are regular functions (simply take $V = W = \mathbb{A}_{\mathbb{C}}^1$ and the conjugation map). That is because continuity in the Zariski topology is not enough to preserve the algebraic relations. In particular, any point $a \in V \subseteq \mathbb{A}_k^n$ satisfying

$$f_1(a) = \dots = f_n(a) = 0$$

must have an image that satisfies:

$$g_1(w) = \dots = g_m(w) = 0 \quad \text{i.e.} \quad g_1(\varphi(a)) = \dots = g_m(\varphi(a)) = 0$$

Example 1.2.19: Coordinate Ring and Regular Mapping

1. Let $V = Z(xy - 1)$, which represents the hyperbola $y = 1/x$ in $\mathbb{R}^2$. Then the coordinate ring $\mathbb{R}[V] = \mathbb{R}[x, y]/(xy - 1)$. In $\mathbb{R}[V]$, $f(x, y) = x$ and $g(x, y) = x + xy - 1$ are the same function since $f - g \in \mathcal{I}(V)$. In $\mathbb{R}[V]$, $\overline{xy} = \overline{1}$, giving us that $\mathbb{R}[V] \cong \mathbb{R}[x, 1/x]$. Thus, for any $f \in \mathbb{R}[x, y]$ and $(a, b) \in V$, $\overline{f}(a, b) = f(a, 1/a)$ for any f that maps to $\overline{f}$.

Take the regular map $\varphi : V \rightarrow \mathbb{A}_k^1$ mapping $(x, y) \mapsto x$. The image of this map is $\mathbb{A}_k \setminus \{0\}$,

which is certainly *not* a closed set (can the reader see why). It is an open set, being the compliment of the closed set $\{0\}$. However, if we take the map to be $\varphi' : V \rightarrow \mathbb{A}_k^2$ mapping $(x, y) \mapsto (x, 0)$, then the image is no longer an open set, and hence it is possible for the images of regular maps to be neither open or closed. Further exploration the properties of the image of regular maps shall be done section section 1.6.3.

2. Take $k = \mathbb{F}_p$ and consider $\mathbb{A}_{\mathbb{F}_p}^n$. Then the map:

$$\varphi(a_1, \dots, a_n) = (a_1^p, \dots, a_n^p)$$

is a regular function. If $V \subseteq \mathbb{A}_{\mathbb{F}_p}^n$ is an algebraic set, notice that φ will always take V to itself, since if f is one of the polynomials that define V so that $f(a) \equiv 0 \pmod{p}$, then:

$$f(a_1^p, \dots, a_n^p) = (f(a_1, \dots, a_n))^p \equiv (0)^p \equiv 0 \pmod{p}$$

The regular function φ is called the *Frobenius mapping*

3. Take $t \mapsto (t^3, t^2)$. This is a regular function and maps $\mathbb{A}_{\mathbb{R}}^1$ to the curve given by $x^3 = y^2$ in $\mathbb{A}_{\mathbb{R}}^2$.
4. Take $V = \mathcal{Z}(xy - 1)$ and $W = \mathcal{Z}(v - u^2)$. We can define a map between global sections:

$$\varphi : \frac{k[u, v]}{(v - u^2)} \rightarrow \frac{k[x, y]}{(xy - 1)} \quad u \mapsto \frac{x + y}{2}$$

and extend k -linearly (note that $v \mapsto \left(\frac{x+y}{2}\right)^2$). As the left hand side is a free k -algebra up to the quotient relation, let's insure the relation is preserved, and indeed:

$$\varphi(0) = \varphi(v - u^2) = \frac{(x + y)^2}{2} - \frac{(x + y)^2}{2} = 0$$

showing that the relation is preserved, and hence φ is well-defined. The associated map on the varieties is given by:

$$(a, 1/a) \mapsto \left(\frac{a + 1/a}{2}, \left(\frac{a + 1/a}{2} \right)^2 \right)$$

Note the image is 2-to-1. If $k = \mathbb{R}$, the image is *not* surjective (ex. nothing maps to 0). However if $i \in k$ (where $i^2 = -1$), then $i + 1/i = i - i = 0$, which shows the map is surjective and 2-to-1. This demonstrates part of the importance of working over an algebraically closed field.

(give examples, and then show in the examples that the representation of a variety is “coordinate dependent”, and so we will define coordinate rings in a “coordinate-free” way)

Theorem 1.2.20: Regular Map Algebraic Correspondence

Let $V \subseteq \mathbb{A}^n$ and $W \subseteq \mathbb{A}^m$. then there is a bijective correspondence between the morphisms between V and W and the associated k -algebra homomorphisms, more precisely

$$\{\text{regular functions from } V \text{ to } W\} \leftrightarrow \{k\text{-algebra homomorphisms from } k[W] \text{ to } k[V]\}$$

$$\left\{ \begin{array}{c} \text{regular functions} \\ \varphi : V \rightarrow W \end{array} \right\} \xrightleftharpoons[\psi^a]{f \circ \varphi} \left\{ \begin{array}{c} k\text{-algebra homomorphism} \\ \varphi^\# : k[W] \rightarrow k[V] \end{array} \right\}$$

More generally, if $k\text{-}\mathbf{Aff}$ is the category whose objects are algebraic sets and morphism are regular functions and $k\text{-}\mathbf{Alg}_0$ is the category whose objects are coordinate rings (reduced, commutative, finite-type k -algebras) and morphisms are k -homomorphisms, then $k\text{-}\mathbf{Aff}^{\text{op}} \cong k\text{-}\mathbf{Alg}$.

Proof:

it as an exercise

Thus, if $f : X \rightarrow Y$, then $(f^\#)^a = f$ and if $\varphi : k[Y] \rightarrow k[X]$, then $(\varphi^a)^\# = \varphi$. Similarly to the dual theory provided in section ??, the dual map gives information about our original map:

Proposition 1.2.21: Regular Map Properties

Let $\varphi : X \rightarrow Y$ be a regular map with $\varphi^\# : k[Y] \rightarrow k[X]$ the associated k -algebra homomorphism. Then:

1. The kernel of $\tilde{\varphi}$ is $\mathcal{I}(\varphi(X))$: $\ker \tilde{\varphi} = \mathcal{I}(\varphi(X))$. Furthermore, the Zariski closure of $\varphi(X)$ is $\mathcal{Z}(\ker(\tilde{\varphi}))$: $\overline{\varphi(X)} = \mathcal{Z}(\ker(\tilde{\varphi}))$.
2. As a consequence, the homomorphism $\tilde{\varphi}$ is injective if and only if $\varphi(X)$ is Zariski dense in Y , or equivalently if $\overline{f(X)} = Y$
3. $\varphi(v) = w$ if and only if $\varphi^\#(\mathcal{I}(v)) = \mathcal{I}(w)$

Proof:

exercise.

Exercise 1.2.2

1. Show that $k[X \times Y] \cong k[X] \otimes_k k[Y]$ for X, Y algebraic sets. Show this is another way of showing that $\mathbb{A}_k^n \times \mathbb{A}_k^m \cong_{\mathbf{Reg}} \mathbb{A}_k^{n+m}$
2. Show that $X \times Y$ for algebraic sets X, Y is unique up to isomorphism, that is if $\varphi : X \rightarrow X'$ and $\psi : Y \rightarrow Y'$ are isomorphism, there is a unique isomorphism from $X \times Y$ to $X' \times Y'$ commuting with φ and ψ .
3. We mentioned that $\mathbb{A}_k^n$ is independent of the origin. Let $T : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$ be a linear isomorphism where $T(p) = q$. Show that $T^\# : \mathcal{O}_Q(\mathbb{A}_k^n) \rightarrow \mathcal{O}_p(\mathbb{A}_k^n)$ is an isomorphism. If V is a variety,

show that this induces an isomorphism $\mathcal{O}_Q(V) \rightarrow \mathcal{O}_P(V^T)$.

1.2.4 Closed Sets and Radical Ideals

As we've seen in proposition 1.1.7, given some polynomial f or collection of polynomials $S \subseteq k[x_1, \dots, x_n]$, the set of zero's of these polynomials does not change if we take higher powers of the polynomials (ex. $f, f^2, f^3, \dots$ all produce the same zero set). Thus, studying the set $\mathcal{Z}(I^n)$ is equivalent to studying $\mathcal{Z}(I)$. This property inspires the following definition:

Definition 1.2.22: Radical Ideal

Let $I \subseteq R$ be an ideal of a commutative ring R . Then:

1. The *radical of I* in R is the collection elements for which some power of them lie in I , that is:

$$\sqrt{I} = \text{rad}(I) := \{a \in R \mid a^k \in I \text{ for some } k \in \mathbb{N}\}$$

here $\text{rad}(I)$ stands for the radical of I . It is often written as $\sqrt{I}$.

2. The radical of the zero ideal ($\text{rad}(0) = \sqrt{(0)}$) is called the *nilradical* of R
3. An ideal I is called a *radical ideal* if $I = \text{rad}(I)$ ($I = \sqrt{I}$)

Note that the nilradical by definition contains all the nilpotent elements of R , and that $\text{rad}(\text{rad}(I)) = \text{rad}(I)$.

Proposition 1.2.23: $\text{rad}(I)$ Is An Ideal

Let $I \subseteq R$ be an ideal of a commutative ring. Then $\text{rad}(I)$ is indeed an ideal (containing I). Furthermore, $\text{rad}(I)/I$ is the nilradical of R/I .

The last statement shows us that R/I has no nilpotent elements if and only if $I = \text{rad}(I)$.

Proof:

Exercise

A type of ideal in which we've seen a similar property is the prime ideal, in which $ab \in I$ if and only if $a \in I$ or $b \in I$. One important way of looking at radical is in how they correspond to prime ideals. The prime ideals in a sense "generate" the radical ideals:

Proposition 1.2.24: Radical Through Prime Ideals

Let $I \subsetneq R$ be a proper ideal of a commutative ring and $\text{rad}(I)$ the radical ideal of I . Then $\text{rad}(I)$ is the intersection of all prime ideals containing I . In particular, the nilradical is the intersection of all prime ideals of R .

Proof:

Since I is the intersection of all prime ideals containing I if and only if (0) in R/I is the intersection of all prime ideals of R/I (by proposition 1.2.23), it suffices to prove it in the case of $I = (0)$ (i.e. that the nilradical is the intersection of all prime ideals of R). Let

$$N' = \bigcap_{\substack{P \subseteq R \\ P \text{ prime}}} P$$

Let N represent the nilradical.

($N \subseteq N'$) Let $x \in N$ so that there exists some n such that $x^n = 0$. If P is any prime, then $0 = x^n \in P$. Since P is prime, then either $x \in P$ or $x^{n-1} \in P$. If we assume $x^{n-1} \in P$, we can continue this process until we get $x^2 \in P$, so it must be that $x \in P$. Since P and x were arbitrary, we get that $x \in N'$.

($N \supseteq N'$) We'll use Zorn's lemma and show that if $a \notin N$, then $a \notin N'$. Given a is not an element of the nilradical ideal, define the family of ideals $\mathcal{S}$ that does not contain any power of a :

$$\mathcal{S} = \{I \subseteq R \mid \forall k \in \mathbb{N}_{>0}, a^k \notin I\}$$

This collection is nonempty since $(0) \in \mathcal{S}$ (since $a \notin N$). Furthermore, if a^k (for any k) is not contained in any ideal in a chain $I_1 \subseteq I_2 \subseteq \dots$, then it is not contained in the union of those elements. Thus, chains in $\mathcal{S}$ have upper bounds. By Zorn's lemma, $\mathcal{S}$ must contain a maximal element, say P . I claim that P is in fact a prime ideal. Suppose for contradiction that $xy \in P$ but $x, y \notin P$. By maximality of P , it must be that the larger ideals $(x) + P$ and $(y) + P$ contains a power of a , say $a^n \in (x) + P$ and $a^m \in (y) + P$. But then:

$$a^{n+m} \in (xy) + P = P$$

contradicting the fact that $P \in \mathcal{S}$. Thus, P is a prime ideal that does not contain a , and so $a \notin N'$, as we sought to show.

Corollary 1.2.25: Prime Or Maximal Then Radical

Let $I \subseteq R$ be a prime ideal. Then I is a radical ideal

Proof:

Take

$$P = \sqrt{P}$$

Example 1.2.26: Radical Ideals

1. Let R be a UFD. Given an ideal, find its radical.
2. The ideal $(x^3 - y^2)$ is prime in $k[x, y]$, and hence radical

3. If $l_1, l_2, \dots, l_m$ is a collection of linear polynomials in $k[x_1, \dots, x_n]$, then $I = (l_1, l_2, \dots, l_m)$ is either $k[x_1, x_2, \dots, x_n]$ or a prime ideal, and hence radical.

In the context of this chapter, since we are working over polynomial rings, we are working over Noetherian rings. Due to this, the elements of a radical ideal $\text{rad}(I)$ can be related back to elements of I quite naturally:

Proposition 1.2.27: Radical Ideals In Noetherian Rings

Let R be a noetherian ring and $I \subseteq R$ an ideal of R . Then there exists a $k \in \mathbb{N}$ such that $I^k \subseteq \text{rad}(I)$. In particular, if N is the nilradical ideal of R , then $N^k = 0$ for some $k \in \mathbb{N}$ (i.e. N is a nilpotent ideal)

Proof:

Let $I \subseteq R$ be any ideal and $\text{rad}(I)$ its corresponding radical ideal. Since R is Noetherian, $\text{rad}(I)$ is finitely generated, say $\text{rad}(I) = (a_1, a_2, \dots, a_n)$. By definition of $\text{rad}(I)$, for each a_i there exists a k_i such that $a_i^{k_i} \in I$. Take $k = \max(k_i)$. Then each element of $(\text{rad}(I))^{kn}$ is generated by elements of the form $a_1^{d_1} \cdots a_n^{d_n}$ where $d_1, d_2, \dots, d_n = kn$ and at least one factor must have $d_i \geq k$. Then by construction, for that factor, $a_i^{d_i} \in I$, and hence each generator of $(\text{rad}(I))^{kn}$ is in I , showing that

$$(\text{rad}(I))^{kn} \subseteq I$$

as we sought to show.

Now, returning to the fact that $\mathcal{Z}(I) = \mathcal{Z}(I^k)$, we now have that $\mathcal{Z}(I) = \mathcal{Z}(\sqrt{I})$. Remember how we were trying to characterize the ideals for which $\mathcal{Z}$ and $\mathcal{I}$ are inverses, i.e. what are the ideals $\mathcal{I}(\mathcal{Z}(I))$? We can now state one direction of this:

Proposition 1.2.28: Radical and Algebraic Sets

Let $I \subseteq R$ be an ideal where R is a polynomial ring. Then

$$\sqrt{I} \subseteq \mathcal{I}(\mathcal{Z}(I))$$

Proof:

exercise

It would be great if we could get $\sqrt{I} = \mathcal{I}(\mathcal{Z}(I))$, however this is not true in general: consider $I = (x^2 + 1) \subseteq \mathbb{R}[x]$. Since $x^2 + 1$ is irreducible, it is a maximal ideal, and so $I = \sqrt{I}$. Then as we are over the reals, $\mathcal{Z}(I) = \emptyset$, so $\mathcal{I}(\mathcal{Z}(I)) = \mathbb{R}[x]$. But $I \not\subseteq \mathbb{R}[x]$, showing equality doesn't hold. The blocker here was that $x^2 + 1$ doesn't factor in $\mathbb{R}[x]$: if we were working over $\mathbb{C}$, then we would get that $(x^2 + 1) = ((x + i)(x - i))$, telling us that $\mathcal{Z}(I) = \{i, -i\}$, which by the basic properties of $\mathcal{I}$ we have that $\mathcal{I}(\mathcal{Z}(I)) = ((x - i)(x + i)) = (x^2 + 1) = I$, showing us in this case that $I = \sqrt{I}$. In fact, the underlying field k being algebraically closed is the *only* other condition we need for equality to hold; this will be theorem 1.2.46 (Hilbert's Nullstellensatz). It is tricky to show this, since it is in a sense a very simple condition: All we need to characterize the zero sets of the polynomials (over a closed field) $\mathcal{Z}(I)$ in terms of an ideal is to take the radical of I .

1.2.5 Affine Variety and Prime Ideals

We now return to the earlier fact that $\mathcal{Z}(I) = \mathcal{Z}(f_1, \dots, f_n) = \mathcal{Z}(f_1) \cap \dots \cap \mathcal{Z}(f_n)$. This tells us that we can decompose algebraic sets into smaller components. We will therefore take a moment to see if we can find “decomposition” properties of algebraic sets, and see if there is a corresponding property the coordinate ideals will pose

Definition 1.2.29: [Affine] Variety

let $V \subseteq \mathbb{A}^n$ be an algebraic set. The set V is reducible if for $V_1, V_2 \subseteq V$ where $V_1, V_2 \neq V$

$$V = V_1 \cup V_2$$

If there does not exist such sets, we say that V is *irreducible*, or more commonly an *algebraic variety*

1.2.30 Remark Note that this notion of irreducibility is in fact a topological notion, we simply choose the Zariski topology. In particular, X is irreducible if $X = X_1 \cup X_2$ for some closed subsets X_1 and X_2 , then either $X = X_1$ or $X = X_2$. For those who are curious, it may be shown that irreducibility implies connected. Furthermore, the condition translates to open sets to say that if

$$\emptyset = U_1 \cap U_2$$

where U_1, U_2 open, then either $U_1 = \emptyset$ or $U_2 = \emptyset$. It follows that it is equivalent to say that every open set is *dense* in an irreducible space. This contrasts heavily with any Hausdorff space, since by definition a Hausdorff space *must* have disjoint open subsets if the space is not empty or a singleton.

The name *algebraic variety* comes from HERE. Note that this condition automatically implies that V_1 and V_2 are non-empty (do you see why?). If V is an irreducible algebraic set, then the corresponding ideal has a very nice property:

Proposition 1.2.31: Coordinate Ideal of Algebraic Variety

A set $V \subseteq \mathbb{A}^n$ is an algebraic variety if and only if $I(V)$ is a prime ideal

Proof:

($\Rightarrow$) Taking the contrapositive, let's assume $I(V)$ is not prime. Then there exists $a, b \in k[x_1, \dots, x_n]$ such that $ab \in I(V)$ but $a \notin I(V)$ and $b \notin I(V)$. But then:

$$V = (V \cap \mathcal{Z}(a)) \cup (V \cap \mathcal{Z}(b))$$

with $V \cap \mathcal{Z}(a) \subsetneq V$, $V \cap \mathcal{Z}(b) \subsetneq V$, showing that V is in fact reducible

($\Leftarrow$) Doing the contrapositive again, let's assume V is reducible, so $V = V_1 \cup V_2$, where both $V_i \subsetneq V$. Then by the contra-positive correspondence of inclusion between algebraic sets and ideals, $I(V) \subsetneq I(V_i)$, $i \in \{1, 2\}$. Since the inclusion is proper, choose $a \in I(V_1)$ and $b \in I(V_2)$ such that $a, b \notin I(V)$. Then since a and b are polynomials which are zero on

their respective algebraic sets and $V = V_1 \cup V_2$, $ab \in I(V)$. However, $a, b \notin I(V)$, showing that $I(V)$ is not prime

Corollary 1.2.32: Equivalent characterization

Let $V \subseteq \mathbb{A}^n$ be an algebraic set. Then V is an algebraic variety if and only if $k[V]$ is an integral domain

This means there is a certain correspondence between being a prime and being irreducible as a variety. To deepen the connection, let's prove that every algebraic set can be decomposed into algebraic varieties:

Theorem 1.2.33: Irreducible Decomposition

Let $V \subseteq \mathbb{A}^n$ be an algebraic set. Then there exists a unique set of algebraic varieties $V_1, \dots, V_n$ such that

$$V = \bigcup_i V_i \quad V_i \not\subseteq V_j \quad i \neq j$$

Proof:

First, recall that since $k[x_1, \dots, x_n]$ is Noetherian, any collection of ideals contains a maximal ideal. By the correspondence between ideals and algebraic sets, there exists a corresponding minimal algebraic set which is not contained in any other algebraic set (or else its corresponding ideal would contain the maximal ideal, a contradiction).

Now, Let $S = \{V \subseteq \mathbb{A}^n \mid V \text{ is not the union of finitely many algebraic varieties}\}$. If S were empty, we're done, so for the sake of contradiction let's assume it's not. Then the corresponding set of ideals must contain a maximal ideal, so S contains a corresponding minimal algebraic set. Since $V \in S$, V cannot be a variety, and so there exists a decomposition $V = V_a \cup V_b$. Since V_a and V_b are not in S (or else that would contradict minimality), they both admit a finite decomposition into irreducibles, say:

$$V_a = V_{1,1} \cup \dots \cup V_{1,n_1} \quad V_b = V_{2,1} \cup \dots \cup V_{2,n_2}$$

But then

$$V = \bigcup_{i,j} V_{i,j}$$

showing that V is in fact decomposable into irreducibles, a contradiction to the fact that $V \in S$. Thus, S must be empty.

If it happens that $V_{i,j} \subseteq V_{i',j'}$ for appropriate indices, simply discard the smaller one, so that the remaining $V_1, \dots, V_n$ will have the property of $V_i \not\subseteq V_j$ for $i \neq j$, fulfilling the 2nd condition.

For uniqueness, suppose that there was a second decomposition $V = W_1 \cup \dots \cup W_m$. Then by some set theory manipulation, we see that:

$$V_i = \bigcup_j (W_j \cap V_i)$$

so $V_i \subseteq W_{j(i)}$ for some appropriate $j(i)$. Symmetrically, we have that $W_{j(i)} \subseteq V_k$, implying $V_i \subseteq V_k$. But since no proper-subset exists, it must be that $i = k$ and $V_i = V_k$, so $V_i = W_{j(i)}$,

meaning each V_i equals to some $W_{i(j)}$. Symmetrically, each W_j is equal to some $V_{i(j)}$, completing the proof.

Definition 1.2.34: Irreducible Decomposition

Let V be an algebraic set. Then the *irreducible decomposition* of V is

$$V = V_1 \cup \cdots \cup V_n \quad V_i \not\subseteq V_j \text{ if } i \neq j$$

for appropriate algebraic varieties V_i , $1 \leq i \leq n$

1.2.35 Remark In a more general topological setting, it can be shown that if X is a Noetherian space (i.e. satisfies A.C.C. on open sets, so D.C.C. on closed sets), X can be decomposed into irreducible sets just liked was done in theorem 1.2.33. It then follows easily that any closed subset of X has this property too.

Example 1.2.36: Affine Varieties

1. All linear polynomials produce affine varieties. Thus, the study of linear algebra can be thought as the study of varieties with multiplicity 1.
2. $\mathcal{Z}(xy - 1) \subseteq \mathbb{A}_{\mathbb{R}}^2$ is an affine variety, as $\mathbb{R}[x, y]/(xy - 1) \cong \mathbb{R}[x, 1/x]$. Note that $xy - 1$ has two branches, however the variety is still irreducible and connected in the Zariski topology. Show that there is no variety that is only one branch of $xy - 1$ (show that if two affine varieties agree on an open subset, they agree everywhere).

Irreducible sets and Primary Decomposition

let us now parallel the discussion of irreducible decomposition using rings.

(Note how this would imply the intersection of two primary ideals is not primary, and that is indeed the case: $(x), (y) \subseteq R[x, y]$ are prime ideals if R is integral, however $(x) \cap (y) = (xy)$, but (xy) is clearly not prime (x and y is not in (xy)). However, this does imply the intersection of radical ideals is radical)

In the integers any $n \in \mathbb{Z}$ can be factored into $n = p_1^{n_1} \cdots p_k^{n_k}$. for primes p_i . Unique factorization does not hold in general integral domains, however unique factorization of ideals holds in Noetherian rings! The generalization of prime power is a *primary ideal*.

Definition 1.2.37: Primary Ideal

Let R be a ring and $I \subseteq R$ an ideal. Then I is said to be a *primary ideal* if

$$xy \in I \Rightarrow x \in I \quad \text{or} \quad y^n \in I \text{ for some } n > 0$$

Translated into quotients, this is saying

$$I \text{ is primary} \quad \Leftrightarrow \quad R/I \neq 0 \text{ and every zero-divisor in } R/I \text{ is nilpotent}$$

Certainly every prime ideal is primary, and the contradiction of a primary ideal is primary since if $f : A \rightarrow B$ is a homomorphism and I is a primary ideal of B , then $A/I^c \cong B/I$.

Lemma 1.2.38: Radical Of Primary Ideal

let $I \subseteq R$ be a primary ideal. Then $\sqrt{I}$ is the smallest prime ideal containing I

Proof:

It suffices to show that $\sqrt{I}$ is prime. Let $xy \in \sqrt{I}$ so that $(xy)^m \in I$. Then for some $m > 0$, either $x^m \in I$ or $y^m \in I$ for some $n > 0$. But that's the same as either $x \in \sqrt{I}$ or $y \in \sqrt{I}$, completing the proof.

Example 1.2.39: Primary Ideals

1. In $\mathbb{Z}$, the primary ideals are (0) and (p^n) for prime p .
2. In $A = k[x, y]$ take $q = (x, y^2)$. Then $A/q \cong k[y]/(y^2)$, showing q is primary. The radical of q is (x, y) . Notice that $\text{rad}(I)^2 \subseteq q \subseteq \text{rad}(I)$, showing that primary ideals are not necessarily a prime-power.
3. Note that if P is a prime ideal, P^n is *not* necessarily a primary ideal, even though its radical is the prime ideal p . For example, let $A = k[x, y, z]/(xy - z^2)$ and $\bar{x}, \bar{y}, \bar{z}$ are the images of x, y, z . Then $p = (\bar{x}, \bar{y})$ is prime ($A/p \cong k[y]$ is an integral domain) and $\bar{x}\bar{y} = \bar{z}^2 \in p^2$, but $\bar{x} \notin p^2$ and $\bar{y} \notin \sqrt{p^2} = p$. Thus, p^2 is not primary.

We do get a partial converse if our ideal is maximal

Proposition 1.2.40: Radical Maximal, Then Primary

Let $I \subseteq R$ be an idealst $\sqrt{I}$ is maximal. Then I is primary. Hence, the power of a maximal ideal is primary

Proof:

Let $\sqrt{I} = m$. Then the image of m in A/I is the nilradical of A/I , hence A/I has only one prime ideal. Then every element of A/I is either a unit or nilpotent, so every zero-divisor of A/I is nilpotent, completing the proof.

Theorem 1.2.41: Lasker-Noether Theorem

Let A be a Noetherian domain of dimension 1. Then every non-zero ideal $\mathfrak{a} \subseteq A$ can be uniquely expressed as a product of primary ideals whose radicals are all distinct

Proof:

exercise

Corollary 1.2.42: Prime Ideals In Noetherian Rings

Let R be a Noetherian ring, $I \subseteq R$ be an ideal, $P \subseteq R$ a prime ideal. Then:

1. $I \subseteq P$ if and only if P contains one of the associated primes of I . In other words, $I \subseteq P$ if and only if P contains one of the isolated prime of I , i.e., the isolated primes of I that are precisely the minimal elements in the set of all prime ideals containing I . In particular, there are only finitely many minimal elements among the prime ideals containing I
2. The radical of I is the intersection of the associated prime ideal of I , and so also the intersection of the isolated prime ideals of I
3. There exists prime ideals $P_1, \dots, P_n$ (not necessarily distinct) containing I such that

$$P_1 P_2 \cdots P_n \subseteq I$$

Proof:

1. here
2. here
3. For the sake of contradiction, let's say there exist ideal(s) which do not contain a product of prime ideals. Then such a set of ideals, say $\mathcal{S}$ is non-empty and so it has a maximal member M . Then M cannot be prime since then it's trivially be a product of prime ideals, and so there must exist $r, s \in R \setminus M$ such that $rs \in M$ but $r, s \notin M$. Then the ideal $M + (r)$ and $M + (s)$ are strictly bigger than M and so contain a product of prime ideals say $P = p_1, p_2, \dots, p_n$. Hence, so does the product of these ideals. However:

$$P \subseteq (M + (r))(M + (s)) \subseteq M$$

a contradiction, hence every ideal contains a product of prime ideals.

Exercise 1.2.3

1. Show that the product of two irreducible sets is irreducible.
2. Show that the intersection of two irreducibles sets need not be irreducible (hint: consider the varieties given by $x^2 - yw = 0$ and $xy - zw = 0$; show their intersection gives a twisted cubic curve and a line).

1.2.6 Computation of Algebraic Sets and k-algebras

That title is straight up from Dummit and Foote. It basically uses Gröbner bases to find such values. This feels like something to eventually understand.

1.2.7 Hilbert's Nullstellensatz

In this section, we shall make precise the correspondence between which ideals of the coordinate ring of an affine set correspond to closed subsets of the affine set. In this exploration, we shall also produce a new tower condition on ring extensions via Noether's Normalization theorem which builds upon the theme of separating an extension to "property" and "not property" (ex. separable and inseparable, algebraic and transcendental¹⁹) Recall if $a_1, a_2, \dots, a_n \in S$ for a k -algebra S , these elements are called *algebraically independent* if

$$k[x_1, x_2, \dots, x_n] \rightarrow k[a_1, a_2, \dots, a_n] \quad x_i \mapsto a_i$$

is an isomorphism. Noether's Normalization theorem is the generalization of the idea that any field extension can be split into a transcendental extension, followed by an algebraic extension:

Theorem 1.2.43: Noether's Normalization Theorem

Let k be a field and suppose that $A = k[r_1, \dots, r_n]$ is a finitely generated k -algebra. Then for some q , $0 \leq q \leq n$, there are algebraically independent elements $y_1, y_2, \dots, y_q \in A$ such that A is integral over $k[y_1, y_2, \dots, y_q]$

Proof:

If $r_1, r_2, \dots, r_n$ are algebraically independent, then A is an integral extension of itself. Otherwise, if $r_1, \dots, r_n$ are algebraically *dependent*, then there exists some non-trivial polynomial $f(x_1, \dots, x_n) \in k[x_1, x_2, \dots, x_n]$ such that $f(r_1, r_2, \dots, r_n) = 0$. f is a multinomial, monomials of the form $ax_1^{m_1}x_2^{m_2}\dots x_n^{m_n}$ which have degree $m_1 + \dots + m_n$. The degree of f is taken to be the maximum of the degree of all the monomials. Renumbering the polynomial if necessary, we assume that f is a non-constant polynomial in x_n (non-constant since $r_1, r_2, \dots, r_n$ are algebraically dependent) with coefficients in $k[x_1, \dots, x_{n-1}]$

Now, we will perform a sort of change of variables to f that "normalizes" f into a *monic polynomial* in x_n (giving the namesake to this theorem). The coefficients of this f as a polynomial in x_n will form a subring A generated over k by $m-1$ elements, at which point we shall imply induction (the base case will be trivial).

First, define the integers $\alpha_i = (1+d)^i$. Using these, define the variables $X_i = x_i - x_n^{\alpha_i}$, $1 \leq i \leq m-1$. Define a new polynomial:

$$g(X_1, \dots, X_{m-1}, x_n) = f(X_1 + x_n^{\alpha_1}, \dots, X_{m-1} + x_n^{\alpha_{m-1}}, x_n)$$

so that $g \in k[X_1, \dots, X_{m-1}, x_n]$. The key property of this polynomial is the following: for each monomial term in f , it contributes a single term of the form a constant (in $k[x_1, \dots, x_{m-1}]$) times x_n^e to g . Importantly, the choice of α_i insured that distinct monomials in f give different values of e ; we can view the degrees of the monomials in the new variables as integers expressed in base $d+1$. Let's say N is the highest power of x_n . Then we get that:

$$g = cx_n^N + \sum_{i=1}^{N-1} h_i(X_1, \dots, X_{m-1})x_n^i$$

¹⁹Another such example will appear in the next chapter, and shall be ramified and unramified

for some nonzero $c \in k$. Returning to the original fact that $f(r_1, \dots, r_n) = 0$, letting $s_i = r_i - r_n^{\alpha_i}$, we get:

$$\frac{1}{c}g(s_1, \dots, s_{n-1}, r_n) = \frac{1}{c}f(r_1, \dots, r_{n-1}, r_n) = 0$$

showing that r_n is integral over $B = k[s_1, \dots, s_{n-1}]$. Furthermore, each r_i is integral over $B[r_n]$ since r_i is the root of the monic polynomial $x - (s_i + r_n^{\alpha_i})$, thus A integral over $B[r_n]$. By transitivity of integrality, A is integral over B . Since B is a k -algebra generated by $n-1$ elements, by the principle of induction (HERE) (Dummit simply concludes here)

Noether's Normalization can be thought of as a very geometric result: we shall see that any finitely generated k -algebra shall be an n -dimensional affine set along with some extra important information given by the integral extension, more on this in section 1.4. One very important is that if we have an extension F/k that is a finite-type k -algebra, is finitely generated if we allow for both addition and multiplication, it is finitely generated by *just* allowing addition!!

Theorem 1.2.44: Zariski's Lemma

Let F/k be a field extension, and assume F is a finite-type k -algebra. Then F/k is a finite (and hence algebraic) extension.

Proof:

Since F is a finite-type k -algebra, $F = k[r_1, r_2, \dots, r_n]$. By Noether's Normalization Theorem, there exists a $q \in \mathbb{N}$ (possibly 0) enumerating variables $y_1, \dots, y_q \in F$ such that F is integral over $k[y_1, y_2, \dots, y_q]$, i.e. F is a finitely generated $k[y_1, \dots, y_q]$ -module. We can re-phrase this by saying we have a finite integral extension $F/k[y_1, y_2, \dots, y_q]$. By ??, since F is an integral domain, F is a field if and only if $k[y_1, y_2, \dots, y_q]$ is a field. But then, $k[y_1, y_2, \dots, y_q]$ is a field if and only if there are no indeterminate variables, i.e. $q = 0$. Thus, we get the finite integral extension F/k , which is equivalent to saying we get a finite field extension, as we sought to show.

The last two theorems are true no matter what field we are working over. The following theorem requires k to be algebraically closed.

Theorem 1.2.45: Hilbert's Nullstellensatz - Weak Version

Let k be an algebraically closed field. Then M is a maximal ideal of the polynomial ring $k[x_1, \dots, x_n]$ if and only if

$$M = (x_1 - a_1, x_2 - a_2, \dots, x_n - a_n)$$

for any $a_1, a_2, \dots, a_n \in k$. As a result, the maps:

$$\{\text{Points in } \mathbb{A}^n\} \xrightleftharpoons[\mathcal{Z}]{\mathcal{I}} \{\text{Maximal ideals in } \Gamma(\mathbb{A}^n)\}$$

are in bijective correspondence, and if $I \subseteq \Gamma(\mathbb{A}^n)$ is a proper ideal, then $\mathcal{Z}(I) \neq \emptyset$

Proof:

Certainly, $(x_1 - a_1, \dots, x_n - a_n)$ is a maximal ideal in $k[x_1, x_2, \dots, x_n]$ since its quotient is k . Conversely, suppose M is a maximal ideal in $k[x_1, x_2, \dots, x_n]$. Consider $F = k[x_1, x_2, \dots, x_n]/M$. Then since $k \subseteq F$, F/k is a field extension that is also a finitely generated k -algebra (generated by $\overline{x_1}, \dots, \overline{x_n}$). By Zariski's lemma, F/k is a finite field extension, and hence is algebraic over k .

Now, we take advantage of the fact that k is algebraically closed, meaning there does not exist a non-trivial algebraic extension of k (??), implying that $F \cong_k k$, in particular we $\overline{x_i} \mapsto a_i$ for some $a_i \in k$. Thus, this is a k -algebra homomorphism, and $\overline{a_i} \in F$, $\overline{x_i} - \overline{a_i} \mapsto a_i - a_i = 0$, which since the two fields are isomorphic, they are in bijective correspondence, implying $\overline{x_i} - \overline{a_i} = \overline{0}$, i.e. $x_i - a_i \in M$. Thus, we must have $x_i - a_i \in M$ for every x_i , implying that $(x_1 - a_1, \dots, x_n - a_n) \subseteq M$, which by maximality tells us that $(x_1 - a_1, \dots, x_n - a_n) = M$.

As a consequence, if I is any ideal of $k[x_1, x_2, \dots, x_n]$, then $I \subseteq M$ for some maximal ideal M . By what we've just shown, for some $a_1, \dots, a_n$, $M = (x_1 - a_1, \dots, x_n - a_n)$. Thus, $(a_1, \dots, a_n) \in \mathcal{Z}(I)$, showing that $\mathcal{Z}(I)$ is non-empty.

Theorem 1.2.46: Hilbert's Nullstellensatz

Let k be an algebraically closed field. Then for any ideal $I \subseteq k[x_1, x_2, \dots, x_n]$,

$$\mathcal{I}(\mathcal{Z}(I)) = \sqrt{I} = \text{rad}(I)$$

Furthermore, the maps $\mathcal{I}$ and $\mathcal{Z}$ are order-reversing bijections:

$$\{\text{Affine Varieties}\} = \{\text{Closed Sets}\} \begin{matrix} \xrightarrow{\mathcal{I}} \\ \xleftarrow{\mathcal{Z}} \end{matrix} \{\text{Radical Ideals}\}$$

Proof:

First, we've already shown that $\text{rad}(I) \subseteq \mathcal{I}(\mathcal{Z}(I))$ (proposition 1.2.28), so it remains to show that $\text{rad}(I) \supseteq \mathcal{I}(\mathcal{Z}(I))$. By Hilbert's basis theorem, we have that $I = (f_1, f_2, \dots, f_n)$. Let $g \in \mathcal{I}(\mathcal{Z}(I))$. If we show that $g^N \in I$ for some $N \in \mathbb{N}_{>0}$, this would imply $g \in \text{rad}(I)$. To that end, we employ the following trick known as the Rabinowitsch trick: introduce a new variable x_{n+1} and consider the ideal:

$$(f_1, \dots, f_n, x_{n+1}g - 1) = I' \subseteq k[x_1, \dots, x_n, x_{n+1}]$$

By definition of g , at any point where each $f_1, f_2, \dots, f_n$ vanish, g also vanishes (since $g \in \mathcal{I}(\mathcal{Z}(I))$). But then, $x_{n+1}g - 1$ is nonzero! Thus, $\mathcal{Z}(I') = \emptyset$ in $\mathbb{A}^{n+1}$, and by Hilbert's weak Nullstellensatz, I' cannot be a proper ideal, and so $1 \in I'$. Since $1 \in I'$, we can write it as:

$$1 = a_1 f_1 + \dots + a_n f_n + a_{n+1}(x_{n+1}g - 1) \quad a_i \in k[x_1, \dots, x_{n+1}]$$

For the following steps, it is insightful to expand the equation:

$$1 = \sum_i^n a_i(x_1, \dots, x_{n+1})f_i(x_1, \dots, x_n) + a_{n+1}(x_1, \dots, x_{n+1})(x_{n+1}g(x_1, \dots, x_n) - 1)$$

Now, since x_{n+1} is a free variable, this equation still holds true if we substitute x_{n+1} for another expression (it will simply limit the equation to solutions which also satisfy the substituted expression).

We will choose $x_{n+1} = 1/g(x_1, \dots, x_n)$ so we get the following expression in $K(x_1, \dots, x_n)$:

$$1 = \sum_i^n a_i \left(x_1, \dots, x_n, \frac{1}{g(x_1, \dots, x_n)} \right) f_i(x_1, \dots, x_n)$$

Importantly for us, the only terms in the denominator would be some powers of $g(x_1, \dots, x_n)$. Thus, for suitably large $N \in \mathbb{N}$, we may re-write the right hand side as:

$$1 = \sum_i^n \frac{b_i(x_1, \dots, x_n) f_i(x_1, \dots, x_n)}{g(x_1, \dots, x_n)^N}$$

given appropriately substituted $h_1, \dots, h_n \in k[x_1, \dots, x_n]$. Multiplying both sides by the denominator, we get that:

$$g(x_1, \dots, x_n)^N = \sum_i^n b_i(x_1, \dots, x_n) f_i(x_1, \dots, x_n)$$

Notice the right hand side is an element in the ideal I , and as $g^N \in I$! Thus, by definition, $g \in \text{rad}(I)$. Since g was arbitrary, we see that $\mathcal{I}(\mathcal{Z}(I)) \subseteq \text{rad}(I)$, and so we get $\mathcal{I}(\mathcal{Z}(I)) = \text{rad}(I)$. Furthermore, we want to show the given inverse bijections: If I is a radical ideal, then we've just shown that

$$I = \mathcal{I}(\mathcal{Z}(I))$$

conversely, if $A \subseteq k^n$ is an algebraic set, consider $\mathcal{I}(A)$. We want to check that $\mathcal{I}(A)$ is radical, so let's say $f^n \in \mathcal{I}(A)$, so $f^n(a)$ vanishes on all points of A . But then $f(a)$ vanishes at all points (since the field k has no nilpotent elements). Thus $f \in \mathcal{I}(A)$, showing that all algebraic sets map to radical ideals, as we sought to show.

Corollary 1.2.47: Nullstellensatz in Algebraic Closure

Let k be any field with algebraic closure $\bar{k}$, and $I \subseteq k[x_1, \dots, x_n]$ an ideal. Then:

$$\mathcal{I}_k(\mathcal{Z}_{\bar{k}}(I)) = \sqrt{I}$$

As a particular result, $I = (1)$ if and only if there is no common zeros in $\bar{k}^n$ of the polynomials in I

Proof:
exercise.

More on Algebra-Geometry Dictionary

With these results established, let us recap all the algebraic/geometric correspondences and prove some more. There are some immediate consequence of these result. An immediate easy result is that it can finally be concluded that:

$$\mathcal{I}(A \cap B) = \sqrt{\mathcal{I}(A) + \mathcal{I}(B)}$$

Next, we now have made precise the relation between radical ideals and prime ideals, as we have fully geometrically classified radical and prime ideals. Next, we get the following dictionary between the language of geometry and the language of algebra:

Geometry	Algebra
Affine sets V	coordinate rings $k[V]$
points of V	maximal ideals of $k[V]$
closed subsets of V	radical ideals of $k[V]$
sub-varieties of V	prime ideals of $k[V]$
regular function: $\varphi : V \rightarrow W$	k -algebra homomorphism $\tilde{\varphi} : k[W] \rightarrow k[V]$

We shall keep expanding this dictionary throughout this part of the book (i.e. part ??). As a teaser, we shall show that all PID's correspond to smooth curves (see section ??).

Now that we have established this definition between algebraic sets and it's algebraic counter-part, we can expand some more on the Zariski Topology:

Proposition 1.2.48: Zariski Topology is Compact

Let $\mathbb{A}_k^n$ have the Zariski topology. Then $\mathbb{A}_k^n$ is compact

Proof:

Let $\mathcal{O}$ index an open cover for $\mathbb{A}_k^n$

$$\mathbb{A}_k^n = \bigcup_{\alpha \in \mathcal{O}} U_\alpha$$

By proposition 1.2.4, any open set is the union of sets of the form $D(f)$, so we may re-write this as:

$$\mathbb{A}_k^n = \bigcup_{\alpha \in \mathcal{O}'} D(f_\alpha)$$

appropriately re-indexing. Then taking the compliment, we get that it's equivalent that:

$$\emptyset = \bigcap_{\alpha \in \mathcal{O}'} \mathcal{Z}(f_\alpha) = \mathcal{Z}\left(\sum_{\alpha \in \mathcal{O}'} (f_\alpha)\right)$$

By the Nullstellensatz, we get that it's equivalent that

$$1 \in \sum_{\alpha \in \mathcal{O}'} (f_\alpha)$$

Thus, for appropriate coefficients $g_i \in k$, we get that it's equivalent that:

$$1 = g_1 f_1 + \cdots + g_n f_n \in (f_1, \dots, f_n) = \sum_{i=1}^n (f_i)$$

But then, we get that

$$\emptyset = \bigcap_{i=1}^n \mathcal{Z}(f_i) \Leftrightarrow \mathbb{A}_k^n = \bigcup_{i=1}^n D(f_i)$$

Recall that for any algebraic subset $V \subseteq \mathbb{A}_k^n$, the corresponding coordinate ring $k[V]$ is used to define the subspace topology. Using the Nullstellensatz, we may deduce that it cannot have nilpotent elements:

Corollary 1.2.49: Coordinate Ring Always Reduced

Let $V \subseteq \mathbb{A}_k^n$ be an algebraic subset and $k[V]$ the associated coordinate ring. Then $k[V]$ is has no nilpotent elements

Proof:

Since $k[V] = k[x_1, x_2, \dots, x_n]/I(V)$, and $I(V)$ is a radical ideal by the Nullstellensatz, the quotient cannot contain any nilpotent elements, as we sought to show.

Corollary 1.2.50: Nullstellensatz For Subspaces

Let $V \subseteq \mathbb{A}_k^n$ be an algebraic subset (i.e. a closed subset) with corresponding coordinate ring $k[V]$. Then we can define $\mathcal{Z}$ and $\mathcal{I}$ as:

$$\left\{ \begin{array}{c} \text{affine algebraic subsets} \\ \text{of } V \end{array} \right\} \begin{array}{c} \xrightarrow{\mathcal{I}} \\ \xleftarrow{\mathcal{Z}} \end{array} \left\{ \begin{array}{c} \text{Radical Ideals} \\ \text{of } k[V] \end{array} \right\}$$

Proof:

This follows immediately from the correspondence theorem and the Nullstellensatz

Next, sticking to the case where $k = \bar{k}$ is algebraically closed, from this section we can deduce that any coordinate ring is finitely generated, nilpotent-free (i.e. reduced) and is a k -algebra over an algebraically closed field. We can show that all such rings appear as coordinate rings for an affine set.

We first have the following:

Definition 1.2.51: Reduced Ring

Let R be a ring. Then R is said to be *reduced* if $x^n = 0$ implies $x = 0$

(finish here)

Proposition 1.2.52: Coordinate Ring Correspondence

Let R be a ring. Then if R is:

1. Finitely generated
2. Reduced
3. over an algebraically closed field

then R is isomorphic to a coordinate ring for an appropriate dimension n of $\mathbb{A}_k^n$ and $\mathcal{Z}(S) \subseteq \mathbb{A}_k^n$.

Proof:

exercise. I think I should prob. write it down actually

Notice that through this result, we can define an affine set without the reference of an ambient space, similar to how we may define a manifold without the need an ambient embedding space. This idea shall be greatly generalized in section 1.9 and algebraic geometry more generally (see cite:HERE).

Exercise 1.2.4

1. Perhaps some on the Nullstellensatz over non-algebraically closed fields (In particular, I think if we take $\mathcal{Z}$ over the algebraic closure of k , then we recover the Nullstellensatz)

1.3 Rational Maps: Classifying Varieties

Let V be a variety and $k = \bar{k}$ be an algebraically closed field. In this section, we take advantage of the fact that we may localize $k[x_1, \dots, x_n]/\mathcal{I}(V)$. By Noether's Normalization Theorem (theorem 1.2.43) $k[V]/k$ is a finite transcendental extension followed by an integral extension, and so as $k[V]$ is an integral domain we may take its field of fractions and $k(V)/k$ is a transcendental extension generated by finitely many elements, hence $k(V) \cong k(X_1, \dots, X_q)$ for some generators X_i . Geometrically $k(V)$ shall represent all *rational polynomial functions*, or all rational polynomials functions defined on V . Note that we must be a bit more careful in interpreting elements of $k(V)$ as functions: if $f/g \in k(V)$, then if $g(p) = 0$ for some $p \in V$ then we get the undefined $f(p)/g(p) = f(p)/0$. We shall address this soon.

One of the important uses of rational functional shall be a for a slighter weaker condition than isomorphism to classify varieties. We shall see that being *birationally equivalent* is weaker than being a (regular) isomorphism, giving coarser isomorphism classes of varieties that shall give us more information about algebraic sets. As an example, elliptic curves²⁰ shall require three curve invariants to be classified (the genus, the j -invariant, and the Mordell-Weil group of the curve). However, up to birational map it shall only require the genus and we can without loss of generality work with smooth elliptic curves. Though coarser, it is nonetheless a powerful tool in gathering information about varieties.

Definition 1.3.1: Rational Functions

Let V be an affine variety. Then the fraction field of $k[V]$ is called the field of *rational function on V* , and is denoted $k(V)$. An element of $k(V)$ is called a *rational function*.

The *evaluation* of a regular function f/g at p is defined whenever $g(p) \neq 0$. Two rational functions are equal if they agree on evaluation at every point on which they are defined. The set of points where $g(p) = 0$ is called the *pole set*.

If $\varphi \in k(V)$, then $\varphi = \frac{f}{g}$ where $g \notin \mathcal{I}(V)$ (or else the denominator would be the zero function, contradicting the construction of the field of fractions). We shall say that two rational polynomial f/g and f'/g' represent the same rational function if and only if $fg' - f'g \in \mathcal{I}(V)$. Note that if $k[V]$ is not a UFD, then there can be representatives of $k(V)$ which are not equivalent up to a unit:

²⁰for those familiar with complex geometry, genus 1 curves

Example 1.3.2: Not UFD, Different Polynomial Representatives

Let $V = \mathcal{Z}(xw - yz) \subseteq \mathbb{A}_k^4$. Show that:

$$\frac{\bar{x}}{\bar{y}} = \frac{\bar{z}}{\bar{w}} = f \in k(V)$$

if $y \neq 0$ or $w \neq 0$.

Another way of constructing $k(V)$ is the following: take the collection

$$\mathcal{O}_V = \{\varphi = f/g \in k(x_1, \dots, x_n) \mid g \notin I(V)\}$$

then take the subset $\mathfrak{m}_V = \{\varphi = f/g \in \mathcal{O}_V \mid f \in I(V)\}$. Show that (exercise ref:HERE):

$$k(V) \cong \mathcal{O}_V / \mathfrak{m}_V$$

This shall become the dominant way of interpreting the set of rational functions when working with quasi-projective varieties (see 1.6.3)²¹.

A key difference between regular and rational functions is that a rational function need not be defined at every point. In fact, it *has* to be the case that a rational function is not defined on a finite set of points for it to not be equivalent to a regular function (when working over an algebraically closed field). To see this, consider the following definition:

Definition 1.3.3: Regular Point

Let $\varphi \in k(X)$ be a rational function. Then a *regular point* at $x \in X$ is a point where we can represent φ as f/g , $\varphi(x) = f(x)/g(x)$, and $g(x) \neq 0$. In this case, the element $f(x)/g(x)$ in k is called the *value* of φ at x and is denoted $\varphi(x)$.

For example, take $\varphi = 1/x$ on $\mathbb{A}_k^1$. Then all points except 0 are regular points. If $\varphi = f/g$ is regular at a point x , then it is regular in an open neighborhood of x , namely the neighborhood $D(g) \cap X$. Some authors put that a rational function is regular at a point if it is regular in an open neighborhood.

Proposition 1.3.4: Regular at All Points, then Regular

Let $\varphi \in k(X)$ be a rational function that is regular at all points. Then φ is a regular function

Proof:

Let $\varphi \in k(X)$ be a rational function that is regular at every point. Then for each $x \in X$, we can represent φ as f_x/g_x . Take $\mathfrak{a} = (g_x)_{x \in X}$. Then as $k[X]$ is Noetherian, this is a finitely generated ideal:

$$\mathfrak{a} = (g_{x_1}, \dots, g_{x_n})$$

Note that the g_{x_i} 's cannot have a common zero $x \in X$, or else all the functions in the ideal would vanish at x , however we know that $g_x(x) \neq 0$. Then this gives an equivalent result to the Nullstellensatz (see exercise ref:HERE), namely that:

$$\mathfrak{a} = (1)$$

²¹In Scheme theory, this shall be the rational functions at the point (closed or non-closed) associated to X

In particular, there exists $u_i \in k[X]$ such that

$$\sum_i^n u_i g_i = 1$$

Now, by what we've covered $\varphi = f_i/g_i$ on open neighborhoods U_i that cover X . Multiply both sides of the equation by φ , we get:

$$\varphi = \sum_i^n u_i f_i$$

showing it is a regular function, completing the proof.

The Nullstellensatz is essential for the proof, i.e. that k is an algebraically closed field, or else this result is false!

Example 1.3.5: Rational Function That is Regular

the function

$$\frac{x}{x+100}$$

is regular on $\mathcal{Z}(x^2 + y^2 - 1) \subseteq \mathbb{A}_{\mathbb{R}}^2$ even though it is not a polynomial: there does not exist a point (x, y) satisfying $x^2 + y^2 - 1$ such that $x + 100 = 0$ namely since all points x where $x^2 + y^2 - 1 = 0$ must be within $[-1, 1]$ ^a

^aThis failure is *not* present when working with the scheme $\text{Spec } \mathbb{R}[x, y]/(x^2 + y^2 - 1)$ as $x + 100$ has a zero at the prime ideal $(x - 100, y^2 + 9999)$, and so this “anomaly” is present if we don't work in the right context

Hence we always have points on which a rational function is not defined. This set of points will always be dense:

Lemma 1.3.6: Domain of Definition

Let $\varphi \in k(V)$ be a rational function. Then set of regular points is non-empty, open and dense.

Proof:

Let $\varphi = f/g$ and take $U = \{p \in V \mid g(p) \neq 0\} = D(g) \cap V$. Then in V , this set is certainly open, and as g is not the zero function there is certainly at least one point $p \in V$ such that $g(p) \neq 0$.

To show it's dense, for the sake of contradiction let's assume there is an open set $O \subseteq V$ such that $\overline{U} \cap O = \emptyset$. Then $U \cap O = \emptyset$, so $(U \cap O)^c = U^c \cup O^c = V$. But V is irreducible, hence this is not possible, showing that U must indeed be dense, completing the proof.

Definition 1.3.7: Domain of Definition

Let $\varphi \in k(V)$ a rational function. Then the set of regular point is called the *domain of definition*.

Corollary 1.3.8: Collection of Rational Points

Let $\varphi_1, \dots, \varphi_n \in K(X)$ be rational functions. Then their common domain of definition is open and non-empty

Proof:

The key is to notice that the intersection of finitely many open dense sets is dense.

In fact, as the open sets in the Zariski topology are “huge”, it suffices to define a rational mapping on any open subset:

Proposition 1.3.9: Suffice To Define On Open Set

Let $\varphi \in K(X)$ be a rational function. Then if φ is specified on some non-empty open subset $U \subseteq X$, then there is a unique extension of φ to all of X .

Proof:

Take $\varphi \in K(X)$. If $\varphi = 0$, there is nothing to prove, so assume $\varphi \neq 0$. Say we specify what φ is on an open non-empty subset $U \subseteq X$, call this function $\varphi|_U$. Then let's say that φ' is a rational function where φ' on U is equal to $\varphi|_U$ (where a-priori it is unknown where there is a unique extension, and hence it may be another function). Consider $\varphi - \varphi' = \varphi''$. On U , this function is identically 0. Take the representative $\varphi'' = f/g$ on U so that $f = 0$. Then take $X_1 \subseteq X$ given by $f = 0$. This is not all of X as $\varphi \neq 0$, however we then have $X_1 \cup (X \setminus U) = X$ where both are proper subsets, contradicting irreducibility of X , completing the proof.

We are often interested in all the functions that are defined at a particular point, namely since geometrically much of the information of an object is given by collections of functions at every point (the reader familiar with differential geometry should be thinking of germs). To that end, we box the following

Definition 1.3.10: Stalk At A Point

Let V be an affine variety and $p \in V$. Then $\mathcal{O}_p(V) \subseteq k(V)$ is the collection of rational functions:

$$\mathcal{O}_p(V) = \{f/g \in k(V) \mid g(p) \neq 0\}$$

Proposition 1.3.11: Regular Functions and Stalk

$$k[V] = \bigcap_{p \in V} \mathcal{O}_p(V)$$

Proof:

Certainly $k[V] \subseteq \mathcal{O}_p(V)$ for all $p \in V$ so $k[V] \subseteq \bigcap_{p \in V} \mathcal{O}_p(V)$.

For the converse, let us first show that the poles of a rational function $f/g \in k(V)$ is always an

algebraic subset. Let $F \in k[x_1, \dots, x_n]$ be representatives of f so that $\overline{F} = f \in k[V]$. Define:

$$P_f = \{G \in k[x_1, \dots, x_n] \mid gf \in k[V]\}$$

where g is the residue of G . Then P_f is an ideal of $k[x_1, \dots, x_n]$ containing $\mathcal{I}(V)$ and $\mathcal{Z}(P_f)$ are all there points where f/g is not defined.

Now, consider $f/g \in \bigcap_{p \in V} \mathcal{O}_p(V)$. Then notice that since f/g must be defined at every point, $V(P_f) = \emptyset$, thus by the Nullstellensatz $1 \in P_f$, and so we must have:

$$1 \cdot f = f \in k[V]$$

as we sought to show.

Proposition 1.3.12: Stalks Are Local Noetherian Domains

let V be a variety and $p \in V$. Then $\mathcal{O}_p(V)$ is a local Noetherian domain

Proof:

To show it's local, let's show R has a unique maximal ideal that contains every proper ideal. Let

$$\mathfrak{m}_p = \{f/g \in \mathcal{O}_p(V) \mid f(p) = 0\}$$

This is an ideal as its the kernel of the map $\text{ev}_p : \mathcal{O}_p(V) \rightarrow k$ where $f/g \mapsto f(p)/g(p)$. This ideal is maximal, for all other elements of $\mathcal{O}_p(V)$ are units. Indeed, if $f/g \in \mathcal{O}_p(V)$ where $f(p) \neq 0$, then g/f is well-defined in $\mathcal{O}_p(V)$ as $f(p) \neq 0$ giving us our inverse.

But then $\mathfrak{m}_p$ must also contain all other ideals, showing $\mathcal{O}_p(V)$ is indeed a local domain.

To show it's Noetherian, note that $k[V]$ is Noetherian (being the quotient of a Noetherian domain), and by exercise ref:HERE the localization of a Noetherian domain is Noetherian, completing the proof.

In section 1.5.2, we shall see that when V is a curve and $\mathcal{O}_p(V)$ is a *discrete valuation ring*, then V is *smooth* at p . This requires a lot of build-up, and so we shall leave stalks alone for now.

Definition 1.3.13: Order Function of Stalk

Let $p \in f$ be a simple point of a function f . Then $\text{ord}_p^f : \mathcal{O}_p(F) \rightarrow \mathbb{N}$ is the *order function* given by the discrete valuation of $\mathcal{O}_p(F)$.

I don't like this style of writing with a "throw-away" definition for now. I want to modify this later.

Let us now generalize regular maps:

Definition 1.3.14: Rational Map

Let X, Y be varieties. Then a *rational mapping* $\varphi : X \dashrightarrow Y$ is a map given by

$$\varphi = (\varphi_1, \dots, \varphi_n)$$

where each φ_i is a (regular or rational) function and for each $x \in X$, $(\varphi_1(x), \dots, \varphi_n(x)) \in Y$ where all of the functions φ_i are regular.

The image of the rational map shall be denoted $\varphi(X)$, even though not every point may be regular. Note that a rational map shall necessarily map an open subset $U \subseteq X$ into Y , and hence it is well-defined by proposition 1.3.9. By proposition 1.3.9, an equivalent definition of a rational map is that it is the collection of equivalence relations $[(\varphi, U)]$ with $U \subseteq X$ open and $\varphi : U \rightarrow Y$ where $[(\varphi, U)] \sim [(\psi, V)]$ if and only if

$$\varphi|_{U \cap V} = \psi|_{U \cap V}$$

This equivalent to definition 1.3.14, and is more prominent in the context of algebraic geometry. Its upside is that each representative is a well-defined map and hence can be seen as the appropriate quotient of some hom-sets, while its downside is that a rational mapping is now an equivalence relationship.

The following is the equivalent of being surjective:

Definition 1.3.15: Dominating Rational Map

Let $f : X \dashrightarrow Y$ be a rational map. Then f is said to be *dominating* if $f(U)$ is dense in Y .

Recall that if A, B are local rings where $A \subseteq B$ is a subring, then B dominates A if $\mathfrak{m}_A \subseteq \mathfrak{m}_B$. This vocabulary shall be made clear from the following

Proposition 1.3.16: Properties of Rational Maps

Let $F : X \dashrightarrow Y$ be a dominating rational map, $U \subseteq X$ and $V \subseteq Y$ affine open subsets where $f : U \rightarrow V$ represents F . Then:

1. Let $\varphi : X \dashrightarrow Y$ be a rational mapping. Then $\varphi^\# : k(Y) \rightarrow k(X)$ is a ring isomorphism where $\varphi^\#$ determines an isomorphic embedding of $k(Y)$ into $k(X)$.
2. If $\varphi : X \dashrightarrow Y$ and $\psi : Y \dashrightarrow Z$ are rational mapping with dense images, then $\psi \circ \varphi$ has dense image and $(\psi \circ \varphi)^\# = \psi^\# \circ \varphi^\#$.
3. $f^\# : k[V] \rightarrow k[U]$ is injective, hence extends to an injective map $\kappa(V) \rightarrow \kappa(U)$, and so implies $F^\# : \kappa(Y) \rightarrow \kappa(X)$, i.e. a field extension $\kappa(X)/\kappa(Y)$.
4. If p is in the domain of F , $F(p) = q$, then $\mathcal{O}_p(X)$ dominates $F^\#(\mathcal{O}_q(Y))$. Conversely, if $p \in X$ and $q \in Y$ where $\mathcal{O}_p(X)$ dominates $F^\#(\mathcal{O}_q(Y))$, then p belongs to the domain of F and $F(p) = q$.
5. Any homomorphism $\kappa(Y) \rightarrow \kappa(X)$ induces a unique dominating rational map $X \dashrightarrow Y$.

Proof:

1. exercise.
2. exercise.
3. recall $f^\#(\varphi) = \varphi \circ f$. Suppose $\varphi \circ f = f^\#(\varphi) = f^\#(\psi) = \psi \circ f$. As F is dominating, we see that the image of these two maps agree on a dense set, hence they must be equal: $\varphi = \psi$. Thus, the localization is injective, and hence the function fields is injective. As both U and V are open subsets of varieties, we have that the function fields on X, Y agree with those on U, V .
4. if $f(p) = q$, then it is a matter of pushing around the definition to see the $\mathcal{O}_p(X)$ dominates $F^\#(\mathcal{O}_q(Y))$. Conversely, suppose $\mathcal{O}_p(X)$ dominates $F^\#(\mathcal{O}_q(Y))$. Take an affine neighbourhood V of p and W of q where

$$k[W] = k[y_1, \dots, y_n]$$

and $F^\#(y_i) = \frac{a_i}{b_i}$ for $a_i, b_i \in k[V]$ with $b_i(p) \neq 0$. Take $b = b_1 \cdots b_n$, and take the localization V_b . Then

$$F^\#(k[W]) \subseteq k[V_b]$$

Hence the map $F^\# : k[W] \rightarrow k[V_b]$ induces a unique morphism $f : V_b \rightarrow W$. Then if $g \in k[W]$ vanishes at q , $F^\#(g)$ vanishes at p , but then it must be that $f(p) = q$

5. If $\varphi : k(Y) \rightarrow k(X)$ with $\text{h}\varphi(k[Y]) \subseteq k[X_b]$, for some $b \in k[X]$, φ induces a morphism $f : X_b \rightarrow Y$. Hence $f(X_b)$ is dense in Y as $f^\#$ is injective, giving us the desired result.

Definition 1.3.17: Birational Map

Let $\varphi : X \dashrightarrow Y$ be a rational map. Then φ is said to be *birational* if the image of φ is dense and there exists a rational map $\psi : Y \dashrightarrow X$ with dense image such that $\varphi \circ \psi = \psi \circ \varphi = 1$.

Proposition 1.3.18: Birational Maps and Isomorphisms

Furthermore, φ is birational if and only if φ^* is an isomorphism.

Proof:

exercise

We have yet to define dimension formally (see section 1.4) [write a little bit about it here](#).

Definition 1.3.19: Rational Closed Set

Let V be a variety. Then if V is birational to $\mathbb{A}_k^n$ or $\mathbb{P}_k^n$, then V is called *affine rational* or *projective rational* respectively.

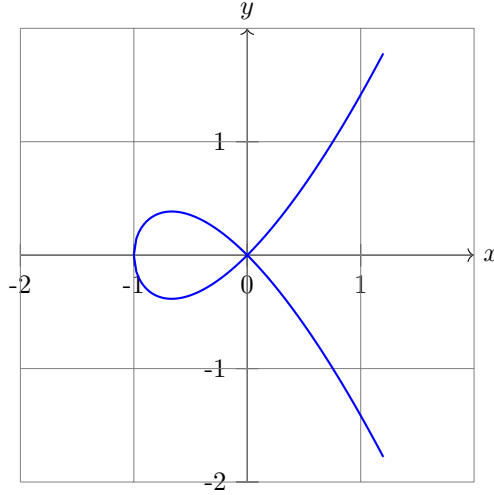
If V satisfies one of the above two conditions and has dimension 1, 2 or codimension 1, then it is a *rational curve*, *rational surface*, or *rational hypersurface* respectively.

Example 1.3.20: Rational Mapping and Birational Maps

1. Naturally, all curves of order 1 (i.e., straight lines) are tautologically rational. More generally any linear subspace is rational.
2. Consider the curve given by

$$y^2 = x^2 + x^3$$

which looks like:



Label this curve C . Then this is a rational curve. To see this, consider that the line given by $y = tx$ passing through the origin intersects the curve at a single point (not counting the origin). Then substituting $y = tx$ into our equation, we get:

$$x^2(t^2 - x - 1) = 0$$

The root $x = 0$ corresponds to the origin, and the other roots $x = t^2 - 1$ corresponds to non-trivial solution. Hence, we get $x = t^2 - 1, y = t(t^2 - 1)$, and the rational map:

$$\varphi(t) = (t^2 - 1, t(t^2 - 1))$$

For the inverse, start with any line for example the line $x = 1$. For any $(x_0, y_0) = p \in C$, make a line passing through the origin O and the point O , this is the line PO . Then this line intersects Q . Then with an argument similar to the above one, we can see that there is a rational map $\psi : C \rightarrow \mathbb{A}_k^1$, which furthermore is the inverse of the above map, giving birationality. Note that this map is certainly not an isomorphism or a homeomorphism.

If we take $k = \mathbb{Q}$, then this gives all rational solutions to the equation $y^2 = x^3 + x^2$. The case is a little more subtle if we were looking for integer solutions to the equation, which is explored in chapter ??.

3. Let us show that all irreducible curves C of order 2 are rational (note that the above example is an order 3 curve, the maximal degree being 3). Let C be given by the equation $f(x, y)$. Take any point $(x_0, y_0) \in C$, and equation of the line through (x_0, y_0) with slope t

$$y - y_0 = t(x - x_0)$$

Label this line L_t . Let us find the points of intersection of the curve C with line L_t . To do so, we substitute:

$$f(x, y_0 + t(x - x_0)) = 0$$

which is equivalent to the mapping $\mathbb{A}_k^1$ to the curve. This mapping is well-defined (notice it is still of degree 2). Then one of the roots is certainly x_0 . Then let a denote the coefficient of x after dividing the equation $f(x, y_0 + t(x - x_0))$ by the coefficient of x^2 . Then we can see that the remaining roots must be:

$$x + x_0 = -a \quad \Rightarrow \quad x = -x_0 - a$$

Then x is a rational function of t , and substituting the expression for x into L_t , we often an expression for y as a rational function of t .

Note that if the coordinates x_0, y_0 of $(x_0, y_0) \in C$ and the coefficients of $f(x, y)$ belong to some subfield $k_0 \subseteq k$, then so do the coefficients of the function that gives the parameterization. In this case, we can find the general form of the solution in rational numbers of a Diophantine equation of degree 2 by knowing *at least* one solution! However, it turns out that knowing there is at least one solution can sometimes be tricky^a

4. More generally, an irreducible hypersurface $X \subseteq \mathbb{A}_k^n$ given by a polynomial of degree 2 $f(x_1, \dots, x_n) = 0$ is rational. The proof generalizes the above and is left as an exercise. The geometric picture is projection the hypersurface to a hyper-plane via a projection that does not pass through some point x . The point x can be any point that is not a “vertex” on X , namely that for at least one variable:

$$\frac{\partial f}{\partial x_i}(x) \neq 0$$

5. The curve order 3 given by $x^3 + y^3 = 1$ is *not* rational. More generally, $x^n + y^n = 1$ is not rational for $n \geq 3$ (if $n \nmid \text{char } k$). To see this, let C_n be the curve given by $x^n + y^n = 1$ and suppose it is rational so that there exists rational functions φ, ψ such that $x = \varphi(t), y = \psi(t)$. Then we may find polynomials p, q, r relatively prime such that

$$\varphi(t) = \frac{p(t)}{r(t)} \quad \psi(t) = \frac{q(t)}{r(t)}$$

where the denominator is the same for simplicity (the same argument follows with more steps by taking p/r and q/s). Then plugging into our equation we get:

$$p(t)^n + q(t)^n - r(t)^n = 0$$

for all $t \in k$. Then differentiating, we get a non-zero solution (as $n \nmid \text{char } k$) and so we have:

$$p(t)^{n-1}p'(t) + q(t)^{n-1}q'(t) - r(t)^{n-1}r'(t) = 0$$

We now have two systems equations. If we assume that $p^{n-1}, q^{n-1}, -r^{n-1}$ are variable, we can consider the above two equations as being a system of linear equations which has the matrix:

$$\begin{pmatrix} p & q & r \\ p' & q' & r' \end{pmatrix}$$

Solving this system of linear equations, we get that $p^{n-1}, q^{n-1}, -r^{n-1}$ are proportional too

$$qr' - rq' \quad rp' - pr' \quad pq' - qp'$$

As p, q, r are relatively prime, we get that:

$$p^{n-1} | (qr' - rq') \quad q^{n-1} | (rp' - pr') \quad r^{n-1} | (pq' - qp')$$

Finally, if a, b, c are the degrees of p, q, r . If $a \geq b \geq c$, then

$$(n-1)a \leq b + c - 1$$

which can be show to be a as $n \geq 3$. A similar result occurs for all other cases, leading to a contradiction.

^asee Legendre theorem in cite:HERE

Birational mappings are weaker than isomorphisms (of varieties). This problem can be translated algebraically to the difference between having $k[X]$ and $k[Y]$ be isomorphic and $k(X)$ and $k(Y)$ be isomorphic. The reader may recall from section 1.2.7 that $k[X]$ and $k[Y]$ must be finitely generated reduced k -algebras, $k(X)$ and $k(Y)$ finite field extensions. As the structure of field extensions is much simpler, we expect birational mappings to be much simpler. In fact, we may even classify all affine variety up to birational isomorphism:

Theorem 1.3.21: Classification of Varieties Up To Birational Map

Every variety X is birationally isomorphic to a hypersurface of some affine space $\mathbb{A}_k^n$.

Thus, birationality of varieties reduces to the classification of hypersurfaces. Note that being birationally isomorphic to a hypersurface does not imply a curve or plane is rational. For example, the curve given by $x^3 + y^3 = 1$ is already a hypersurface in $\mathbb{A}^2$ as it is given by $\mathcal{Z}(x^3 + y^3 - 1)$, however it is not a rational curve as it is not birational to $\mathbb{A}_k^1$.

Proof:

Take the field $k(X) \cong k[x_1, \dots, x_n]/I(X)$ which is a finitely generated k -algebra, namely it can be generated by the functions $t_1, \dots, t_n$. Take the collection of all algebraically independent subsets of $t_1, \dots, t_n$ and (as $k(X)$ is Noetherian) choose a maximal subset, index it as $t_1, \dots, t_d$. Then every $y \in k(X)$ must have the be algebraically dependent on $t_1, \dots, t_d$ for the set $t_1, \dots, t_d, y$ algebraically dependent, that is is that there exists a function f such that

$$f(t_1, \dots, t_d, y) = 0$$

and furthermore, the polynomial representative $f(x_1, \dots, x_d, x_{d+1})$ can be chosen to be irreducible. So choose a polynomial $f(x_1, \dots, x_d, x_{d+1})$ such that $f(t_1, \dots, t_d, t_{d+1}) = 0$. Then we shall show that $k(X)$ is isomorphic to $k(Y)$ where $k(Y)$ is given by the hypersurface defined by f .

Let us start by showing that adjoining t_{d+1} to $k(t_1, \dots, t_d)$ gives a separable extension. To do this, let us first show that $\frac{\partial f}{\partial x_i}(x_1, \dots, x_{d+1}) \neq 0$ for at least on index i . If k has characteristic 0, it is clear as f is irreducible with degree higher than 0. If k has characteristic p , then for the sake of contradiction assume that it were not the case. Then each x_i would occur in f with

degrees that must be a multiple of the characteristic p of k . Hence we can write f as:

$$f = \sum a_{i_1, \dots, i_{d+1}} x_1^{p i_1} x_2^{p i_2} \dots x_{d+1}^{p i_{d+1}}$$

Setting $b_{i_1, \dots, i_{d+1}} = b_{i_1, \dots, i_{d+1}}^p$, we can define a g such that $f = g^p$, contradicting the irreducibility of f .

Now, as $\frac{\partial f}{\partial x_i} \neq 0$, the polynomial f must have monomials with x_i occurring. Then it must be that $t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_{d+1}$ are algebraically independent over k . First, t_i is algebraic over $k(t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_{d+1})$ as f has nonzero terms with x_i (and hence correspondingly t_i). Thus if the elements $t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_{d+1}$ were dependent, the transcendence degree of $k(t_1, \dots, t_{d+1})$ would be less than d , contradicting the algebraic independence of $t_1, \dots, t_d$.

Thus, renumber $x_1, \dots, x_n$ so that $t_1, \dots, t_n$ are independent over k and $\frac{\partial f}{\partial x_{d+1}} \neq 0$. Then t_{d+1} is separable over $k(t_1, \dots, t_d)$. This gives the right circumstances for the primitive root theorem: as t_{d+2} is algebraic over $k(t_1, \dots, t_d)$, we have:

$$k(t_1, \dots, t_d, t_{d+1}, t_{d+2}) = k(t_1, \dots, t_d)(y)$$

Now, repeating this process, we may adjoin the elements $t_{d+1}, \dots, t_n$ and replacing y with the appropriate element, so that we can represent $k(X)$ as $k(z_1, \dots, z_{d+1})$ where $z_1, \dots, z_d$ are algebraically independent over k and:

$$f(z_1, \dots, z_d, z_{d+1}) = 0$$

The polynomial f is irreducible over k , and $\frac{\partial f}{\partial x_{d+1}} \neq 0$. Let Y be the closed set defined by the above f . Then $k(Y)$ is isomorphic to $k(X)$ by construction, which by example 1.3.20 means that X and Y are birationally isomorphic, as we sought to show.

Note that $k(X)/k(z_1, \dots, z_d)$ is a finite separable extension, and that the primitive elements $z_1, \dots, z_{d+1}$ can be chosen as linear combinations of the $x_1, \dots, x_n$ for appropriate c_{ij} :

$$z_i = \sum_j^n c_{ij} x_j$$

Then the mapping $(x_1, \dots, x_n) \rightarrow (z_1, \dots, z_{d+1})$ is a projection from $\mathbb{A}_k^n$ that is parallel to the linear subspace given by:

$$\sum_j^n c_{ij} x_j = 0 \quad 1 \leq i \leq d+1$$

Thus, the birational mapping theorem says that there is an appropriate hypersurface (not necessarily a hyperplane) on which we may project in a rational manner the variety! From this perspective, linear algebra results which are birationally invariant can be translated over to varieties, which we shall soon see gives us some interesting results!

To conclude this section, we define one more important ring of functions that can be thought of as the ring that focuses on a single point. This ring is extremely

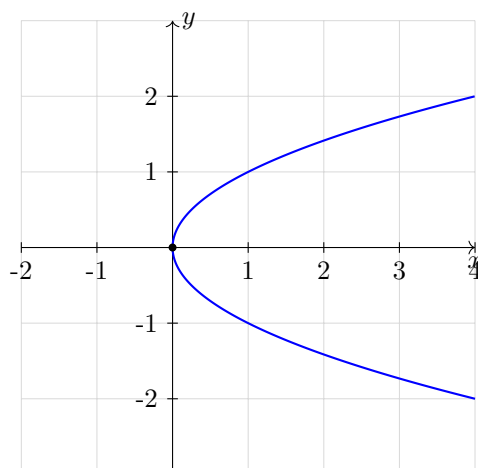
Exercise 1.3.1

1. Complete the proof of proposition 1.3.4

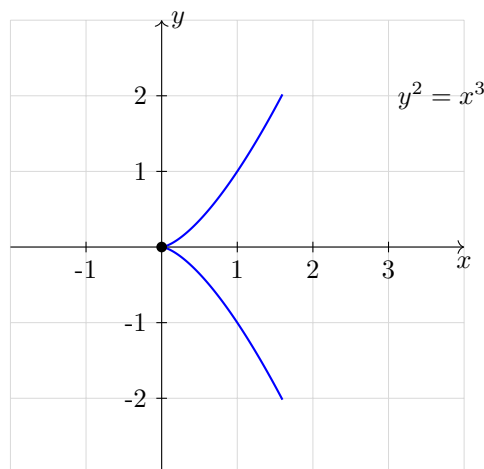
2. Let X be irreducible. Show that any finite collection of rational functions on X has a non-empty common domain of definition
3. Show that the collection of poles of a rational function $f \in k(V)$ is an algebraic set
4. Show that for each $a \in k$, the stalk $\mathcal{O}_a(\mathbb{A}_k^1)$ at a is a DVR with uniforming parameter $x - a$.
5. Define $\mathcal{O}_\infty = \left\{ \frac{f}{g} \mid \deg(f) \leq \deg(g) \right\}$. Show that $\mathcal{O}_\infty$ is a DVR with uniforming parameter $1/x$. (Note: This is foreshadowing that it is natural consider curves in *projective space* rather than affine space; see section 1.6).

1.4 Global Dimension

When looking at $\mathbb{A}_k^n$, we may naturally think of this space as being n -dimensional, and in this special case this matches with our notion of a 3-dimensional vector-space. We may think that varieties also have a notion of dimension, for example the curve with the corresponding coordinate ring $\frac{k[x,y]}{(y^2-x)}$ gives the impression that it should be thought of as a 1-dimensional variety:



This variety should even be thought of as being smooth, as compared to the variety $\frac{k[x,y]}{y^2-x^3}$ which has a cusp at $(x,y) = (0,0)$.



We may hope that the notion of dimension is as simple as it was for vector spaces, however this turns out to be a more subtle problem. For example, continuous maps do not preserve dimension (there is a continuous map from $\mathbb{R}$ onto $\mathbb{R}^2$ giving by Peano space-fill curve). Homeomorphisms between open sets of $\mathbb{R}^n$ and $\mathbb{R}^m$ do force the condition that $n = m$ ²², however we still need to define the notion of dimension so that it can be preserved.

One of the most general definition of dimension can be defined in the topological level and is called the *Lebesgue Covering Dimension* where a space X is dimension at most n if every open cover can be refined to an open cover where each point is covered by at most $n + 1$ sets (see CITE:TOP). However, this definition will fail spectacularly in the Zariski topology as most open sets are huge (in fact, most Zariski topologies will have infinite Lebesgue covering dimension).

There are a few other notion of dimension (to name some notable example: the Brower's-Menger-Urysohn's Boundary Dimension, the Hausdorff Dimension, Deviation of Poset). One to point out that is meaningful that we have seen with the Nullstellensatz is by looking at the transcendental degree of a field extension. For example, if we take $k[V]$ to be some coordinate ring for a variety (so that $k[V]$ is an integral domain), we may take the field of fractions $k(V)$. Then taking the transcendental degree of $k(V)/k$, we may define this to be the *transcendental dimension* of $k[V]$. This works well on smooth shapes, and by exercise ref:HERE we see that all affine and projective varieties have finite maps to $\mathbb{A}_k^n$ or $\mathbb{P}_k^n$ respectively. Furthermore, it is not to difficult to prove that the dimension of these spaces is invariant under homeomorphism (see exercise ref:HERE). However this definition fails when there the coordinate ring is not an integral domain²³. Furthermore, it does not work so well once we try to generalize this notion to spectrum in section 1.9 (for example, we shall show that the geometric object associated to $\mathbb{Z}$ will have transcendence degree 0, while in the definition that we shall settle on it will have dimension 1).

One fundamental limiter is that these notion of dimension is defined in a *global way*; the entire space has some dimension. This turns out to be a misnomer, as the example of the curve $y^2 = x^3$ mentioned earlier demonstrates (it *drops* a dimension at $(x, y) = (0, 0)$). To anyone who has done differential geometry, they may already be comfortable with the fact that the dimension of a smooth manifold is given by the vector space dimension of the tangent space at a point (which for a smooth-manifold

²²see [2]

²³It also fails as an adequate definition when localizing a ring or taking it's completion, which are two important tools that we shall often use on shapes

should be invariant to the choice of point, and hence a well-defined global notion). Then the reader well-versed in the equivalent definition's of tangent spaces of a manifold may recall that the tangent space can be defined as the dual of $\mathfrak{m}/\mathfrak{m}^2$ where $\mathfrak{m}$ is the maximal ideal of C_a^∞ (the set of germs of C^∞ on a manifold at the point a). This definition is agnostic of the differential structure, and so can be used on varieties. It shall also capture the local subtleties; to reiterate the same example the tangent space at $(x, y) = (0, 0)$ of $y^2 = x^3$ will be dimension 2, representing the fact that it should be treated differently than the other points. We shall explore this notion starting in section 1.5.1.

1.4.1 Krull dimension and Transcendental Degree

What has been settled on as the most useful notion of global dimension that translates well between algebra and geometry is the following:²⁴

Definition 1.4.1: Topological Krull Dimension

Let X be a topological space. Then the *Krull dimension* of X is the supremum of the length of all chains:

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n$$

of distinct irreducible subsets of X . The Krull dimension of a topological space will be denoted kdim

This definition is not interesting in euclidean topology, for example for $\mathbb{R}^n$ as only points are irreducible we have $\text{kdim}(\mathbb{R}^n) = 0$. If the dimension is finite, then it is equal to the length of the longest chain of irreducible closed subsets. The Krull dimension of $\mathbb{A}_k^1$ is always 1, $\text{kdim}(\mathbb{A}_k^1) = 1$, as the only irreducible closed subsets of $\mathbb{A}_k^1$ are the single point and whole space. As irreducible closed subsets correspond to prime ideals, we have the equivalent notion for primes:

Definition 1.4.2: Height of Prime

Let $\mathfrak{p} \subseteq R$ be a prime ideal. Then the *height* of a prime ideal is the supremum of all integers n such there is a chain of distinct prime ideals:

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{p}$$

The *dimension* of R is the supremum of the heights of all prime ideals.

Fields always have primes of height 0 while Principal Ideal Domains which all have Krull dimension 1.

Proposition 1.4.3: Dimension of Affine Algebraic Set

Let $Y \subseteq \mathbb{A}_k^n$ be an affine algebraic set. Then $\text{kdim}(Y)$ is equal to the dimension of the affine coordinate ring $k[Y]$

²⁴The Krull dimension turns out to be equivalent to the Brouwer-Menger-Uryson definition of dimension, which is useful for cohomological intuitions

Proof:

The closed irreducible subsets of Y correspond to the prime ideals of $k[x_1, \dots, x_n]$ containing $I(Y)$, which correspond to the prime ideals of $k[Y]$. But then $\text{kdim}(Y)$ is the height of the longest prime ideals in $k[Y]$, as we sought to show.

When covering Noether's Normalization (theorem 1.2.43), it was observed that this can be used to define the dimension of a ring. The following definition captures this idea:

Definition 1.4.4: Transcendental Dimension

Let X be a quasi-projective irreducible variety. Then the *transcendental dimension* of X (over k) is the transcendental degree of the field $k(X)$. It shall be denoted $\dim X$.

Proposition 1.4.5: Transcendental and Krull Dimension

Let k be a field and R a finitely generated k -algebra. Then

$$\text{kdim}(R) = \text{trdeg}_k(R)$$

Furthermore, if $\mathfrak{p} \subseteq R$, then:

$$\text{height}(\mathfrak{p}) + \text{kdim}(R/\mathfrak{p}) = \text{kdim}(R)$$

Proof:

Hint: use the normalization theorem (either Noether's normalization or its geometric counter-part seen in near the end section 1.6.4).

The conditions are necessary for the equality of both equations:

Example 1.4.6: Different Krull and Transcendental Dimension

1. Take $R = k(x)$. Then it has Krull dimension 0, but transcendental dimension 1 (over k).
2. In general the Krull dimension equalling the transcendental dimension is well-defined because of the finite generation: Let R be a ring with transcendence dimension n , and let the following be a chain of prime ideals

$$\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_m = \mathfrak{p}$$

so that the height of $\mathfrak{p}$ is m . Choose $p_i \in \mathfrak{p}_i \setminus \mathfrak{p}_{i-1}$, let $B = k[p_1, \dots, p_n]$ be the subalgebra $B \subseteq R$ generated by these elements. Define the primes $\mathfrak{q}_i = \mathfrak{p}_i \cap B$ which still form a chain as $p_i \in \mathfrak{q}_i \setminus \mathfrak{q}_{i-1}$. Then $\text{Frac}(B) \subseteq \text{Frac}(R)$, hence the transcendence dimension of $\text{Frac}(B)$ is at most n . As B is finitely generated, the Krull dimension must be at most n (or else we can create a longer chain). But then $m \leq n$.

For those who know some scheme theory, we can think of this phenomenon as localizing at a non-closed point. The $k(x)$ is the localization at the generic point of $\mathbb{A}_k^1$, while the algebraic dimension is a field.

Corollary 1.4.7: Dimension of Affine Space

The Krull dimension of affine space $\mathbb{A}_k^n$ is n .

Proof:

apply proposition 1.4.5.

Codimension**Definition 1.4.8: Codimension**

Let $Y \subseteq X$ be a closed subvariety of X . Then

$$\text{codim} Y = \dim X - \dim Y$$

Note that dimension is thus determined by any open subset of a variety, as $k(U) = k(X)$ implies $\dim U = \dim X$. Verify that planar curves have dimension 1, finite sets have dimension 0, $\mathbb{A}_k^n$ and $\mathbb{P}_k^n$ have dimensions n , and that $\dim(X \times Y) = \dim(X) + \dim(Y)$.

Definition 1.4.9: Curves and Surfaces

Let X be an algebraic variety. Then:

1. If X is 1-dimensional, it is called a *curve*
2. If X is 2-dimensional, it is called a *surface*
3. If X is 1-codimensional, it is called a *hypersurface*.

Proposition 1.4.10: Dimension Between Irreducible Varieties

Let $Y \subseteq X$. Then $\dim Y \leq \dim X$. If Y is closed and $\dim(Y) = \dim(X)$, then $X = Y$

There are lots of great theorem from Shafarevich that I will put down when I have the time
Shafarevich p.54

Main idea's that I want to have down:

1. irred components of hypersurface have codim 1, and variety whose components have codim 1 is hypersurface (their ideals are also principal).
2. any subvariety $X \subseteq \prod \mathbb{P}_k^{n_i}$ whose components all have codim 1 is given by a singel equation, homogeneous in each groups of variables.
3. Look at intersecting hypersurfaces.
4. Non-vanishing form on irred proj var will add a single dimension of restriction: $\dim X_F = \dim X - 1$

5. results relating $\dim$ and codim and forms, incl under intersections
6. results on fibers!
7. Classifying lines on $\mathbb{P}_k^3$.
8. Chow coordinates of proj variety

Exercise 1.4.1

1. Let X, Y be irreducible varieties. Show that $\dim(X \times Y) = \dim(X) + \dim(Y)$ (hint: use the transcendental degree definition)

1.5 Local Dimension

here

1.5.1 Hilbert Function

As mentioned earlier, there are flaws in the other ways of defining dimension; there is a strong dependence on the structure of the ambient space or the chains, both of which turn out to have flaws (put here the flaws that were in ALGEO? I think it would be really instructive)

The proper way of defining dimension is by using *Hilbert Functions*. This notion shall become more important in [1], however the key concepts are useful in a more nuanced interpretation of dimensions of projective varieties. It shall turn out to have important ties to cohomological explorations, the “genus” of a curve, and how the graded components of a projective space grow. Overall, it shall capture a lot of valuable information.

As this is an optional section, it shall dive a little more into the generalities in preparation for the results in [1]. For almost all cases, we need only consider projective varieties and their corresponding projective coordinate ring. The best example to keep in mind is a hypersurface of degree m embedded in projective space of dimension n .

The dimension function satisfies the following important property:

Definition 1.5.1: Additive Function on Collection

Let C be a class of R -modules and $\lambda : C \rightarrow \mathbb{Z}$ a function. Then λ is *additive* if for each short exact sequence

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$$

with the R -modules belonging to C we have

$$\lambda(M') - \lambda(M) + \lambda(M'') = 0 \quad \text{equiv.} \quad \lambda(M') + \lambda(M'') = \lambda(M)$$

Example 1.5.2: Additive Function

1. Let $R = k$ and let C be all finite-dimensional k -vector spaces. Then $V \mapsto \dim V$ is an additive function on C .
2. Let C be the collection of all Artinian ring (recall they are also Noetherian). Then the length is well-defined, and in fact $\ell(-)$ is an additive function.

Proposition 1.5.3: Additive Function on Exact Sequence

Let

$$0 \rightarrow M_0 \rightarrow M_1 \rightarrow \cdots \rightarrow M_n \rightarrow 0$$

be an exact sequence of R -modules. Then for any additive function λ on C ,

$$\sum_{i=0}^n (-1)^i \lambda(M_i) = 0$$

Proof:

Recall any exact sequence can be split into short exact sequences:

$$0 \rightarrow N_i \rightarrow M_i \rightarrow N_{i+1} \rightarrow 0$$

where $N_0 = N_{n+1} = 0$. Then $\lambda(M_i) = \lambda(N_i) + \lambda(N_{i+1})$, and taking the alternating sum of the $\lambda(M_i)$ will simply insure everything cancels out.

Hilbert Function

Let $R = \bigoplus_{n=0}^{\infty} R_n$ be a Noetherian graded ring. Then R_0 is Noetherian, and say that R is generated by $r_1, \dots, r_s$ as an R_0 -algebra, which without loss of generality we may take to be homogeneous elements of (strictly positive) degrees $k_1, \dots, k_s$. The ring I keep in mind is any homogenized curve, for example:

$$\frac{k[x_0, x, y]}{(x_0 x^2 - y^3)}$$

Let M be a finitely generated graded R -module so that M can be generated by a finite number of homogeneous elements $m_1, \dots, m_t$. This can be thought of as a fiber bundle on our curve (ex. a vector bundle, line bundle, or tangent bundle). By ??, the following is well-defined:

Definition 1.5.4: Poincaré Series

The *Poincaré Series* of M with respect to λ is the generating function of $\lambda(M_n)$, that is the power series:

$$P(M, t) = \sum_{n=0}^{\infty} \lambda(M_n) t^n \in \mathbb{Z}[[x]]$$

Proposition 1.5.5: Poincaré Series is Rational Function

Let $P(M, t)$ be the Poincaré series for M with respect to λ , where M is a finitely generated R -module over a graded ring R that is a finitely generated R_0 -algebra generated by s elements each with homogeneous degree k_i . Then $P(M, t)$ is a rational function in t of the form:

$$P(M, t) = \frac{f(t)}{\prod_{i=1}^s (1 - t^{k_i})} \quad f(t) \in \mathbb{Z}[t]$$

The polynomial $f(t)$ can be thought of as holding information about the lack of freeness.

Proof:

We shall proceed by induction on s . Starting with $s = 0$ so that $R = R_0$ and $R_n = 0$ for all $n > 0$. Then M is a finitely generated R_0 -module, and so the generators have at most some degree n_0 , and hence $M_n = 0$ for $n > n_0$. But then $P(M, t)$ is a polynomial, as $\lambda(0) = 0$ (which can be verified using short exact sequences). Its coefficients will be the dimensions of the M_n 's.

Now assume the result holds for any R_0 -algebra R generated by $s - 1$ elements and let R be generated by s elements, $r_1, \dots, r_s$, each with degree k_i (which without loss of generality we may take are > 0). We shall find how to reduce to $s - 1$ elements to apply the inductive hypothesis. To that end, consider the R -module homomorphism $M \xrightarrow{r_s} M$ that will map $M_n \rightarrow M_{n+k_s}$. This induces an exact sequence:

$$0 \rightarrow r_s M_n \rightarrow M_n \xrightarrow{r_s} M_{n+k_s} \rightarrow M_{n+k_s}/r_s M_n \rightarrow 0$$

Let $r_s M_n = K_n$ and $M_{n+k_s}/r_s M_n = L_{n+k_s}$ for each $n \geq 0$. Now, applying the additive function λ to the above short exact sequence, we get by proposition 1.5.3 that for all n :

$$\lambda(K_n) - \lambda(M_n) + \lambda(M_{n+k_s}) - \lambda(L_{n+k_s}) = 0$$

which re-arranged we get:

$$\lambda(M_{n+k_s}) - \lambda(M_n) = \lambda(L_{n+k_s}) - \lambda(K_n)$$

Take $K = \bigoplus K_n$ and $L = \bigoplus L_n = M/r_s M$. Then these are both finitely generated, being a submodule and a quotient of a finitely generated R -module M . As they are both annihilated by r_s , we have that they are both $R_0[r_1, \dots, r_{s-1}]$ -modules. This shall let us apply the induce

hypothesis to the following chain of equalities:

$$\begin{aligned}
(1 - t^{k_s})P(M, t) &= \sum_{n=0}^{\infty} \lambda(M_n) t^n - \sum_{n=0}^{\infty} \lambda(M_n) t^{n+k_s} \\
&= g(t) + \sum_{n=0}^{\infty} (\lambda(M_{n+k_s}) - \lambda(M_n)) t^{n+k_s} \\
&= g(t) + \sum_{n=0}^{\infty} (\lambda(L_{n+k_s}) - \lambda(K_n)) t^{n+k_s} \\
&= P(L, t) - t^{k_s} P(K, t) \\
&\stackrel{\text{I.H.}}{=} \frac{f_L(t) - t^{k_s} f_K(t)}{\prod_{i=1}^{s-1} (1 - t^{k_i})} \\
&= \frac{f(t)}{\prod_{i=1}^{s-1} (1 - t^{k_i})}
\end{aligned}$$

where

$$g(t) = \sum_{n=0}^{k_s-1} \lambda(M_n) t^n \in \mathbb{Z}[x]$$

Rearranging, we get:

$$P(M, t) = \frac{f(t)}{\prod_{i=1}^s (1 - t^{k_i})} \quad f(t) \in \mathbb{Z}[t]$$

as we sought to show.

Hence, $P(M, t)$ has a pole at $t = 1$. Let $d(M)$ denote the order of the pole; this pole shall in many ways give a measure of the size of M relative to λ .

For the rest of this section, let the degree of the zero polynomial be -1 and $\binom{n}{-1} = 0$ for all $n \geq 0$ and if $n = 1$, then $\binom{-1}{-1} = -1$.

Corollary 1.5.6: Hilbert Polynomial of M

If $k_i = 1$ for each i , then for sufficiently large n , $\lambda(M_n)$ is a polynomial in n (with rational coefficients) of degree $d - 1$

Proof:

Assuming $k_i = 1$, from proposition 1.5.5, we get that each $\lambda(M_n)$ is the coefficient of t^n in the expansion of:

$$\frac{f(t)}{(1 - t)^s}$$

If $f(t)$ has $(1 - t)$ factors, we may cancel them out so that $s = d$, let $g(t)$ be the polynomial such that

$$\frac{f(t)}{(1 - t)^s} = \frac{g(t)}{(1 - t)^d}$$

where $g(1) \neq 0$. Let $g(t) = \sum_{k=0}^N a_k t^k$. Now, power-series of $(1-t)^{-d}$ is:

$$(1-t)^{-d} = \sum_{k=0}^{\infty} \binom{d+k-1}{d-1} t^k$$

Grouping terms we get for all terms larger than the highest degree of g :

$$\lambda(M_n) = \sum_{k=0}^{\infty} a_k \binom{d+n-k-1}{d-1} \quad \forall n \geq N$$

Then distributing the right hand side, we get that it is a polynomial in n , its leading term being

$$\frac{\sum a_k}{(d-1)!} n^{d-1}$$

showing it is of degree $d-1$, completing the proof.

Note that $f(n)$ may be an integer for all integers n even though the coefficients of f are not all integers, for example $\frac{1}{2}x(x+1)$. In fact, all integer-valued polynomials will be an integer linear combination of the terms (see exercise ref:HERE):

$$\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \dots$$

Definition 1.5.7: Hilbert Function

Let R be a graded and finitely generated R_0 algebra where R_0 is Artinian. Let M be a finitely generated graded R -module. Then the *Hilbert function* of M is given by:

$$H_M(n) := \ell_{R_0}(M_n)$$

for all $n \in \mathbb{N}$

Proposition 1.5.8: Non-Zero Divisor and Order of Pole

Let $x \in R_k$ be an element that is *not* a zero-divisor in M . Then:

$$d(M/xM) = d(M) - 1$$

Proof:

As x is not a zero-divisor we have that $xm = 0$ implies $m = 0$. Then in the exact sequence:

$$0 \rightarrow K_n \rightarrow M_n \xrightarrow{x} M_{n+k_s} \rightarrow L_{n+k_s} \rightarrow 0$$

we have $K_n = 0$, and $L = M/xM$. Then by corollary 1.5.6:

$$d(L) = d(M) - 1$$

which and as $L = M/xM$, we have $d(M/xM) = d(M) - 1$, as we sought to show.

Example 1.5.9: Additive Function and Hilbert Polynomial

An example the reader may want to try is with A_0 being an Artin ring (or a field) and $\lambda(M)$ is the length of M ; $\ell(M)$. By ?? the length is additive. For a concrete example:

1. Let $R = R_0[r_1, \dots, r_s]$, where R_0 is an Artinian ring (say a field k) and each r_i is an indeterminate (the simplest case scenario). Then R_n is a free R_0 -module generated by monomials $r_1^{m_1}, \dots, r_s^{m_s}$ with $\sum m_i = n$. There are

$$\binom{s+n-1}{s-1}$$

such monomials, and hence by proposition 1.5.8

$$P(R, t) = (1 - t)^{-s}$$

For high enough degree, the Hilbert polynomial is:

$$\begin{aligned} h_R(t) &= \frac{1}{N!}(x+N)(x+N-1)\cdots(x+1) \\ &= \frac{1}{N!}x^N + \frac{N-1}{2(N-1)!}x^{N-1} + \cdots + 1 \end{aligned}$$

Proposition 1.5.10: Hilbert Function On Local Ring

Let R be a Noetherian local ring, $\mathfrak{m}$ its associated maximal ideal, $\mathfrak{q}$ an $\mathfrak{m}$ -primary ideal, M a finitely generated R -module, and (M_n) a stable $\mathfrak{q}$ -filtration of M . Then:

1. M/M_n is of finite length for each $n \geq 0$
2. for all sufficiently large n , this length is a polynomial $g(n)$ of degree $\leq s$ in n , where s is the least number of generators of $\mathfrak{q}$
3. The degree and leading coefficient of $g(n)$ depends only on M and $\mathfrak{q}$, not on the chosen filtration

Proof:

1. Take $G(R) = \bigoplus_n \mathfrak{q}^n / \mathfrak{q}^{n+1}$ and $G(M) = \bigoplus_n M_n / M_{n+1}$. Then by ?? $G_0(A) = A/\mathfrak{q}$ is an Artinian ring. By ??, $G(R)$ is Noetherian and $G(M)$ is a finitely generated graded $G(R)$ -module. As each $G_n(M) = M_n / M_{n+1}$ is a Noetherian R -module annihilated by $\mathfrak{q}$, it is a Noetherian $R/\mathfrak{q}$ -module, and hence is of finite length (as $R/\mathfrak{q}$ is Artinian). It follows that M/M_n is of finite length and :

$$\ell_n = \ell(M/M_n) = \sum_{r=1}^n \ell(M_{r-1}/M_r)$$

2. Let $x_1, \dots, x_s$ generate $\mathfrak{q}$, and take $\bar{x}_i \in \mathfrak{q}/\mathfrak{q}^2$, which generate $G(R)$ as an $R/\mathfrak{q}$ -algebra. As each $\bar{x}_i$ has degree 1, we have by corollary 1.5.6

$$\ell(M_n/M_{n+1}) = f(n)$$

where $f(n)$ is a polynomial in n of degree $\leq s - 1$ for large n . Then by (1) we have $\ell_{n+1} - \ell_n = f(n)$, we get that ℓ_n is a polynomial $g(n)$ of degree $\leq s$ for large n

3. Let $(\tilde{M}_n)$ be another stable $\mathfrak{q}$ -filtration of M and $\tilde{g}(n) = \ell(M/\tilde{M}_n)$. Then by ??, the two have bounded difference, meaning there exists an integer n_0 such that

$$M_{n+n_0} \subseteq \tilde{M}_n \quad \text{and} \quad \tilde{M}_{n+n_0} \subseteq M_N$$

for all $n \geq 0$. Hence:

$$g(n + n_0) \geq \tilde{g}(n) \quad \text{and} \quad \tilde{g}(n + n_0) \geq g(n)$$

as $g, \tilde{g}$ are polynomials for all large n , we get:

$$\lim_{n \rightarrow \infty} \frac{g(n)}{\tilde{g}(n)} = 1$$

and so g and $\tilde{g}$ have the same degree and leading coefficients, completing the proof.

Definition 1.5.11: Characteristic Polynomial of $\mathfrak{m}$ -primary Ideal

Let R be a local ring with maximal ideal $\mathfrak{m}$, $\mathfrak{q}$ a $\mathfrak{m}$ -primary ideal, and $g(n)$ be the polynomial corresponding to the filtration $(\mathfrak{q}^n M)$. Then it is denoted by $\chi_{\mathfrak{q}}^M(n)$ and is called the *characteristic polynomial* of the $\mathfrak{m}$ -primary ideal $\mathfrak{q}$, and we have:

$$\chi_{\mathfrak{q}}^M(n) = \ell(M/\mathfrak{q}^n M)$$

If $M = R$, then we often abbreviate to $\chi_{\mathfrak{q}}(n) := \chi_{\mathfrak{q}}^R(n)$.

Corollary 1.5.12: Characteristic Polynomial For Local Ring

Let R be a local ring and $\mathfrak{q}$ a $\mathfrak{m}$ -primary ideal. Then for all large n , $\chi_{\mathfrak{q}}(n) = \ell(R/\mathfrak{q}^n R)$ is a polynomial of degree $\leq s$, where s is the least number of generators of $\mathfrak{q}$

Proof:

exercise (apply proposition proposition 1.5.10)

Proposition 1.5.13: Degree of Characteristic Polynomial of Local Ring

Let R be a local ring with maximal ideal $\mathfrak{m}$ and $\mathfrak{q}$ a $\mathfrak{m}$ -primary ideal. Then:

$$\deg \chi_{\mathfrak{q}}(n) = \deg_{\mathfrak{m}} \chi(n)$$

Proof:

By definition of $\mathfrak{m}$ -primary ideals, $\mathfrak{m} \supseteq \mathfrak{q} \supseteq \mathfrak{m}^r$ for some r . Then $\mathfrak{m}^n \supseteq \mathfrak{q}^n \supseteq \mathfrak{m}^{rn}$ and hence:

$$\chi_{\mathfrak{m}}(n) \leq \chi_{\mathfrak{q}}(n) \leq \chi_{\mathfrak{m}}(rn)$$

| for all large n . Letting $n \rightarrow \infty$ gives the desired result.

We hence have:

$$\deg \chi_{\mathfrak{q}}(n) = d(R) = d(G_{\mathfrak{m}}(R))$$

Let us apply this to projective varieties. Recall from exercise ref:HERE that if $k[X] = S/I(X)$ is the projective coordinate ring with grading:

$$k[X]_n = \frac{S_n + I(X)}{I(X)}$$

with $S_0 = k$ is a field, and hence an Artinian ring which allows us to use the above results on projective varieties. Note too that $\ell_k(k[X]_n) = \dim_k \left(\frac{S_n + I(X)}{I(X)} \right)$ as we can see that $k[X]_n$ is a vector space and hence a (and hence the) composition series can be given via the usual filtration of a vector-space (in this case, each S_n is generated by the homogeneous elements of degree n given from the s generators). This leads to the following:

Definition 1.5.14: Hilbert Function On Projective Variety

Let $X \subseteq \mathbb{P}_k^n$ be a projective variety with corresponding projective coordinate ring $k[X] = S/I(X)$ where $S = k[x_0, \dots, x_n]$ with k algebraically closed and $I(X)$ the homogeneous radical ideal. Then the *Hilbert Function* for all $n \in \mathbb{N}$ of X is:

$$H_X(n) = H_{k[X]}(n) = \ell_k(k[X]_n) = \dim_k \left(\frac{S_n + I(X)}{I(X)} \right)$$

for all $n \in \mathbb{N}$. For sufficiently large n , this agrees with the *Hilbert polynomial* of X given by:

$$h_X(t) := h_{k[X]}(t) \in \mathbb{Q}[x]$$

1.5.2 Application: Dimension of Noetherian Local Ring

Dimension is a local property, that is to be extracted from a local ring. In this section let R be a Noetherian local ring, with $\mathfrak{m}$ its maximal ideal. Let

$$\begin{aligned} \delta(R) &= \text{least number of generators of an } \mathfrak{m}\text{-primary ideal of } R \\ d(R) &= d(G_{\mathfrak{m}}(R)) = \text{pole at } t = 1 \text{ of the Hilbert function of } G_{\mathfrak{m}}(R) \end{aligned}$$

We shall show that:

$$\delta(R) = d(R) = \text{kd} R$$

by showing a sequence of inequalities starting and ending with $\delta(R)$. By corollary 1.5.12 and proposition 1.5.13, we already have:

$$\delta(R) \geq d(R)$$

Let us next prove an analogous version of proposition 1.5.8

Proposition 1.5.15: Degree Diminish by Quotient

Let R be a local ring with maximal ideal $\mathfrak{m}$ and $\mathfrak{q}$ a $\mathfrak{m}$ -primary ideal. Let M be a finitely generated R -module, and $x \in R$ a non-zero-divisor in M . Take $M' = M/xM$. Then:

$$\deg \chi_{\mathfrak{q}}^{M'} \leq \deg \chi_{\mathfrak{q}}^M - 1$$

Proof:

Take $N = xM$ so that $N \cong_R M$ (as x is not a zero divisor). Take $N_n = N \cap \mathfrak{q}^n M$ we have :

$$0 \rightarrow N/N_n \rightarrow M/\mathfrak{q}^n M \rightarrow M'/\mathfrak{q}^n M' \rightarrow 0$$

Hence, given $g(n) = \ell(N/N_n)$ we have:

$$g(n) - \chi_{\mathfrak{q}}^M(n) + \chi_{\mathfrak{q}}^{M'}(n) = 0$$

for large n . Then by Artin-Rees lemma, (N_n) is a stable $\mathfrak{q}$ -filtration of N . As $N \cong_R M$, by proposition 1.5.10(3) we have that $g(n)$ and $\chi_{\mathfrak{q}}^M(n)$ have the same leading terms, completing the proof.

Corollary 1.5.16: Non-zero Divisor and Order of Pole for Local Ring

Let R be a Noetherian local ring, x a non-zero-divisor in R . Then:

$$d(R/xR) \leq d(R) - 1$$

Proof:

Take $M = R$ in proposition 1.5.8.

Proposition 1.5.17: Pole and Dimension of Local Ring

$$d(R) \geq \text{kdim}(R)$$

Proof:

We proceed by induction. Let $d = d(R)$. If $d = 0$, then $\ell(R/\mathfrak{m}^n)$ is constant for all large n , and so $\mathfrak{m}^N = \mathfrak{m}^{n+1}$, and so by Nakayama's lemma $\mathfrak{m}^n = 0$. Thus, R is an Artinian ring, and hence $\text{kdim}(R) = 0$.

Now suppose $d > 0$ and that

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_r$$

is any chain of prime ideals in R . Take $x \in \mathfrak{p}_1$ where $x \notin \mathfrak{p}_0$ and $R' = R/\mathfrak{p}_0$. Then $\bar{x} \neq \bar{0} \in R'$. As R' is an integral domain, by corollary 1.5.16:

$$d(R'/x'R') \leq d(R') - 1$$

Furthermore, if $\mathfrak{m}'$ is the maximal ideal of R' , then $R'/(\mathfrak{m}')^n$ is the image of $R/\mathfrak{m}^n$, and so

$$\ell(R/\mathfrak{m}^n) \geq \ell(R'/(\mathfrak{m}')^n) \quad \Rightarrow \quad d(R) \geq d(R') \quad \Rightarrow \quad d(R'/x'R') \leq d(R) - 1 = d - 1$$

Thus, by the inductive hypothesis, the length of the chain of prime ideals in $R'/x'R'$ is less than or equal to $d - 1$ ($\leq d - 1$). As the image of $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ in $R'/x'R'$ forms a chain of length $r - 1$, we have

$$r - 1 \leq d - 1 \quad \text{iff} \quad r \leq d \quad \text{iff} \quad \text{kdim}(R) \leq d$$

as we sought to show.

Corollary 1.5.18: Finite Dimension of Local Noetherian Ring

Let R be a local Noetherian ring. Then R is finite

Proof:

apply the above result.

Corollary 1.5.19: Height and Dimension

$$\text{height}(\mathfrak{p}) = \text{kdim} R_{\mathfrak{p}}$$

Proof:

Apply the above, noting that the height is the supremum of the length of the prime ideals;

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \dots \subsetneq \mathfrak{p}_r = \mathfrak{p}$$

Corollary 1.5.20: D.C.C. on Prime Ideals of Noetherian Ring

Let R be a Noetherian ring. Then each prime ideal has finite height, and the set of prime ideals satisfies the descending chain condition.

Proof:

exercise.

Proposition 1.5.21: Generation of Local Noetherian Ring and Dimension

Let R be a Noetherian local ring of dimension d . Then there exists an $\mathfrak{m}$ -primary ideal in R generated by d elements $r_1, \dots, r_d$, hence:

$$\dim(R) \geq \delta(R)$$

Proof:

We shall cleverly construct $r_1, \dots, r_d$ inductively in such a way so that every prime ideal containing $(r_1, \dots, r_i)$ has height $\geq i$ for each i .

Suppose $i > 0$ and $r_1, \dots, r_{i-1}$ has been constructed. Let $\mathfrak{p}_j$ with $1 \leq j \leq s$ be the minimal prime ideals of $(r_1, \dots, r_{i-1})$ (if any such minimal prime ideals exist) which has height exactly $i - 1$. As $i - 1 < d = \text{kdim} R = \text{height}(\mathfrak{m})$, we have

$$\mathfrak{m} \neq \mathfrak{p}_j \quad (1 \leq j \leq s) \quad \xrightarrow{\text{cor. 1.5.18}} \quad \mathfrak{m} \neq \bigcup_{j=1}^s \mathfrak{p}_j$$

Now, choose $r_i \in \mathfrak{m} \setminus \bigcup \mathfrak{p}_j$. Let $\mathfrak{q}$ be any prime containing $(r_1, \dots, r_i)$. Then $\mathfrak{q}$ contains some minimal prime ideal $\mathfrak{p}$ of $(r_1, \dots, r_{i-1})$. If $\mathfrak{p} = \mathfrak{p}_j$ for some j , we get $r_i \in \mathfrak{q}$ and $f_i \notin \mathfrak{p}$, and so

$$\mathfrak{q} \supsetneq \mathfrak{p}$$

implying that the height of $\mathfrak{q}$ is $\geq i$. If $\mathfrak{p} \neq \mathfrak{p}_j$ (for $1 \leq j \leq s$) then $\text{height}(\mathfrak{p}) \geq i$, and so $\text{height}(\mathfrak{q}) \geq i$. Thus, every prime ideal containing $(r_1, \dots, r_i)$ has height $\geq i$.

Now, consider $(r_1, \dots, r_d)$. If $\mathfrak{p}$ is a prime ideal of this ideal, then $\mathfrak{p}$ has height $\geq d$, and hence it must be that $\mathfrak{p} = \mathfrak{m}$ or else $\mathfrak{p} \subsetneq \mathfrak{m}$ implying $\text{height}(\mathfrak{p}) < \text{height}(\mathfrak{m}) = d$, which is impossible. Hence, the ideal $(r_1, \dots, r_s)$ is $\mathfrak{m}$ -primary, completing the proof.

Summing up all these results:

Theorem 1.5.22: Dimension Theorem

Let R be a local Noetherian ring. then the following are equivalent:

1. $\text{kdim}(R)$
2. $\deg \chi_{\mathfrak{m}}(n) = \ell(R/\mathfrak{m}^n)$
3. the least number of generator of an $\mathfrak{m}$ -primary ideal of R

Corollary 1.5.23: Dimension and Tangent Space

$$\text{kdim}(R) \leq \dim_k(\mathfrak{m}/\mathfrak{m}^2)$$

Proof:

Take $r_i \in \mathfrak{m}$ ($1 \leq i \leq s$) such that their image in $\mathfrak{m}/\mathfrak{m}^2$ form basis. Then the r_i generate $\mathfrak{m}$, hence:

$$\dim_k(\mathfrak{m}/\mathfrak{m}^2) = s \stackrel{!}{\geq} \text{kdim}(R)$$

Where the $\stackrel{!}{\geq}$ inequality comes from proposition 1.5.21.

Corollary 1.5.24: Height of Minimal Ideals belonging to Set

Let R be a Noetherian ring and let $r_1, \dots, r_s \in R$. Then every minimal ideal $\mathfrak{p}$ belonging to $(r_1, \dots, r_s)0$ has height $\leq s$

Proof:

Take $(r_1, \dots, r_s)0$ in $R_{\mathfrak{p}}$, which becomes a $\mathfrak{p}^e$ -primary ideal. Hence:

$$s \geq \dim R_{\mathfrak{p}} = \text{height}(\mathfrak{p})$$

Corollary 1.5.25: Krull Principal Ideal Theorem

Let R be a Noetherian ring and let $x \in R$ be an element which is neither a zero-divisor nor a unit. Then every minimal prime ideal $\mathfrak{p}$ of (x) has height 1

Proof:

By corollary 1.5.24, $\text{height}(\mathfrak{p}) \leq 1$. If $\mathfrak{p} = 0$, then $\mathfrak{p}$ is a prime ideal belonging to 0, but then every element of $\mathfrak{p}$ is a zero-divisor, contradicting our assumption.

Corollary 1.5.26: Quotient of Local Noetherian Ring

Let R be a local Noetherian ring, $x \in \mathfrak{m} \subseteq M$ an element of the maximal ideal which is not a zero-divisor. Then:

$$\text{kdim} R/xR = \text{kdim} R - 1$$

Proof:

Let $d = \text{kdim}(R/xR)$. By corollary 1.5.16 and theorem 1.5.22 we get

$$d \leq \text{kdim}(R) - 1$$

On the other hand, if r_i ($1 \leq i \leq d$) are the elements of $\mathfrak{m}$ whose image in R/xR generate an $(\mathfrak{m}/xR)$ -primary ideal, then the ideal $(x, r_1, \dots, r_d)$ in R is $\mathfrak{m}$ -primary, and so

$$d + 1 \geq \text{kdim}(R)$$

Corollary 1.5.27: Dimension of Completion

Let $\hat{R}$ be the $\mathfrak{m}$ -adic completion of the local Noetherian ring R . Then

$$\text{kdim}(R) = \text{kdim}(\hat{R})$$

Proof:

As $R/\mathfrak{m}^n \cong \hat{R}/\hat{\mathfrak{m}}^n$, $\chi_{\mathfrak{m}}(n) = \chi_{\hat{\mathfrak{m}}}(n)$, and so apply the dimension theorem.

The following will be important for an equivalent definition of regular local ring

Proposition 1.5.28: System of Parameters

Let $r_1, \dots, r_d$ be a system of parameters for R and let $\mathfrak{q} = (r_1, \dots, r_d)$ be the $\mathfrak{m}$ -primary ideal generated by them. Let $f(x_1, \dots, x_d)$ be a homogeneous polynomial of degree s with coefficient in R . Assume that :

$$f(r_1, \dots, r_d) \in \mathfrak{q}^{s+1}$$

Then all the coefficients of f lie in $\mathfrak{m}$

Proof:

Take the surjective homomorphism:

$$\alpha : (R/\mathfrak{q})[x_1, \dots, x_d] \rightarrow G_{\mathfrak{q}}(R)$$

where $x_i \mapsto \bar{r}_i$. Then by assumption, $f \in \ker(\alpha)$. Then as $\bar{f}$ can is not a zero-divisor, we have:

$$\begin{aligned} d(G_{\mathfrak{q}}(R)) &\leq d((R/\mathfrak{q})[x_1, \dots, x_d]/(\bar{f})) \\ &= d((R/\mathfrak{q})[x_1, \dots, x_d]) - 1 \\ &= d - 1 \end{aligned}$$

But we know that $d(G_{\mathfrak{q}}(R)) = d$ by the Dimension Theorem, a contradiction.

If R contains a field k mapping isomorphically onto the residue field $R/\mathfrak{m}$, we get the following

Corollary 1.5.29: Residue Field and Algebraic Independence

Let $k \subseteq R$ be a field mapping isomorphically onto $R/\mathfrak{m}$ and let $r_1, \dots, r_d$ be a system of parameters. Then $r_1, \dots, r_d$ are algebraically independent over k

Proof:

For the sake of contradiction, assume $f(r_1, \dots, r_d) = 0$ where f is a polynomial with coefficients in k . If $f \neq 0$, then $f = f_s + g$ where g has some higher terms and f_s is homogeneous of degree s . Then $f_s \neq 0$. By proposition 1.5.28, f_s has all coefficients in $\mathfrak{m}$. As f_s has coefficients in k we get

$$f_s \equiv 0$$

but we can't both have $f_s \equiv 0$ and $f_s \neq 0$, a contradiction. Hence, no such f can exist, and these are algebraically independent.

1.5.3 Smoothness: Regular Local Rings

One of the fundamental properties of many geometric object is the idea of *smoothness* or *nonsingular*. In this section, we shall introduce an intuitive way of defining smoothness given an embedding into an affine/projective space. We shall then show a more intrinsic definition that will not require any embedding.

Definition 1.5.30: Nonsingular Variety - Embedding Variant

Let $X \subseteq \mathbb{A}_k^n$ be an affine variety, $f_1, \dots, f_n \in k[X]$ be generators, and $\text{kdim}(X) = r$. Then X is *nonsingular* at a point p if

$$\text{rank} \left(\frac{\partial f_i}{\partial x_j} \right) = n - r$$

We say X is *nonsingular* if it is nonsingular at every point.

If the reader knows differential geometry, it should be clear that this is saying that the tangent space must be “full rank” at every point. The reader should check that this definition is independent of choice of generators (see exercise ref:HERE). The matrix is called the *Jacobian Matrix*. This definition is geometrically well-motivated, but is not intrinsic to the variety: it is dependent on the “global” information of the embedding space and generators. We shall immediately work towards the intrinsic definition of nonsingular variety. The key is that we at each point p , the local ring $\mathcal{O}_{X,p}$ has a singular maximal ideal $\mathfrak{m}_p$ representing where functions evaluate to 0, and $\mathfrak{m}_p/\mathfrak{m}_p^2$ “remembers” only linear terms, making it the best analogy to the cotangent space at a point. Recall that $\dim_k \mathfrak{m}_p/\mathfrak{m}_p^2 \geq \text{kdim} k[X]$ by corollary 1.5.23. This is because the curve is singular at a point (not smooth), then its tangent space (cotangent space) shall increase dimension. We thus seek conditions where the local ring at each point shall have the same dimension everywhere:

Theorem 1.5.31: Regular Local Ring Equivalent Conditions

Let R be a local Noetherian ring of dimension d with maximal ideal $\mathfrak{m}$ and residue field $k = R/\mathfrak{m}$. Then the following are equivalent:

1. $G_{\mathfrak{m}}(R) \cong k[t_1, \dots, t_d]$ where the t_i are independent and indeterminate
2. $\dim_k(\mathfrak{m}/\mathfrak{m}^2) = d$
3. $\mathfrak{m}$ can be generated by d elements

Proof:

These are all straight-forward. (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (3) are immediate (use the above corollaries). For (3) $\Rightarrow$ (1), let $\mathfrak{m} = (r_1, \dots, r_d)$. Then by proposition 1.5.28 the map

$$\alpha : k[r_1, \dots, r_d] \rightarrow G_{\mathfrak{m}}(R)$$

is an isomorphism of graded rings.

Definition 1.5.32: Regular Local Ring

Let R be a local Noetherian ring. Then if R satisfies any of the conditions in theorem 1.5.31, R is called a *regular local ring*.

Proposition 1.5.33: Jacobson Matrix and Regular Local Ring

Let $X \subseteq \mathbb{A}_k^n$ be a variety. Then X is nonsingular at p iff the local ring at p is regular.

Proof:

Let $p = (a_1, \dots, a_n) \in \mathbb{A}^n$, let $\mathfrak{a}_p = (x_1 - a_1, \dots, x_n - a_n)$ be the corresponding maximal ideal in $R = k[x_1, \dots, x_n]$. Define a linear map $\theta_p : A \rightarrow k^n$ by

$$\theta_p(f) = \left\langle \frac{\partial f}{\partial x_1}(p), \dots, \frac{\partial f}{\partial x_n}(p) \right\rangle$$

It is clear that $\theta_p(x_i - a_i)$, $i = 1, \dots, n$, forms a basis of k^n , and $\ker \theta_p = \mathfrak{a}_p^2$. Thus θ_p induces an isomorphism $\theta'_p : \mathfrak{a}_p / \mathfrak{a}_p^2 \rightarrow k^n$.

Now let $\mathfrak{b}$ be the ideal of X in A , and let $f_1, \dots, f_t$ be a set of generators of $\mathfrak{b}$. Then the rank of the Jacobian matrix $J = (\partial f_i / \partial x_j)(p)$ is just the dimension of $\theta_p(\mathfrak{b})$ as a subspace of k^n . Using the isomorphism θ'_p , this is the same as the dimension of the subspace $(\mathfrak{b} + \mathfrak{a}_p^2) / \mathfrak{a}_p^2$ of $\mathfrak{a}_p / \mathfrak{a}_p^2$. On the other hand, the local ring $\mathcal{O}_p$ of p on X is obtained from A by dividing by $\mathfrak{b}$ and localizing at the maximal ideal $\mathfrak{a}_p$. Thus if $\mathfrak{m}$ is the maximal ideal of $\mathcal{O}_p$, we have

$$\mathfrak{m} / \mathfrak{m}^2 \cong \mathfrak{a}_p / (\mathfrak{b} + \mathfrak{a}_p^2).$$

Counting dimensions of vector spaces, we have $\dim \mathfrak{m} / \mathfrak{m}^2 + \text{rank}(J) = n$.

But this gives what we want. If $\dim X = r$, then $\mathcal{O}_p$ is a local ring of dimension r , so $\mathcal{O}_p$ is regular if and only if $\dim_k \mathfrak{m} / \mathfrak{m}^2 = r$. But this is equivalent to $\text{rank}(J) = n - r$, which says that p is a nonsingular point of X , completing the proof.

From this, we can extend the definition naturally to any quasiprojective variety.

Definition 1.5.34: Nonsingular Variety

Let X be a quasi-projective variety. Then X is *nonsingular* or *regular* if the local ring at every point is regular.

The set of points that are singular are usually “measure 0”. We shall not go so far to prove this statement, but the following simpler result that a variety cannot *only* consist of singular points:

Proposition 1.5.35: Singular Points of Variety

Let X be a variety. Then the set of singular points of a variety from of proper closed subset

The fact that its closed should be “obvious” as it is essentially the pre-image of the closed set 0 for the right continuous function, the key take-away is that it must be proper.

Proof:

Let $\text{Sing}(X)$ represent the set of singular points of X . Let us start by showing $\text{Sing}(X)$ is closed. By taking some open covering $X = \bigcup X_i$ of X , it suffices to show that $\text{Sing} X_i$ is closed for each i . Hence may assume that X is affine. Then as the rank of the Jacobian matrix is always $\leq n - r$, the set of singular points is the set of points where the rank is $< n - r$. Thus $\text{Sing}(X)$ is the algebraic set defined by the ideal generated by $I(X)$ together with all determinants of $(n - r) \times (n - r)$ sub-matrices of the matrix $\|\partial f_i / \partial x_j\|$. From these, we can see that $\text{Sing} X$ is closed.

Next, let's show that $\text{Sing}(X)$ is a proper subset of X . By slightly modifying theorem 1.3.21 for quasi-projective varieties, we see that X is birational to a hypersurface in $\mathbb{P}^n$. Since birational varieties have isomorphic open subsets, we can reduce to the case of a hypersurface. Then it is enough to consider any open affine subset of X , so we may assume that X is a hypersurface in $\mathbb{A}^n$, defined by a single irreducible polynomial $f(x_1, \dots, x_n) = 0$.

Now, $\text{Sing}(X)$ is the set of points $p \in X$ such that $(\partial f / \partial x_i)(p) = 0$ for $i = 1, \dots, n$. If $\text{Sing} X = X$, then the functions $\partial f / \partial x_i$ are zero on X , and hence $\partial f / \partial x_i \in I(X)$ for each i . But $I(X)$ is the principal ideal generated by f , and $\deg(\partial f / \partial x_i) \leq \deg f - 1$ for each i , so we must have $\partial f / \partial x_i = 0$ for each i .

In characteristic 0 this is already impossible, because if x_i occurs in f , then $\partial f / \partial x_i \neq 0$. So we must have $\text{char } k = p > 0$, and then the fact that $\partial f / \partial x_i = 0$ implies that f is actually a polynomial in x_i^p . This is true for each i , so by taking p th roots of the coefficients (which is possible since k is algebraically closed), we get a polynomial $g(x_1, \dots, x_n)$ such that $f = g^p$. But this contradicts the hypothesis that f was irreducible, and so $\text{Sing} X < X$ as we sought to show.

1.5.4 Relating local dimension and transcendental dimension

Example 1.5.36: Infinite Global Dimension When There is A Singularity

Take $R = k[x, y] / (x^2 - y^3)$. This corresponds to the cusp singularity at the origin. Let us compute the global dimension by finding the projective dimension of the residue field $k \cong R / (x, y)$ (where $\mathfrak{m} = (\bar{x}, \bar{y})$). We begin a free resolution of k :

$$R \xrightarrow{d_1} k \rightarrow 0$$

where $d_1 = \begin{pmatrix} x & y \end{pmatrix}$. The kernel of this map consists of relations between x and y . In the polynomial ring, the only relation is trivial ($yx - xy = 0$). However, in our quotient ring R , we have the relation defining the ring: $x^2 - y^3 = 0$. Thus, the kernel is generated by two elements:

1. The “Koszul” relation: $y \cdot (x) - x \cdot (y) = 0$
2. The “Curve” relation: $x \cdot (x) - y^2 \cdot (y) = 0$

So we have a map $R^2 \xrightarrow{d_2} R$ given by the matrix $\begin{pmatrix} y & x \\ -x & -y^2 \end{pmatrix}$. (Check: $\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} y & x \\ -x & -y^2 \end{pmatrix} = \begin{pmatrix} xy - yx & x^2 - y^3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix}$).

Now we need the kernel of d_2 . Calculating the determinant of the matrix, we find $\det(d_2) = -y^3 + x^2 = 0$. Because the determinant vanishes, the columns are linearly dependent, and the kernel is non-trivial. In fact, the resolution becomes **periodic**. The resolution continues forever with alternating matrices:

$$\dots \xrightarrow{d_3} R^2 \xrightarrow{d_2} R^2 \xrightarrow{d_1} R \rightarrow k \rightarrow 0$$

where $d_{\text{odd}} = \begin{pmatrix} y & x \\ -x & -y^2 \end{pmatrix}$ and $d_{\text{even}} = \begin{pmatrix} y^2 & x \\ -x & -y \end{pmatrix}$. Since the resolution never terminates (it never reaches a 0 module), the projective dimension of k is infinite. Therefore, the global dimension of R is infinite.

Before stating the main theorem on regular local rings, we need a homological invariant²⁵:

Definition 1.5.37: Projective and Global Dimension

Let R be a ring and M an R -module. The *projective dimension* of M , denoted $\text{pd}_R(M)$, is the minimal length n of a projective resolution of M :

$$0 \rightarrow P_n \rightarrow \cdots \rightarrow P_0 \rightarrow M \rightarrow 0$$

If no finite resolution exists, $\text{pd}_R(M) = \infty$.

The *global dimension* of R , denoted $\text{gldim}(R)$, is the supremum of the projective dimensions of all R -modules:

$$\text{gldim}(R) = \sup\{\text{pd}_R(M) \mid M \in \text{Mod}_R\}$$

For a local ring $(R, \mathfrak{m})$ with residue field k , a fundamental result of Auslander and Buchsbaum states that $\text{gldim}(R) = \text{pd}_R(k)$. Thus, the global dimension is determined entirely by the resolution of the residue field.

Theorem 1.5.38: Serre's Theorem

A local Noetherian ring $(R, \mathfrak{m})$ has finite global dimension if and only if it is a Regular Local Ring.

Proof:

We shall take for granted the result of ?? that for a local ring,

$$\text{gldim}(R) = \text{pd}_R(k)$$

Thus, we must show that R is regular if and only if k has a finite free resolution.

($\Leftarrow$) Suppose R is a regular local ring of dimension n . Then $\mathfrak{m}$ is generated by n elements $x_1, \dots, x_n$. Since R is regular, these elements form a *regular sequence* (meaning x_i is not a zero divisor in $R/(x_1, \dots, x_{i-1})$). The *Koszul Complex* $K_\bullet(x_1, \dots, x_n)$ associated to this sequence provides an explicit free resolution of k :

$$0 \rightarrow R^{\binom{n}{n}} \rightarrow \cdots \rightarrow R^{\binom{n}{1}} \rightarrow R \rightarrow k \rightarrow 0$$

This resolution has finite length n . Thus $\text{pd}_R(k) = n < \infty$, so R has finite global dimension.

($\Rightarrow$) We proceed by induction on the dimension $d = \text{kdim}(R)$. We assume $\text{gldim}(R) < \infty$.

Case $d = 0$: If $\text{dim}(R) = 0$, then R is an Artinian ring. A local Artinian ring with finite global dimension must be a field (a known result of Auslander-Buchsbaum: if $\text{pd}_R(k) < \infty$ for Artinian R , then $\text{pd}_R(k) = 0$, so R is free over itself, i.e., a field). A field is clearly a regular local ring (dimension 0, generated by 0 elements).

Case $d > 0$: Since $\text{gldim}(R) < \infty$, we can find an element $x \in \mathfrak{m} \setminus \mathfrak{m}^2$ that is *not* a zero-divisor (if all elements in $\mathfrak{m}$ were zero-divisors or in $\mathfrak{m}^2$, the projective dimension would be infinite). Consider the quotient ring $\overline{R} = R/(x)$. Since x is not a zero divisor and $x \notin \mathfrak{m}^2$, we have:

²⁵These terms will be central in section ?? and will be re-introduced there

1. $\dim(\overline{R}) = d - 1$ (by corollary 1.5.26)
2. $\text{gldim}(\overline{R}) = \text{gldim}(R) - 1 < \infty$ (a standard homological identity for non-zero divisors).

By the inductive hypothesis, $\overline{R}$ is a regular local ring of dimension $d - 1$. Therefore, its maximal ideal $\overline{\mathfrak{m}} = \mathfrak{m}/(x)$ can be generated by $d - 1$ elements, say $\overline{y}_1, \dots, \overline{y}_{d-1}$. Lifting these back to R , the ideal $\mathfrak{m}$ is generated by $\{x, y_1, \dots, y_{d-1}\}$. Thus $\mathfrak{m}$ is generated by $(d - 1) + 1 = d$ elements. Since $\dim(R) = d$ and $\mathfrak{m}$ is generated by d elements, R is a regular local ring.

1.5.5 Nonsingular Curves and Surfaces

Let us now take a closer at other properties regular local rings must have, especially in low dimension. In one dimension, this shall produce a very restrictive result that will be very powerful, and in higher dimensions we shall find an important algebraic conditions that shall eventually allows us to interpret ring of integers (defined in the next chapter) as nonsingular $\mathbb{Z}$ -curves. For curves in particular, we shall be able to show that in each birational equivalence class of curves, there is a unique nonsingular projective variety that represents the class. We shall thus be able to say that a field extension K/k of transcendental degree one has *the* curve $C_{K/k}$ associated to it.

We first requires the following technical lemma:

Lemma 1.5.39: Regular Local Ring Always Integral Domain

Let R be a ring and $\mathfrak{a}$ an ideal of R such that

$$\bigcap_n \mathfrak{a}^n = 0$$

Suppose that $G_{\mathfrak{a}}(R)$ is an integral domain. Then R is an integral domain

Proof:

Let $x, y \in R$ be non-zero elements. As $\bigcap_n \mathfrak{a}^n = 0$, there exists $r, s \geq 0$ such that

$$\begin{aligned} x &\in \mathfrak{a}^r & x &\notin \mathfrak{a}^{r+1} \\ y &\in \mathfrak{a}^s & y &\notin \mathfrak{a}^{s+1} \end{aligned}$$

Let $\overline{x}, \overline{y}$ be the images of x, y in $G_r(R)$ and $G_s(R)$ respectively. Then $\overline{x} \neq 0$ and $\overline{y} \neq 0$. Hence $\overline{x}\overline{y} \neq 0$, and so $xy \neq 0$, completing the proof.

Corollary 1.5.40: Regular Local Rings of Dimension 1

Let R be a local Noetherian ring of dimension 1. Then R is a regular local ring if and only if R is a DVR.

Proof:

Let $(R, \mathfrak{m})$ be a regular local ring of dimension 1. Then $\text{kdim} R = 1 = \dim \mathfrak{m}/\mathfrak{m}^2$. Then $\mathfrak{m}/\mathfrak{m}^2$ is a 1 dimensional $R/\mathfrak{m}$ -vector space. Then as a consequence of Nakayama's lemma (??), we can lift and conclude that $\mathfrak{m}$ is a 1-dimensional free R -module, namely $R \cong \mathfrak{m}$. But then there is some m that 1 must map to, hence $(m) \cong \mathfrak{m}$, showing that $\mathfrak{m}$ is principal. Then by exercise ref:HERE, all of the other ideals are principal (namely they are $(m)^p$ for some power of p), and so by ?? R is a DVR.

Conversely, if R is a DVR, it is a local ring as it has a single prime (and maximal) ideal (m) which is principal. Then $\text{kdim} R = 1$. Certainly, $(m)/(m)^2$ is a 1-dimensional R/m -vector space, hence we have the equality, DVR's are local Noetherian rings of dimension 1, as we sought to show.

This gives a nice condition for 1-dimensional regular local rings. It will sometimes be useful to identify regular local rings in different context:

Proposition 1.5.41: Equivalences of 1 Dimensional Regular Local Rings

Let $(R, \mathfrak{m})$ be a Noetherian local domain of dimension 1. Then the following are equivalent:

1. R is a regular local ring
2. R is a DVR
3. R is normal (integrally closed)
4. $\mathfrak{m}$ is principal.

Proof:

exercise.

For higher-dimensions, it is more subtle in ways that go beyond this book (see cite algoGeo, Divisors), but we still get the following:

Corollary 1.5.42: Regular Local Ring, then Normal

Let R be a regular local Noetherian ring. Then R is normal.

Proof:

Let R be a local ring. If $G_{\mathfrak{m}}(R)$ is a normal domain, then we can show that R is a normal domain, and hence regular local ring is normal, completing the proof.

The converse does not hold in general in dimension > 1 , namely a normal domain need not be regular (see CITE:ALG-GEO), and so normalness is insufficient of a condition to find nonsingular varieties.

Proposition 1.5.43: Completion and Regularity

Let R be a Noetherian local ring. Then R is regular iff $\overline{R}$ is regular

Proof:

This is a matter of putting together the right results. Show that $\overline{R}$ is a Noetherian local ring of the same dimension as R with maximal ideal $\overline{\mathfrak{m}}$. Then assert that $G_{\mathfrak{m}}(R) \cong G_{\overline{\mathfrak{m}}}(\overline{R})$ to complete the proof.

In this case, it must be that $\overline{R}$ is also an integral domain. For the geometric interpretation, see CITE:ALG-GEO.

Exercise 1.5.1

1. (Compare Gelfand-Kirillov Dimension to Krull dimension in Commutative case; the former can be defined more generally and can give fractional dimensions)
2. Define the *depth* of a prime $\mathfrak{p}$ to be the chains of ideals which start at $\mathfrak{p}$. Show that $\text{depth}(\mathfrak{p}) = \dim R/\mathfrak{p}$ if R is Noetherian, and give an example where this is infinite. Show this is always finite if R is a local ring.
3. Let R be a Noetherian Ring. Show that $\text{kdim} R[x] = \text{kdim} R + 1$

1.6 Projective Variety

So far, all of the algebraic sets were called *affine* algebraic sets, as they were all a subset of $\mathbb{A}_k^n$. These offer great geometrical intuition, however they are lacking in certain algebraic qualities, namely rational functions need not be defined at all points, and not all algebraic curves intersect. As an example of the former point, it is simplest to resort to an analogy from calculus. The reader may recall that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = 1/x$ is famously undefined at $x = 0$, as we cannot divide by 0.

However, we can enlarge $\mathbb{R}$ to make this function well-defined by adding a *point at infinity*. Namely, we may take $\mathbb{R}_{\infty} = \mathbb{R} \cup \{\infty\}$ where (for the sake of limits)²⁶ we extend the arithmetic's of $\mathbb{R}$ to $\mathbb{R}_{\infty}$ by doing:

$$a + \infty = \infty + a = \infty \quad b \cdot \infty = \infty \cdot b = \infty$$

where $a \in \mathbb{R}$ and $0 \neq b \in \mathbb{R}_{\infty}$. We then say that if (x_n) diverges, then

$$\lim_{n \rightarrow \infty} x_n = \infty$$

In this case, we now have that the function $1/x$ has a well-defined value of ∞ at 0, and even better under the right frame-work (provided in differential geometry), the function $f(x) = 1/x$ can be shown to be smooth on $\mathbb{R}_{\infty}$. In this way, we have insured that we cannot have an undefined rational polynomials on $\mathbb{R}_{\infty}$! The space $\mathbb{R}_{\infty}$ is called the *projective line*. This “closure” on the set of definable rational polynomials makes the projective line an algebraically nicer space to work with, in a similar way that $\mathbb{C}$ is the better space to work in as all polynomials have roots.

For the latter interpretation (the one about there always being an intersection), consider a polynomial $f(z)$ defined over $\mathbb{C}$ so that there is no worry about roots not existing. Let us consider $\mathbb{C}$ has a $\mathbb{R}^2$

²⁶The arithmetic defined here certainly doesn't satisfy of a ring; instead it's a structure called a *wheel*; we shall not study these in this textbook

with coordinates x, y so that $f(z) = f(x, y)$. Consider the line $L_{(a,b)}$ which has angle α given by (a, b) :

$$L_{(a,b)} : t \mapsto (at, bt)$$

Then either $f = L$, in which case their intersection is an entire line, or f and $L_{(a,b)}$ intersect where these two equations agree:

$$f(at, bt) = f_n(a, b)t^n + \cdots + f_1(a, b)t + f_0(a, b)$$

where $f_n(a, b)$ represents the constant in front of the t -term. if $f_n(a, b) \neq 0$, then by we know there are n roots, and hence n points of intersections. However, if $f_n(a, b) = 0$, then there must be a smallest m where

$$f_n(a, b) = 0, f_{n-1}(a, b) = 0, \dots, f_{m+1}(a, b) = 0, f_m(a, b) \neq 0$$

or else $f = L_{(a,b)}$, which we are taking to not be the case. But then we would have a number of intersections lower than the number of roots. There is a remedy to this, which is to consider the remaining $(n - m)$ points to be the intersection of L with f at a point at infinity. To see this, consider the substitution $s = 1/t$. Then:

$$f_n(a, b) + f_{n-1}(a, b)s + \cdots + f_0(a, b)s^n = 0$$

Notice that if $f_n(a, b) \neq 0$, then at $s = 0$ (conceptually can be thought of as $1/t = 0$) there is no root. But if there are $n - m$ leading terms that are zero, then this polynomial has a root of order $n - m$ at $s = 0$. Thus, we recover the intersection of L with f as an invariance! In particular, for each line $L_{(a,b)}$ we can think of there being an extra point at infinity. Notice that by combining the coordinates s, t we get a function:

$$f_n(a, b)t^n + f_{n-1}t^{n-1}s + \cdots + f_0(a, b)s^n = 0$$

which allows us to find all points of intersections at once. This motivates the use of homogeneous functions and the linear equivalence construction that shall be at the start of this section.

Why is the space called projective? This is understood from the following construction of projective space that extends to higher dimensions. Let us start with one dimensions. We may define $\mathbb{R}_\infty$ as the

$$\mathbb{RP}^1 := \frac{\mathbb{R}^2 \setminus \{(0, 0)\}}{\{(x, y) \sim (\lambda x, \lambda y), \forall \lambda \in \mathbb{R}_{\lambda \neq 0}\}}$$

by taking the representative of each equivalence class to have norm 1, notice that we may associate all the points of $\mathbb{RP}^1$ to S^1 , and we can associate points of $\mathbb{R}$ with the equivalence class where $y \neq 0$. The equivalence class $[(\lambda x, 0)]$ can be thought of as the “point at infinity” that we’ve added, hence we may think of $\mathbb{RP}^1$ as

$$\mathbb{RP}^1 = \mathbb{R}^1 \sqcup \mathbb{R}^0 \tag{1.5}$$

The term “projective” come from the idea that parallel lines will meet at the “point at infinity”. In terms of “closure” that we talked about earlier, the reader may recall that in classical geometry, two non-parallel lines will always have a solution. In $\mathbb{RP}^1$, we have added the extra property that there is one point at which parallel lines meet; they also have a solution. In terms of linear algebra, every system of linear equations has a solution.

We may extend this further, and take $\mathbb{R}^2$ and add “points at infinity”. In this case, there are many possible pairs of parallel lines. We may consider that up to scaling, we have many new points that we may add. Formally, let:

$$\mathbb{RP}^2 := \frac{\mathbb{R}^3 \setminus \{(0,0,0)\}}{\{(x,y,z) \sim (\lambda x, \lambda y, \lambda z), \forall \lambda \in \mathbb{R}_{\lambda \neq 0}\}}$$

We may imagine that this insures that any rational functions in two variables always has solution. Adding multiple points at infinity insures that the function $1/y$, $1/x$, $1/xy$ and so forth are all considered distinct when evaluating ($1/y$ is zero when evaluated at one infinity but not another). In terms of parallel lines, we may think that parallel lines in any angle will have a unique solution (note that the distance between lines not matter, they still “meet” at the same point at infinity). The shape $\mathbb{RP}^2$ can be thought of as $\mathbb{R}^2$ with a circle of points at infinity added where opposites are glued. Can the reader use this intuition to generalize equation (1.5) to:

$$\mathbb{RP}^2 = \mathbb{R}^2 \sqcup \mathbb{R}^1 \sqcup \mathbb{R}^0$$

The reader familiar with differential geometry will know that $\mathbb{RP}^2$ is a two dimensional smooth manifold, and is one of the simplest example of a shape that is 2-dimensional but cannot be embedded in 3-dimensions without self-intersection. The following image from Wikipedia may help in trying to visualize (note that this is an *immersion*, meaning it is an approximation of the shape by embedding it in a space in which there is not enough dimensions to visualize it without intersection²⁷)

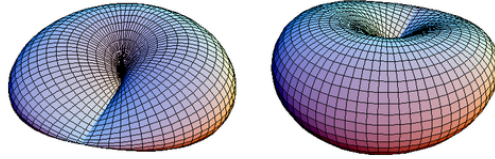


Figure 1.2: Wikipedia: Immersion of $\mathbb{RP}^2$

We may continue this process and define $\mathbb{RP}^n$, or more generally $\mathbb{P}_k^n$ for some field k . These shall have a larger class of functions that we may define on them (we shall eventually say that there are an “ample” amount of function²⁸). Thus, we shall treat $\mathbb{P}_k^n$ as an extension of $\mathbb{A}_k^n$, and consider the equivalence classes of $\mathbb{P}_k^n$ as *points*.

An important technicality that we may encounter is that we must convert a function that is defined on $\mathbb{A}_k^n$ to a function defined on $\mathbb{P}_k^n$, namely the points of $\mathbb{P}_k^n$ are now equivalence relations and hence a function whose domain is $\mathbb{P}_k^n$ must be independent of representative. This shall form an important class of rings that are “good representatives” to extend polynomials to $\mathbb{P}_k^n$. As we are extending results from $\mathbb{A}_k^n$, this class of rings shall have a similar dictionary as that provided between affine varieties and radical ideals of coordinate rings, namely we shall show that polynomials solutions over $\mathbb{P}_k^n$ shall correspond to *graded ideals of graded rings*.

²⁷See CITE:DIFF-GEO

²⁸More technically speaking, we shall say that $\mathbb{P}_k^n$ has *ample line bundles*, a technicality that shall be elucidated in CITE:ALG-GEO

1.6.1 Definitions and Properties

Let us first formally define projective space

Definition 1.6.1: Projective Space

Let k be a field. Then *projective space* over k of dimension n is:

$$\mathbb{P}_k^n = \frac{k^{n+1} \setminus \{0\}}{(x \sim \lambda x, \forall x \in k^{n+1})}$$

The points of $\mathbb{P}_k^n$ are often denoted $[x_1 : x_2 : \cdots : x_{n+1}]$.

The first thing we must do is define the topology on this space which parallels the Zariski topology on $\mathbb{A}_k^n$. Polynomials function need not give well-defined zero-sets on projective space, for example $f(x, y) = x^2 + y$ does not map two elements in the same equivalence class to zero. We must limit to polynomials that maps the representatives of an equivalence class to the same point. These are precisely the homogeneous polynomials:

Definition 1.6.2: Homogeneous Polynomial (Algebraic Form)

Let $f \in k[x_1, \dots, x_n]$ be a polynomial. Then f is said to be a *homogeneous polynomial* of degree d or an *algebraic d -form* if

$$f(\lambda x_1, \dots, \lambda x_n) = \lambda^d f(x_1, \dots, x_n)$$

The term *algebraic form* is a bit out-dated, but is often still used for the degree two case, where such algebraic forms are called *quadratic forms*. This definition can be defined on more general class of functions. The reader comfortable with some analysis will see in exercise 1.6.1 that any (at least) d -times differential degree d homogeneous function is in fact a homogeneous polynomial²⁹.

Example 1.6.3: Homogeneous Polynomials

All homogeneous polynomials $f \in k[x_1, \dots, x_n]$ are polynomials for which all monomials have the same degree (exercise). For example:

1. $f(x, y, z) = x^3 + xy^2 - 2y^3$ is homogeneous
2. $f(x, y) = x^2 + y^3$ is not homogeneous

Notice that if $f_1, \dots, f_n$ are all homogeneous polynomial of the same degree k , then $(f_1 + \dots + f_n)^m$ is a homogeneous polynomial of degree mk . Hence, homogeneous polynomials behave well under addition and multiplication.

If $(a_1, \dots, a_n) \in k^n$ is a solution to a homogeneous polynomials, then all points on the line passing through this point and the origin is also a solution, hence homogeneous polynomials are the “natural” polynomials for the zero set of projective space, as projective space is an equivalence relation on lines. It is tempting conclude the collection of homogeneous polynomials form the set $k[\mathbb{P}_k^n]$ where the set is either equal to $\text{Hom}_{\mathbf{Reg}}(\mathbb{P}_k^n, \mathbb{A}_k^1)$ or $\text{Hom}_{\mathbf{Reg}}(\mathbb{P}_k^n, \mathbb{P}_k^1)$ and we get the same theory as with affine

²⁹homogeneity does not imply differentiability or continuity, can you think of a counter-example?

spaces. However, neither of these sets is the collection of homogeneous polynomials. If we consider $\text{Hom}_{\mathbf{Reg}}(\mathbb{P}_k^n, \mathbb{P}_k^1)$, we have the problem that $[0 : 0] \notin \mathbb{P}_k^1$ and so we lose the fundamental property that the topology is defined by the zero-sets. The reader can further check that if $f : \mathbb{P}_k^n \rightarrow \mathbb{A}_k^1$ is homogeneous, then extending to $\mathbb{P}_k^1$ by mapping to $[f(x_1, \dots, x_n) : 1]$ is not well-defined³⁰ (we shall see that $\mathbb{A}_k^1$ is dense in $\mathbb{P}_k^1$ and the given extension is the *only* continuous extension, and it does not work).

For the first option; even when $n = 1$ there is a problem of well-definedness. Consider $f : \mathbb{P}_k^1 \rightarrow \mathbb{A}_k^1$ given by $f([x : y]) = f\left(\left[\frac{x}{y} : 1\right]\right) = \frac{x}{y}$, i.e. the closest we can do to define the identity. The reader may immediately see the problem: when $y = 0$, x/y is not defined (the codomain is $\mathbb{A}_k^1$). Even if we “ignore” this problem by simply not considering where $y = 0$, we have that f is not well-defined:

$$\frac{x}{y} = f\left(\left[\frac{x}{y} : 1\right]\right) = f([x : y]) = f\left(\left[1 : \frac{y}{x}\right]\right) = 1$$

But then f is not well-defined as it is not the case that for all $x, y \in k$, $x/y = 1$. If we instead try $f([x : y]) = xy$ then

$$xy = f([x : y]) = f\left(\left[\frac{x}{y} : 1\right]\right) = \left(\frac{x}{y}\right) \cdot 1 = \frac{x}{y}$$

but it is not always the case that $xy = \frac{x}{y}$ (ex. let $x = y = 2$), and hence it is once again not well-defined. The reader shall find that any such attempt to define a (non-constant) polynomial shall not be well-defined, namely since $f(\lambda x, \lambda y) \neq f(x, y)$.

This may unsettle the reader, as a “regular function” can’t be well-defined unless the polynomial is constant. As it turns out, we can’t extend (non-constant) polynomials from affine space to projective in a well-defined manner (see section 1.6.2) unless it is a constant, which the above exploration is a first demonstration on why this is the case.

However, we will be able to treat homogeneous polynomials as regular functions whenever we limit to one of the variables x_i to be nonzero. In particular, if $f \in k[x_1, \dots, x_{n+1}]$ is homogeneous, and $x_i \neq 0$, then we can interpret f to be a regular function on $U_i = \{x \in \mathbb{P}_k^n \mid x_i \neq 0\}$:

$$f([x_1 : \dots : x_i : \dots : x_{n+1}]) = f\left(\left[\frac{x_1}{x_i} : \dots : 1 : \dots : \frac{x_{n+1}}{x_i}\right]\right) = f(x'_1, \dots, x'_{n+1})$$

where we write $x'_j = \frac{x_j}{x_i}$ as $x_i \neq 0$. Notice that since the i th position can be fixed to 1 by dividing by x_i , we have chosen a representative of this equivalence relation, which is why we dropped the square brackets in the above equation. For example, for the homogeneous polynomial $x \in k[x, y]$, when $y \neq 0$, we can think of it as defining the function:

$$f(x) = x$$

and when $x \neq 0$, we can think of it as defining the function:

$$f(y) = 1$$

This mirrors how $f([x : y]) = f([x/y : 1]) = f([x' : 1]) = x'$ and $f([x : y]) = f([1 : y/x]) = f([1 : y']) = 1$. Notice that it was important to define two functions and how each behaves when $y \neq 0$ and

³⁰There shall be well-defined regular maps from $\mathbb{P}_k^n$ to $\mathbb{P}_k^1$, see section 1.6.3, but these are not extensions of affine regular maps

$x \neq 0$ respectively. If we want to recover the homogeneous polynomial that defined these functions but only defined it on $x \neq 0$, we see that the $y \neq 0$ case can be λx or simply λ for any nonzero λ , showing we require the function to be defined in both situations (we shall see that having the function defined for each $x_i \neq 0$ and is compatible via “transition mappings” is enough to recover the homogeneous polynomial).

This also means that homogeneous polynomials on $\mathbb{P}_k^n$ act less like functions and more like vector-fields which “locally” are given by functions (i.e. are “local trivializations”). For the reader familiar with Differential Geometry, they may recognize that this vocabulary is used to define *vector bundles*, and that homogeneous polynomials can be seen as the sections of this vector bundle. This is in fact exactly what is going on, but it is beyond the scope of this book to get into the details of this realization (see [1] for a full exploration).

For now, we shall use homogeneous polynomial to define the closed subsets of $\mathbb{P}_k^n$. To that end, let’s examine ideals generate by homogeneous elements. For each $d \in \mathbb{N}$, we can take the collection of homogeneous polynomials $R_d \subseteq k[x_1, \dots, x_{n+1}]$. Then these form a k -vector space (exercise 1.6.1(1)). Furthermore, $R_d R_e \subseteq R_{d+e}$, and so these form an $\mathbb{N}$ -graded algebra over $R_0 = k$ (recall section ??). By ?? and a bit of work, we may consider $k[x_1, \dots, x_{n+1}]$ as:

$$k[x_1, \dots, x_{n+1}] = R_0 \oplus R_1 \oplus \dots \oplus R_d \oplus \dots$$

with the $\mathbb{N}$ -graded structure given by homogeneity.

Definition 1.6.4: Projective Algebraic Set

Take $R \subseteq k[x_1, x_2, \dots, x_{n+1}]$ with the homogeneous $\mathbb{N}$ -graded structure. Let $I \subseteq R$ be a homogeneous ideal. Then a *projective algebraic set* is:

$$\mathcal{Z}_{\mathbb{P}}(I) = \{p \in \mathbb{P}_k^n \mid f(p) = 0, \forall f \in I\}$$

If it is clear from context, the $\mathbb{P}$ subscript will be omitted and it shall be written as $\mathcal{Z}(I)$.

Note that we could have taken an arbitrary homogeneous set $S \subseteq R$, and then taken the homogeneous closure ${}^h\langle S \rangle = I$. By ??, we get that homogeneous ideals are closed under ideal addition, multiplication, intersection, and taking the radical:

$$I + J \quad IJ \quad I \cap J \quad \sqrt{I}$$

Furthermore, to check that a homogeneous ideal I is prime, it suffices to check the condition $fg \in I$ iff $f \in I$ or $g \in I$ with homogeneous polynomials (see ??). The addition can naturally be extended to arbitrary sums, allowing us to define the Zariski topology on $\mathbb{P}_k^n$ when focusing on the homogeneous ideals of a $k[x_1, \dots, x_{n+1}]$:

Definition 1.6.5: Zariski Topology on Projective Space

Letting the projective algebraic sets to be the closed subsets of $\mathbb{P}_k^n$, we get the [projective] Zariski topology on $\mathbb{P}_k^n$.

We can further define $I_{\mathbb{P}}(V)$ in a similar way as we did for affine algebraic sets, namely

$$I_{\mathbb{P}}(V) = \{f \in R \mid f \text{ is homogeneous and } f(p) = 0, \forall p \in V\}$$

Definition 1.6.6: Projective Coordinate Ring

Let $V \subseteq \mathbb{P}_k^n$ be a projective algebraic set. Then a *projective coordinate ring* is defined as:

$$\Gamma_{\mathbb{P}}(V) = k_{\mathbb{P}}[V] = \frac{R}{(\mathcal{I}_{\mathbb{P}}(V))}$$

where we drop the $\mathbb{P}$ subscript when unambiguous.

Calling these the coordinate ring is a little miss-leading as the elements are not functions on V , but the similar algebraic properties between affine and projective coordinate rings shall warrant this nomenclature.

With a topology, we may define all of the topological and geometric notion such as compactness, connectedness, dimension, irreducibility, and so forth. The irreducibility condition in projective space becomes a condition on homogeneous prime ideals, from which we define

Definition 1.6.7: Projective Variety

A *projective variety* is an irreducible projective algebraic set.

As mentioned in the introduction of this section, projective space and projective algebraic sets extend affine space. Let us now make this precise. Let us first consider an algebraic subset $V = \mathcal{Z}_{\mathbb{P}}(I) \subseteq \mathbb{P}_k^n$. Then as $\mathbb{P}_k^n = \mathbb{A}_k^{n+1} / \sim$, we see that we can consider $\mathcal{Z}_{\mathbb{A}}(I) \subseteq \mathbb{A}_k^{n+1}$:

Definition 1.6.8: Affine Cone

Let f be a homogeneous polynomial. Then its zero-set in $\mathbb{A}^{n+1}$ is called an *affine cone*.

More generally, if $V \subseteq \mathbb{P}_k^n$ is a projective algebraic subset, then $C(V) \subseteq \mathbb{A}_k^{n+1}$ is the affine cone of V .

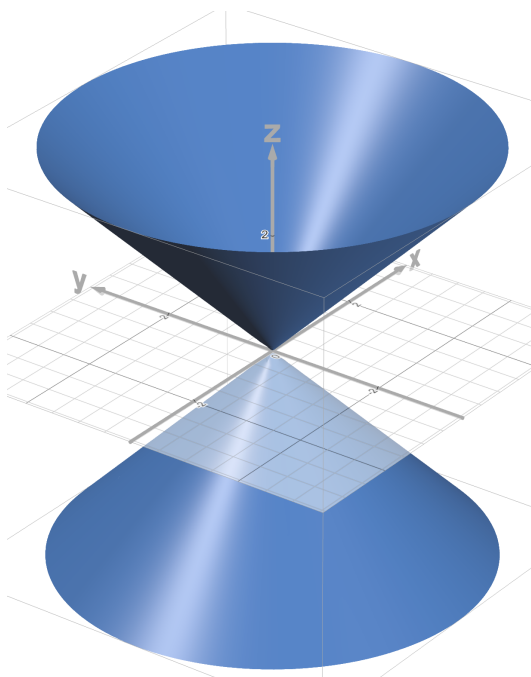
The reader should verify that if $\emptyset \neq V \subseteq \mathbb{P}_k^n$ then

$$\mathcal{I}_{\mathbb{A}}(C(V)) = \mathcal{I}_{\mathbb{P}}(V)$$

and if I is a homogeneous ideal where $\mathcal{Z}_{\mathbb{P}}(I) \neq \emptyset$ then:

$$C(\mathcal{Z}_{\mathbb{P}}(I)) = \mathcal{Z}_{\mathbb{A}}(I)$$

Let us consider a concrete example. The affine cone of the (homogeneous) polynomial $x^2 + y^2 - z^2$ is the zero-set in $\mathbb{A}_k^{2+1}$ that looks like:

Figure 1.3: zero set of $x^2 + y^2 - z^2$

When $z \neq 0$, This equation in $\mathbb{P}_{\mathbb{R}}^2$ would resemble a circle, which we can see by intersecting the above zero-set with the plane $z = 1$, and hence the usual solution to $x^2 + y^2 = 1$ shall be in this zero set. By setting $z = 0$, we see that $x^2 + y^2 = 0$ if and only if $x = y = 0$, but $[0 : 0 : 0] \notin \mathbb{P}_{\mathbb{R}}^2$, and so all the points of the zero-set $x^2 + y^2 - z^2 = 0$ are precisely the points given by lines passing through $x^2 + y^2 - 1 = 0$

Hence, we see that $x^2 + y^2 - z^2$ becomes the equation of a circle in projective space $\mathbb{P}_{\mathbb{R}}^2$. This matches how the circle has no points going “off to infinity”, and so we should expect recovering the same set! However, if we instead consider $xy - z^2$, we see that when $z \neq 0$, we can take the plane where $z = 1$ and get the set $xy = 1$, which is the graph of $y = 1/x$. This has a point at infinity, and indeed if we let $z = 0$, we get the equation $xy = 0$ which is true if either x or y is zero (not that both x and y can both be zero as that would imply $[0 : 0 : 0] \in \mathbb{P}_{\mathbb{R}}^2$). Hence, we gained two additional points corresponding to the two direction the points go off to infinity (see example 1.6.10 for a visual).

The process by which we extend transform a polynomial to a homogeneous polynomial to extend the zero-set is given a name:

Definition 1.6.9: Homogenization

Let $f \in k[x_1, \dots, x_n]$ be a polynomial. Then the *homogenization* of f over degree d is the polynomial ${}^h f \in k[x_1, x_1, \dots, x_{n+1}]$ satisfying:

$${}^h f(x_1, x_1, \dots, x_{n+1}) = x_{n+1}^d f\left(\frac{x_1}{x_{n+1}}, \dots, \frac{x_n}{x_{n+1}}\right)$$

Conversely, for a homogeneous polynomial $f(x_1, \dots, x_{n+1})$. We may *dehomogenize* it with respect to x_i by setting $x_i = 1$

$$f_*(x_1, \dots, x_{n+1}) = f(x_1, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_{n+1})$$

Example 1.6.10: Homogenizing Polynomials

1. Take $x^2 + y^2 - 1$. Homogenizing (with degree 2) and labelling $x_3 = z$, we get $x^2 + y^2 - z^2$. If we homogenizing in degree n , we would get:

$$z^{n-2}x^2 + z^{n-2}y^2 = z^n$$

Note that homogenizing at different degrees will not effect the shape. Homogenization in a degree greater than the “base degree” shall always have an extra solution, namely all points of the form $(x, y, 0)$, however these are “artificially added” points which appear when de-homogenizing in the non- z variables as an extra line of points, and so we shall only homogenize to insure the parity of all degree’s of the monomials with the highest degree^a.

In projective coordinates, the “classical” solutions of the form $x_0^2 + y_0^2 = 1$ become:

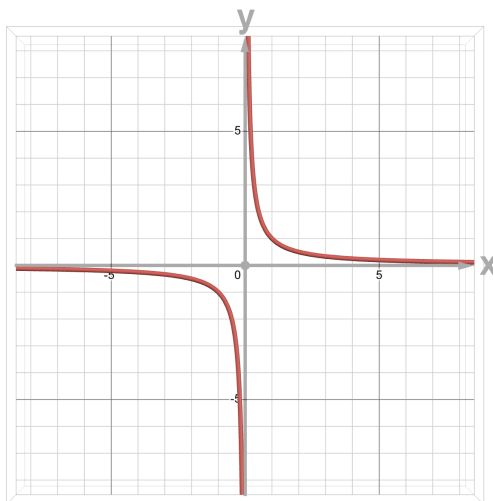
$$[x_0^2, y_0^2, 1]$$

To see if there are new solutions, we may set the last coordinate to 0 (symbolizing going “off to infinity”), in which case we could ask for x_0, y_0 such that

$$[x_0^2 : y_0^2 : 0]$$

is a solution. This is only possible if $x_0 = y_0 = 0$, but $[0 : 0 : 0] \notin \mathbb{P}^2$, hence there is no added points when homogenizing.

2. Take $xy - 1$. It’s shape looks like:

Figure 1.4: zero-set of $xy = 1$

Homogenizing, we get $xy - z^2$. A solution $x_0y_0 - 1 = 0$ in projective coordinates is of the form:

$$[x_0 : y_0 : 1]$$

setting $z = 0$, we can ask if there are any points where:

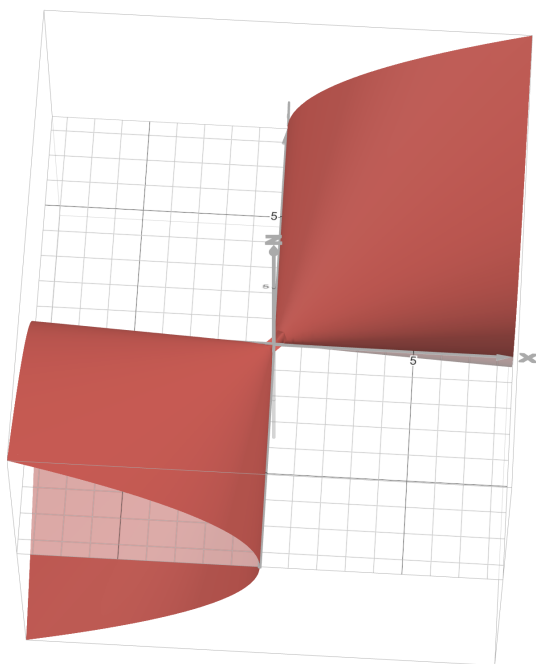
$$[x_0 : y_0 : 0]$$

is a solution. And indeed, if $x_0 = 0$ or $y_0 = 0$, we get a solution, since

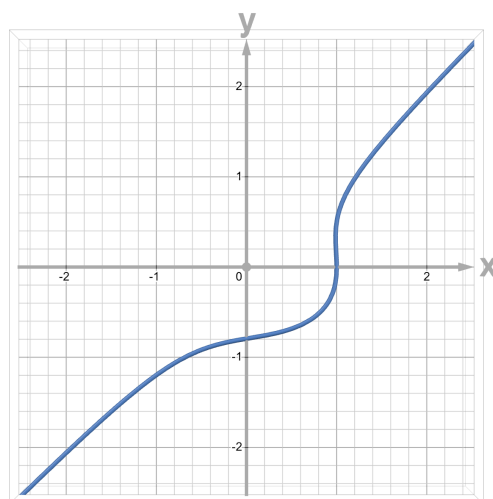
$$0y_0 - 0 = 0 \quad x_00 - 0 = 0$$

We can see by visualizing the zero set that there seems to be two extra point added at infinite, one for the y -axis and one for the x -axis (remember that opposite points are considered the same).

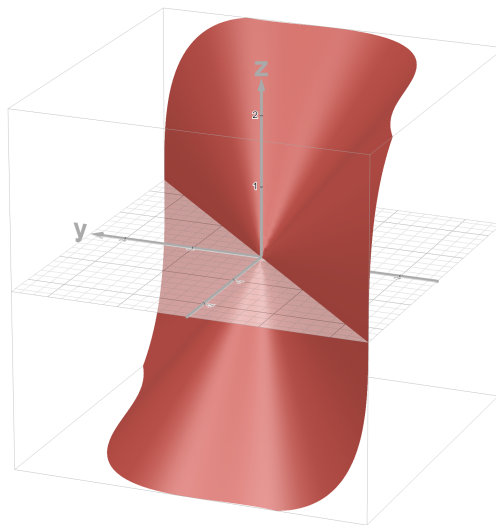
If we de-homogenize at x , we get $y - z^2 = 0$, equivalently $y = z^2$, and de-homogenizing at y we get $x - z^2 = 0$, equivalently $x = z^2$. In both of these cases, we see that $x = z = 0$ and $y = z = 0$ is a solution. Hence, by de-homogenizing at each variable, we can find the vanishing set of a homogeneous polynomial over projective space. This can be seen more clearly when looking at the affine cone of the homogenized polynomial:

Figure 1.5: affine cone: $xy - z^2$

3. Take $x^3 + xy^2 - 2y^3 = 1$. It's shape looks like:

Figure 1.6: $x^3 + xy^2 - 2y^3 = 1$

When homogenizing, the affine cone gives:

Figure 1.7: affine cone of $x^3 = xy^2 - 2y^3 = z^3$

are there any added points, and if so what are they?

4. Take $f(x) = x^2$. Its vanishing set is the single point 0. Then ${}^h f(x, x_\infty) = x^2$ is the degree 2 homogenization. Then this has a single zero, namely $[0 : 1]$, which correspond to the single root of x^2 in $\mathbb{A}_k^1$
5. Take $f(x, y) = x^2 - y$ whose zero set is the graph of the usual quadratic. Then ${}^h f(x, y, x_\infty) = x^2 - x_\infty y$. Then it has the roots:

$$[1 : y_0^2, y_0]$$

As well as:

$$[0 : 1 : 0]$$

in particular, we added a single point at infinity. This is the point where the parabola “meets”. Notice that it wasn’t that there was two points at infinite but just 1. The idea is that the “angle” with which the function approaches infinity tends towards $\pi/2$, and hence the edges of the graph shall meet at infinity. This shows that parabola’s can be thought of as infinitely large ellipsoids^b.

Can the reader see different sorts of connections between circles, parabolas, ellipses, and hyperbolas through projective space?

^aIn [1], we shall see there is an isomorphism between $\widetilde{S(n)}$ and $\widetilde{S(n+1)}$ motivating the lack of difference when homogenizing at different degrees

^bfor those comfortable with some Riemann geometry, can the reader see why projective space can be thought of as having positive curvature?

Proposition 1.6.11: Properties of Homogenization

Let $R = k[x_1, \dots, x_n]$. Let $f^* \in k[x_1, \dots, x_{n+1}]$ be the homogenization, and f_* be the dehomogenization at x_{n+1} . If F, G homogeneous polynomials and f, g are polynomials, then:

1. $(FG)_* = F_*G_*$, $(fg)^* = f^*g^*$
2. $(F + G)_* = F_* + G_*$, $x_{n+1}^t(f + g)^* = x_{n+1}^r f^* + x_{n+1}^s g^*$ where $r = \deg(g)$, $s = \deg(f)$, and $t = r + s - \deg(f + g)$
3. If $F \neq 0$, r is the highest power of x_{n+1} such that $x_{n+1}^r | F$, $x_{n+1}^r (F_*)^* = F$. If $r = 0$, then $(F_*)^* = F$.

Proof:
exercise

Corollary 1.6.12: Factorization of Homogeneous Polynomials

Let $F \in k[x_1, \dots, x_{n+1}]$ be a homogeneous polynomial. Then up to a power of x_{n+1} , factorizing F is the same as factoring F_* .

In particular, if $F \in k[x, y]$, then F factors into a product of linear factors.

Proof:

The first result is immediate from proposition 1.6.11. For the second, as we are assuming k is algebraically closed in this section we get that F_* must factor into a product of linear terms,

$$F_* = c \prod (x - \lambda_i)$$

Then by homogenizing we get:

$$F = cy^r \prod (x - \lambda_i y)$$

as we sought to show.

From this, we are justified in decomposing projective algebraic sets by their behaviour when dehomogenizing elements of the generating ideal:

Example 1.6.13: Zero Sets of Projective Space

Example 1.6.10 contained multiple examples of zero sets. All of those examples were from a single homogeneous polynomial.

1. Let us show that every point $[a_1 : \dots : a_{n+1}] \in \mathbb{P}_k^n$ is the zero set of homogeneous polynomials. First, we need that

$$f(x_1, \dots, x_{n+1}) = \lambda(a_1, \dots, a_{n+1})$$

for some $\lambda \neq 0$. This is the case when each 2×2 minor of the matrix

$$\begin{pmatrix} x_1 & x_2 & \cdots & x_{n+1} \\ a_1 & a_2 & \cdots & a_{n+1} \end{pmatrix}$$

vanish, namely that

$$\begin{vmatrix} x_i & x_j \\ a_i & a_j \end{vmatrix} = a_j x_i - a_i x_j = 0$$

This creates $\binom{n}{2}$ equations, however n of these equations suffices (one for each variable). In other words, we get a point via linear homogeneous polynomials with the appropriate coefficients. This shows how “linear” projective space is. These equations can be thought of as the intersection of all the hyper-planes passing through $[a_0 : \dots : a_n]$. Equivalently, if you know some differential geometry, it shows that we can view $\mathbb{P}_k^n$ as the Grassmanian $G(1, n+1)$.

The coordinate ring associated to a point would be:

$$\frac{k[x_1, \dots, x_{n+1}]}{\langle a_i x_i - a_j x_j \mid 0 \leq i, j \leq n \rangle} \cong k[x_{n+1}]$$

That is, the coordinate ring is *not* just k at a single point! The reader may think of this as the grading condition for the coordinate rings.

2. In $\mathbb{P}_k^1$, the only closed sets are points, the empty set, and the entire space. We shall see why when we show that any closed subset of projective space can be restricted to a closed subset of affine space, in which these are the only possibilities
3. In $\mathbb{P}_k^2$ has many more closed sets just like $\mathbb{A}_k^2$. Any line is given by $a_1 x_1 + a_2 x_2 + a_3 x_3$ and any conic by $ax_1^2 + bx_1 x_2 + cx_2^2 + dx_1 x_3 + ex_2 x_3 + fx_3^2$. A famous set of curves is the Fermat curves given by $x_1^n + x_2^n + x_3^n$. There are many more equations.

To give one example of a coordinate ring, the coordinate ring for a line is isomorphic to $k[s, t]$ with $\deg(s) = \deg(t) = 1$.

4. In $\mathbb{P}_k^3$, we have the first non-rational smooth curves given by the set of three equations

$$x_1 x_3 - x_2^2, x_1 x_4 - x_2 x_3, x_2 x_4 - x_3^2$$

The coordinate ring of the twisted cubic is:

$$\frac{k[x_1, x_2, x_3, x_4]}{(x_1 x_3 - x_2^2, x_1 x_4 - x_2 x_3, x_2 x_4 - x_3^2)} \cong k[s^3, s^2 t, s t^2, t^3] \subseteq k[s, t]$$

Notice this is the binary cubic form (i.e. 2-element degree 3 form).

Another example in $\mathbb{P}_k^3$ are the quadratic surfaces given by $\sum_{i,j} a_{ij} x_i x_j$.

As we are considering the case where k is algebraically closed, we may naturally want the Nullstellensatz to apply between projective space and homogeneous polynomials. In particular, if $I \subseteq k[x_1, \dots, x_n]$ then $Z(I) = \emptyset$ if and only if $I = (1)$. However, this is not the case:

Example 1.6.14: Nullstellensatz False in Projective Space

Take $I_1 = (x_1, \dots, x_n) \subseteq k[x_1, \dots, x_n]$. Then notice on projective space these ideals have no common zeros (as $0 \notin \mathbb{P}_k^n$), hence $Z(I_s) = \emptyset$. More generally, let $I_s \subseteq k[x_1, \dots, x_n]$ denote the ideal consisting of all polynomials which only have terms of degree at least s . Then as each of these ideals contain

x_i^s , $1 \leq i \leq n$, it can only vanish at the origin, which isn't present in $\mathbb{P}_k^n$.

Fortunately, the above ideals are the only ideals for which this could happen, namely because these ideals correspond to ideals that contain the origin in affine space:

Lemma 1.6.15: Characterizing Irrelevant Ideals

Let k be algebraically closed and $I \subseteq k[x_1, \dots, x_n]$ be a homogeneous ideal. Then $Z(I) = \emptyset$ if and only if $I_s \subseteq I$ for some $s > 0$ where $I_s = (x_1^s, \dots, x_n^s)$.

Proof:

Certainly any ideal containing I_s give the empty zero set, so conversely assume $Z(I) = \emptyset$. As $k[x_1, \dots, x_n]$ is Noetherian, let

$$I = (f_1, \dots, f_n)$$

where each f_i is homogeneous. First, as $Z(I) = \emptyset$, I claim the polynomials

$$f_i(1, y_1, \dots, y_n) \quad \left\{ y_i = \frac{x_i}{x_{n+1}} \right.$$

have no common roots. For if $(\alpha_1, \dots, \alpha_n)$ was a common root of the homogeneous polynomials, $(1, \alpha_1, \dots, \alpha_n)$ would be a common root of $f_1, \dots, f_n$ and hence a non-empty zero set.

Now, by Hilbert's theorem, there must exist polynomials g_i such that

$$\sum f_i(1, y_1, \dots, y_n) g_i(y_1, \dots, y_n) = 1$$

Substituting $y_j = x_j/x_{n+1}$ and multiplying by the common denominator (which is $x_{n+1}^{m_{n+1}}$, we get that:

$$x_{n+1}^{m_{n+1}} \in I$$

By de-homogenizing at different points, we can similarly get $x_i^{m_i} \in I$. Finally $m = \max(m_0, \dots, m_n)$, and $d = \max(\deg(f_i))$, letting $s = (m-1)(d+1) + 1$, then in any monomial $x_{n+1}^{a_{n+1}} \cdots x_n a_n$ where $\sum a_i \geq s$, at least one of the s_i must occur with an exponent $a_i \geq m$, and as $s_i^{m_i} \in I$, we have that this term is contained in I , and so $I_s \subseteq I$, as we sought to show.

Definition 1.6.16: Irrelevant Ideal

Let $I_s \subseteq k[x_1, \dots, x_n]$ be the collection of all homogeneous polynomials for which each term has degree at least s . Then I_s is called the *irrelevant ideal* of degree s

1.6.2 Properties and Quasi-Projective Varieties

A common characterization of $\mathbb{P}_k^n$ is by showing that it can be constructed by gluing together $n+1$ copies of $\mathbb{A}_k^n$. Let $R = k[x_1, x_2, \dots, x_{n+1}]$, and let H_i be the zero set of the (homogeneous) polynomial x_i for each i , giving us $H_1, H_2, \dots, H_{n+1}$. These sets certainly cover $\mathbb{P}_k^n$ (their union is $\mathbb{P}_k^n$) and are closed as

$$H_i = Z(x_i)$$

If we consider the map $\mathbb{A}_k^n \rightarrow \mathbb{P}_k^n$ mapping $(a_1, \dots, a_n) \mapsto [1 : a_1 : \dots : a_n]$, then the hyperplane H_0 is sometimes denotes H_0 and is called the *hyperplane at infinity* as it is the hyperplane where $x_{n+1} = 0$ (i.e. it contain the points at infinity). For any hyperplane H_i , we have a map

$$H_i \rightarrow \mathbb{P}_k^{n-1} \quad [a_1, \dots, a_{i-1}, 0, a_{i+1}, \dots, a_n] \rightarrow [a_1, \dots, a_{i-1}, a_{i+1}, \dots, a_n]$$

Once we define regular maps, the reader can verify that $H_i \cong \mathbb{P}_k^{n-1}$. Let

$$D(x_i) = U_i = \mathbb{P}_k^n - H_i$$

Then these sets are open and cover $\mathbb{P}_k^n$, as H_i is closed, and they contain all points where $x_i \neq 0$. Define a map:

$$\varphi_i : U_i \rightarrow \mathbb{A}_k^n \quad [a_0 : \dots : a_n] \mapsto \left(\frac{a_0}{a_i}, \dots, \frac{a_n}{a_i} \right)$$

where we omit $\frac{a_i}{a_i}$. Notice how this is similar to de-homogenization. Verify that the map is well-defined, namely that it is independent of representative of points $U_i \subseteq \mathbb{P}_k^n$. Then:

Proposition 1.6.17: Projective Space Affine Cover

Let φ_i be defined as above. Then φ_i is a homeomorphism:

$$U_i \cong \mathbb{A}_k^n$$

Proof:

First, φ_i is certainly a bijection: if

$$[x_1 : \dots : 1_i : \dots : x_n] = \varphi_i(x_1, \dots, x_n) = \varphi(y_1, \dots, y_n) = [y_1 : \dots : 1_i : \dots : y_n]$$

these two are equal if each $x_j = \lambda y_j$ for some $\lambda \in k$. Indeed, since $1_i = \lambda 1_i$, we must have $\lambda = 1$, which tell us that $x_j = y_j$ for each coefficient, giving injectivity. for surjectivity, given any $[x_1, \dots, x_i, \dots, x_n]$ where $x_i \neq 0$, take:

$$[x_1 : \dots : x_i : \dots : x_n] = \left[\frac{x_1}{x_i} : \dots : 1_i : \dots : \frac{x_n}{x_i} \right]$$

We now see that $(\frac{x_1}{x_i}, \dots, \frac{x_n}{x_i})$ map to this point, giving surjectivity, and hence bijectivity.

For continuity, notice that $\varphi_i = \pi \circ \psi_i$ where $\psi_i : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^{n+1} \setminus \{0\}$ that is the natural embedding with i th-coordinate shifted by 1, and π is the projection. The map π is continuous as $\mathbb{P}_k^n$ is given the natural quotient from $\mathbb{A}_k^n$, and ψ_i is continuous as any pre-image of closed subsets of $\mathbb{A}_k^{n+1} \setminus \{0\}$ is given by the dehomogenization of the polynomials at the i th coordinate.

Furthermore, the map is open, since ψ is open (it's image is a shifted-by-1 copy of $\mathbb{A}_k^n$, so any open set in $\mathbb{A}_k^n$ is open in $V(y_i - 1) \cap \mathbb{A}_k^{n+1} \setminus \{0\}$) and π is open since for any open subset $U \subseteq \mathbb{A}_k^{n+1} \setminus \{0\}$ the set

$$\bigcup_{\lambda \in k^*} \lambda U$$

is open, since the map $x \mapsto \lambda x$ is a homeomorphism (as is quickly verifiable). This is the pre-image of $\pi(U)$ and so since π is a quotient map we have that $\pi(U)$ is also open. Thus, the composition is open, making φ open.

Combining all the results, we get that φ_i is a homeomorphism, as we sought to show.

Given $H_i \cong \mathbb{P}_k^{n-1}$, this implies that we have a decomposition:

$$\mathbb{P}_k^n = \mathbb{A}_k^n \sqcup \mathbb{P}_k^{n-1}$$

For example, if $n = 2$, then:

$$\mathbb{P}_k^2 = \mathbb{A}_k^2 \sqcup \mathbb{P}_k^1$$

For this reason, we often think of $\mathbb{P}_k^2$ to be $\mathbb{A}_k^2$ along with a “line at infinity” (the reader should take the time to internalize this idea). This decomposition can be extended further, breaking down $\mathbb{P}_k^{n-1}$ into an affine and projective space. Repeating inductively, we get:

$$\mathbb{P}_k^n = \mathbb{A}_k^n \sqcup \mathbb{A}_k^{n-1} \sqcup \dots \sqcup \mathbb{A}_k^1 \sqcup \mathbb{A}_k^0$$

If $n = 2$, we get: Another “immediate” result from our above discussion is that projective space $\mathbb{P}_k^n$ can be covered with $n + 1$ copies of $\mathbb{A}_k^n$. This shall become the dominant way of viewing projective space in modern algebraic geometry (similar to how smooth manifolds are viewed as patches of euclidean space glued together), making this result important enough to be a theorem:

Theorem 1.6.18: Projective Variety Affine Cover

Let Y be a projective variety. Then Y is covered by open set $Y \cap U_i$ where each U_i is homeomorphic to $\mathbb{A}_k^n$

Proof:

simple, but important exercise.

For the reader’s awareness, we box the following³¹:

Corollary 1.6.19: Transition Between Affine Cover

For each $D(x_i), D(x_j)$ there exists a transition map $\varphi_{ij} : D(x_i) \cap D(x_j) \rightarrow D(x_i) \cap D(x_j)$ given by:

$$\begin{aligned} (x_1, \dots, \hat{x}_i, \dots, x_{n+1}) &\mapsto [x_1, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_{n+1}] \\ &\mapsto \left[\frac{x_1}{x_j}, \dots, \frac{x_{i-1}}{x_j}, \frac{1}{x_j}, \frac{x_{i+1}}{x_j}, \dots, \frac{x_j}{x_j}, \dots, \frac{x_{n+1}}{x_j} \right] \\ &\mapsto \left(\frac{x_0}{x_j}, \dots, \frac{x_{i-1}}{x_j}, \frac{1}{x_j}, \frac{x_{i+1}}{x_j}, \dots, \hat{1}_j, \dots, \frac{x_{n+1}}{x_j} \right) \end{aligned}$$

where the hats represents omission of the variable, and it was assumed $i < j$ without loss of generality.

Proof:

immediate.

Let us now formalize extending closed sets of $\mathbb{A}_k^n$ to $\mathbb{P}_k^n$, and conversely restrict closed sets of $\mathbb{P}_k^n$ to any $U_i \cong \mathbb{A}_k^n$. Let us first take a closed subset $V \subseteq \mathbb{P}_k^n$, and consider its restriction to $U_i \cong \mathbb{A}_k^n$.

³¹this shall be the transition maps to verify the gluing is well-defined

As $U_i = D(x_i)$ is open, $V \cap U_i = W_i$ is open in V (with the subset topology and Zariski topology of U_i)³². Let us show that W_i is closed in $U_i \cong \mathbb{A}_k^n$, that is it is the zero set of the appropriate polynomials. If $f_1, \dots, f_n$ are the homogeneous polynomials defining V ³³ so that

$$f_1 = \dots = f_n = 0$$

with $\deg f_i = n_i$, then on U_i each f_j is converted to:

$$x_i^{n_j} f_j \left(\frac{x_1}{x_j}, \dots, 1_j, \dots, \frac{x_{n+1}}{x_j} \right)$$

For example $f(x, y, x_3) = xy - x_3^2$ would be converted to $xy - 1$ in U_3 . Then we can see that $V = \cup W_i$, that is each V can be covered by closed subsets from each U_i .

Conversely, if $W \subseteq \mathbb{A}_k^n$ is a closed subset, we may take the closure in $\mathbb{P}_k^n$, $\overline{W}$, which is the intersection of all closed sets containing W where we interpret W as being set in one of the U_i 's. Some closed W are also closed in $\mathbb{P}_k^n$. For example $x^2 + y^2 - 1$ over $\mathbb{C}[x, y]$ when embedded becomes the zero set of $x^2 + y^2 - z^2$ whose zero's are all of the form $[\cos(a) : \sin(a) : 1]$ for $a \in \mathbb{C}^*$. On the other hand, if we take $xy - 1$, then then consider $xy - z^2$, we get that the traditional zero's become:

$$\left[a : \frac{1}{a} : 1 \right]$$

and we now get a new 0 when $z = 0$ giving two more points:

$$[1 : 0 : 0] \quad [0 : 1 : 0]$$

which we can think of as adding the two points at infinity of the graph of $y = 1/x$. Another example would be $y - x^2$. When extending to $\mathbb{P}_k^2$, we get $yz - x^2$. By setting $x = 0$, we get that $y = 0$ is a solution (which already exists in $\mathbb{A}_k^2$) as well as the new solution:

$$[0 : 0 : 1]$$

which is a new point at infinity that makes the parabola shape akin to an infinite ellipse.

Summarizing our discussion and the pertinent consequences:

³²see exercise ref:HERE to show they are equivalent

³³Recall that we are working over Noetherian rings, and hence each ideal is finitely generated

Proposition 1.6.20: Closure of Affine Variety

Let $I \subseteq k[x_1, \dots, x_n]$ be a (not necessarily homogeneous) ideal, $I^* = \{^h f \mid f \in I\}$, $V^* = \mathcal{Z}(I^*) \subseteq \mathbb{P}_k^n$. Let $\mathbb{P}_k^n$ be projective space where we consider the last coordinate to be the coordinate at infinity so that $\mathbb{A}_k^n$ is considered as the extension of $\mathbb{P}_k^n$ via the map $\iota_{n+1} : \mathbb{A}_k^n \rightarrow U_{n+1} \subseteq \mathbb{P}_k^n$. Let $(-)_*$ be the dehomogenization operation so that if $V = \mathcal{Z}(J) \subseteq \mathbb{P}_k^n$, then $I_* = \{f_* \mid f \in J\}$ where f_* is the dehomogenization at the coordinate at infinity, and $V_* = \mathcal{Z}(I_*)$. Let H_∞ represent the hyperplane at infinity (i.e. $\mathcal{Z}(x_{n+1})$). Then:

1. $\iota_{n+1}(V) = V^* \cap U_{n+1}$
2. $(V^*)_* = V$
3. If $V \subseteq W \subseteq \mathbb{A}_k^n$, then $V^* \subseteq W^* \subseteq \mathbb{P}_k^n$
4. If $V \subseteq W \subseteq \mathbb{P}_k^n$, then $V_* \subseteq W_* \subseteq \mathbb{A}_k^n$
5. If V is irreducible in $\mathbb{A}_k^n$, then V^* is irreducible in $\mathbb{P}_k^n$
6. If $V = \cup_i V_i$ is the irreducible decomposition of V in $\mathbb{A}_k^n$, then $V^* = \cup_i V_i^*$ is the irreducible decomposition of V^* in $\mathbb{P}_k^n$
7. If $V \subseteq \mathbb{A}_k^n$, then V^* is the closure of $\iota_{n+1}(V) \subseteq \mathbb{P}_k^n$ in the Zariski topology (i.e. it is the smallest algebraic set in $\mathbb{P}_k^n$ containing $\iota_{n+1}(V)$)
8. If $\emptyset \neq V \subsetneq \mathbb{A}_k^n$, then no components are a subset of H_∞ : $V_i^* \not\subseteq H_\infty$ (though it may intersect it)
9. If $V \subseteq \mathbb{P}_k^n$ and no components $V_i \subseteq H_\infty$ (though it may intersect it), then $V_* \subsetneq \mathbb{A}_k^n$ and $(V_*)^* = V$.

Proof:

Most of these points were proven, the rest is left as an exercise.

Definition 1.6.21: Projective Closure

Let $V \subseteq \mathbb{A}_k^n$ be an algebraic set. Then $V^* \subseteq \mathbb{P}_k^n$ as defined in proposition 1.6.20 is called the *projective closure*.

Note that the homogenization of an ideal is *not* the homogenization of its generators

Example 1.6.22: Homogenization of Ideal and Generators

Let $I = (y - x^2, z - x^3)$ and $V = \mathcal{Z}(I)$ i.e. the twisted cubic in $\mathbb{A}_k^3$. Show that

1. $zw - xy \in I(V)^*$
2. $zw = xy \notin ((y - x^2)^*, (z - x^3)^*)$

In other words, the ideal $(f_1^*, \dots, f_n^*)$ is too small to capture all the new relations (in particular, it shall usually add a line at infinity).

Next, we define $k_{\mathbb{P}}(V)$ to be the localization of $k_{\mathbb{P}}[V]$. Note that if $f/g \in k(V)$ where $\deg(f) = \deg(g) = d$, then f/g is in fact a well-defined function on V , since:

$$\frac{f(\lambda x)}{g(\lambda x)} = \frac{\lambda^d f(x)}{\lambda^d g(x)} = \frac{f(x)}{g(x)}$$

there are thus many more rational function defined on V than regular functions. The set of well-defined functions shall be given a name:

Definition 1.6.23: Function Field

Let V be a projective variety. Then:

$$\kappa(V) = \left\{ \frac{f}{g} \in k_{\mathbb{P}}(V) \mid \deg(f) = \deg(g) = d, d \in \mathbb{N} \right\}$$

Elements of $\kappa(V)$ are called the *rational functions* of V .

Proposition 1.6.24: Rational Functions over Projective and Affine Variety

Let $V \subseteq \mathbb{A}_k^n$ be an affine variety. Then:

$$k(V) \cong \kappa(V^*)$$

Similarly, if $V \subseteq \mathbb{P}_k^n$ is a projective variety not contained in the hyperplane at infinity. Then:

$$\kappa(V) \cong k(V_*)$$

Proof:

Define:

$$\alpha : k(V) \rightarrow \kappa(V^*) \quad \alpha\left(\frac{F}{G}\right) = \frac{F_*}{G_*}$$

Proposition 1.6.11 shows that $((-)_*)^*$ and $((-)^*)_*$ is almost a ring homomorphism (with appropriate checking that the map is well-defined on representatives of the closest). The problem is that homogenization may add an extra x_{n+1}^r term, in particular the map on $k[V]$ need not be injective. However, as we may without loss of generality choose F, G such that $\gcd(F, G) = c$ (some constant), we in fact have a unique homogenization degree, giving us our desired isomorphism.

On the one hand, it may seem counter-intuitive that though we are restricting to degree 0 rational polynomials, we somehow get as many rational functions than when we allowed for any degree $r - s$ rational polynomial. On the other hand, this should not be too strange for the reader: the domain of definition is a dense open subset, and we shall show that V^* is the closure of V when embedded in $\mathbb{P}_k^n$.

We can define the *stalk* at p to be

$$\mathcal{O}_p(V) = \{f/g \in \kappa(V) \mid g(p) \neq 0\}$$

The reader should check (exercise ref:HERE) that $\mathcal{O}_p(V)$ is a local Noetherian domain.

Proposition 1.6.25: Stalk Defined Locally

Let $V = \mathcal{Z}_{\mathbb{P}}(f)$ is a hypersurface, f_* is the dehomogenization of f , and $V_* = \mathcal{Z}_{\mathbb{A}}(f_*)$, then:

$$\mathcal{O}_p(V) \cong \mathcal{O}_p(V_*)$$

This shows that the properties of a stalk are inherently “local”, depending only on the information around the point p .

Proof:

restrict the isomorphism α defined in proposition 1.6.24.

Next, notice there is more than one natural set of coordinates on projective space. It is natural to consider the coordinates of $\mathbb{P}_k^n$ to be given by degree 1 homogeneous functions, namely each point $x \in \mathbb{P}_k^n$ is of the form $[x_1 : \cdots : x_{n+1}]$ where $[\lambda x_1 : \cdots : \lambda x_{n+1}] = \lambda[x_1 : \cdots : x_{n+1}]$. However, we may also just as naturally choose to represent each point in any degree of homogeneity, namely:

$$[\lambda w_1 : \cdots : \lambda w_{n+1}] = \lambda^k[w_1 : \cdots : w_{n+1}]$$

where we can have $w_i = (x_i)^k$.

Finally, as all homogeneous polynomials must be of the same degree, we can give a rough classification of all hypersurfaces of projective space $\mathbb{P}_k^n$ of degree d . A hypersurface is defined by a single homogeneous equation. As there are $n+1$ variable, the reader should check that there are $\binom{n+d}{d} = N$ possible degree d monomials. Labeling them $M_1, \dots, M_N$, any degree d homogeneous polynomial is of the form:

$$\sum_i^N a_i M_i$$

These hypersurface are naturally defined up to multiplication of a constant $c \in k$. But then, the coefficients determine the hypersurface up to a constant, meaning they can be associated to a point in $\mathbb{P}_k^N$, or said differently P_k^N parameterizes the degree d hypersurfaces of $\mathbb{P}_k^n$ or is the *moduli space* for the degree d hypersurfaces of $\mathbb{P}_k^n$.

Note that some points classify the same hypersurface, for example $x^2y = 0$ and $xy^2 = 0$ in $\mathbb{P}_k^n$ for $n \geq 3$. The difference is in the multiplicity, information that would be distinguishable when working with *closed subschemes* instead of closed subsets.

Quasi-Projective Variety

As we saw in the above, affine and projective varieties are very natural linked to each other. Many results that are proven in classical algebraic geometry apply both to projective and affine varieties. Furthermore, the open subset of projective variety (for example $\mathbb{P}_k^1$ minus the point at infinity and zero) have natural algebraic structure but are not the vanishing set of polynomials (they are the opposite, they are the non-vanishing set of polynomials). The properties of such sets often are strongly tied to varieties, or are even varieties themselves (for example $\mathbb{P}_k^1$ without the point at infinity is the closed set $\mathbb{A}_k^1$ in $\mathbb{A}_k^1$, but is an open set in $\mathbb{P}_k^1$). The definition of quasi-projective varieties aims to unify all of these.

If $X \subseteq \mathbb{P}_k^n$ is closed and $X_i = X \cap U_i$ is open in X , they are in fact *closed* in $U_i \cong \mathbb{A}_k^n$. This can be seen by de-homogenizing: if X is given by equations $f_0, \dots, f_n$ where $\deg f_i = n_i$, then U_i is given by:

$$x_i^{-n_j} f_j = f(x_1, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_{n+1}) = 0$$

If we have $X_i \subseteq U_i$ which is closed, then we may consider the closure $\overline{X_i} \subseteq \mathbb{P}_k^n$ by taking the polynomials that generate X_i and homogenizing them, namely if f is one such polynomial where $\deg f = d$, then the equations describing $\overline{X_i}$ are of the form

$$x_i^d f(x_1/x_i, \dots, x_{n+1}/x_i) = 0$$

Which implies that:

$$X_i = \overline{X_i} \cap U_i$$

Hence, if $X \subseteq \mathbb{P}_k^n$, then an open subset of X will correspond to a closed subset of $\mathbb{A}_k^n$. As X is an open subset of itself, it corresponds to an open subset of a closed projective algebraic set. Thus, open subsets of closed projective sets encapsulate affine and projective varieties:

Definition 1.6.26: Quasi-projective Variety

A *quasi-projective variety* is an open subset of a closed projective variety.

A closed subset $C \subseteq U$ of a quasi-projective variety is the intersection of C with a closed subset $\mathbb{P}_k^n$. It should be checked that the irreducible composition given in theorem 1.2.33 applies to quasi-projective varieties (see exercise ref:HERE). If $Y, Y \subseteq X \subseteq \mathbb{P}_k^n$ is a quasi-projective variety in $\mathbb{P}_k^n$, then Y is said to be a subvariety of X .

It must be noted that the collection of quasi-projective varieties is larger than the collection affine and projective varieties. For example $\mathbb{A}^2 \setminus \{(0, 0)\}$ is a quasi-projective variety, but is not affine or projective.

Let X be a quasi-projective variety and $U \subseteq X$ an open subset. Define $\Gamma(U, \mathcal{O}_X) \subseteq \kappa(X)$ to be the collection of all rational functions on X that are defined at each $p \in U$. Equivalently:

$$\Gamma(U, \mathcal{O}_X) = \bigcap_{p \in U} \mathcal{O}_p(X)$$

Note that if $U = X$, then by proposition 1.3.11 and proposition 1.6.25,

$$\Gamma(X, \mathcal{O}_X) = k[X] = \Gamma(X)$$

showing this notation is consistent with our earlier definitions.

Next, let's see that checking that a quasi-projective variety is closed is a open-local condition:

Lemma 1.6.27: Closedness of Quasi-projective Variety is Open-local

Let $Y \subseteq X$ be a quasi-projective variety. Then Y is closed if and only if $X = \bigcup_{\alpha} U_{\alpha}$ where U_{α} is open and $Y \cap U_{\alpha}$ is closed for every α . Namely, closedness is open-local for quasi-projective varieties

Proof:

Let $X = \bigcup_{\alpha} U_{\alpha}$ with U_{α} open and $Y \cap U_{\alpha}$ is closed. As U_{α} is open,

$$U_{\alpha} = X - Z_{\alpha}$$

where Z_{α} is closed, and by definition:

$$U_{\alpha}Y = U_{\alpha} \cap T_{\alpha}$$

If $Y = \bigcap_{\alpha} (Z_{\alpha} \cup T_{\alpha})$, then Y is closed. But this is a routine check, completing the proof.

The following condition shall show that any type of variety is fundamentally locally affine:

Lemma 1.6.28: Varieties Are Locally Affine

Let X be a quasi-projective variety. Then every $x \in X$ has an open neighborhood that is isomorphic to an affine variety

Proof:

Take $X \subseteq \mathbb{P}_k^n$. If $x \in U_{n+1}^n$, then $x \in X \cap U_{n+1}^n$ and by definition of a quasi-projective variety:

$$X \cap U_{n+1}^n = Y - Y_1$$

where Y_1, Y are closed subsets of U_{n+1}^n . As $x \in Y$, there exists a polynomial f with coordinates in U_{n+1}^n where $f = 0$ on Y_1 and $f(x) \neq 0$. Let take the ideal (f) which corresponds to all points on Y where f is 0. By treating the ideal like set of point, $D(f) = Y \setminus (f)$ is a neighborhood of x .

I claim this neighborhood is isomorphic to an affine variety. If $(f_1, \dots, f_m)$ is the ideal associated to $Y \subseteq U_{n+1}^n$, then define the variety in $\mathbb{A}_k^{n+1}$ via:

$$f_1(x_1, \dots, x_n) = \dots = f_m(x_1, \dots, x_n) = 0 \quad f(x_1, \dots, x_n)x_{n+1} = 1$$

i.e.

$$Z = Z(f_1, \dots, f_m, f \cdot x_{n+1} - 1)$$

Then the mapping

$$\varphi(x_1, \dots, x_{n+1}) \rightarrow (x_1, \dots, x_n)$$

gives a regular mapping from Z into $D(f)$, and

$$\psi(x_1, \dots, x_n) \rightarrow (x_1, \dots, x_n, f(x_1, \dots, x_n)^{-1})$$

is a regular mapping of $D(f)$ into Z where $\varphi \circ \psi = \psi \circ \varphi = \text{id}$, as we sought to show.

This result also showed that principal open sets (definition 1.2.5) are naturally associated to an affine algebraic set, and that the regular function on $D(f)$ are:

$$k[D(f)] = k[X][1/f]$$

In particular, f is nonzero on $D(f)$, this is well-defined. Overall, when checking if a mapping between quasi-projective is regular, it suffices to check that any affine variety in the codomain has a close pre-image.

1.6.3 Regular Maps

Let us now define regular mappings on quasi-projective varieties. This definition should extend that of affine varieties where we had a polynomials function for each coordinate.

Recall that a rational map on $\mathbb{P}_k^n$ is the quotient of two homogeneous polynomials, with degree $r - s$ where r, s is the degree of the numerator and denominator respectively. A regular point x of a rational function f would be a point where f can be represented by p/q with $q(x) \neq 0$. When working over projective varieties, we shall have that most rational polynomials are not rational functions; we shall require that $r = s$, that is it must be of degree 0.

Definition 1.6.29: Regular Point [on Quasi-projective Variety]

Let X be a quasi-projective variety, and $f = p/q : X \rightarrow \mathbb{A}_k^1$ a regular homogeneous rational function or map of degree 0. If $q(x) \neq 0$ (with codomain k) for some point x , then it is nonzero in some open neighborhood of x (namely $D(q)$) and we say that f is *regular* in a neighborhood of x .

If f is regular at all points, then f is said to be *regular* and shall be denoted $k[X]$.

When we defined a regular function before, it was an element of the coordinate ring. We have now defined it to be locally regular at all points. When k is algebraically closed, these two are in fact the same. If X is irreducible, then this is the result of proposition 1.3.4. For the general case, say $f = p_x/q_x$ where $q_x \neq 0$ on U_x so that on U_x :

$$q_x f = p_x \tag{1.6}$$

Then this equation holds on all of X . To see this, choose a regular functions that vanishes on $X \setminus U_x$ but not at x (recall that U_x is dense in X), and multiply it by $p_x q_x$. Then we see that equation (1.6) must hold outside of U_x as both sides of the equality will vanish. Then like the irreducible case, we can now find points $x_1, \dots, x_n$ and regular functions $h_1, \dots, h_n$ such that

$$\sum_i^n q_x h_i = 1$$

Then for each x_i , multiplying by h_i and adding up we get:

$$f = \sum_i^n p_x h_i$$

and hence f is regular in the same sense given in definition 1.2.17. Recall that it was mentioned that if $X = \mathbb{P}_k^n$ that $k[\mathbb{P}_k^n]$ consists of only constants functions. We shall soon show that if $X = V \subseteq \mathbb{P}_k^n$ is any projective variety that $k[V]$ is the collection of constant functions.

Definition 1.6.30: Regular Map [On Quasi-projective Variety]

Let $f : X \rightarrow Y$ be a mapping of quasi-projective varieties and $Y \subseteq \mathbb{P}_k^m$. Then the mapping is called *regular* if for every $x \in X$ and every open affine set $U \subseteq U_i$ containing the point $f(x)$, there exists a neighborhood U of x such that $f(U) \subseteq U_i$ and the mapping $f : U \rightarrow U_i^m$ is regular (see definition 1.3.3)

The collection of regular function is denoted $k[X]$, $\mathcal{O}_X(X)$, or $\Gamma(X)^a$, and the collection of regular mapping between two quasi-projective varieties is denoted $\text{Hom}_{\mathbf{Reg}}(X, Y)$.

^aThis notation shall become more prominent in modern algebraic geometry

This definition of regular function is well-defined and is independent of choice of U_i : if $f(x) = (y_0, \dots, \hat{1}_i, \dots, y_m) \in U_i$ (where the hat indicates omission), it is also contained in U_j when $y_j \neq 0$, and the coordinates of this point in U_i are of the form:

$$(y_0/y_j, \dots, 1/y_j, \dots, \hat{1}_j, \dots, y_m/y_j)$$

Hence, given the mapping $f : U \rightarrow U_i$ is given by

$$(f_0, \dots, \hat{1}_i, \dots, f_m)$$

then $f : U \rightarrow U_j^m$ is given by $(f_0/f_j, \dots, 1/f_j, \dots, \hat{1}_j, \dots, f_m/f_j)$. As $f_j(x) \neq 0$ and the set U' of points of U where $f_j \neq 0$ is open. Then on U' , the functions $f_0/f_j, \dots, 1/f_j, \dots, \hat{1}_j, \dots, f_m/f_j$ are regular, and so $f : U' \rightarrow U_j^m$ is regular.

If we have an affine closed set, by theorem 1.2.20 the regular mapping $f : X \rightarrow Y$ is determined by the mapping $f^* : k[Y] \rightarrow k[X]$. This is not the case if X, Y are projective, for as mentioned a few times already we shall have $k[X]$ and $k[Y]$ consist of only constant functions. However, it must be noted that for a general quasi-projective variety, $k[X]$ could be quite large³⁴. If X is isomorphic to a closed subset of an affine space, it shall be called an *affine variety*, and if it is isomorphic to a closed subset of projective space, it shall be called *projective variety*. Note all quasi-projective varieties are affine or projective (for example $\mathbb{A}_k^2 \setminus \{(0, 0)\}$).

Let us now characterize regular maps from quasi-projective varieties to projective space³⁵. To understand such a map, start with a regular mapping f on X and a point $x \in X$ where $f(x) \in \mathbb{A}_k^m = U_0^m$ so that there is an open neighborhood where $f = (f_1, \dots, f_m) : U \rightarrow U_i^m$. Then as it is regular we can write $f_i = p_i/q_i$ where p_i, q_i are of the same degree in homogeneous coordinates and $q_i(x) \neq 0$. Taking the l.c.m. of such fractions we get that $f_i = F_i/F_0$ where $F_0, \dots, F_m$ are forms of the same degree and $F_0(x) \neq 0$; overall we can think of the point $f(x) \in U_i^m$ as the point:

$$f(x) = [F_0(x) : \dots : F_m(x)] \in \mathbb{P}_k^m$$

in projective space. Note that there is more than one representative for such a regular function, namely if

$$F(x) = [F_0(x) : \dots : F_m(x)] \quad G(x) = [G_0(x) : \dots : G_m(x)]$$

then they are the same point if and only if $F_i G_j = F_j G_i$ on X for all $0 \leq i, j \leq m$. This gives us an explicit condition for when $Y = \mathbb{P}_k^m$:

³⁴Shafarevich mentions that Rees and Nagata constructed an example where $k[X]$ is not finitely generated

³⁵This is in fact a larger exploration that shall culminating in understanding line bundles on projective scheme, see cite:ALG-GEO

Proposition 1.6.31: Equivalent Regular Function

Let $f : X \rightarrow \mathbb{P}_m^k$ be a set function from an irreducible quasi-projective variety. Then it is a regular mapping if and only if f can be characterized as a collection of maps:

$$[F_0 : \cdots : F_m]$$

all of the same degree in homogeneous coordinates where two mappings are identical if

$$f_i g_j = f_j g_i \text{ on } X \text{ for all } 0 \leq i, j \leq m.$$

Example 1.6.32: Regular Mappings

1. *Projection:* Let $E \subseteq \mathbb{P}_k^n$ be a d -dimensional subspace given by $n - d$ linearly independent linear equations $(L_1, \dots, L_{n-d})$. In homogeneous coordinates, the equations L_i are called *linear forms*. The mapping:

$$\pi(x) = (L_1(x) : \cdots : L_{n-d}(x))$$

is well-defined and is called the projection with center E . Notice this set is regular on $\mathbb{P}_k^n \setminus E$ as the L_i 's do not vanish simultaneously. We for any $X \subseteq \mathbb{P}_k^n$ where $X \cap E = \emptyset$ and X is closed, we have a regular mapping

$$\pi : X \rightarrow \mathbb{P}_k^{n-d-1}$$

Geometrically, we can interpret the mapping as follows. Take any $(n - d - 1)$ -dimensional subspace $V \subseteq \mathbb{P}_k^n$ such that $H \cap E = \emptyset$. Through any point $x \in \mathbb{P}_k^n \setminus E$ and the space E , there passes a unique $(d + 1)$ -dimensional subspace E_x . Then this subspace intersects H at a unique point, namely $\pi(x)$. If X intersects E , but does not contain it, then this is a rational mapping.

We have seen the case where $d = 0$ before (think of the hyperbola to $\mathbb{A}_k^1$)

2. Take $V = \mathcal{Z}(x - y^2, z - x^3)$ (i.e. the twisted cubic). This is a rational curve with parameterization:

$$t \mapsto (t, t^2, t^3)$$

Let us now homogenize this variety. The homogenization of this map is:

$$[s : t] \mapsto \left[1 : \frac{t}{s} : \frac{t^2}{s^2} : \frac{t^3}{s^3} \right] = [s^3 : ts^2 : t^2s : t^3]$$

3. Take $V = \mathcal{Z}(x^2 + y^2 - z^2)$; the projectivization of the circle. Define the map:

$$f : \mathbb{P}_k^1 \rightarrow V \quad [s : t] \rightarrow [s^2 - t^2 : 2st : s^2 + t^2]$$

Show that this is an isomorphism with inverse $[x : y : z] \mapsto [x + z : y]!$ Hence, $\mathbb{P}_k^1$ is isomorphic to a circle!

Show that this is *not* the case for $\mathbb{P}_k^n$ for $n \geq 2$; in differential geometry it can be shown that these surfaces are all non-orientable, while the sphere S^n is always orientable (see [5]).

4. *Veronese Mapping:* Consider the collection of all projective hypersurfaces on $\mathbb{P}_k^n$. **return to this, Shafarevich p.40**
5. **Silverman example with circle of diff radii not being iso because need $\mathbb{Q}(\sqrt{3})$**

Rational Functions on Quasi-projective Varieties

When working with affine varieties, we had the quotient of two regular functions. However, mentioned a few times, on $\mathbb{P}_k^n$ there are only constant regular functions, and a rational function, being the ratio of two regular functions, would still be constant.

The solution is to recall the definition of $\mathcal{O}_X$, namely:

$$\mathcal{O}_X = \{\varphi = f/g \in k[x_1, \dots, x_n] \mid g \notin I(X)\}$$

where f and g are of the same degree. Then take the subset $\mathfrak{m}_X = \{\varphi = f/g \in \mathcal{O}_X \mid f \in I(X)\}$.

Definition 1.6.33: Rational Function On quasi-projective Variety

Let X be a quasi-projective variety. Then the set of rational functions on X is given by:

$$k(X) \cong \mathcal{O}_X / \mathfrak{m}_X$$

By a reasoning similar to proposition 1.3.9, a homogeneous function vanishes on X if and only if it vanishes on an open subset $U \subseteq X$. Hence $k(X) = k(U)$. As a consequence:

$$k(X) = k(\overline{X})$$

Due to this, when studying rational functions on quasi-projective spaces, we may reduce to studying rational functions to affine or projective varieties, which often leads most results to focus on the irreducible case. The set of regular points shall be called the *domain of definition*. If $f : X \rightarrow \mathbb{P}_k^m$, then it is determined by specifying $m+1$ form $[F_0 : \dots : F_m]$ of $n+1$ homogeneous coordinates of projective space $\mathbb{P}_k^n$ containing X (where at least one form must not vanish on X). Furthermore, if f can be given by functions $[f_0 : \dots : f_m]$ where each f_i are regular at all $x \in X$ and not all vanish at this point, then the map is regular at x , and hence determines a regular mapping of some neighborhood x into $\mathbb{P}_k^m$.

We saw before that birational maps are weaker than isomorphisms. However, rational maps are regular on an open dense set (as the reader should verify with quasi-projective varieties). This leads to the following result:

Proposition 1.6.34: Reduction To Open Subset

Let X, Y be irreducible varieties. Then X and Y are birationally isomorphic if and only if the contains isomorphic open subsets $U \subseteq X$ and $V \subseteq Y$.

Proof:

Let $f : X \rightarrow Y$ be a birational isomorphism so that $g : Y \rightarrow X$ is it's inverse. Let $U_1 \subseteq X$ and $V_1 \subseteq Y$ be the domains of regularity for f, g respectively (domains in which f, g are regular). As $f(U_1)$ is dense in Y , $f^{-1}(V_1) \cap U_1$ is non-empty, and is open. Letting $U = f^{-1}(V_1) \cap U_1$, $V = g^{-1}(U_1) \cap V_1$, then we can see that $f(U) = V$, $g(V) = U$, $f \circ g = g \circ f = \text{id}$, completing the proof.

1.6.4 Products

In exercise ref:HERE, the reader showed that the (direct) product algebraic sets $X \subseteq \mathbb{A}_k^n$ and $Y \subseteq \mathbb{A}_k^m$ is algebraic, and lives in the space $\mathbb{A}_k^{n+m}$. If $X' \subseteq \mathbb{A}_k^p$ and $Y' \subseteq \mathbb{A}_k^q$ are algebraic sets st $X \cong X'$ and $Y \cong Y'$, then $X \times Y \cong X' \times Y' \subseteq \mathbb{A}_k^{p+q}$, showing that this definition is embedding-independent. Though not as immediate as affine space, there does exist an N such that $\mathbb{P}_k^n \times \mathbb{P}_k^m \cong \mathbb{P}_k^N$, though $N \neq n + m$. In this section, we shall start by looking into finding the right N . Defining the product of projective spaces shall also allow us to HERE.

To start, recall that $\mathbb{A}_k^n \times \mathbb{A}_k^m \cong \mathbb{A}_k^{n+m}$. A first attempt at understanding $\mathbb{P}_k^n \times \mathbb{P}_k^m$ would be to consider $\mathbb{P}_k^{n+m}$. However, this space is dependent on the representation of the homogeneous points in $\mathbb{P}_k^n$ and $\mathbb{P}_k^m$, for example if $[2 : 0] = [1 : 0] \in \mathbb{P}_k^1$, and $[3 : 0] = [1 : 0] \in \mathbb{P}_k^1$, then:

$$[1 : 1 : 0] \neq [2 : 3 : 0] \in \mathbb{P}_k^2$$

Furthermore, if we choose $[x_1 : x_2] \in \mathbb{P}_k^1$ and $[y_1 : y_2] \in \mathbb{P}_k^1$, which point in $\mathbb{P}_k^2$ ought to represent this point? To resolve this, we shall “twist-multiply”

$$[x_1 y_1 : x_1 y_2 : x_2 y_1 : x_2 y_2] \in \mathbb{P}_k^{(1+1)(1+1)-1}$$

Notice that for $[\lambda x_1 : \lambda x_2]$ and $[\mu y_1 : \mu y_2]$ then:

$$[\lambda \mu x_1 y_1 : \lambda \mu x_1 y_2 : \lambda \mu x_2 y_1 : \lambda \mu x_2 y_2] = [x_1 y_1 : x_1 y_2 : x_2 y_1 : x_2 y_2] \in \mathbb{P}_k^3$$

and hence is independent of representation. Generalizing, this, we may define for any homogeneous

$$x = [x_0 : \cdots : x_n], y = [y_0 : \cdots : y_m] \quad w = [w_{ij}] \in \mathbb{P}_k^{(n+1)(m+1)-1}, \quad w_{ij} = x_i y_j$$

Consider now the embedding $\varphi : \mathbb{P}_k^n \times \mathbb{P}_k^m \rightarrow \mathbb{P}_k^{(n+1)(m+1)-1}$ given by the above. The image is indeed a closed subset, for it satisfies the equations:

$$w_{ij} w_{kl} - w_{ji} w_{lk} = 0 \tag{1.7}$$

Let us show that this construction is natural by showing that it is covered by hyperplanes H_{ij} satisfying the same property as projective space. Consider how the sets H_i get transformed under the embedding φ . To keep track of the dimensions better, let us assume we are working over a field k and write $\mathbb{A}_k^n$ as $\mathbb{A}^n$. Then let $\mathbb{A}_i^n = H_i$, and:

$$\varphi(\mathbb{A}_i^n \times \mathbb{A}_j^m) = W_{ij} = \mathbb{A}_{ij}^{(n+1)(m+1)-1} \cap \varphi(\mathbb{P}_k^n \times \mathbb{P}_k^m)$$

where $\mathbb{A}_{ij}^{(n+1)(m+1)-1}$ is the set of all points where $w_{ij} \neq 0$. Let $N = (n+1)(m+1) - 1$. Consider now the set $\mathbb{A}_{00}^N$. Take a point in projective space where $w_{00} \neq 0$ so that it corresponds to a point in $\mathbb{A}_{00}^N$:

$$[w_{ij}] = \varphi(x, y) \in \mathbb{P}_k^N$$

Take $z_{ij} = w_{ij}/w_{00}$, $x_i = u_i/u_0$, $y_j = v_j/v_0$ as the inhomogeneous coordinates. Then:

$$z_{i0} = x_i \quad z_{0j} = y_j \quad z_{ij} = x_i y_j = z_{i0} z_{0j}$$

But then $\mathbb{A}_0^N \cong \mathbb{A}^{n+m}$ with coordinates $(x_1, \dots, x_n, y_1, \dots, y_m)$ with the isomorphism given by φ ! Hence φ is indeed the natural choice for the embedding of $\mathbb{P}_k^n \times \mathbb{P}_k^m$. This may feel more natural if we

recall that each point of projective space $[a_0 : \cdots : a_n] \in \mathbb{P}_k^n$ is closed and given by a system of linear equations

$$a_i x_j - a_j x_i = 0$$

mirroring equation (1.7). Notice that we may interpret the point $[w_{ij}]$ as an $(n+1) \times (m+1)$ matrix given by multiplying the points x and y as a $(n+1, 1)$ matrix and a $(1, m+1)$ matrix respectively (note the resulting rank of the matrix will be 1). From this, we may define the product of two quasi-projective varieties

Definition 1.6.35: Serge Embedding: Product of Projective Space

Let $V \subseteq \mathbb{P}_k^n, W \subseteq \mathbb{P}_k^m$ be two quasi-projective varieties. Then:

$$V \times W \subseteq \varphi(\mathbb{P}_k^n \times \mathbb{P}_k^m) \subseteq \mathbb{P}_k^{(n+1)(m+1)-1}$$

is the product of the quasi-projective varieties V, W .

Example 1.6.36: Product of Projective Lines

Let $V = W = \mathbb{P}_k^1$ and consider $\varphi(\mathbb{P}_k^1 \times \mathbb{P}_k^1) \subseteq \mathbb{P}_k^3$. This set satisfies the equation:

$$w_{11}w_{00} - w_{01}w_{10} = 0$$

which is a (non-degenerate) quadratic Q in $\mathbb{P}_k^3$. Let us consider the set

$$\varphi(\{\alpha\}, \mathbb{P}_k^1)$$

then given $\alpha = [\alpha_0 : \alpha_1]$, these are all points in $\mathbb{P}_k^3$ such that

$$\alpha_1 w_{00} - \alpha_0 w_{10} = 0 \quad \alpha_1 w_{01} - \alpha_0 w_{11} = 0$$

These equations give lines in $\mathbb{P}_k^3$. Similarly for $\varphi(\mathbb{P}_k^1, \{\beta\})$. The following visual from Hartshorne is an excellent visual:

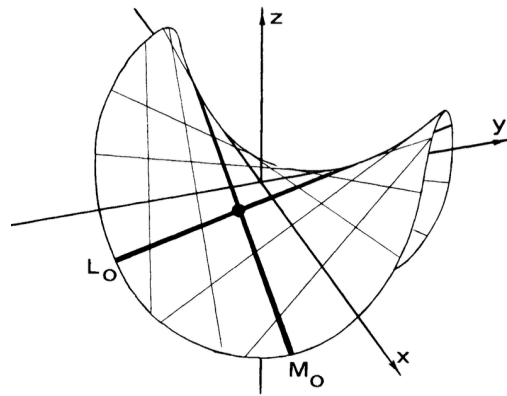


Figure 1.8: Serge Embedding of Product of Projective Lines

The closed subsets of $\text{im}\varphi \subseteq \mathbb{P}_k^{(n+1)(m+1)-1}$ correspond well with the closed subsets of $\mathbb{P}_k^n \times \mathbb{P}_k^m$. Let us start with seeing which subset of $\mathbb{P}_k^n \times \mathbb{P}_k^m$ are algebraic subsets under the Serge Embedding. Let $X \subseteq \mathbb{P}_k^{(n+1)(m+1)-1}$ be a sub-variety so that it is defined by homogeneous equations $f_i(w_{00}, \dots, w_{nm})$. Then these induce the homogeneous polynomials $g_i(u_0, \dots, u_n, v_0, \dots, v_m)$ on $\mathbb{P}_k^n \times \mathbb{P}_k^m$ where the u_i, v_j are the same homogeneous degree. Conversely, if there are two systems of homogeneous equations taking variables of the same homogeneous degree in u_i, v_j , then there exists homogeneous polynomials taking variables in $u_i v_j$. If the degree's of the variables u_i and v_j differ, say the former are of degree r and the latter of degree s , then we can create the system of homogeneous equations:

$$v_j^{r-s} f_i(u_0, \dots, u_n, v_0, \dots, v_m) = 0$$

assuming $r > s$. One important consequence of this is that we can directly check for a set $Z \subseteq \mathbb{P}_k^n \times \mathbb{P}_k^m$ being closed by looking for a single polynomial in $\mathbb{P}_k^{(n+1)(m+1)-1}$ instead of looking at for two polynomials in each separate projective space and verifying the set is the direct product of those two closed sets. Let us box this for future reference:

Proposition 1.6.37: Serge Embedding is Closed

Let $\varphi : \mathbb{P}_k^n \times \mathbb{P}_k^m \rightarrow \mathbb{P}_k^{(n+1)(m+1)-1}$ be the Serge embedding. Then $\text{im}\varphi$ is closed. Furthermore, $X \subseteq \mathbb{P}_k^n \times \mathbb{P}_k^m$ is closed if and only if $\varphi(X) \subseteq \text{im}\varphi$ is closed.

Proof:

formally put together the above argument.

This is actually rather special: as we saw in example ref:HERE (in particular, consider $X = V(xy - 1)$, $Y = \mathbb{A}_k^1$, and $\pi : (a, b) \mapsto a$) the image of an regular map need not be closed, but we have there that we have a condition for which the image is closed. However, the image of a projective variety will always be projective! If the reader has done some analysis this should remind the reader of the fact that the image of compact sets are compact. Though all affine and projective algebraic sets are compact (they are closed subsets of the compact space $\mathbb{A}_k^n$ or $\mathbb{P}_k^n$), projective varieties behave more like compact sets. In Algebraic Geometry, cite:HERE, this distinction shall be made by showing that projective sets having the property of being *proper*³⁶. To see this, we require a few lemmas:

Lemma 1.6.38: Graph of Regular Mapping

Let $\varphi : X \rightarrow Y$ be a regular mapping between quasi-projective varieties. Then the graph

$$\Gamma(\varphi) = \{(x, \varphi(x)) \mid x \in X\}$$

is closed in $X \times Y$

If the Zariski topology was Hausdorff, then we could have concluded using classical topological results, however some more care has to be done as the topology is not Hausdorff.

³⁶In analysis, a map is proper is the pre-image of a compact set is compact

Proof:

First, we may take Y to be a projective space; for an appropriate choice of m , $Y \subseteq \mathbb{P}_k^m$, so that $X \times Y \subseteq X \times \mathbb{P}_k^m$. Then by proposition 1.3.9, f determine a mapping $\bar{f} : X \rightarrow \mathbb{P}_k^m$ and $\Gamma_f = \Gamma_{\bar{f}} \cap (X \times Y)$. As $X \times Y$ is closed (as shown in exercises and proposition 1.6.37), if $\Gamma_{\bar{f}}$ is closed then Γ_f is closed.

Next, define $(f, i) : X \times \mathbb{P}_k^m \rightarrow \mathbb{P}_k^m \times \mathbb{P}_k^m$ where i is the identity map. Then Γ_f is the inverse image of Γ_i under (f, i) . As this is a continuous map, the inverse image of a closed set is closed, and hence it suffices to show that Γ_i is closed.

Now, Γ_i are points $(x, y) \in \mathbb{P}_k^m \times \mathbb{P}_k^m$ of the form

$$[u_0 : \cdots u_m], [v_0 : \cdots v_m]$$

where $u_i v_j = u_j v_i$. By proposition 1.6.37, Γ_i is closed, and $\Gamma_{\bar{f}}$ is closed, and so so Γ_f is closed, completing the proof.

Proposition 1.6.39: Projection of Projective Variety

Let X be a projective variety and Y a quasi-projective variety. Then the projection map:

$$\pi_2 : X \times Y \rightarrow Y$$

is a closed map (the image of a closed set is closed)

Hence, the projection of a projective variety can always be written as a system of polynomial equations! This is not true for affine varieties, consider $X = V(xy - 1)$ and $Y = \mathbb{A}_k^1$

Proof:

Let us first reduce the problem: as $X \subseteq \mathbb{P}_k^n$ is closed, $X \times Y \subseteq \mathbb{P}_k^n \times Y$ is a closed subset, hence if $Z \subseteq X$ is closed, it is closed in $Z \times Y \subseteq \mathbb{P}_k^n \times Y$, and so it is enough to show it for the case of $X = \mathbb{P}_k^n$. Next, As being closed is affine local, it suffices to cover Y by affine open subsets U_i and prove the theorem on each of these. Thus, it suffices to assume that Y is an affine variety. Finally, as Y is closed in $\mathbb{A}_k^m$, $\mathbb{P}_k^n \times Y \subseteq \mathbb{P}_k^n \times \mathbb{A}_k^m$ is closed, and so we can reduce further to $Y = \mathbb{A}_k^m$.

Thus, let us characterize closed subsets $Z \subseteq \mathbb{P}_k^n \times \mathbb{A}_k^m$. By definition of the Zariski topology, and by a simple generalization of proposition 1.6.37, we see that Z is closed if and only if it is the zero set of the collection of polynomials

$$g_i([u_0 : \cdots : u_n], y_1, \dots, y_m) \quad 1 \leq i \leq t$$

In particular, we do not need to create two zero sets for each component and check that the their product produces Z : we can directly work with a set of polynomials. Let us write $g_i(u; y)$ to represent these polynomials with the two different inputs.

Now, take any $y_0 \in \mathbb{A}_k^m$. Then $\pi^{-1}(y_0)$ consists of all solutions (whether empty or non-empty) to the system of equations

$$g_i(u; y_0) = 0 \quad 1 \leq i \leq t$$

Take the collection of y_0 such that the system of polynomials equations $g_i(u; y_0) = 0$ has a nonzero solution; call this collection $T \subseteq \mathbb{A}_k^m$. We want to show that T is closed, for then the

projection π_2 is a closed map. By lemma 1.6.15, the system of polynomial equations $g_i(u; y_0) = 0$ has a nonzero solution if and only if:

$$I_s \not\subseteq (g_1(u, y_0), \dots, g_t(u, y_0)) \quad \forall s \in \mathbb{N}$$

where I_s is the irrelevant ideal of degree $s \geq 1$. Thus, we shall that the set T_s of points $y_0 \in \mathbb{A}_k^m$ for which $I_s \not\subseteq (g_1(u, y_0), \dots, g_t(u, y_0))$ is closed, in particular if we find polynomial equations whose zero set is T_s , then as $T = \bigcap_s T_s$, T is closed which completing the proof.

To that end, let k_i be the homogeneous degree of $g_i(u; y)$ in the u_j variables. Let $(M^\alpha)_\alpha$ be the monomials of degree s in $u_0, \dots, u_n$ with some total ordering (note this set is finite). Then $I_s \subseteq (g_1(u, y_0), \dots, g_t(u, y_0))$ if and only if each monomial M^α can be expressed as:

$$M^\alpha = \sum_{i=1}^t g_i(u, y_0) F_{i,\alpha}(u) \quad (1.8)$$

As each g_i has homogeneous degree k_i , it must be that $F_{i,\alpha}$ has degree $s - k_i$ or $F_{i,\alpha} = 0$ if $k_i > s$. Then we may replace $F_{i,\alpha}$ with elements in $(N_i^\beta)_\beta$ which are monomial of degree $s - k_i$ written in some total ordering. Then equation (1.8) holds if and only if M^α is a linear combination of the polynomials $g_i(u, y_0) N_i^\beta$. As this is true for each M^α , this is equivalent to saying that the polynomials $g_i(u, y_0) N_i^\beta$ span the entire vector space S of homogeneous polynomials of degree s in $u_0, \dots, u_n$. As $I_s \not\subseteq (g_1(u, y_0), \dots, g_t(u, y_0))$, we have that $g_i(u, y_0) N_i^\beta$ do not span S .

Now, take the matrix $(a_{\alpha,\beta})$ that represents the coefficients of the M^α terms as a linear combination of the $g_i(u, y_0) N_i^\beta$. As not all M^α can be written as such terms, this matrix cannot be full rank. Hence all $\sigma \times \sigma$ minors (where $\sigma = \dim(S)$) are polynomials in the coefficients of the polynomials $g_i(u, y_0)$, in particular they are polynomials in the coordinates of the point y_0 . But then we get a system of polynomials each point y_0 , showing that T_s is closed!

Thus, since the arbitrary intersection of closed sets is closed $T = \bigcap_s T_s$ is closed, completing the proof.

For those interested in algebraic geometry, the above proof is in part in proving that projective varieties are *proper*.

Corollary 1.6.40: Image of Projective Variety is Closed

Let $f : X \rightarrow Y$ be a regular map where X is a projective variety. Then $\text{im} f$ is closed

Proof:

Let $\pi : X \times Y \rightarrow Y$ be the natural projection. Consider $\Gamma_f \subseteq Y$. Then certainly:

$$\text{im} f = \pi(\Gamma_f)$$

and hence by proposition 1.6.39 $\text{im} f$ is closed, completing the proof.

We can finally rigorously show that the collection of regular functions on projective space, and more generally projective varieties, are only constant functions:

Corollary 1.6.41: Regular Function on Projective Variety

Let φ be a regular function non an irreducible projective variety. Then $\varphi \in k$, i.e. φ is constant.

Proof:

Take $\varphi : X \rightarrow \mathbb{A}_k^1$. As $\mathbb{A}_k^1$ is dense in $\mathbb{P}_k^1$, there is a unique extension to a regular map $\bar{\varphi} : X \rightarrow \mathbb{P}_k^1$. As X is a projective variety, by corollary 1.6.40 $\bar{\varphi}(X) \subseteq \mathbb{P}_k^1$ is closed. As $\bar{\varphi}(X) \subseteq \mathbb{A}_k^1$, $x_\infty \notin \bar{\varphi}(X)$, and so by irreducibility of $\mathbb{P}_k^1$ it must be that $\varphi(X) = \mathbb{A}_k^1$ or some finite set $S \subseteq \mathbb{A}_k^1$. It cannot be the latter as $\bar{\varphi}(X) = \mathbb{A}_k^1$ needs to be closed in $\mathbb{P}_k^1$ but $\mathbb{A}_k^1$ is not closed. Hence $\varphi(X)$ is a collection of points. Then as these points are distinct and points are closed, their pre-image would make X the non-trivial union of closed sets, contradicting its irreducibility, and hence it must be that the image is a single point, in other words a constant, as we sought to show.

Corollary 1.6.42: Regular Map on Projective Variety

Let $f : X \rightarrow Y$ be a regular map from a projective variety to an affine variety. Then $\text{im} f$ is a point

Proof:

Take $f(x) = (\varphi_1(x), \dots, \varphi_m(x))$ where each φ_i is a regular function. Then by corollary 1.6.41 these are all constant, but then

$$f(X) = (\alpha_1, \dots, \alpha_m)$$

completing the proof.

One final somewhat surprising application is that we can determine whether a set of homogeneous polynomials is irreducible by just looking at their coefficients:

Proposition 1.6.43: Parameterizing Reducible Sets

The collection of points $\zeta \in \mathbb{P}_k^{\nu_{n,m}}$ where $\mathbb{P}_k^{\nu_{n,m}}$ is the Veronese embeddings space, which correspond to reducible polynomials form a closed subset.

Proof:

Let Z the set of points $\zeta \in \mathbb{P}_k^{\nu_{n,m}}$ corresponding to reducible polynomials, and by $Z_j (j = 1, \dots, m-1)$ the set of all points corresponding to polynomials Z that split into factors of degree j and $m-j$. Clearly $Z = \bigcup Z_j$, hence need only prove that each Z_j is closed.

Consider the projective spaces $\mathbb{P}_k^{\nu_{n,j}}$ and $\mathbb{P}_k^{\nu_{n,m-j}}$. The multiplication of two polynomials of degree j and $m-j$ determines a mapping

$$f : \mathbb{P}_k^{\nu_{n,j}} \times \mathbb{P}_k^{\nu_{n,m-j}} \rightarrow \mathbb{P}_k^{\nu_{n,m}}$$

which is regular. Naturally, $Z_j = f(\mathbb{P}_k^{\nu_{n,j}} \times \mathbb{P}_k^{\nu_{n,m-j}})$. Then as the product of projective spaces is a projective variety, it follows from that Z_j is closed, as we sought to show.

As an example, in the case where $m = n = 2$, then f is reducible iff the determinant of the 2×2 is zero: $\det(a_{ij})_{i,j} = 0$.

1.6.5 Finite Maps and Normalization

These maps shall be the mirror of integral extensions, and shall be one of the easiest examples of *proper morphisms* which is a somewhat complicated but fundamental type of morphism in scheme theory.

Definition 1.6.44: Finite Map [on Affine Varieties]

Let $f : X \rightarrow Y$ be a regular map between affine varieties where $f(X)$ is dense in Y and $f^* : k[Y] \hookrightarrow k[X]$ is the induced map. Then if $k[X]$ is integral over $k[Y]$, then f is called a *finite map*.

The composition of finite maps is finite. If $f : X \rightarrow Y$ is finite, then its fibers must be finite (see exercise ref:HERE). A reasonable assumption would be that a map is finite if and only if its fibers are finite, but that is not the case: an example of a non-finite map would be $\pi_1 : V(xy - 1) \rightarrow \mathbb{A}_k^1$ as $k \hookrightarrow k[x, 1/x]$ is not an integral extension. Instead, one way of thinking of a finite map $f : X \rightarrow Y$ is that given an element $y \in Y$, as y is moved around in Y and we look at $f^{-1}(y)$, none of the roots could tend to infinity, that is none can disappear. This intuition can be captured by the following proposition:

Proposition 1.6.45: Finite Maps Are Surjective

Let $f : X \rightarrow Y$ be a finite map between affine varieties. Then f is surjective.

The key result that we shall employ is lying-over property of integral extensions (see ??), namely that if B/A is an integral extensions, if $\mathfrak{a} \subseteq A$ then $\mathfrak{a}B \subseteq B$. In particular, if $\mathfrak{a} \not\subseteq A$, then $\mathfrak{a}B \not\subseteq B$.

Proof:

Let $f : X \rightarrow Y$ be a finite map between affine varieties. Take any $y \in Y$ and let $\mathfrak{m}_y \subseteq k[Y]$ be the functions that take the value 0 at y : if $f \in \mathfrak{m}_y$, then $f(y) = 0$. Let $x_1, \dots, x_n$ be coordinate functions on Y and $y = (\alpha_1, \dots, \alpha_n)$ so that

$$\mathfrak{m}_Y = (x_1 - \alpha_1, \dots, x_n - \alpha_n)$$

Then the equations of the variety $f^{-1}(y)$ have the form:

$$f^*(x_1) - \alpha_1, \dots, f^*(x_n) - \alpha_n$$

By the Nullstellensatz, $f^{-1}(y) = \emptyset$ iff the equations $f^*(x_i) - \alpha_i$ generate the trivial ideal, that is:

$$(f^*(x_1) - \alpha_1, \dots, f^*(x_n) - \alpha_n) = k[X]$$

Viewing $k[Y]$ as a subring of $k[X]$, we see that the above is equivalent to:

$$(x_1 - \alpha_1, \dots, x_n - \alpha_n) = k[X]$$

In other words, $\mathfrak{m}_y k[X] = k[X]$. Then since $k[X]$ is integral over $k[Y]$, we get the result by the going-up property, which completes the proof.

Corollary 1.6.46: Finite Maps Are Closed Maps

Let $f : X \rightarrow Y$ be a finite map between affine varieties. Then f is a closed map.

Proof:

It suffices to check on irreducible closed subsets $Z \subseteq X$. By the above theorem the restriction of f to Z , $f|_Z : Z \rightarrow \overline{f(Z)}$, is a finite map between affine varieties, and so by surjectivity we get $f(Z) = \overline{f(Z)}$, but then $f(Z)$ is closed, completing the proof.

Let us now build-up to defining finite maps for quasi-projective varieties. For that, we start by showing that finiteness is a local property:

Lemma 1.6.47: Finiteness is Local

Let $f : X \rightarrow Y$ be a regular map of affine varieties. Then if for every point $y \in Y$, there is an affine neighborhood U of y such that $V = f^{-1}(U)$ is affine and $f : V \rightarrow U$ is finite, then f is finite.

Proof:

Let $k[Y] = A$ and $k[X] = B$. Take the principal open set around each point of Y , and as Y is compact take a finite subcover, label these $D(g_\alpha)$ so that $Y = \bigcup_\alpha D(g_\alpha)$. Then:

$$V_\alpha = f^{-1}(D(g_\alpha)) = D(f^*(g_\alpha))$$

and

$$k[D(g_\alpha)] = A[1/g_\alpha] \quad k[V_\alpha] = B[1/g_\alpha]$$

As the map is finite, $B[1/g_\alpha]$ has a finite $A[1/g_\alpha]$ -basis, say $w_{i,\alpha}$. Then we can assume that $w_{i,\alpha} \in B$, since if the basis consisted of elements $w_{i,\alpha}/g_\alpha^{m_i}$ where $w_{i,\alpha} \in B$ then the elements $w_{i,\alpha}$ would also be a basis.

Now, take the union of all bases $w_{i,\alpha}$. I claim these form an A -basis for B . To see this, take any $b \in B$. Then for each α we can write it as :

$$b = \sum_i \frac{a_{i,\alpha}}{g_\alpha^{n_\alpha}} w_{i,\alpha}$$

As $g_\alpha^{n_\alpha}$ generate the entire A (as the union of the distinguished sets $D(g_\alpha)$ is Y), there exists $h_\alpha \in A$ such that

$$\sum_\alpha g_\alpha^{n_\alpha} h_\alpha = 1$$

Thus:

$$b = b \sum_\alpha g_\alpha^{n_\alpha} h_\alpha = \sum_i \sum_\alpha a_{i,\alpha} h_\alpha w_{i,\alpha}$$

as we sought to show.

Thus, we can define finite maps for quasi-projective varieties:

Definition 1.6.48: Finite Map [on Quasi-Projective Varieties]

Let $f : X \rightarrow Y$ be a regular map between quasi-projective varieties. Then f is said to be *finite* if any point $y \in Y$ has an affine local neighborhood V such that $U = f^{-1}(V)$ is affine and $f : U \rightarrow V$ is a finite map between affine varieties.

Naturally, $f^{-1}(y)$ is finite for every $y \in Y$, and it follows easily that any finite map is surjective. Note that the density condition is now a local condition. One consequence of these results that we shall not prove is that if $f : X \rightarrow Y$ between quasi-projective is a regular map (not necessarily finite) and $f(X)$ dense in Y , then $f(X)$ contains an open subset of Y . This is *not* true in analysis, the canonical example being the wrapping of the real line around the torus with an irrational period which will be a dense map with empty interior. For a proof of the existence of an open subset in a dense mapping, see Shafarevich cite:HERE.

Normalization

An important culminating result is that every irreducible projective variety is in some way only a “finite extension” from projective space. These results shall allow us to diminish many arguments about irreducible varieties to arguments about irreducible varieties into arguments about affine/projective spaces that are closed under finite extension! These ideas shall come back more in modern algebraic geometry (see [1]) We first need a technical lemma:

1.6.49 Normalization and Singularities Note that this is *not* to be confused with normalization of singularities! Rather confusingly, normalization of singularities shall be the map in the *opposite direction* to the normalization map that will be defined here. Both of these used the word normal differently, and it is not the case that a normal resolution of singularities results in a map $\mathbb{A}_k^n \rightarrow X$ or $\mathbb{P}_k^n \rightarrow X$ for a variety X (where the normalization map we shall do will be map of the form $X \rightarrow \mathbb{A}_k^n$ or $X \rightarrow \mathbb{P}_k^n$).

Lemma 1.6.50: Projection is Finite Mapping

let X is closed in $\mathbb{P}^n$ and $X \subset \mathbb{P}^n - E$, where E is a d -dimensional linear subspace. Then the projection $\pi : X \rightarrow \mathbb{P}^{n-d-1}$ with center at E determines a finite mapping $X \rightarrow \pi(X)$.

Proof:

Let $y_0, \dots, y_{n-d-1}$ be homogeneous coordinates in $\mathbb{P}^{n-d-1}$ and let π be given by the formulae $y_j = L_j(x)$, $x \in X$ ($j = 0, \dots, n-d-1$). Naturally, $U_i = \pi^{-1}(\mathbb{A}^{n-d-1}) \cap X$ is given by the condition $L_i(x) \neq 0$ and is an affine open subset of X . We want to show that $\pi : U_i \rightarrow \mathbb{A}^{n-d-1} \cap \pi(X)$ is a finite mapping. To that end, first note that every function $g \in k[U_i]$ is of the form $g = G_i(x_0, \dots, x_n)/L_i^r$, where G_i is a form of degree m . Consider the mapping $\pi_1 : X \rightarrow \mathbb{P}^{n-d} : z_j = L_j^m(x)$ ($j = 0, \dots, n-d-1$), $z_{n-d} = G_i(x)$, where $z_0, \dots, z_{n-d}$ are homogeneous coordinates in $\mathbb{P}^{n-d}$. It is a regular mapping and its image $\pi_1(X)$ is closed in $\mathbb{P}^{n-d}$. Let $F_1 = \dots = F_s = 0$ be its equations. Since $X \subset \mathbb{P}^n - E$, the forms L_i ($i = 0, \dots, n-d-1$) do not vanish simultaneously on X . Hence, the point $0 = [0 : \dots : 1]$ is not contained in $\pi_1(X)$, in other words, that the equations $z_0 = \dots = z_{n-d-1} = F_1 = \dots = F_s = 0$ have no solution in

$\mathbb{P}^{n-d}$. Hence, for some $l > 0$:

$$(z_0, \dots, z_{n-d-1}, F_1, \dots, F_s) \supset T_l$$

in particular, $z_{n-d}^l \in (z_0, \dots, z_{n-d-1}, F_1, \dots, F_s)$. This means that

$$z_{n-d}^l = \sum_{j=0}^{n-d-1} z_j H_j + \sum_{j=1}^s F_j P_j,$$

where H_j and P_j are polynomials. Denote by $H^{(q)}$ the homogeneous component of H of degree q . Then:

$$\Phi(z_0, \dots, z_{n-d}) = z_{n-d}^l - \sum z_j H_j^{(l-1)} = 0 \quad \text{on } \pi_1(X) \quad (1.9)$$

The homogeneous polynomial Φ is of degree l and as a polynomial in z_{n-d} it has the leading coefficient 1:

$$\Phi = z_{n-d}^l - \sum_{j=0}^{l-1} A_{l-j}(z_0, \dots, z_{n-d-1}) z_{n-d}^j. \quad (1.10)$$

Substituting the transformation formulae for π_1 in equation (1.9) we find that $\Phi(L_0^m, \dots, L_{n-d-1}^m, G_i) = 0$ on X , with Φ of the form equation (1.10). Dividing this relation by L_i^m we obtain the required relation

$$g^l + \sum_{j=0}^{l-1} A_{l-j}(x_0, \dots, 1, \dots, x_{n-d-1}) g^j = 0,$$

where $x_r = y_r/y_i$ are coordinates in $\mathbb{A}^{n-d-1}$. But then we have that the mapping is finite, completing the proof.

Theorem 1.6.51: Normalization of Projective Variety

Let X be an irreducible projective variety. Then there exists a finite mapping $\varphi : X \rightarrow \mathbb{P}_k^m$ for appropriate m .

Proof:

Take $X \subseteq \mathbb{P}_k^n$. If $X = \mathbb{P}_k^n$ we're done so take $X \neq \mathbb{P}_k^n$. Take $x \in \mathbb{P}_k^n \setminus X$ and take the projection from x , $\varphi_x : X \rightarrow \mathbb{P}_k^{n-1}$. This map is regular by example 1.6.32, and the image is closed, hence projective. This map is furthermore finite by lemma 1.6.50. If $\varphi(X) \neq \mathbb{P}_k^{n-1}$. We can repeat this argument. All the same properties will remain, as the composition of closed/finite maps is closed/finite. This process shall terminate since if $n - \ell = 1$ then it must be that $\varphi(X) = \mathbb{P}_k^1$. Hence we get that for appropriate m a finite mapping

$$\varphi(X) = \mathbb{P}_k^m$$

completing the proof.

Theorem 1.6.52: Normalization of Affine Variety

Let X be an irreducible affine variety. Then there exists a finite mapping $\varphi : X \rightarrow \mathbb{A}_k^m$ for appropriate m .

Note that this is the analog of Noether's Normalization theorem for rings that are finitely generated k -algebras without zero-divisors, and in fact can be proven using it, see exercise 1.6.1. This proof is more generalizable to more general rings, as is important for modern algebraic geometry (see cite:HERE algGeo, or Hartshorne's chapter on Curves and Surfaces).

Proof:

Take $X \subseteq \mathbb{A}_k^n$. If $X = \mathbb{A}_k^n$ we're done so take $X \neq \mathbb{A}_k^n$. Consider $\mathbb{A}_k^n \cong U_i$, X is open in $\mathbb{P}_k^n$ so take $\overline{X}$. Then choose a point $x \in \mathbb{P}_k^n \setminus \mathbb{A}_k^n$ where $x \notin \overline{X}$ (which exists as $X \neq \mathbb{A}_k^n$). Then take the same projection $\varphi : \overline{X} \rightarrow \mathbb{P}_k^{n-1}$. Then it is not hard to see that X is projected onto $\mathbb{A}_k^{n-1}$. We may again repeat this process like in theorem 1.6.51 a surjective map:

$$\varphi(X) = \mathbb{A}_k^m$$

completing the proof.

Thus, though not all varieties are rational, all varieties are a finite cover of affine or projective space, and hence the study of which varieties are not rational will naturally lead to the question of how far away are they from being rational³⁷! More on this in the exercises.

Example 1.6.53: Finite Maps and Normalization

Consider the curve $f : \mathbb{A}_k^1 \rightarrow \mathbb{A}_k^2$ given by $f(t) = (t^2, t^3)$. This is associated to the map of global section $f^\# : k[x, y] \rightarrow k[t]$, $f(x, y) \mapsto f(t^2, t^3)$. Taking the quotient of the domain we get $k[x, y]/(x^3 - y^2) \cong k[t^2, t^3] \hookrightarrow k[t]$ which is not injective since nothing can map to t . As $(x^3 - y^2)$ is prime, we take the field of fraction and extend the map, which has $t^3/t^2 = t$ we have that $\text{Frac}(k[t^2, t^3]) = k(t)$ giving a field isomorphism.

This is a 1-to-1 map. Define an n -to-1^a map $f : \mathbb{A}_k^1 \rightarrow \mathbb{A}_k^2$ given by $t \mapsto (t^{2n}, t^{3n})$. Notice the image is the same. The global section's are also isomorphic as $k[t^2, t^3] \cong k[t^{2n}, t^{3n}]$. Thus the field of fractions are isomorphic, $k(t) \cong k(t^n)$. Working with a specific n -to-1 map, the field of fraction of the global section of the image is $k(t^n)$ which has index n in the field of fractions of the domain, namely $k(t)$. This reflects the geometric nature of finite maps in algebraic properties.

Notice that $x^2 - y^3$ forms a curve with a cusp; it has a singularity at $(0, 0)$. This curve maps onto

^anote it is important to assume that either k is algebraically closed, or at minimum that k contains all the roots of unity

Exercise 1.6.1

1. Let $R_d \subseteq k[x_1, \dots, x_n]$ be the vector space of homogeneous polynomials of degree d . Show that $\dim_k R_d = \binom{d+n-1}{n-1}$
2. Let $T : \mathbb{A}_k^{n+1} \rightarrow \mathbb{A}_k^{n+1}$ be a linear isomorphism.

³⁷for those studying algebraic geometry or following [1], the Chevalley's Theorem can be thought of as the best generalization of this result for general schemes

- a) Show that V is a variety if and only if $V^T = T^{-1}(V)$ is a variety.
- b) Show that $V \cong V^T$.
3. Let $V = \mathbb{P}_k^1$, $\Gamma_h(V) = k[x, y]$, and $t = x/y$. Show that $\kappa(V) \cong k(t)$.
 - a) show there is a natural correspondence between the points of $\mathbb{P}_k^1$ and the DVR's with quotient field $k(V)$ containing k . What's the DVR for the point at infinity?
 - b) Show that T induces an isomorphism on homogeneous coordinate rings.
4. Show that any two lines in projective space $\mathbb{P}_k^n$ intersect, that is if L_1, L_2 are two equation of lines and we set $L_1 = 0, L_2 = 0$, then these two zero-sets must have a common point.
5. if $\frac{\partial f}{\partial x_i}$ is the formal partial derivative applied to a function f , then show that if f is a homogeneous polynomial of degree d that

$$df = \sum_{i=1}^n x_i \frac{\partial f}{\partial x_i}$$

6. Let R be a graded ring so that $1 = \sum_{k=0}^{\infty} r_k$. Show that r_0 is the multiplicative identity of R_0 .
7. Check the decomposition theorem for quasi-projective varieties.
8. (Calculus exercise) Let f be a k -times differentiable function degree k homogeneous function. Show that f is a degree k homogeneous polynomial. Hence homogeneity characterizes homogeneous polynomials.
9. Good exercise!
10. Since Noether's Normalization theorem was already proven earlier in this chapter, I think a good exercise would be converting the result from Shafarevich into some guided exercises here (theorem 7-10 chapter 1 section 5) [here](#)
11. Prove the affine normalization theorem using Noether's Normalization Theorem.

1.7 Application: Curves

Arguably one of the most important results on [classical] projective curves is the fact that we can extend the fundamental theorem of algebra to its fullest extent when working in projective space. The fundamental Theorem of algebra states that a polynomial of degree n over an algebraically closed fields has n roots. This can also be thought of as the polynomial f and the line $x = 0$ intersecting at n locations (counting multiplicity). Generalizing this, we shall see that we can have two polynomials f, g of degree n and m that, when embedded into projective space, intersect at nm places (counting multiplicity). In affine space, this is not the case; consider $x^2 - y + 1$ and $y^2 + x + 1$ in $\mathbb{A}_{\mathbb{C}}^2$.

This section shall strongly follow the lead of Fulton from [6] to be a quick exposition on the topic.

Defining our Terms

For the rest of this section, we shall be only working with *curves*, which are the one-dimensional varieties, that are over an algebraically closed field $k = \bar{k}$. We shall cover dimension extensively in section 1.4, for this section we can define an *affine curve* to be a hypersurface in $\mathbb{A}_k^2$. Taking its closure in $\mathbb{P}_k^2$ gives a *projective curve*. This is the homogenization of the hypersurface in $\mathbb{A}_k^3$. The homogenization can be thought of as containing all scalar multiples of the curve. We can thus define the following:

Definition 1.7.1: Projective Plane Curve

A projective plane curve is an equivalence class of nonconstant homogeneous polynomial in $k[x, y, z]$, where $f \sim g$ if and only if $f = \alpha g$ for some $\alpha \in k^*$. The degree of a curve is defined to be the degree of the defining homogeneous polynomial.

If $f = \prod_i^n f_i^{e_i}$ for some curves f_i , then we shall say that f is *irreducible* if $n = 1$. We shall call f_i *simple* if $e_i = 1$, and *multiple* or *ramified* otherwise. Note that the multiplicity information is lost when working with $V(f)$ ³⁸.

When looking at the intersection of curves, we shall often require to have an algebraic object that holds intersection information. We shall see keeping track of all functions that are defined at a point will be the right definition. The following is defined for varieties in general:

Definition 1.7.2: Recall: Stalk At A Point

Let V be a variety and $p \in V$ a point. Then the *stalk* at p is the set:

$$\mathcal{O}_p(V) = \{f/g \in k(V) \mid g(p) \neq 0\}$$

The intuition is that $\mathcal{O}_p(V)$ contains all functions which are defined at p , and so we can take all rational functions whose denominator is nonzero at p .

Lemma 1.7.3: Properties of Stalk

Let V be a variety, $p \in V$ a point, and $\mathcal{O}_p(V)$ the stalk at p . Then $\mathcal{O}_p(V)$ is a local ring with a single maximal ideal defined by $\mathfrak{m}_V = \{f/g \in \mathcal{O}_p(V) \mid f(p) = 0\}$.

Proof:

First show that $\mathfrak{m}_V$ is maximal. Then show that if $x \notin \mathfrak{m}_V$, then x is a unit, and hence there can only be one maximal ideal.

The next concept we introduce will allow us to simplify many of our computations by translating our points of intersection of curves to the origin:

³⁸In [1], we shall see that schemes remember multiplicity information, giving us another reason to expand the algebraic tools in modern algebraic geometry

Definition 1.7.4: Linear Coordinate Change

Let $\mathbb{A}_k^n$ be affine space with coordinate ring $k[x_1, \dots, x_n]$. Then a bijective affine transformation $T : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$ is called a *linear change in coordinate*

If $T = (T_1, \dots, T_n)$, then this maps $f(x_1, \dots, x_n) \mapsto f(T_1, \dots, T_n)$. Such a transformation can be decomposed into a translation and a bijective linear transformation. As an example of this application, we shall define multiplicity at a point by first translating it to 0:

Definition 1.7.5: Multiplicity

Let $f \in k[x, y]$. Then the *multiplicity* of $p \in V(f)$ to be the number of times the curve f intersects the point p . This number is denoted $m_p(f)$.

We shall usually translate our curve (or more generally variety) and consider the multiplicity at $p = (0, 0)$.

By using the derivative, f is nonsingular at p if and only if $m_p(f) = 1$. Let $f = \sum_{i=m}^n f_i$ be the homogeneous decomposition of f after showing f so that $p = (0, 0)$. Then $m_p(f) = m$. Notice that p is simple (i.e. nonsingular) if and only if $m_p(f) = 1$.

We now define tangent spaces. Let $x \in V$ be a point in a variety. Without loss of generality, change the coordinate so that $x = O = (0, 0, \dots, 0)$. Then all the lines passing through 0 are of the form:

$$L_a = \{ta \mid t \in k^*\}$$

for any $a \in \mathbb{A}_k^n$. These lines naturally intersect X at O . To see find the rest of the intersections $L_a \cap X$, we see where the solutions of polynomials match, namely if $V = \mathcal{Z}(f_1, \dots, f_n)$, then we are looking for all t where:

$$f_1(ta) = \dots = f_n(ta) = 0$$

These are now all polynomials in one variable t , and so their common roots will be the g.c.d. Let f be the polynomial representing the common intersection points (i.e., the vanishing set of the intersection, *counting* multiplicity):

$$f_a(t) = \alpha_0 \prod (t - \alpha_i)^{l_i}$$

By the coordinate change, 0 is always a root and hence this polynomial is always nonzero.

Coming back to the multiplicity at p , by exercise ref:HERE,

$$f_m = \prod_j L_j^{r_j}$$

These lines are called tangent lines. The lines L_i are simple if $r_i = 1$. If f has m distinct tangent lines, we shall say that p is an *ordinary multiple point*. An ordinary double point is called a *node*. If a line passing through p is not tangent, we shall say it has multiplicity 0.

If $f = \prod f_i^{e_i}$ is the irreducible decomposition, then:

$$m_p(f) = \sum_i e_i m_p(f_i)$$

If L is a tangent line to f_i with multiplicity r_i , then L is tangent to f with multiplicity $\sum_i e_i r_i$ as the lowest degree term of f is given by the product of the lowest degree term of each f_i . Then a

point is simple if and only if it only 1) belongs to one irreducible component of f , 2) f_i is simple if f , and 3) p is a simple point of f_i .

Definition 1.7.6: Intersection Multiplicity

Let X be a variety and L a line. Then the *intersection multiplicity* of L at a point in $\alpha_i \in L \cap X$ is the degree of the term $(t - \alpha_i)$ in the above polynomial.

If the polynomials $f_i(ta)$ all vanish identically, then we would say the intersection multiplicity is ∞ ³⁹. It should be noted that $f_a(t) = \gcd\{f(ta) \mid f \in \mathcal{I}(X)\}$ and so this notion is independent of choices of functions.

Next, we shall need to distinguish the “smooth” and “singular” points. We shall specialize to the case of an irreducible curve;

Definition 1.7.7: Tangent Line

Let $C = V(F) \subseteq \mathbb{P}_k^2$ and $p \in C$. Then p is called *nonsingular* or *simple* if $\partial_x F \neq 0$ or $\partial_y F \neq 0$ (i.e. the partials is full rank). Then, given $p = (a, b)$ the *tangent line* is the set of points satisfying the equation:

$$\frac{\partial f}{\partial x}(p)(x - a) + \frac{\partial f}{\partial y}(p)(y - b) = 0$$

A non-simple point is called *singular*.

Next, let us show how the multiplicity can be determined by the local ring $\mathcal{O}_p(F)$ at a point:

Theorem 1.7.8: Multiplicity and Local Ring

Let $p \in f$ be a point of an irreducible curve $V(f) \subseteq \mathbb{A}_k^2$. Then for n sufficiently large:

$$m_p(f) = \dim_k \left(\frac{\mathfrak{m}_p(f)^n}{\mathfrak{m}_p(f)^{n+1}} \right)$$

Thus, the multiplicity of f at p can be completely determined by the local ring $\mathcal{O}_p(f)$

Proof:

Let $\mathcal{O} = \mathcal{O}_p(f)$ and $\mathfrak{m} \subseteq \mathcal{O}$ be the maximal ideal. Then it is a quick exercise to show that the following is a short exact sequence:

$$0 \rightarrow \mathfrak{m}^n / \mathfrak{m}^{n+1} \rightarrow \mathcal{O} / \mathfrak{m}^{n+1} \rightarrow \mathcal{O} / \mathfrak{m}^n \rightarrow 0$$

These are all k -vector spaces, and dimension is additives (namely, the dimension of the middle is the sum of the dimension of the left and right, see exercise ref:HERE). Then it suffices to show that:

$$\dim_k (\mathcal{O} / \mathfrak{m}^n) = nm_p(f) + s$$

for some natural number $s \in \mathbb{N}$ and all $n \geq m_p(f)$. To simplify computation, apply the appropriate linear transformations so that $p = (0, 0)$. Then $\mathfrak{m}^n = I^n = (x, y)^n \subseteq k[x, y]$. As

³⁹in algebraic geometry, we would rather say that it intersects a non-closed points, but this is a discussion for [1]

$V(I^n) = \{p\}$, by exercise ref:HERE

$$\frac{k[x, y]}{(I^n, f)} \cong \frac{\mathcal{O}_p(\mathbb{A}_k^2)}{(I^n, f)\mathbb{A}_k^2} \cong \frac{\mathcal{O}_p(f)}{I^n \mathcal{O}_p(f)} \cong \mathcal{O}/\mathfrak{m}^n$$

Hence, we are reduced to calculating the dimension of $\frac{k[x, y]}{(I^n, f)}$. Let $m = m_p(f)$ be the multiplicity of f at p . We shall setup another short exact sequence and take advantage of the additivity of dimension. First, notice that if $fg \in I^n$, then $g \in I^{n-m}$ (as $m_p(f) = m$). Then let $\varphi : k[x, y]/I^n \rightarrow k[x, y]/(I^n, f)$ be the natural projection map and :

$$\psi : \frac{k[x, y]}{I^{n-m}} \rightarrow \frac{k[x, y]}{I^n} \quad \psi(\bar{g}) = \overline{fg}$$

Now, The dimension of $k[x, y]/(I^k)$ is readily computable: $1 + 2 + \cdots + k = \frac{k(k-1)}{2}$ (see ref:HERE). Then by additivity we get:

$$\dim_k(\mathcal{O}/\mathfrak{m}^n) = mn - \frac{m(m-1)}{2} = nm_p(f) + s$$

for all $n \geq m$, as we sought to show.

Corollary 1.7.9: Simple Points and Multiplicity

let $p \in f$ be a point. Then p is simple if and only if $\mathcal{O}_p(f)$ is a DVR. In this case, if $L \subseteq \mathbb{A}_k^2$ is a line through p that is not tangent at p , then the image $\bar{L}$ in $\mathcal{O}_p(f)$ is a uniforming parameters (a generator for the maximal ideal).

Proof:

Let $p \in f$ be a simple point, L a line through P not tangent to f at p . Doing the usual change of coordinates, assume $p = (0, 0)$ Y is the tangent line, $L = X$. Then it suffices to show that $\mathfrak{m}_p(f)$ is generated by x . To that end, note that $\mathfrak{m}_p(f) = (x, y)$; these are the values for which any function evaluates to 0 at $p = (0, 0)$. We shall show that $y \in (x)$. First, note that by the Taylor theorem f can be decomposed into $f = y + p(x, y)$ the coefficient of y is 1 as y is tangent to f at p , and $p(x, y)$ contains all terms degree ≥ 2 . Let us group all terms in f that contain a y , and notice the rest of the terms must have an x^2 factor, letting us write $f = yg + x^2h'$ for $g = 1 + p(x, y)$. Without loss of generality, we can take $h = -h'$ and write $f = yg - x^2h$. Then in $k[V(f)]$ we have:

$$yg = x^2h$$

Since $g(p) \neq 0$, it is invertible in $\mathcal{O}_p(f)$, and so we have:

$$y = x^2hg^{-1} \in (x)$$

where the inclusion comes by being closed under multiplication. But then we have $\mathfrak{m}_p(f) = (x, y) = (x)$, and so by ?? it is a DVR. The converse comes from theorem 1.7.8, completing the proof.

This immediately shows that any non-tangent line passing through p , as an element of $k[x, y]$ must

have $\nu_p(L) = 1$, and tangent lines must have order at least 2 for:

$$\nu_p(y) = \nu_p(x^2 h g^{-1}) = \nu_p(x^2) + \nu_p(h g^{-1}) \geq 2$$

Next, we require an important technical lemma for describing coordinate rings for a set of finite points:

Lemma 1.7.10: Coordinate Ring For Finite Points

Let $I \subseteq k[x_1, \dots, x_n]$ such that $V(I) = \{p_1, \dots, p_m\}$. Letting $\mathcal{O}_i = \mathcal{O}_{p_i}(\mathbb{A}_k^n)$. Then:

$$\frac{k[x_1, \dots, x_n]}{I} \cong \prod_i^m \frac{\mathcal{O}_i}{I\mathcal{O}_i}$$

Proof:

For notational simplicity, let $I_i \subset k[x_1, \dots, x_n]$ be the (distinct) maximal ideals containing I , $R = k[X_1, \dots, X_n]/I$, and $R_i = \mathcal{O}_i/I\mathcal{O}_i$. Let φ_i be the homomorphism from R to R_i , their combination inducing a homomorphism φ from R to $\prod_{i=1}^m R_i$. Let $\text{Rad}(I) = I(\{P_1, \dots, P_m \mid \}) = \bigcap_{i=1}^m I_i$, so that by exercise ref:HERE, for some d :

$$(\bigcap I_i)^d \subset I$$

Since $\bigcap_{j \neq i} I_j$ and I_i are co-maximal (exercise ref:HERE), it follows that

$$\bigcap I_i^d = (I_1 \cdots I_m)^d = (\bigcap I_j)^d \subset I.$$

Now, choose $f_i \in k[x_1, \dots, x_n]$ such that $f_i(p_j) = 0$ if $i \neq j$, $f_i(p_i) = 1$; such polynomials exist by exercise ref:HERE. Let

$$e_i = 1 - (1 - f_i^d)^d$$

Note that $e_i = f_i^d d_i$ for some d_i , so $e_i \in I_j^d$ if $i \neq j$, and $1 - \sum_i e_i = (1 - e_j) - \sum_{i \neq j} e_i \in \bigcap I_i^d \subset I$. Furthermore, $e_i - e_i^2 = e_i(1 - f_i^d)^d$ is in $\bigcap_{j \neq i} I_j^d \cdot I_i^d \subset I$. If we let $\overline{e_i}$ be the residue of e_i in R , then we have

$$\overline{e_i}^2 = \overline{e_i}, \quad e_i e_j = 0 \text{ if } i \neq j, \quad \sum e_i = 1$$

I claim that if $g \in k[x_1, \dots, x_n]$, and $g(p_i) \neq 0$, then there is a $t \in R$ such that $t\overline{g} = \overline{e_i}$. Indeed, we may first assume that $g(p_i) = 1$. Let $h = 1 - g$. Then:

$$(1 - h)(e_i + h e_i + \cdots + h^{d-1} e_i) = e_i - h^d e_i$$

Hence $h \in I_i$, so $h^d e_i \in I$. Therefore

$$\overline{g(e_i + h e_i + \cdots + h^{d-1} e_i)} = \overline{e_i}$$

as desired.

From this, we can show that φ is an isomorphism. Indeed, if $\varphi(f) = 0$, then for each i there is a g_i with $g_i(p_i) \neq 0$ and $\overline{g_i f} \in I$. Let $\overline{t_i g_i} = \overline{e_i}$. Then $\overline{f} = \sum \overline{e_i f} = \sum \overline{t_i g_i f} = 0$, showing injectivity. For surjectivity, since $e_i(p_i) = 1$, $\varphi_i(\overline{e_i})$ is a unit in R_i since

$$\varphi_i(\overline{e_i}) \varphi_i(\overline{e_j}) = \varphi_i(\overline{e_i e_j}) = 0$$

if $i \neq j$, $\phi_i(\bar{e}_j) = 0$ for $i \neq j$. Therefore $\phi_i(\bar{e}_i) = \phi_i(\sum \bar{e}_j) = \phi_i(1) = 1$. Now, suppose

$$z = (a_1/s_1, \dots, a_m/s_m) \in \prod R_i$$

By the earlier claim, find $\bar{t}_i$ so that $\bar{t}_i \bar{s}_i = \bar{e}_i$. Then $\bar{a}_i/\bar{s}_i = \bar{a}_i \bar{t}_i \in R_i$, so

$$\phi_i(\sum \bar{t}_j \bar{a}_j \bar{e}_j) = \phi_i(\bar{t}_i \bar{a}_i) = \bar{a}_i/\bar{s}_i$$

and $\phi(\sum \bar{t}_j \bar{a}_j \bar{e}_j) = z$, showing surjectivity and completing the proof.

Corollary 1.7.11: Dimension of Coordinate Ring of Finite Points

Let $R = k[x_1, \dots, x_n]/I$ be the coordinate ring for finitely many points. Then:

$$\dim_k \frac{k[x_1, \dots, x_n]}{I} = \sum_i^n \dim_k(\mathcal{O}_i/I\mathcal{O}_i)$$

Proof:

immediate

Corollary 1.7.12: Coordinate Ring Single Point

Let $R = k[x_1, \dots, x_n]/I$ be the coordinate ring for a single point. Then:

$$k[x_1, \dots, x_n]/I \cong \frac{\mathcal{O}_p(\mathbb{A}_k^n)}{I\mathcal{O}_p(\mathbb{A}_k^n)}$$

Proof:

immediate

Let us conclude this section by pointing out some important facts about rational maps where the domain is a curve C .

Lemma 1.7.13: Rational Map On Smooth Curves

let C be a curve, $V \subseteq \mathbb{P}_k^n$ be a variety, and $\varphi : C \rightarrow V$ a rational map. If $p \in C$ is a smooth point, then φ is regular at p . If C is a smooth curve, then φ is a regular morphism.

Proof:

Let $\varphi = [f_0, \dots, f_n]$ be a representation of the map. As p is smooth, let $t \in k(C)$ be the uniformizer of the DVR associated to p . Define:

$$m = \min_{0 \leq i \leq n} (\text{ord}_p(f_i))$$

Then:

$$\text{ord}_p(t^{-m} f_i) \geq 0 \quad \forall i \quad \text{ord}_p(t^{-m} f_j) = 0 \text{ for some } j$$

Then we see that $t^{-m}\varphi$ is regular at p . Then as

$$[t^{-m}f_0, \dots, t^{-m}f_n] = [f_0, \dots, f_n]$$

we see that φ is regular. If C is smooth at every point, we see that φ must be regular, completing the proof.

Exercise 1.7.1

1. Let $p = (0, \dots, 0) \in \mathbb{A}_k^2$. Show that the maximal ideal $\mathfrak{m} \subseteq \mathcal{O}_p(\mathbb{A}_k^2)$ is generated is $(x_1, \dots, x_n)$.
2. Show that $\dim_k \left(\frac{k[x, y]}{\mathfrak{m}^n} \right) = \frac{n(n+1)}{2}$ where $I = (x, y)$.
3. Let $V \subseteq \mathbb{A}_k^n$ be a variety given by $I = I(V) \subseteq k[x_1, \dots, x_n] = R$. Let $J \subseteq R$ such that $I \subseteq J$. Show that $\mathcal{O}_p(\mathbb{A}_k^n)/J\mathcal{O}_p(\mathbb{A}_k^n) \cong \mathcal{O}_p(V)/\bar{J}\mathcal{O}_p(V)$. If $I = J$, then $\mathcal{O}_p(\mathbb{A}_k^n)/I\mathcal{O}_p(\mathbb{A}_k^n) \cong \mathcal{O}_p(V)$.

1.7.1 Intersection Number

Let $f, g \subseteq k[x, y]$ be curves and $p \in \mathbb{A}_k^2$. Let us now define the *intersection number* of p on $f \cap g$: $I(p, f \cap g)$. As we may have multiple new behaviors for polynomials over several variables (as compared to linear polynomials over several variables), more care must be put in insuring a consistent definition. One way to approach this problem is by giving a set of conditions the intersection number should satisfy. The following condition are given by Fulton (see [6, section 3.3]); for another great perspective that leans on using a bit of complex analysis and a more “zoomed in” approach relying on the intuition introduced at the beginning of section ??, see [3]. The following conditions will not be minimal, in fact the reader will see when computing intersection numbers that we don’t usually use the full force of these axioms. It is however convenient to prove the intersection number satisfies these stronger properties:

1. If f, g contains no common components that both contains p , we shall say that f, g *intersect properly* at p . Then $I(p, f \cap g)$ is a non-negative integer for any curves f, g and point $p \in \mathbb{A}_k^2$, and $I(p, f \cap g) = \infty$ is f, g *do not* intersect properly at p .
2. $I(p, f \cap g) = 0$ if and only if $p \notin f \cap g$.
3. $I(p, f \cap g)$ depend only the components that pass through p
4. $I(p, f \cap g) = 0$ if either f or g is constant
5. the intersection number is invariant under linear transformation
6. If two curves intersect transversally (they intersect properly, p is a simple point of both curves, and all of the tangent lines intersect properly), then $I(p, f \cap g) = 1$. More generally, $I(p, f \cap g) \geq m_p(f)m_p(g)$ with equality occurring is they share no tangent lines. In general, $I(p, f \cap g) \geq m_p(f)m_p(g)$.
7. If $f = f_i^{r_i}, g = g_i^{s_i}$, then:

$$I(p, f \cap g) = \sum_{i,j} r_i s_j I(p, f_i \cap g_j)$$

8. (Invariance of representation)

$$I(p, f \cap g) = I(p, f \cap (g + hf)) \quad \forall h \in k[x, y]$$

Theorem 1.7.14: Intersection Number

Let f, g be curves and $p \in \mathbb{A}_k^2$. Then there is a unique intersection number $I(p, f \cap g)$ defined given the above conditions where:

$$I(p, f \cap g) = \dim_k \left(\frac{\mathcal{O}_p(\mathbb{A}_k^2)}{(f, g)} \right)$$

Proof:

Let us first show that the above conditions give an algorithm for calculating $I(p, f \cap g)$ which shall *a-fortiori* show uniqueness. Let's say that $I(p, f \cap g)$ satisfies the 8 properties. By (1)-(2), we may assume $I(p, f \cap g)$ is finite, and by (3) we may assume $p = (0, 0)$. If $I(p, f \cap g) = 0$, then by (3) the curves do not intersect, so let's assume $0 < I(p, f \cap g) < \infty$. We shall show by (strong) induction that this number must be computable and unique: assume $I(p, f \cap g) = n$ and $I(p, a \cap b)$ can be computed whenever $I(p, a \cap b) < n$. Take:

$$f(x, 0), g(x, 0) \in k[x]$$

Assume these are of degree r, s respectively. By (5), we may assume $r \leq s$. Let us show this number is computed by lower-degree terms:

1. (case $r = 0$) If $r = 0$, then $y|f$ so $f = yh$. By (7):

$$I(p, f \cap g) = I(p, y \cap g) + I(p, h \cap g)$$

Then if $g(x, 0) = x^m(a_0 + a_1x + \dots)$ with $a_0 \neq 0$, then the above properties allow us to deduce:

$$I(p, y \cap g) = I(p, y \cap g(x, 0)) = I(p, y \cap x^m) = m$$

As $p \in G$, $m > 0$, so $I(p, h \cap g) < n$, allowing us to conclude via induction

2. (case $r > 0$) First, Without loss of generality let $f(x, 0), g(x, 0)$ be monic (multiply by a constant, which shall not change the intersection number). Let $h = g - x^{s-r}f$ so that $\deg(h(x, 0)) = t < s$. Then by (8):

$$I(p, f \cap g) = I(p, f \cap h)$$

Repeating this process, interchanging the order of f and h if $t < r$, we shall eventually reach a pair of curves a, b satisfying $r = 0$, and we get

$$I(p, f \cap g) = I(p, a \cap b)$$

giving us the desired result

Thus, we have an algorithmic procedure to compute the intersection number, which also gives its uniqueness.

Let us show that this number must equal the equation of the theorem:

$$I(p, f \cap g) = \dim_k \frac{\mathcal{O}_p(\mathbb{A}_k^2)}{(f, g)}$$

and satisfies the 8 properties. By the above result we shall both get a method of computation and uniqueness. First, certainly $I(p, f \cap g)$ depends only on the ideal, hence property (3), (5), and (8) are satisfied, (2) is immediate, and (4) is satisfied as affine isomorphisms certainly preserve local ring isomorphisms (in particular, maximal ideals map into maximal ideals). Thus, we may without loss of generality assume $p = (0, 0)$ and all components of f, g pass through p . As we may assume $p = (0, 0)$, we shall simplify notation and write $\mathcal{O} := \mathcal{O}_p(\mathbb{A}_k^2)$. What's left to show is (1), (6), and (7).

For (1), if f, g have no common components, $V((f, g))$ contains finitely many points and hence by corollary 1.7.11 $I(p, f \cap g)$ is finite. On the other hand, if $h \subseteq f \cap g$ is a common component, then $(f, g) \subseteq (h)$. Hence, there is a natural surjective homomorphism

$$\frac{\mathcal{O}}{(f, g)} \twoheadrightarrow \frac{\mathcal{O}}{(h)}$$

Implying $I(p, f \cap g) \geq \dim_k \mathcal{O}/(h)$. But $\mathcal{O}/(h) \cong \mathcal{O}_p(h) \supseteq k[V(h)]$, where $k[V(h)]$ is infinite dimensional, hence we have $I(p, f \cap g) = \infty$, showing (1).

For (7), by induction and symmetry of intersection it suffices to show that

$$I(p, f \cap gh) = I(p, f \cap g) + I(p, f \cap h)$$

By (1), we may assume f, g, h have no common components. Let $\varphi : \mathcal{O}/(f, gh) \rightarrow \mathcal{O}/(f, g)$ be the natural surjective homomorphism and $\psi : \mathcal{O}/(f, h) \rightarrow \mathcal{O}/(f, gh)$ be the k -linear map $\varphi(\bar{x}) = \bar{G}x$. As dimension is additive (exercise ref:HERE), it is enough to show that:

$$0 \rightarrow \frac{\mathcal{O}}{(f, h)} \xrightarrow{\psi} \frac{\mathcal{O}}{(f, gh)} \xrightarrow{\varphi} \frac{\mathcal{O}}{(f, g)} \rightarrow 0$$

is exact. φ is certainly surjective, and it is almost immediate that $\text{im } \psi = \ker \varphi$. Hence, let's show injectivity of ψ . If $\psi(\bar{z}) = \bar{0} = \bar{G}z$, then $Gz = uf + vgh$ ($u, v \in \mathcal{O}$). Then we take advantage of invertibility in the local ring: choose $s \in k[x, y]$ where $s(p) \neq 0$. Let:

$$A = su \quad B = sv \quad C = sz$$

Then $g(C - Bh) = Af$, and as f, g have no common factors f must divide $C - Bh$, hence $C - Bh = Df$ for some D . Then:

$$z = \frac{B}{s}h + \frac{D}{s}f \quad \text{i.e.} \quad \bar{z} = \bar{0}$$

showing injectivity.

Finally, let us show (6): $I(p, f \cap g) \geq m_p(f)m_p(g)$ with equality if the intersection is transversal. Let $m = m_p(f)$, $n = m_p(g)$, and $I = (x, y) \subseteq k[x, y]$. Consider the following diagram with linear transformation as morphisms:

$$\begin{array}{ccccccc} \frac{k[x, y]}{I^n} \times \frac{k[x, y]}{I^m} & \xrightarrow{\psi} & \frac{k[x, y]}{I^{n+m}} & \xrightarrow{\varphi} & \frac{k[x, y]}{(I^{n+m}, f, g)} & \longrightarrow & 0 \\ & & & & \downarrow \alpha & & \\ \frac{\mathcal{O}}{(f, g)} & \xrightarrow{\pi} & \frac{\mathcal{O}}{(I^{n+m}, f, g)} & \longrightarrow & 0 & & \end{array}$$

where φ, π, α are the natural maps and $\psi(\bar{a}, \bar{b}) = \overline{af + bg}$. Now, the top row is exact (note we need not ψ to be injective), and α is an isomorphism as the dimension matches by corollary 1.7.11. Thus:

$$\dim_k \frac{k[x, y]}{I^n} + \dim_k \frac{k[x, y]}{I^m} \geq \dim_k \ker \varphi$$

where there is equality iff ψ is injective and

$$\dim_k \frac{k[x, y]}{(I^{m+n}, f, g)} = \dim_k \frac{k[x, y]}{I^{m+n}} - \dim_k \ker \varphi$$

Putting all these results together:

$$\begin{aligned} I(p, f \cap g) &= \dim_k \frac{\mathcal{O}}{(f, g)} \\ &= \dim_k \frac{\mathcal{O}}{(I^{n+m}, f, g)} \\ &= \dim_k \frac{k[x, y]}{(I^{n+m} f, g)} \\ &\geq \dim_k \frac{k[x, y]}{I^{n+m}} - \dim_k \frac{k[x, y]}{I^n} - \dim_k \frac{k[x, y]}{I^m} \\ &= mn \end{aligned}$$

Hence $I(p, f \cap g) \geq mn$. We get equality if the two respective inequalities are equalities, namely if π is an isomorphism (implying $I^{n+m} \subseteq (f, g)\mathcal{O}$) and ψ is injective. Both of these properties shall be the consequence of the following lemma (lemma 1.7.15), which completes the proof.

Lemma 1.7.15: Technical Lemma - Intersection and Tangents

Let the variable be as in theorem 1.7.14. Then

1. If f, g have no common tangents at p , then $I^t \subseteq (f, g)\mathcal{O}$ where $t \geq m + n - 1$
2. ψ is injective if and only if f, g have no common tangents at p (i.e. if they intersect transversally and no tangents of f (resp. g) are the same).

Proof:

1. Let $L_1, \dots, L_m$ and $M_1, \dots, M_n$ be the tangent lines of f and g at p respectively: these are all distinct by assumption. Then these can be used to form a basis for I^t . To define this basis, let $L_i = L_m$ for $i > m$ and $M_j = M_n$ for $j > n$, and take

$$A_{ij} = L_1 L_2 \cdots L_i M_1 M_2 \cdots M_j$$

where A_{i0}, A_{0j} both signify omitting any L or M tangents respectively, and $A_{00} = 1$. Then the set $\{A_{ij} \mid i + j = t\}$ forms a k -basis for I^t (how does the fact f, g have no common tangents come into play here?), and hence it suffices to show that $A_{ij} \in (f, g)\mathcal{O}$ for all $i + j \geq m + n - 1$. Since $i + j \geq m + n - 1$, it must be that either $i \geq m$ or $j \geq n$; say $i \geq m$. Then we can write:

$$A_{ij} = A_{m0} B$$

where $\deg B = i + j - m$. Since A_{m0} contains all the tangents of f , we can write:

$$f = A_{m0} + f'$$

where f' contains all the terms of degree $\geq m + 1$. Multiplying both sides of the above equation by B and re-arranging, we get:

$$A_{ij} = Bf - Bf'$$

where each term of Bf' is of degree $\geq i + j + 1$. As $f \in (f, g)\mathcal{O}$, $Bf \in (f, g)\mathcal{O}$, hence if we show that all forms of degree $i + j + 1$ are in $(f, g)\mathcal{O}$, we have $-Bf' \in (f, g)\mathcal{O}$, which shall give $A_{ij} \in (f, g)\mathcal{O}$, and complete the proof.

To that end, it is enough to show that $I^t \in (f, g)\mathcal{O}$ for sufficiently large t . Let

$$V((f, g)) = \{p, q_1, \dots, q_s\}$$

Then by exercise ref:HERE, choose a polynomial h such that $h(q_i) = 0$ and $h(p) \neq 0$. Then certainly:

$$xh, yh \in I(V((f, g))) \quad \text{thus} \quad (xh)^N, (yh)^N \in (f, g)$$

Now, as h^N is a unit in $\mathcal{O}$, x^N, y^N are in $(f, g)\mathcal{O}$, and so $I^{2N} \subseteq (f, g)\mathcal{O}$, and so for $t \geq 2N$ we have $I^t \subseteq I^{2N} \subseteq (f, g)\mathcal{O}$, giving the desired result.

2. Let

$$\psi(\overline{A}, \overline{B}) = \overline{Af + Bg} = 0$$

and assume all tangents are distinct. We will show that $(\overline{A}, \overline{B}) = 0$. Note that $Af + Bg$ consists entirely of degree $\geq n + m$. Decompose A, B like so:

$$A = A_r + \text{higher terms} \quad B = B_s + \text{higher terms}$$

where r, s are the lowest degrees of A, B respectively. Then we can write:

$$Af + Bg = A_r f_m + B_s g_n + \text{higher terms}$$

where f_m, g_n are the lowest degree terms of f, g respectively. For the sake of contradiction, let's assume $r < m$ and $s < n$ which would imply $A_r f_m + B_s g_n = 0$ as we assumed that the image of ψ can only have terms of degree $\geq m + n$ and so these must cancel out, namely $A_r f_m = -B_s g_n$. This implies $m + s = n + r$ (or else they wouldn't be equal up to a constant). But now, f_m, g_n have no common factors, so $f_m | B_s$ and $g_n | A_r$, implying $s \geq m$ and $r \geq n$. But then, as the domain of ψ is $\frac{k[x, y]}{I^n} \times \frac{k[x, y]}{I^m}$, this implies $(\overline{A}, \overline{B}) = (\overline{0}, \overline{0})$.

Conversely, If f, g have a common tangent, say L , then write $f_m = Lf'_{m-1}$, $g_n = Lg'_{n-1}$, so

$$\psi(\overline{g'_{n-1}}, -\overline{f'_{m-1}}) = 0$$

showing injectivity fails, completing the proof.

From this, we get the rest of the proof of theorem 1.7.14.

Example 1.7.16: Intersection Number

Let

$$E = (x^2 + y^2)^2 + 3x^2y - y^3 = x^4 + 2x^2y^2 + 3x^2y + y^4 - y^3$$

$$F = (x^2 + y^2)^3 - 4x^2y^2 = x^6 + 3x^4y^2 + 3x^2y^4 - 4x^2y^2 + y^6$$

and $p = (0, 0)$. Their graph in $\mathbb{A}_{\mathbb{R}}^2$ is:

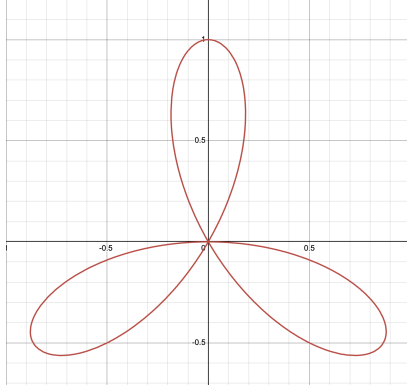


Figure 1.9: Three-Leaf Curve

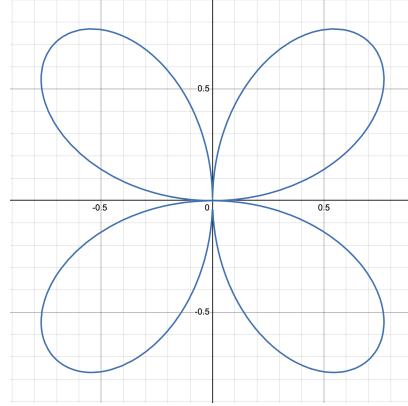


Figure 1.10: Four-Leaf Curve

By (8), we may replace F with $F - (x^2 + y^2)E = y((x^2 + y^2)(y^2 - 3x^2) - 4x^2y) = yG$ for some G . We thus reduce to:

$$I(p, E \cap F) = I(p, E \cap yG) \stackrel{(7)}{=} I(p, E \cap y) + I(p, E \cap G)$$

To compute $I(p, E \cap G)$ we shall eliminate the x -term: replace $G = (x^2 + y^2)(y^2 - 3x^2) - 4x^2y = -3x^4 - 2x^2y^2 - 4x^2y + y^4$ by $G + 3E$ which gives:

$$G + 3E = y(5x^2 - 3y^2 + 4x^3 + 4x^2y) = yH$$

We then get:

$$I(p, E \cap F) \stackrel{(7)}{=} 2I(p, E \cap y) + I(p, E \cap H)$$

Now, By applying(8) to $E \cap y$, we can eliminate all the y terms and get:

$$I(p, E \cap y) \stackrel{(8)}{=} I(p, x^4 \cap y) \stackrel{(7)}{=} 4I(p, x \cap y) \stackrel{(6)}{=} 4 \cdot 1$$

Then for $I(p, E \cap H)$, as they do not intersect transversally (the point p is not simple) and they share no tangent lines we get:

$$I(p, E \cap H) \stackrel{(6)}{=} m_p(E)m_p(H) = 3 \cdot 2 = 6$$

Thus, we get:

$$I(p, E \cap F) = 2 \cdot 4 + 6 = 14$$

Corollary 1.7.17: Intersection Point - Simple Points and Dimension

Let the variable be as in theorem 1.7.14. Then

9. If p is a simple point of f , then

$$I(p, f \cap g) = \text{ord}_p^f(g)$$

10. If f, g have no common components so that $f \cap g$ is finite, then:

$$\sum_p I(p, f \cap g) = \dim_k \frac{k[x, y]}{(f, g)}$$

Proof:

1. Without loss of generality assume f is irreducible. Then it is easy to see that:

$$\text{ord}_p^f(g) = \dim_k \frac{\mathcal{O}_p(f)}{(\bar{g})}$$

Show that $\mathcal{O}_p(f)/(\bar{g}) \cong \mathcal{O}_p(\mathbb{A}_k^2)/(f, g)$ to conclude the result

2. Corollary 1.7.11.

1.7.2 Bézout Theorem

We now move to the characterizing property of projective space. When working with curves, we shall get that they *always* intersect, and their intersection number is given by the degrees of the respective curve. Note that this intersection property is specific to curves, not all projective varieties need to intersect non-trivially (ex. $\mathcal{Z}(x, y)$ and $\mathcal{Z}(z)$, i.e. The point at infinity and the line $z = 0$, don't intersect). +

Theorem 1.7.18: Bézout's Theorem

Let $C = Z_{\mathbb{P}}(f)$ and $D = Z_{\mathbb{P}}(g)$ be projective plane curves that do not have a common component. Then, if C has degree m and D has degree n , C and D intersect in mn points counting multiplicity.

Proof:

Let f, g be the projective curves with no common components. Then $f \cap g$ is finite, and so by a change of coordinates we may assume none of the points of $f \cap g$ lie on the line at infinity $z = 0$; let f_* and g_* represent the de-homogenization of these curves. Then by property 10 of the intersection number:

$$\sum_p I(p, f \cap g) = \sum_p I(p, f_* \cap g_*) = \dim_k \frac{k[x, y]}{(f_*, g_*)}$$

To diminish notational clutter, let:

$$\Gamma_* = \frac{k[x, y]}{(f_*, g_*)} \quad \Gamma = \frac{k[x, y, z]}{(f, g)} \quad R = k[x, y, z]$$

Let Γ_d and R_d be the vector space with basis given by the monomial of degree d in Γ (resp. R).

If we can show:

$$\dim \Gamma_* \stackrel{1}{=} \dim \Gamma_d \stackrel{2}{=} mn$$

for appropriate d , we are done.

Let us start with $\stackrel{2}{=}$. Let $d \geq m + n$. We shall use an exact sequence argument. define:

$$\begin{aligned} \psi : R &\rightarrow R \times R & \psi(c) &= (gc, -fc) \\ \varphi : R \times R &\rightarrow R & \varphi(a, b) &= af + bg \\ \pi : R &\rightarrow \Gamma & \pi(d) &= \bar{d} \end{aligned}$$

As f, g have no common components, we get an exact sequence:

$$0 \rightarrow R \xrightarrow{\psi} R \times R \xrightarrow{\varphi} R \xrightarrow{\pi} \Gamma \rightarrow 0$$

These maps are well-defined when restricting to the right degrees. If we restrict to:

$$0 \rightarrow R_{d-m-n} \xrightarrow{\psi} R_{d-m} \times R_{d-n} \xrightarrow{\varphi} R_d \xrightarrow{\pi} \Gamma \rightarrow 0$$

As $\dim_k R_d = \frac{(d+1)(d+2)}{2}$, we can use the same additive trick to see that for $d \geq m + n$ that:

$$\dim_k \Gamma_d = mn$$

giving us $\stackrel{2}{=}$.

For $\stackrel{1}{=}$, we shall first show that for $d \geq m + n$, there is an isomorphism from Γ_d to Γ_{d+1} . Define:

$$\alpha : \Gamma \rightarrow \Gamma \quad \alpha(\bar{h}) = \overline{zh}$$

This map is injective. Let $\overline{zh} = \bar{0}$ so that $zh = af + bg$. Then if $h = a'f + b'g$ for some $a', b' \in R$ we are done for then $\bar{h} = \bar{0}$. First, let $h_0 = h(x, y, 0)$ denote the polynomial which evaluates the z -term to 0. As f, g, z have no common zeros, f_0, g_0 must be relatively prime in $k[x, y]$. Now as $zh = af + bg, 0 = a_0f_0 + b_0g_0$ so $-a_0f_0 = b_0g_0$ and so for some $c \in k[x, y]$

$$b_0 = f_0c \quad a_0 = -g_0c$$

Define $a_1 = a + cg, b_1 = b + cg$. Then $(a_1)_0 = (b_1)_0 = 0$, so $a_1 = za'$ and $b_1 = zb'$ for some a', b' , but then $h = a'f + b'g$, so $\bar{h} = \bar{0}$. Using this, we can show that $\alpha : \Gamma_d \xrightarrow{\sim} \Gamma_{d+1}$ is an isomorphism, since it is an injective map between two vector-spaces of the same dimension (as $d \geq m + n$).

Now, keeping $d \geq m + n$, choose $\beta_1, \dots, \beta_{mn} \in R_d$ such that their residues form a basis for Γ_d . Let $(\beta_i)_* = \beta_i(x, y, 1) \in k[x, y]$ be the de-homogenization, and $(\beta_i)_i = b_i$ be the residue in Γ_* . I claim that $b_1, \dots, b_{mn}$ is a basis for Γ_* . First, as α is an isomorphism we have that

$$z^r b_1, \dots, z^r b_{nm}$$

is a basis for Γ_{d+r} . Let's use this to show that the b_i generate Γ_* . Let $\bar{h} \in \Gamma_*$. Then $z^N({}^h h)$ is a form of degree $d+r$, so:

$$z^N({}^h h) = \sum_{i=1}^{mn} \lambda_i z^r \beta_i + bf + cg$$

for some constants $\lambda_i \in k$ and $b, c \in k[x, y, z]$. Then dehomogenizing we get:

$$h = (z^N({}^h h))_* = \sum_i \lambda(\beta_i)_* + b_* f_* + c_* g_*$$

and so $\bar{h} = \sum_i \lambda_i b_i$, showing they generate. To show they are linearly independent, assume:

$$\sum_i \lambda_i b_i = 0 \quad \Rightarrow \quad \sum_i \lambda_i (\beta_i)_* = bf_* + cg_*$$

Thus, for some natural numbers r, s, t :

$$z^r \sum_i \lambda_i \beta_i = z^s({}^h b)f + z^t({}^h c)g$$

which implies $\sum_i \lambda_i \overline{z^r \beta_i} = \bar{0} \in \Gamma_{d+r}$, and as the forms $\overline{z^r \beta_i}$ form a basis for Γ_{d+r} it must be that $\lambda_i = 0$ for all i . But then we have a basis, and hence the dimensions match, and so:

$$\dim_k \Gamma_* = mn$$

which gives the desired result, completing the proof.

Corollary 1.7.19: Degree and Multiplicity

If f, g have no common factors, then:

$$\sum_p m_p(f)m_p(g) \leq \deg(f)\deg(g)$$

Proof:

immediate

Corollary 1.7.20: Intersection and Simple Points

If f, g meet at mn distinct points (where $m = \deg(f)$ and $n = \deg(g)$), then these are all simple points of f and g

Proof:

immediate

Corollary 1.7.21: Intersection and Common Component

If two curves f, g of degree m, n respectively have more than mn points in common, they have a common component.

Proof:

immediate.

Finally, we can use the following for some useful classification of curves of low degree:

Corollary 1.7.22: Bounding Multiplicity

Let f be an irreducible curve of degree n . Then:

$$\sum_p \frac{m_p(f)(m_p(f) - 1)}{2} \leq \frac{n(n - 1)}{2}$$

Proof:

By applying exercise ref:HERE, define:

$$r = \frac{(n - 1)(n - 1 + 3)}{2} - \sum_p \frac{(m_p - 1)m_p}{2} \geq \frac{(n - 1)n}{2} - \sum_p \frac{(m_p - 1)m_p}{2} \geq 0$$

where p are the points of multiplicity with multiplicity p . As r is positive, we may choose choose simple points $Q_1, \dots, Q_r \in F$. By exercise ref:HEREof, choose a curve G of degree $n - 1$ such that $m_P(G) \geq m_P - 1$ for all P , and $m_{Q_i}(G) \geq 1$.

Now, since f is irreducible and has no common components with g , by corollary 1.7.19, we get:

$$n(n - 1) \geq \sum m_P(m_P - 1) + r$$

By substituting the value for r , we get the desired inequality, completing the proof.

Note that this is a tight bound: the curve $x^n + y^{n-1}z$ has has multiplicity $n - 1$ at $[0 : 0 : 1]$.

1.7.3 Divisors and Groups on Cubic Curves

Definition 1.7.23: Zero-cycle

A formal sum $\sum_p n_p p$ where $p \in \mathbb{P}_k^2$ where all but finitely many $n_p = 0$ is called a *zero-cycle*.

Defining summing formally gives an abelian group. The degree of a zero-cycle is the sum $\sum_p n_p$. Define an order $\sum_p n_p p \leq \sum_p m_p p$ if $n_p \leq m_p$ for each p . Let f, g be two projective curves with no common components of degree n, m respectively. Then we can define the following multiplication

called the *intersection cycle*:

$$f \cdot g = \sum_{p \in \mathbb{P}_k^2} I(p; f \cap g) p$$

Note that:

1. $f \cdot g = g \cdot f$
2. $f \cdot gh = c \cdot g + f \cdot h$
3. If A is a form of degree $\deg(g) - \deg(a)$, then $f \cdot (g + af) = f \cdot g$

One of the most fundamental questions we can ask is if every zero-cycle can be realised as an intersection cycle. The reader should check in exercise ref:HERE that this is indeed the case. This question becomes more non-trivial when working over points in a projective variety of various rather than $\mathbb{P}_k^2$, and shall be the beginning of the study of *divisors*. Another insight that the name suggests is that these give interesting cohomological questions. In this section, we shall explore one aspect of it: let's say

$$h \cdot f \leq h \cdot g$$

Then when is there a curve b such that is the difference:

$$b \cdot f = h \cdot f - h \cdot g$$

From the enumerated points above, we can deduce that b must be of degree $\deg(h) - \deg(g)$, and that it suffices to find forms a, b such that

$$h = af + bg$$

since then $h \cdot f = bg \cdot f = b \cdot f + g \cdot f$. The following gives a condition for this to be the case:

Theorem 1.7.24: Max Noether's Fundamental Theorem

Let f, g, h be projective plane curves where f, g have no common components. Then the equation

$$h = af + bg$$

with forms a, b of degree $\deg(h) - \deg(f)$ and $\deg(h) - \deg(g)$ respectively exists iff for each $p \in f \cap g$:

$$h_* \in (f_*, g_*) \in \mathcal{O}_p(\mathbb{P}_k^2)$$

Notice that this links the global existence of an h to the local properties of the points; such results are often called *local-to-global conditions*.

Proof:

Certainly if $h = af + bg$, then $h_* = a_* f_* + b_* g_*$ at any p .

For the converse, without loss of generality assume $V(F, G, Z) = \emptyset$. As usual let $f_* = f(x, y, 1)$, $g_* = g(x, y, 1)$, and $h_* = h(x, y, 1)$. Noether's conditions say that the residue of h_* in $\mathcal{O}_P(\mathbb{P}_k^2)/(f_*, g_*)$ is zero for each $p \in f \cap g$. Then from lemma 1.7.10 the residue of h_* in $k[x, y]/(f_*, g_*)$ is zero, i.e., $h_* = \alpha f_* + \beta g_*$, $\alpha, \beta \in k[x, y]$. Then $z^r h = af + bg$ for some r, a, b

Now, in the proof of Step 2 of Bézout's Theorem we saw that multiplication by z on $k[x, y, z]/(f, g)$ is one-to-one, so $h = a'f + b'g$ for some a', b' . If $a' = \sum a'_i$, $b' = \sum b'_i$, a'_i, b'_i forms of degree i ,

then $h = a'_s f + b'_t g$, $s = \deg(h) - \deg(f)$, $t = \deg(h) - \deg(g)$, completing the proof.

Proposition 1.7.25: Noether's Criterion

Let f, g, h be projective curves and $p \in f \cap g$. Then the following are sufficient conditions for Noether's criterion:

1. f, g meet transversally at p and $p \in h$
2. p is a simple point of f and $I(p; h \cap f) \geq I(p; g \cap f)$
3. f, g have distinct tangent lines at P , and $m_p(h) \geq m_p(f) + m_p(g) - 1$

Proof:

For (2), assume $i(p, h \cap f) \geq i(p; g \cap f)$. Then this implies $\text{ord}_p^f(h) \geq \text{ord}_p^f(g)$, so

$$\bar{h}_* \in (\bar{g}_*) \subseteq \mathcal{O}_p(f)$$

Then as $\mathcal{O}_p(f)/(\bar{g}_*) \cong \mathcal{O}_p(\mathbb{P}_k^2)/(f_*, g_*)$, we have that the residue of h_* in $\mathcal{O}_p(\mathbb{P}_k^2)/(f_*, g_*)$ is zero, showing Noether's condition.

For (3), so let $m_p(h_*) \geq m_p(f_*) + m_p(g_*) - 1$ and without loss of generality we may take $p = [0 : 0 : 1]$. Letting $(x, y) = I$, we have (as shown in the equivalent definition of intersection number):

$$h_* \in I^t \subseteq (f_*, g_*) \subseteq \mathcal{O}_p(\mathbb{P}_k^2) \quad t \geq m + n - 1$$

giving us the desired result.

Notice that (1) is a special case of (2) or (3).

Corollary 1.7.26: Sufficient Condition For Max Noether Theorem

If either:

1. f, g meet in $\deg(f) \deg(g)$ distinct points, and h passes through these points
2. all the points of $f \cap g$ are simple points of f with $h \cdot f \geq g \cdot f$

Then there is a curve b such that

$$b \cdot f = h \cdot f - g \cdot f$$

Proof:

combine theorem 1.7.24 and proposition 1.7.25.

Addition Law on Cubics

Let C be a nonsingular cubic curve. As it is nonsingular, any line L passing through two points $p, q \in C$ (where it is not necessarily the case that $p \neq q$ must satisfy by Bézout's Theorem:

$$L \cdot C = p + q + r$$

If we choose any $p, q \in C$, we have a unique r associated to it. Define: $\varphi : C \times C \rightarrow C$ given by

$$\varphi(p, q) = r$$

This function defines an abelian binary operation. It does not define an abelian group operation there would need to be an identity an inverse, and associativity would further have to be proven. For the identity, we can pick any $O \in C$, and define a new binary operation:

$$p \oplus q = \varphi(O, \varphi(p, q))$$

This is indeed a well-defined identity since:

$$O \oplus p = \varphi(O, \varphi(O, p)) = \varphi(O, q) = p = \varphi(O, q) = \varphi(O, \varphi(p, O)) = p \oplus O$$

We also have an inverse and associativity is well-defined, and that the choice of identity does not change the group-structure. We first require a lemma for associativity⁴⁰

Lemma 1.7.27: Cubic Point Counting

Let C be an irreducible cubic and C', C'' cubics. Suppose $C' \cdot C = \sum_{i=1}^9 P_i$, where the P_i are simple (not necessarily distinct) points on C , and $C'' \cdot C = \sum_{i=1}^8 P_i + Q$. Then

$$Q = P_9$$

That is, having 8 matching points determines the 9th.

Proof:

Let L be a line through P_9 that doesn't pass through Q ; $L \cdot C = P_9 + R + S$. Then $LC'' \cdot C = C' \cdot C + Q + R + S$, so there is a line L' such that $L' \cdot C = Q + R + S$. But then $L' = L$ and so $P_9 = Q$.

Lemma 1.7.28: Sum of Colinear Points

Let $P, Q, R \in C$ be colinear points. Then

$$P \oplus Q \oplus R = O$$

Proof:

compute this directly.

⁴⁰If you know Riemann-Roch's Theorem, you can use that instead

Proposition 1.7.29: Group Law For Cubics

With $\oplus$ defined as above on the cubic C over k . Then we have group structure that is independent of choice of identity point. Label this group $E_k(C)$

Proof:

The above proved the existence of an identity. For the inverse of P , take the point $\varphi(P, O)$, for

$$P \oplus \varphi(P, O) = \varphi(O, \varphi(P, \varphi(P, O))) = \varphi(O, O) \stackrel{!}{=} O$$

To show $\stackrel{!}{=}$, by lemma 1.7.28 we know the sum of three colinear point is o . If we let $Q = \varphi(O, O)$ then we must have:

$$O = O \oplus O \oplus Q = O \oplus (O \oplus Q) = O \oplus Q = Q$$

giving us $\varphi(O, O) = O$, hence $\varphi(P, O)$ is the inverse.

For associativity, suppose $P, Q, R \in C$. Let $L_1 \cdot C = P + Q + S'$, $M_1 \cdot C = O + S' + S$, $L_2 \cdot C = S + R + T'$.

Let $M_2 \cdot C = Q + R + U'$, $L_3 \cdot C = O + U' + U$, $M_3 \cdot C = P + U + T''$. Since $(P \oplus Q) \oplus R = \varphi(O, T')$, and $P \oplus (Q \oplus R) = \varphi(O, T'')$, it suffices to show that $T' = T''$. Let $C' = L_1 L_2 L_3$, $C'' = M_1 M_2 M_3$, and apply lemma 1.7.27.

Finally, let us show that for any choice of points $O, O' \in C$, which give groups $(C, O, \oplus)$ and $(C, O', \oplus')$ that are isomorphic. Indeed, let $Q = \varphi(O, O')$ and take $\alpha : (C, O, \oplus) \rightarrow (C, O', \oplus')$ given by:

$$\alpha(P) = \varphi(Q, P)$$

First,

$$\alpha(A \oplus B) = \varphi(Q, A \oplus B) = \varphi(Q, \varphi(O, \varphi(A, B)))$$

and

$$\alpha(A) \oplus' \alpha(B) = \varphi(Q, A) \oplus' \varphi(Q, B) = \varphi(O', \varphi(\varphi(Q, A), \varphi(Q, B)))$$

To show equality, we must do a bunch of co-linearity arguments. First, notice that if A, B, C are co-linear, then:

$$A \oplus B \oplus C = O$$

since $A \oplus B = \varphi(O, \varphi(A, B)) = \varphi(O, C) = -C$ so $-C \oplus C = O$. We shall use this extensively soon. For denote:

1. $D = \varphi(A, B)$ (the third point on the line through A and B)
2. $E = \varphi(O, D)$ (so $A \oplus B = E$)
3. $A' = \varphi(Q, A)$ (so $\alpha(A) = A'$)
4. $B' = \varphi(Q, B)$ (so $\alpha(B) = B'$)
5. $D' = \varphi(A', B')$ (the third point on the line through A' and B')
6. $E' = \varphi(O', D')$ (so $A' \oplus' B' = E'$)

Using this notation, to show the equality of both sides reduces to showing:

$$\varphi(Q, E) = E'$$

Using co-linearity, we get the following relationships:

1. $D = -A - B$
2. $E = -O - D = -O + A + B$
3. $A' = -Q - A$
4. $B' = -Q - B$
5. $D' = -A' - B' = -(-Q - A) - (-Q - B) = 2Q + A + B$
6. $E' = -O' - D' = -O' - (2Q + A + B)$

Now, as:

$$Q \oplus E \oplus \varphi(Q, E) = O$$

we get the following

$$\varphi(Q, E) = -Q - E \stackrel{1.}{=} -Q - (-O \oplus A \oplus B) = -Q \oplus O - A - B$$

If we show $E' = -O' - (2Q \oplus A \oplus B)$ equals the above, we are done. Indeed, first as $O \oplus O' \oplus Q = O$ we get $O' = -O - Q$. Substituting, we get:

$$E' = -O' - (2Q \oplus A \oplus B) = -(-O - Q) - (2Q \oplus A \oplus B) = O \oplus Q - 2Q - A - B = O - Q - A - B$$

But that is equality our equation, hence:

$$\alpha(a \oplus b) = \alpha(a) \oplus' \alpha(b)$$

as we sought to show.

Exercise 1.7.2

1. Show that a nonsingular projective curve must be irreducible. Show this is not necessarily the case for nonsingular affine irreducible curves.

1.8 Blowing Up

Let C be a curve. We shall show how to “resolve singularities” by a process known as a *blow up*.

Build up

Definition 1.8.1: Local Ring of Field

Let K/k be a field extension. Then A is a *local ring of K* if $\text{Frac}(A) = K$ and $k \subseteq A$. If A is a DVR, we shall say that A is a *discrete valuation ring of K* .

The prototypical example of local fields over a ring are the stalks $\mathcal{O}_p(V)$ of $\kappa(V)$ for a variety V .

Theorem 1.8.2: Dominating DVR

Let C be a projective curve, $K = \kappa(C)$, and let L/K be a field extension. Let R be a DVR of L containing k . Assume $K \not\subseteq R$. Then there is a unique point $p \in C$ such that R dominates $\mathcal{O}_p(C)$.

Proof:

First, such a point is unique; if R dominates $\mathcal{O}_p(C)$ and $\mathcal{O}_q(C)$, then choose $f \in \mathfrak{m}_p(C)$ so $1/f \in \mathcal{O}_q(C)$. Then $\text{ord}(f) > 0$ and $\text{ord}(1/f) \geq 0$ – a contradiction.

Thus, let's show it exists. Let $C \subseteq \mathbb{P}_k^n$ be a projective curve (a closed one-dimensional subvariety). Without loss of generality, we may take C to not be strictly contained in any x_i -hyperplane so that $C \cap U_i \neq \emptyset$. Then in each $k_{\mathbb{P}}[C] = k[x_1, \dots, x_n + 1]/I(C) \cong k[u_1, \dots, u_{n+1}]$. Let

$$N = \max_{i,j} (x_i/x_j)$$

As the max is achieved, without loss of generality assume $\text{ord}(x_j/x_{n+1}) = N$ for some j . Then for all i :

$$\text{ord}\left(\frac{u_i}{u_{n+1}}\right) = \text{ord}\left(\frac{(u_j/u_{n+1})}{(u_i/u_j)}\right) = N - \text{ord}(u_j/u_i) \geq 0$$

If C_* is the affine curve corresponding to $C \cap U_{n+1}$, then $\Gamma(C_*)$ can be defined with

$$k\left[\frac{u_1}{u_{n+1}}, \dots, \frac{u_n}{u_{n+1}}\right]$$

hence, $\Gamma(C_*) \subseteq R$.

Now, if $\mathfrak{m}$ is the maximal ideal of R , let $J = \mathfrak{m} \cap \Gamma(C_*)$. Then J is a prime ideal, so J corresponds to a closed subvariety $W \subseteq C_*$. If $W = C_*$, $J = 0$, and every nonzero element of $\Gamma(C_*)$ is a unit of R . But then $K \subseteq R$, contradiction our assumption. So $W = \{p\}$ must be a point. Then R certainly dominates $\mathcal{O}_p(C_*)$, and hence dominates $\mathcal{O}_p(C)$, completing the proof.

Corollary 1.8.3: Rational Map of Curves

Let f be a rational map from C' to a projective curve C . Then the domain of f must include every simple point of C' . If C' is nonsingular, f is a morphism.

Proof:

If F is not dominating, it is constant (as shown in corollary 1.6.41), hence its domain must be all of C . Thus, assume $F^\#$ embeds $K = \kappa(C)$ as subfield of $L = \kappa(C')$. Let P be a simple point of C' , $R = \mathcal{O}_P(C')$. Then it is enough to show that $K \not\subseteq R$. Suppose that $K \subseteq R \subseteq L$. The L is a finite algebraic extension of K , so R is in fact a field. But DVR's are by definition *not* a field.

Corollary 1.8.4: Function Field and Morphism Correspondence

Let C be a projective curve, C' a nonsingular curve. Then there is a natural one-to-one correspondence between dominant morphisms $f : C' \rightarrow C$ and homomorphism $f^\# : \kappa(C) \rightarrow \kappa(C')$.

Proof:

immediate

Corollary 1.8.5: Characterizing Nonsingular Projective Curves

Two nonsingular projective curves are isomorphic if and only if their function fields are isomorphic if and only if they are birationally equivalent.

Proof:

immediate

Corollary 1.8.6: Function Field associated to Curve

Let C be a nonsingular projective curve over k , $K = \kappa(C)$. Then there is a natural one-to-one correspondence between the points of C and the discrete valuation rings of K , namely

$$p \in C \text{ correspond to } k \subseteq \mathcal{O}_p(C) \subseteq K$$

Proof:

Certainly $\mathcal{O}_p(C)$ is a DVR in K . On the other hand if R is any such DVR, then we saw that R dominates a unique $\mathcal{O}_p(C)$, and as these are both DVR's of K , $R = \mathcal{O}_p(C)$, as we sought to show.

From this, notice that we can create the Zariski topology on the collection of DVRs: Let X be the set of all DVRs of $\kappa(C)/k$. Give the cofinite topology on X . Then the correspondence $p \mapsto \mathcal{O}_p(C)$ is a homeomorphism! We shall use this observation when looking at “smooth $\mathbb{Z}$ -curves” in section ?? which shall show how to interpret Dedekind domains as smooth curves!

Blow up

Here

Exercise 1.8.1

1. Let X be a curve. Then $k[X] = \bigcap_{p \in X} \mathcal{O}_p(X)$.
2. Let $\mathcal{Z}(I) = \{p_1, \dots, p_n\}$ contains finitely many points. Then if $\mathcal{O}_{p_i}(\mathbb{A}_k^n) = \mathcal{O}_i$, Show that $k[X]/I \cong \prod_{p \in \mathcal{Z}(I)} \mathcal{O}_i/I\mathcal{O}_i$.
3. Prove that a curve can have at most finitely many multiple points.
4. Let f be a curve of degree n and assume $\frac{\partial f}{\partial x} \neq 0$. Prove that $\sum_p m_p(f)(m_p(f) - 1) \leq n(n - 1)$ where p goes over all multiple points
5. Show that line and irreducible conics must be nonsingular (either directly or via one of the corollaries).
6. Let C, C' be cubics, $C' \cdot C = \sum_i^9 p_i$. Suppose Q is a conic where $Q \cdot C = \sum_i^t \sum_i p_i$. Assume $p_1, \dots, p_6$ are simple points on C . Then p_7, p_8, p_9 lie on a straight line
7. Using Max Noether's Theorem, we can deduce some classical geometrical results:
 - a) (Pascals Theorem) If a hexagon is inscribed in an irreducible conic, then the opposite sides meet in co linear points
 - b) (Pappus Theorem) Let L_1, L_2 be two lines with $p_1, p_2, p_3 \in L_1$ and $q_1, q_2, q_3 \in L_2$ (with non of these points in $L_1 \cap L_2$). Let L_{ij} be the line line between p_i and q_j . For each i, j, k , with $\{i, j, k\} = \{1, 2, 3\}$, let $R_k = L_{ij} \cdot L_{ji}$. Then $R - 1, R_2, R_2$ are co-linear.

1.9 *Prelude to Algebraic Geometry

We have now created a dictionary translating properties of algebraic sets to coordinate rings. In particular, we have:

$$\{\text{Affine Varieties}\} \xrightleftharpoons[\varphi]{V(S)} \left\{ \begin{array}{c} \text{Finitely Generated,} \\ \text{Reduced rings} \\ \text{over algebraically closed fields} \end{array} \right\} \quad (1.11)$$

This allows for a certain subset of commutative rings to be interpreted as geometric objects, namely zero sets of polynomials. The question arises then if this duality can be extended to general commutative rings? In particular, does there exist some topological space such that:

$$\left\{ ??? \right\} \xrightleftharpoons[V(S)]{R[x]/V} \left\{ \text{Commutative Ring} \right\}$$

Note the necessity of commutativity, since many of the theorem on prime ideals such as ref:HERE, ref:HERE, and ref:HERE rely on commutativity⁴¹.

In this section, we build-up the object that should should be put on the other side, known as an *affine scheme*. An affine scheme will be defined in such a way that if we restricted to finitely generated reduced rings over an algebraically closed field, the affine scheme be an affine variety, making the following diagram the correct generalization of equation (1.11)

⁴¹The theory of “non-commutative geometry” using non-commutative rings can be explored in *Spectral Theory* in a class on functional analysis (see EYT NKA functional analysis for more details)

$$\left\{ \text{Affine Scheme} \right\} \begin{array}{c} \xrightarrow{R[x]/V} \\ \xleftarrow{V(S)} \end{array} \left\{ \text{Commutative Ring} \right\} \quad (1.12)$$

Why do this? This abstraction is not simply for abstractions-sake. It is because the dictionary between algebra and geometry that has been built up to is incredibly useful, and will fix some of the problems of algebraic varieties, for example:

1. There are many algebras over more general rings that have geometric interpretations. For a rather trivial but illustrative example, the coordinate ring of a line whose points are in some arbitrary ring k (the coordinate ring being $k[x]$) currently does not have a good representation since the Nullstellensatz limits to Algebraically closed fields. Another example that may at first seem un-geometric but turns out to have many strong geometric properties are rings such as $\mathbb{Z}$, $\mathbb{Z}[i]$ (the Gaussian integers), and more generally *algebraic number rings* (explored in the chapter ??). Mathematicians have found that these corresponds naturally to curves in a plane (as we shall see in chapter ??, section ??).
2. As we saw in section 1.4, localization gives us powerful tools to explore the behavior of a variety at a point. For example, if we take the coordinate ring of an affine line, $k[x]$, takes the closure $k(x)$, or study $k[x]_{(x)}$, we love the fact that we are finitely generated. However, as we saw these certainly still have interpretations as algebraic correspondences to geometric objects.
3. Finally, varieties loose multiplicity information. Saw we have the variety over $\mathbb{C}$ defined by the equations

$$y = x^2 \quad y = 0$$

then if we take their intersection, we get a point $(0,0)$. The corresponding affine variety is given by taking the union of quotient of the two coordinate rings, namely:

$$\frac{k[x, y]}{(y, x^2 - y)} = \frac{k[x]}{(x^2)}$$

Usually, we would say omit the x^2 term and just use $k[x]/(x)$ for the coordinate ring, however this eliminate the multiplicity information, namely that the variety $y = x^2$ “doubly intersects” the variety $y = 0$ the variety $y = 0$. The ring $k[x]/(x^2)$ keeps track that we have a “double-point”, while $k[x]/(x)$ does not. This is achieved by the fact that the ring has nilpotent elements.

In this section, we shall build up the definition of an affine scheme. We shall then for the first time properly venture into the realm of geometry and define a new object known as an *scheme* that shall will mirror the properties of a (smooth) manifold. (TBD)

After having presented schemes, we shall give a few notable examples, but the full exploration of scheme theory and algebraic geometry deserves a book on its own, and so they shall be relegated to future readings after that point. There is still a point in introducing schemes since the concept shall come a few times later in this book (notably in number theory).

Topology of Affine Schemes

(Mention also how Schemes take away the need to have an ambient affine or projective space, that is where the)

A natural first place to start is by interpreting the set of maximal ideals of R as the “points” on which we can define a topology (analogous to the Zariski topology). If this were the case, we would want that the functor translating a ring homomorphism $\varphi : R \rightarrow S$ has an associated continuous function mapping the maximal ideal $M \subseteq S$ to a maximal ideal $\varphi^{-1}(M)$ (recall that regular function corresponds contra-variantly to k -algebra homomorphisms). The problem is that the pre-image of maximal ideals need not be maximal (ex. take $\mathbb{Z} \hookrightarrow \mathbb{Q}$. Then (0) is maximal in $\mathbb{Q}$, but certainly not maximal in $\mathbb{Z}$). Fortunately, if we loosen our requirement a bit and consider prime ideals, then it is indeed the case that $\varphi^{-1}(\mathfrak{p})$ is prime. This inspires the following definition:

Definition 1.9.1: Spectrum Of A Ring

Let R be a commutative ring. Then the *spectrum* or *prime spectrum* of the ring R is the collection of all prime ideals of R , and is denoted $\text{Spec}(R)$:

$$\text{Spec}(R) := \{\mathfrak{p} \subseteq R \mid \mathfrak{p} \text{ is a prime ideal of } R\}$$

Similarly, the collection of all maximal ideals is called the *maximal spectrum* of R and is denoted $\text{mSpec}(R)$.

Example 1.9.2: Spec Of Rings

1. For any field k , $\text{Spec}(k) = \text{mSpec}(k) = \{(0)\}$
2. Let $R = \mathbb{Z}$. Then $\text{Spec}(\mathbb{Z}) = \{(p) \mid p \text{ is prime}\}$ and $\text{mSpec}(\mathbb{Z}) = \text{Spec}(\mathbb{Z}) \setminus \{(0)\}$
3. $\text{Spec}(\mathbb{Z}[x])$ consists of (really cool explanation by Richard Borcherd at the end of his video on spectra of a ring!!)

Proposition 1.9.3: Spec(−) Is Functorial

Let R, S be rings and $\varphi : R \rightarrow S$ a ring homomorphism. Then $\text{Spec}(-)$ induces

$$\varphi^* : \text{Spec}(S) \rightarrow \text{Spec}(R) \quad p \mapsto \varphi^{-1}(p)$$

Making it a contravariant functor $\text{Spec}(-) : \mathbf{Ring} \rightarrow \mathbf{Set}$

Soon, we shall see we may induce a natural topology on $\text{Spec}(R)$ for any ring R , making it a functor to **Top**

Proof:

The key is to recall (or re-prove) that if p is a prime ideal, then $\varphi^{-1}(p)$ is a prime ideal.

Next, when we defined the algebraic sets, they were the zero sets of some collection of functions. When the ring we worked over was $k[x]$, we already had a good notion of function that was readily available to us, letting us define a map from $k[x]$ to $\text{Hom}_{\text{Set}}(V, k)$ where:

$$f \mapsto (p \mapsto f(p))$$

However, an arbitrary element of a ring no longer has notion of being “evaluatable”. We thus must

come up with an analogous idea of “evaluating an element at a point” for an arbitrary ring. As a start, we already have that our domain of our function would be $\text{Spec}(R)$ instead of $\mathbb{A}^n$: Any algebraic set (i.e. closed set of $\mathbb{A}^n$) is the union algebraic varieties, and so we would want that any closed set of $\text{Spec}(R)$ (once we define a suitable topology) also be the union of prime ideals. But how would an element $x \in R$ then be “evaluated” at a “point” $\mathfrak{p} \in \text{Spec}(R)$?

We generalize this to the following: let’s first consider when $f \in k[x]$. If $\text{Spec}(k[x])$ is the spectrum of $k[x]$, then the *value* of f at $p \in \text{Spec}(R)$ is the element of k representing f in the quotient $K[V]/\mathcal{I}(p) \cong k$. Thus, the “value of f ” in $k[V]$ at $p \in V$ can be viewed as the element $\bar{f} \in k[V]/\mathcal{I}(p) \cong k$. Using this intuition, we can generalize the notion of evaluating a function:

$$R \rightarrow \text{Hom}_{\text{Set}} \left(\text{Spec}(R), \coprod_{\mathfrak{p} \in \text{Spec}(R)} R/\mathfrak{p} \right) \quad f \mapsto (\mathfrak{p} \rightarrow f \bmod \mathfrak{p})$$

Notice how this generalizes equation (1.1) seen at the beginning of section 1.1.

Definition 1.9.4: Value Of Element

Let $f \in R$. Then the “value” of f at the “point” $\mathfrak{p} \in \text{Spec}(R)$ is the element:

$$f(\mathfrak{p}) := \bar{f} \in R/\mathfrak{p}$$

Next, we see that we can define the maps $\mathcal{Z}$ and $\mathcal{I}$ as follows. For $\mathcal{Z}$, we get:

$$\mathcal{Z}(A) = \{\mathfrak{p} \in \text{Spec}(R) \mid A \subseteq \mathfrak{p}\} \subseteq \text{Spec}(R)$$

That is, $\mathcal{Z}(A)$ is the collection of all prime ideals containing A . Clearly, $\mathcal{Z}(A) = \mathcal{Z}(I)$ where $I = (A)$, so we only need to consider ideals. To see that this relates to the notion of a zero-locus as we’ve seen with polynomial rings, notice that if $p \in \mathcal{Z}(I)$, then $I \subseteq \mathfrak{p}$, so $f \in \mathfrak{p}$ for every $f \in I$. If we view $f \in R$ as a “function” on $\text{Spec}(R)$, we see that $\mathfrak{p} \in \mathcal{Z}(I)$ if and only if

$$f(\mathfrak{p}) = f \bmod \mathfrak{p} = 0 \in R/\mathfrak{p}$$

for all $f \in I$, showing that, in a sense, $\mathcal{Z}(I)$ consists of points in $\text{Spec}(R)$ at which all the “functions” of I are zero.

For the other direction, the analogous concept would be:

$$\mathcal{I}(Y) = \bigcap_{\mathfrak{p} \in Y} \mathfrak{p}$$

Proposition 1.9.5: Properties Of Correspondent Maps on Spectrum

Let R be a ring. Then:

1. For any ideal $I \subseteq R$, $\mathcal{Z}(I) = \mathcal{Z}(\text{rad}(I)) = \mathcal{Z}(\mathcal{I}(\mathcal{Z}(I)))$ and $\mathcal{I}(\mathcal{Z}(I)) = \text{rad}(I)$
2. For any ideal $I, J \subseteq R$ $\mathcal{Z}(I \cap J) = \mathcal{Z}(IJ) = \mathcal{Z}(I) \cup \mathcal{Z}(J)$
3. If $\{I_j\}$ is an arbitrary collection of prime ideals of R , then $\mathcal{Z}(\cup_j I_j) = \cap_j \mathcal{Z}(I_j)$

Proof:

Dummit p. 732

1. If P is a prime ideal containing I , then $\mathfrak{p}$ contains $\text{rad}(I)$, hence $\mathcal{Z}(I) = \mathcal{Z}(\text{rad}(I))$. Since $\text{rad}(I)$ is the intersection of all prime ideals containing I , the definition of $\mathcal{I}(I)$ gives $\mathcal{Z}(\text{rad}(I)) = \mathcal{Z}(\mathcal{I}(I))$. Similarly:

$$\mathcal{I}(\mathcal{Z}(I)) = \mathcal{I} \bigcap_{\mathfrak{p} \in \mathcal{Z}(I)} \mathfrak{p} = \bigcap_{I \subseteq \mathfrak{p}} \mathfrak{p} = \text{rad}(I)$$

2. essentially immediate
3. essentially immediate

This shows that we are back in the condition's for the Nullstellensatz: every $\mathcal{Z}(I)$ in $\text{Spec}(R)$ occurs as a radical ideal, which is unique since $\mathcal{I}(\mathcal{Z}(I)) = \text{rad}(I)$. Furthermore, the 2 other statements shows that we have also kept the same topology on $\text{Spec}(R)$ as we had for polynomial rings, in particular, we can define a topology on

$$\mathcal{T} = \{\mathcal{Z}(I) \mid I \text{ is an ideal of } R\}$$

Definition 1.9.6: Zariski Topology

Let R be a ring and $\text{Spec}(R)$ the corresponding spectrum. Then the collection of $\mathcal{Z}(I)$ defines the closed sets of the *Zariski topology* on $\text{Spec}(R)$.

Definition 1.9.7: Closed Points

Let $\mathfrak{m} \subseteq R$ be a maximal ideal of R and $\text{Spec}(R)$ be the spectrum of R . Then the maximal ideals of R in $\text{Spec}(R)$ are called the *closed points* of R .

It is easy to see that maximal ideals are closed, since the Zariski closure of them is the set of all prime ideals containing the maximal ideal, which since the ideal is maximal can only be itself, and hence is closed. Thus, the Zariski topology on $\text{Spec}(R)$ is still T_1 , and again rarely Hausdorff.

Definition 1.9.8: Basic Open Set

Let R be a ring and $f \in R$. Then $D(f)$ or X_f denotes the collection of all prime ideals in $\text{Spec}(R)$ that do not contain f , i.e. $D(f)$ is the set of points in $\text{Spec}(R)$ for which the value of $f \in R$ is nonzero.

I would suggest you take a moment to go back and see that the properties of the Zariski topology we have shown for $k[x]$ generalize well for the general Zariski topology on $\text{Spec}(R)$ (including irreducibility). One notable one to point out is the generalization of the Nullstellensatz:

Proposition 1.9.9: Nullstellensatz On Spectrum

Let R be a ring and $\text{Spec}(R)$ the spectrum with the Zariski topology. Then the maps $\mathcal{Z}$ and $\mathcal{I}$ define order-reversing bijections:

$$\left\{ \begin{array}{c} \text{Zariski closed subset of} \\ \text{Spec}(R) \end{array} \right\} \xrightleftharpoons[\mathcal{Z}]{\mathcal{I}} \{\text{Radical Ideals of } R\}$$

Proof:

Use proposition 1.9.5.

Example 1.9.10: Zariski Topology

1. on $R = \mathbb{Z}$
2. on $R = \mathbb{Z}[x]$
3. Let $R = k[V] = k[x_1, \dots, x_n]/I(V)$ be a coordinate ring. Then we have set up the generalization so that all the same properties work for $\text{Spec}(R)$.

The 3rd example given can be generalized:

Definition 1.9.11: Affine k -Algebra

Let R be an algebra over a field k . Then if R is finitely-generated, k is algebraically closed, R has no nilpotent elements, then it is called an *affine k -algebra*

The lack of nilpotence is important for the last part of the Nullstellensatz to correspond exactly. (Words on showing that affine k -algebra are exactly the ring arising as the ring of functions on affine algebraic sets over algebraically closed fields)

(the corresponding k -algebra homomorphism maps maximal ideals to maximal ideals. In general, this is not the case, which is why we have $\text{Spec}(R)$ have prime ideals. The following generalizes the notion of regular function)

Proposition 1.9.12: Spectrum Map

Let $\varphi : R \rightarrow S$ be a ring homomorphism. Then there is an induced map $\varphi^* : \text{Spec}(S) \rightarrow \text{Spec}(R)$ mapping

$$\varphi^*(P) = \varphi^{-1}(P)$$

This map is continuous with respect to the Zariski Topology

Proof:

exercise

(many examples)

Proposition 1.9.13: Properties Of Zariski Topology

Let $f \in R$, $X = \text{Spec}(R)$, and X_f be the principal open set. Then

1. $X_f = X$ if and only if f is a unit, and $X_f = \emptyset$ if and only if f is nilpotent
2. $X_f \cap X_g = X_{fg}$
3. $X_f \subseteq X_{g_1} \cup \dots \cup X_{g_n}$ if and only if $f \in \text{rad}(g_1, \dots, g_n)$. In particular, $X_f = X_g$ if and only if $\text{rad}(f) = \text{rad}(g)$
4. The principal open sets form a basis for the Zariski topology on X
5. The natural map from R to R_f induces a homeomorphism from $\text{Spec}(R_f)$ to X_f where R_f is the localization of R at f
6. X is Compact
7. If $\varphi : R \rightarrow S$ is a ring homomorphism, with corresponding $\varphi^* : \text{Spec}(S) \rightarrow \text{Spec}(R)$, then letting $Y = \text{Spec}(S)$, the pre-image of X_f is $Y_{\varphi(f)}$ in Y

Proof:

should be done

(now define localization for Zariski topology. Build up intuition)

Definition 1.9.14: Presheaf

Let $X = \text{Spec}(R)$ and $U \subseteq X$ be an open subset. If $U = \emptyset$, define $\mathcal{O}(U) = 0$. Otherwise, define $\mathcal{O}(U)$ to be the set of functions $s : U \rightarrow \coprod_{Q \in U} R_Q$ from U to the disjoint union of the localizations R_Q for $Q \in U$ with the following two properties:

1. $s(Q) \in R_Q$ for every $Q \in U$
2. For every $P \in U$, there exists an open neighbourhood $X_f \subseteq U$ containing P and an element a/f^n in the localization R_f defining s on X_f , i.e. $s(Q) = a/f^n \in R_Q$ for every $Q \in X_f$.

Notice that for $s, t \in \mathcal{O}(U)$, $s + t, st \in \mathcal{O}(U)$, showing that it is a ring.

Definition 1.9.15: Sheaf

Let R be a commutative ring, $X = \text{Spec}(R)$.

1. The collection of rings $\mathcal{O}(U)$ for the collection of Zariski open sets of X together with the restriction maps $\mathcal{O}(U) \rightarrow \mathcal{O}(U')$ for $U' \subseteq U$ is called the *[structure] sheaf* on X , and is denoted $\mathcal{O}$ (or $\mathcal{O}_X$)
2. The elements s of $\mathcal{O}(U)$ are called *sections* of $\mathcal{O}$ over U . The elements of $\mathcal{O}(X)$ are called *global sections* of $\mathcal{O}$

(a proposition on global section's of sheaf)

Definition 1.9.16: Stalk

Let $(\text{Spec}(R), \mathcal{O})$ be a structure sheaf on $\text{Spec}(R)$. Then a stalk at $p \in \text{Spec}(R)$ is

$$\mathcal{O}_p = \varinjlim F(\mathcal{O})$$

Definition 1.9.17: Affine Scheme

Let R be a commutative ring. Then the pair $(\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$ is called an *affine scheme*.

(define scheme morphism, and show every ring homomorphism induce scheme morphisms)

(a word on how a *scheme* is a locally ringed space in which each point is in a neighbourhood which is isomorphic to an affine scheme)

(Also, incorporate this: it will make the definition of a morphism seem more topological)

- map $\phi: W \rightarrow V$ that pre-composes a linear functional $W \rightarrow \mathbb{K}$ with ϕ to obtain a linear functional $V \xrightarrow{\phi} W \xrightarrow{\omega} \mathbb{K}$.
- (iii) The functor $\mathcal{O}: \mathbf{Top}^{\text{op}} \rightarrow \mathbf{Poset}$ that carries a space X to its poset $\mathcal{O}(X)$ of open subsets is contravariant on the category of spaces: a continuous map $f: X \rightarrow Y$ gives rise to a function $f^{-1}: \mathcal{O}(Y) \rightarrow \mathcal{O}(X)$ that carries an open subset $U \subset Y$ to its preimage $f^{-1}(U)$, which is open in X ; this is the definition of **continuity**. A similar functor $\mathcal{C}: \mathbf{Top}^{\text{op}} \rightarrow \mathbf{Poset}$ carries a space to its poset of closed subsets.
 - (iv) There is a contravariant functor $\text{Spec}: \mathbf{CRing}^{\text{op}} \rightarrow \mathbf{Top}$ that sends a commutative ring R to its set $\text{Spec}(R)$ of prime ideals given the **Zariski topology**. The closed subsets in the Zariski topology are those subsets $V_I \subset \text{Spec}(R)$ of prime ideals containing a fixed ideal $I \subset R$. This construction is contravariantly functorial: for any ring homomorphism $\phi: R \rightarrow S$ and prime ideal $\mathfrak{p} \subset S$, the inverse image $\phi^{-1}(\mathfrak{p}) \subset R$ defines a prime ideal of R , and the inverse image function $\phi^{-1}: \text{Spec}(S) \rightarrow \text{Spec}(R)$ is continuous with respect to the Zariski topology.
 - (v) For a generic small category $\mathbf{C}$, a functor $\mathbf{C}^{\text{op}} \rightarrow \mathbf{Set}$ is called a (set-valued) **presheaf** on $\mathbf{C}$. A typical example is the functor $\mathcal{O}(X)^{\text{op}} \rightarrow \mathbf{Set}$ whose domain is the poset $\mathcal{O}(X)$ of open subsets of a topological space X and whose value at $U \subset X$ is the set of continuous real-valued functions on U . The action on morphisms is by restriction. This presheaf is a *sheaf*, if it satisfies an axiom that is introduced in Definition 3.3.4.
 - (vi) Presheaves on the category Δ , of finite non-empty ordinals and order-preserving

(Given a topological space X , we can define an algebra by taking the set of all continuous functions $C(X)$ with the operations being pointwise. Now, given just the information $C(X)$, does there exist a X such that $C(X)$ is isomorphic to A ? In other words, can we recover X ? The answer is yes if X is a locally compact topological space! This is the algebra/topology duality. The same is true if X is a Riemann manifold, though you need more information, something called the *spectral triple*. Later, Grothendieck generalized this and we got to create from any algebra A a topological space $\text{Spec}(A)$!!! Given the appropriate topological properties on $\text{Spec}(A)$, we can recover A !! In other words, there is this way to go back and forth between algebra and geometry. This same exists in analysis too: the dual of $C_c(X)$ for a LCH X is intimately related to Radon measure, and the dual of $C_c(X)$ with the appropriate conditions let's us generalize the notion of derivatives to locally integrable functions!!!)